QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

bitbox02-firmware-en @ ae548ef4e40f · BitBox (Shift Crypto) — public repository (passive review)

Organization	QSI
Report date	2026-11-06 13:05 UTC
Report version	1
Audit run	686851a8-c120-482f-86ce-0890074cc565 · Document audit · 2026-11-06 13:05 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`cf3c23870e4b4c35feb21d303d5082fae1730b014e64646954327f37e2becf0c`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

79	1	11	58	3
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	1
Known vulnerable component	32
Confirmed configuration error	1
Potential cryptographic risk	3
Long-term post-quantum risk	40
Improvement recommendation	1
Insufficient data	1

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-17854	Critical	certifi 2022.12.7 is affected by GHSA-6q48-6c7w-9gw2 (PYSEC-2026-4320)	54.2
QSI-2026-17830	High	TLS certificate verification disabled in scripts/generate_version_headers.py	70.4
QSI-2026-17855	High	certifi 2022.12.7 is affected by GHSA-xqr8-7jwr-rhp7 (CVE-2023-37920, PYSEC-2023-135)	50.2
QSI-2026-17859	High	jinja2 3.1.2 is affected by GHSA-gmj6-6f8f-6699 (CVE-2024-56201, PYSEC-2026-1472)	50.2
QSI-2026-17862	High	jinja2 3.1.2 is affected by GHSA-q2x7-8rv6-6q7h (CVE-2024-56326, PYSEC-2026-1475)	50.2

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 15 confirmed, 5 not applicable, 0 dismissed as false positives.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/BitBoxSwiss/bitbox02-firmware @ ae548ef4e40f548a17542ba457f138db4fa0069d — evidence links open the lines at this commit.

Supplied inputs

2645 files, 52428220 bytes in total, 19 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	2556	external/CMSIS/Include/cmsis_armclang.h: 707 lines scanned with 16 rule patterns and 266 registry patterns; external/CMSIS/Include/cmsis_clang.h: 708 lines scanned with 16 rule patterns and 266 registry patterns; external/CMSIS/Include/cmsis_compiler.h: 292 lines scanned with 16 rule patterns and 266 registry patterns
chain_map	checked	3	messages/btc.proto: Bitcoin; messages/cardano.proto: Cardano; messages/eth.proto: Ethereum and EVM chains
dependencies	checked	48	py/pyproject.toml: 0 dependencies parsed (0 pinned); py/requirements.txt: 8 dependencies parsed (0 pinned); src/rust/Cargo.lock: 135 dependencies parsed (135 pinned)
certificates_and_keys	checked	23	releases/pubkeys/JuliusWiedemann.asc: certificates/keys parsed; releases/pubkeys/JunghoonLee.asc: certificates/keys parsed; releases/pubkeys/My1.asc: certificates/keys parsed
keys	Insufficient data	1	releases/pubkeys/psteiger.asc line 1: OpenPGP algorithm ID 18 is not in the algorithm registry
secrets	checked	9	src/rust/bitbox-secp256k1/src/lib.rs: value(s) at line(s) 375 (test_dleq), 407 (test_secp256k1_sign), 441 (test_secp256k1_sign_zero_host_nonce_differs_from_no_host_nonce_sign), 456 (test_secp256k1_sign_no_host_nonce_deterministic), 491 (test_secp256k1_nonce_commit) are declared inside a fixture and treated as test data, although the file itself is not test code; src/rust/bitbox-secp256k1/src/lib.rs: 5 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name); src/rust/bitbox02-rust/src/hww/api/change_password.rs: value(s) at line(s) 196 (test_process_confirm_password_mismatch) are declared inside a fixture and treated as test data, although the file itself is not test code
tls_code	Insufficient data	1	tools/atecc608/go.mod: go 1.19 — before Go 1.24 crypto/tls offers no X25519MLKEM768 by default; relevant if the module uses TLS

certificates	error (not checked)	1	external/cryptoauthlib/lib/atcacert/atcacert_pem.h line 37: certificate could not be parsed (ValueError)
tls_key_exchange_groups	Insufficient data	4	external/lvgl/docs/doxygen_xml.py line 349: not in the algorithm registry, not assessed: {}; external/lvgl/docs/src/details/main-modules/indev/groups.rst line 4: not in the algorithm registry, not assessed: =====; external/lvgl/scripts/generate_lv_conf.py line 67: not in the algorithm registry, not assessed: re.search(r'([A-Z0-9_]+)\s+(.+)'
tls_configuration	checked	4	external/lvgl/docs/doxygen_xml.py: openssl TLS directives parsed; external/lvgl/docs/src/details/main-modules/indev/groups.rst: openssl TLS directives parsed; external/lvgl/scripts/generate_lv_conf.py: openssl TLS directives parsed
entropy	checked	1	random-number generation: 2556 code files read with 39724 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)
osv_advisories	not checked	1	3 package(s) are built from git repositories (forks or unreleased code), so registry advisories were not looked up for them: bip32-ed25519, bip39, rtt-target
osv_advisories	checked	1	204 pinned packages queried at OSV.dev; 14 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 2632 (input files: C)

Not checked

- keys: releases/pubkeys/psteiger.asc line 1: OpenPGP algorithm ID 18 is not in the algorithm registry
- tls_code: tools/atecc608/go.mod: go 1.19 — before Go 1.24 crypto/tls offers no X25519MLKEM768 by default; relevant if the module uses TLS
- certificates: external/cryptoauthlib/lib/atcacert/atcacert_pem.h line 37: certificate could not be parsed (ValueError)
- tls_key_exchange_groups: external/lvgl/docs/doxygen_xml.py line 349: not in the algorithm registry, not assessed: {}
- tls_key_exchange_groups: external/lvgl/docs/src/details/main-modules/indev/groups.rst line 4: not in the algorithm registry, not assessed: =====
- tls_key_exchange_groups: external/lvgl/scripts/generate_lv_conf.py line 67: not in the algorithm registry, not assessed: re.search(r'([A-Z0-9_]+)\s+(.+)'
- tls_key_exchange_groups: external/lvgl/scripts/update_version.py line 28: not in the algorithm registry, not assessed: re.search(self.RE_PATTERN
- osv_advisories: 3 package(s) are built from git repositories (forks or unreleased code), so registry advisories were not looked up for them: bip32-ed25519, bip39, rtt-target

Architecture overview

Static review of the source tree bitbox02-firmware-en at ae548ef4e40f548a17542ba457f138db4fa0069d. 2645 files analysed; of 8952 files, 4270 are test, fuzzing, benchmark or example code and were left out, 1518 are not read by any analyzer, 1 exceed the per-file limit and 518 did not fit the total limit. Files no analyzer reads include: .ci/README.md, .ci/check-container-sources-modified, .ci/check-container-version-published, .ci/check-hashes, .ci/check-pep8. Because the total limit bound, this inventory covers 2645 of 8952 candidate files (29%) and is a sample of the tree rather than a reading of it; a different limit gives a different figure. Files it could not fit include: external/optiga-trust-m/external/mbedtls/include/mbedtls/compat-1.3.h, external/optiga-trust-m/external/mbedtls/include/mbedtls/config.h, external/optiga-trust-m/external/mbedtls/include/mbedtls/debug.h, external/optiga-trust-m/external/mbedtls/include/mbedtls/des.h, external/optiga-trust-m/external/mbedtls/include/mbedtls/dhm.h. 15717 files in build-output or dependency-cache directories (.git, .hg, .venv, __pycache__, build, dist, node_modules, target, vendor, venv) were not read. 1543 files come from directories a .gitmodules pins and are read as shipped source rather than skipped as a dependency cache: external/lvgl (1223), external/cryptoauthlib (240), external/optiga-trust-m (67), external/embedded-swd (12). No requests were sent to the owner's systems.

Overall security posture

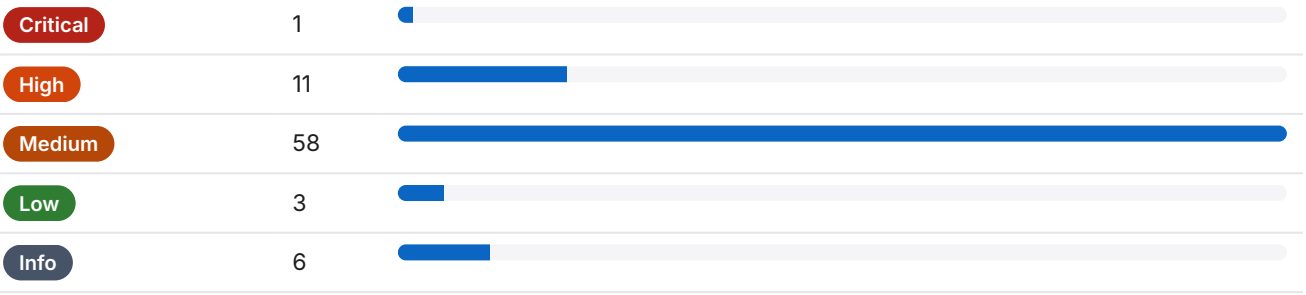
0
Indicative posture index (0–100)

55.9%
Quantum-vulnerable share of inventory

202
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-17854	certifi 2022.12.7 is affected by GHSA-6q48-6c7w-9gw2 (PYSEC-2026-4320)	Known vulnerable component	Critical	Medium
QSI-2026-17830	TLS certificate verification disabled in scripts/generate_version_headers.py	Confirmed configuration error	High	High
QSI-2026-17855	certifi 2022.12.7 is affected by GHSA-xqr8-7jwr-rhp7 (CVE-2023-37920, PYSEC-2023-135)	Known vulnerable component	High	Medium
QSI-2026-17859	jinja2 3.1.2 is affected by GHSA-gmj6-6f8f-6699 (CVE-2024-56201, PYSEC-2026-1472)	Known vulnerable component	High	Medium
QSI-2026-17862	jinja2 3.1.2 is affected by GHSA-q2x7-8rv6-6q7h (CVE-2024-56326, PYSEC-2026-1475)	Known vulnerable component	High	Medium
QSI-2026-17869	urllib3 1.26.15 is affected by GHSA-2xpw-w6gg-jr37 (CVE-2025-66471, PYSEC-2026-1994)	Known vulnerable component	High	Medium
QSI-2026-17871	urllib3 1.26.15 is affected by GHSA-38jv-5279-wg99 (CVE-2026-21441, PYSEC-2026-1996)	Known vulnerable component	High	Medium
QSI-2026-17872	urllib3 1.26.15 is affected by GHSA-8988-9cw3-xx77 (CVE-2026-97687, PYSEC-2026-4175)	Known vulnerable component	High	Medium
QSI-2026-17874	urllib3 1.26.15 is affected by GHSA-gm62-xv2j-4w53 (CVE-2025-66418, PYSEC-2026-1998)	Known vulnerable component	High	Medium
QSI-2026-17878	urllib3 1.26.15 is affected by GHSA-vxq7-64xx-v4gw (CVE-2026-97689, PYSEC-2026-4177)	Known vulnerable component	High	Medium
QSI-2026-17801	ECB block-cipher mode used in external/samd51a-ds/include/component/aes.h	Potential cryptographic risk	High	Medium, preliminary
QSI-2026-17843	ECB block-cipher mode used in external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h	Potential cryptographic risk	High	Medium, preliminary
QSI-2026-17844	Transaction signing for 3 chain(s) relies on elliptic-curve signatures	Long-term post-quantum risk	Medium	High

QSI-2026-17804	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/JuliusWiedemann.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17805	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/JunghoonLee.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17806	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/My1.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17807	Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/My1.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17808	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/TheCharlatan.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17809	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/beerosagos.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17810	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/benma.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17811	Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/brianddk.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17812	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/cstenglein.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17813	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/frknakk.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17814	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/guggerf.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17815	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/jonathancross.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17816	Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/jonathancross.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17817	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/nickez.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17818	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/offbyn.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17819	Quantum-vulnerable Ed25519 key (Ed25519) in releases/pubkeys/psteiger.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17820	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/stadicus.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17821	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/strnbrg.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17822	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/sutterseba.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17823	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/tomasvrba.asc	Long-term post-quantum risk	Medium	High

QSI-2026-17824	Quantum-vulnerable RSA key (3072-bit) in releases/pubkeys/userofreddit44.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17825	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/x1ddos.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17826	Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/x1ddos.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17827	Quantum-vulnerable RSA key (3072-bit) in releases/pubkeys/x1ddos.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17828	Quantum-vulnerable Ed25519 key (Ed25519) in releases/pubkeys/yaziza.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17829	Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/yaziza.asc	Long-term post-quantum risk	Medium	High
QSI-2026-17838	Quantum-vulnerable ECDSA key (secp256r1) in external/cryptoauthlib/app/pkcs11/README.md	Long-term post-quantum risk	Medium	High
QSI-2026-17856	idna 3.4 is affected by GHSA-65pc-fj4g-8rjx (CVE-2026-45409, PYSEC-2026-215)	Known vulnerable component	Medium	Medium
QSI-2026-17857	idna 3.4 is affected by GHSA-jjg7-2v4v-x38h (CVE-2024-3651, PYSEC-2024-60)	Known vulnerable component	Medium	Medium
QSI-2026-17858	jinja2 3.1.2 is affected by GHSA-cpwx-vrp4-4pq7 (CVE-2025-27516, PYSEC-2026-1471)	Known vulnerable component	Medium	Medium
QSI-2026-17860	jinja2 3.1.2 is affected by GHSA-h5c8-rqwp-cp95 (CVE-2024-22195, PYSEC-2026-1473)	Known vulnerable component	Medium	Medium
QSI-2026-17861	jinja2 3.1.2 is affected by GHSA-h75v-3vvj-5mfj (CVE-2024-34064, PYSEC-2026-1474)	Known vulnerable component	Medium	Medium
QSI-2026-17864	pygments 2.14.0 is affected by GHSA-mrwq-x4v8-fh7p (CVE-2022-40896, PYSEC-2023-117)	Known vulnerable component	Medium	Medium
QSI-2026-17865	requests 2.28.2 is affected by GHSA-9hjg-9r4m-mvj7 (CVE-2024-47081, PYSEC-2026-1872)	Known vulnerable component	Medium	Medium
QSI-2026-17866	requests 2.28.2 is affected by GHSA-9wx4-h78v-vm56 (CVE-2024-35195, PYSEC-2026-1873)	Known vulnerable component	Medium	Medium
QSI-2026-17867	requests 2.28.2 is affected by GHSA-gc5v-m9x4-r6x2 (CVE-2026-25645, PYSEC-2026-2275)	Known vulnerable component	Medium	Medium
QSI-2026-17868	requests 2.28.2 is affected by GHSA-j8r2-6x86-q33q (CVE-2023-32681, PYSEC-2023-74)	Known vulnerable component	Medium	Medium

QSI-2026-17870	urllib3 1.26.15 is affected by GHSA-34jh-p97f-mpxf (CVE-2024-37891, PYSEC-2026-1995)	Known vulnerable component	Medium	Medium
QSI-2026-17873	urllib3 1.26.15 is affected by GHSA-g4mx-q9vg-27p4 (CVE-2023-45803, PYSEC-2023-212)	Known vulnerable component	Medium	Medium
QSI-2026-17875	urllib3 1.26.15 is affected by GHSA-pq67-6m6q-mj2v (CVE-2025-50181, PYSEC-2026-1999)	Known vulnerable component	Medium	Medium
QSI-2026-17876	urllib3 1.26.15 is affected by GHSA-qccp-gfcp-xxvc (CVE-2026-44431, PYSEC-2026-141)	Known vulnerable component	Medium	Medium
QSI-2026-17877	urllib3 1.26.15 is affected by GHSA-v845-jxx5-vc9f (CVE-2023-43804, PYSEC-2023-192)	Known vulnerable component	Medium	Medium
QSI-2026-17879	zipp 3.15.0 is affected by GHSA-jfmj-5v4g-7637 (CVE-2024-5569, PYSEC-2026-2074)	Known vulnerable component	Medium	Medium
QSI-2026-17831	Quantum-vulnerable ECDSA: firmware and boot verification (root of trust) (3 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17833	Quantum-vulnerable X25519: key wrapping and data encryption at rest (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17840	Quantum-vulnerable ECDH: key wrapping and data encryption at rest (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17803	Quantum-vulnerable ECDSA: transaction and message signing (14 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17832	Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (5 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17835	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (2 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17836	Quantum-vulnerable Ed25519: transaction and message signing (4 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17802	Quantum-vulnerable ECDSA: role not determined (10 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17834	Quantum-vulnerable Ed25519: authentication and identity (certificates, tokens, device attestation) (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17837	Quantum-vulnerable Schnorr (BIP-340): role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17839	Quantum-vulnerable ECDH: role not determined (4 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17842	Quantum-vulnerable X25519: role not determined (6 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-17852	bytes 1.5.0 is affected by GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007)	Known vulnerable component	Medium	Low

QSI-2026-17841	Non-cryptographic RNG near security-sensitive identifiers in external/lvgl/docs/src/_static/jquery.js	Potential cryptographic risk	Medium	Low, preliminary
QSI-2026-17853	certifi 2022.12.7 is affected by GHSA-248v-346w-9cwc (CVE-2024-39689, PYSEC-2024-230)	Known vulnerable component	Low	Medium
QSI-2026-17863	pygments 2.14.0 is affected by GHSA-5239-wwwm-4pmq (CVE-2026-4539, PYSEC-2026-2987)	Known vulnerable component	Low	Medium
QSI-2026-17848	crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)	Confirmed technical vulnerability	Low	Low
QSI-2026-17849	keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)	Known vulnerable component	Info	Medium
QSI-2026-17850	time 0.3.44 is affected by GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009)	Known vulnerable component	Info	Medium
QSI-2026-17846	anyhow 1.0.31 is affected by RUSTSEC-2026-0190 (no severity rating)	Known vulnerable component	Info	Low
QSI-2026-17851	anyhow 1.0.75 is affected by RUSTSEC-2026-0190 (no severity rating)	Known vulnerable component	Info	Low
QSI-2026-17845	No post-quantum capable cryptographic library detected	Insufficient data	Info	Medium
QSI-2026-17847	bare-metal 0.2.5: informational advisory RUSTSEC-2026-0110 (unmaintained)	Improvement recommendation	Info	Low

Detailed findings

QSI-2026-17854 · DEP-KNOWN-ADVISORY

Critical

certifi 2022.12.7 is affected by GHSA-6q48-6c7w-9gw2 (PYSEC-2026-4320)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-6q48-6c7w-9gw2 (PYSEC-2026-4320) for PyPI package certifi at version 2022.12.7: certifi bundles emSign root CAs with compromised intermediate CA keys. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/certifi@2022.12.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	54.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-6q48-6c7w-9gw2
```

EVIDENCE

tool_output · OSV.dev GHSA-6q48-6c7w-9gw2 · line 13 · SHA-256 87c1ba23d8d1ad68...


```
{"id": "GHSA-6q48-6c7w-9gw2", "aliases": ["PYSEC-2026-4320"], "modified":
"2026-10-26T16:15:35.371128995Z", "summary": "certifi bundles emSign root CAs with compromised
intermediate CA keys", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:H/AT:P/PR:N/UI:N/
VC:H/VI:H/VA:N/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2026.10.18).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-6q48-6c7w-9gw2

QSI-2026-17830 · CODE-TLS-VERIFY-DISABLED

High

TLS certificate verification disabled in scripts/generate_version_headers.py

Confirmed configuration error | Confidence: High

1 occurrence(s) of disabled verification (verify=False), first at line 230.

Affected asset	file:scripts/generate_version_headers.py
Data source	scripts/generate_version_headers.py
Detection method	Regular-expression matching of language-specific API patterns.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	70.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	6.8 — CVSS:3.1/AV:A/AC:H/PR:N/UI:N/S:U/C:H/I:H/A:N

POTENTIAL IMPACT

Connections are open to man-in-the-middle interception.

REPRODUCTION STEPS

```
Open line 230 in the supplied file scripts/generate_version_headers.py and inspect the quoted lines
(see evidence).
```

EVIDENCE

file_excerpt · scripts/generate_version_headers.py · line 230 · SHA-256 14c945f1f3952da4... · File 675bbd250e4d75df...

```
  228 |
  229 | def strict_tag_version(repo_root: str | os.PathLike[str], prefix: str | None) -> str:
  230 |     return compute_tag_version(repo_root, prefix, check_gpg=True, verify=False,
check_semver=True)
  231 |
  232 |
```

LIMITATIONS OF THE CHECK

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Remove the override; use a proper CA bundle or pinning.

RETEST PROCEDURE

Re-scan the source and review the call sites listed in the evidence.

OWASP Application Security Verification Standard

QSI-2026-17855 · DEP-KNOWN-ADVISORY

High

certifi 2022.12.7 is affected by GHSA-xqr8-7jwr-rhp7 (CVE-2023-37920, PYSEC-2023-135)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-xqr8-7jwr-rhp7 (CVE-2023-37920, PYSEC-2023-135) for PyPI package certifi at version 2022.12.7: Removal of e-Tugra root certificate. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/certifi@2022.12.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC

Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:H/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-xqr8-7jwr-rhp7
```

EVIDENCE

tool_output · OSV.dev GHSA-xqr8-7jwr-rhp7 · line 13 · SHA-256 2f94715e09d8ea44...


```
{"id": "GHSA-xqr8-7jwr-rhp7", "aliases": ["CVE-2023-37920", "PYSEC-2023-135"], "modified":
"2026-09-10T03:50:08.653753752Z", "summary": "Removal of e-Tugra root certificate", "severity":
[{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:H/A:N"}],
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2023.7.22).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-xqr8-7jwr-rhp7

QSI-2026-17859 · DEP-KNOWN-ADVISORY

High

jinja2 3.1.2 is affected by GHSA-gmj6-6f8f-6699 (CVE-2024-56201, PYSEC-2026-1472)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-gmj6-6f8f-6699 (CVE-2024-56201, PYSEC-2026-1472) for PyPI package jinja2 at version 3.1.2: Jinja has a sandbox breakout through malicious filenames. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/jinja2@3.1.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	8.8 — CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-gmj6-6f8f-6699
```

EVIDENCE

tool_output · OSV.dev GHSA-gmj6-6f8f-6699 · line 28 · SHA-256 1fc2fcfcf66b7cde...


```
{"id": "GHSA-gmj6-6f8f-6699", "aliases": ["CVE-2024-56201", "PYSEC-2026-1472"], "modified":
"2026-09-10T03:50:56.152882717Z", "summary": "Jinja has a sandbox breakout through malicious
filenames", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/
A:H"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:P/PR:L/UI:P/VC:H/VI:H/VA:H/SC:N/SI:N/
SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.1.5).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-gmj6-6f8f-6699

QSI-2026-17862 · DEP-KNOWN-ADVISORY

High

jinja2 3.1.2 is affected by GHSA-q2x7-8rv6-6q7h (CVE-2024-56326, PYSEC-2026-1475)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-q2x7-8rv6-6q7h (CVE-2024-56326, PYSEC-2026-1475) for PyPI package jinja2 at version 3.1.2: Jinja has a sandbox breakout through indirect reference to format method. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/jinja2@3.1.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	7.8 — CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-q2x7-8rv6-6q7h
```

EVIDENCE

tool_output · OSV.dev GHSA-q2x7-8rv6-6q7h · line 28 · SHA-256 cf69831558e5cfcc...

{"id": "GHSA-q2x7-8rv6-6q7h", "aliases": ["CVE-2024-56326", "PYSEC-2026-1475"], "modified": "2026-09-10T03:50:21.662855250Z", "summary": "Jinja has a sandbox breakout through indirect reference to format method", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:P/PR:L/UI:P/VC:H/VI:H/VA:H/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.1.5).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-q2x7-8rv6-6q7h

QSI-2026-17869 · DEP-KNOWN-ADVISORY

High

urllib3 1.26.15 is affected by GHSA-2xpw-w6gg-jr37 (CVE-2025-66471, PYSEC-2026-1994)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-2xpw-w6gg-jr37 (CVE-2025-66471, PYSEC-2026-1994) for PyPI package urllib3 at version 1.26.15: urllib3 streaming API improperly handles highly compressed data. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-2xpw-w6gg-jr37
```

EVIDENCE

tool_output · OSV.dev GHSA-2xpw-w6gg-jr37 · line 60 · SHA-256 8f549172278134db...

{"id": "GHSA-2xpw-w6gg-jr37", "aliases": ["CVE-2025-66471", "PYSEC-2026-1994"], "modified": "2026-09-25T17:15:05.968189421Z", "summary": "urllib3 streaming API improperly handles highly compressed data", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": true}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.6.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-2xpw-w6gg-jr37

QSI-2026-17871 · DEP-KNOWN-ADVISORY

High

urllib3 1.26.15 is affected by GHSA-38jv-5279-wg99 (CVE-2026-21441, PYSEC-2026-1996)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-38jv-5279-wg99 (CVE-2026-21441, PYSEC-2026-1996) for PyPI package urllib3 at version 1.26.15: Decompression-bomb safeguards bypassed when following HTTP redirects (streaming API). The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-38jv-5279-wg99
```

EVIDENCE

tool_output · OSV.dev GHSA-38jv-5279-wg99 · line 60 · SHA-256 528407f765ddc336…


```
{"id": "GHSA-38jv-5279-wg99", "aliases": ["CVE-2026-21441", "PYSEC-2026-1996"], "modified":
"2026-09-10T03:50:32.562010895Z", "summary": "Decompression-bomb safeguards bypassed when following
HTTP redirects (streaming API)", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/
UI:N/S:U/C:N/I:N/A:H"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:N/VC:N/VI:N/
VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.6.3).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-38jv-5279-wg99

QSI-2026-17872 · DEP-KNOWN-ADVISORY

High

urllib3 1.26.15 is affected by GHSA-8988-9cw3-xx77 (CVE-2026-97687, PYSEC-2026-4175)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-8988-9cw3-xx77 (CVE-2026-97687, PYSEC-2026-4175) for PyPI package urllib3 at version 1.26.15: urllib3: HTTPS proxy TLS configuration may be ignored or overridden. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-8988-9cw3-xx77
```

EVIDENCE

tool_output · OSV.dev GHSA-8988-9cw3-xx77 · line 60 · SHA-256 c51c0d680d885d63...


```
{"id": "GHSA-8988-9cw3-xx77", "aliases": ["CVE-2026-97687", "PYSEC-2026-4175"], "modified":
"2026-10-01T17:55:53.793608550Z", "summary": "urllib3: HTTPS proxy TLS configuration may be ignored
or overridden", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:P/VC:H/
VI:H/VA:N/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.8.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-8988-9cw3-xx77

QSI-2026-17874 · DEP-KNOWN-ADVISORY

High

urllib3 1.26.15 is affected by GHSA-gm62-xv2j-4w53 (CVE-2025-66418, PYSEC-2026-1998)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-gm62-xv2j-4w53 (CVE-2025-66418, PYSEC-2026-1998) for PyPI package urllib3 at version 1.26.15: urllib3 allows an unbounded number of links in the decompression chain. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-gm62-xv2j-4w53
```

EVIDENCE

tool_output · OSV.dev GHSA-gm62-xv2j-4w53 · line 60 · SHA-256 b69c9a68201dc7d1...


```
{"id": "GHSA-gm62-xv2j-4w53", "aliases": ["CVE-2025-66418", "PYSEC-2026-1998"], "modified":
"2026-09-10T03:50:58.741847479Z", "summary": "urllib3 allows an unbounded number of links in the
decompression chain", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:N/
VC:N/VI:N/VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.6.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-gm62-xv2j-4w53

QSI-2026-17878 · DEP-KNOWN-ADVISORY

High

urllib3 1.26.15 is affected by GHSA-vxq7-64xx-v4gw (CVE-2026-97689, PYSEC-2026-4177)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-vxq7-64xx-v4gw (CVE-2026-97689, PYSEC-2026-4177) for PyPI package urllib3 at version 1.26.15: urllib3: HTTPResponse.stream()/read_chunked() buffers an unbounded chunk-size line into memory. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-vxq7-64xx-v4gw
```

EVIDENCE

tool_output · OSV.dev GHSA-vxq7-64xx-v4gw · line 60 · SHA-256 cbf2e0ae22b27747...


```
{"id": "GHSA-vxq7-64xx-v4gw", "aliases": ["CVE-2026-97689", "PYSEC-2026-4177"], "modified":
"2026-10-01T17:55:57.514829071Z", "summary": "urllib3: HTTPResponse.stream()/read_chunked() buffers
an unbounded chunk-size line into memory", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/
AC:L/AT:P/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.8.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-vxq7-64xx-v4gw

QSI-2026-17801 · CODE-ECB-MODE

High

ECB block-cipher mode used in external/samd51a-ds/include/component/aes.h

Potential cryptographic risk | Confidence: Medium | preliminary — requires further verification

1 usage(s) of ECB mode (first at line 81).

Affected asset	file:external/samd51a-ds/include/component/aes.h
Data source	external/samd51a-ds/include/component/aes.h
Detection method	Regular-expression matching of language-specific API patterns.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	38.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

ECB reveals patterns in encrypted data and provides no integrity.

REPRODUCTION STEPS

```
Open line 81 in the supplied file external/samd51a-ds/include/component/aes.h and inspect the quoted lines (see evidence).
```

EVIDENCE

file_excerpt · external/samd51a-ds/include/component/aes.h · line 81 · SHA-256 92ef17166ab022a2… · File aa117523ab12a7fa…

```
   79 | #define   AES_CTRLA_AESMODE_CCM_Val       _U_(0x5)   /**< \brief (AES_CTRLA) CCM mode */
   80 | #define   AES_CTRLA_AESMODE_GCM_Val       _U_(0x6)   /**< \brief (AES_CTRLA) Galois counter
mode */
   81 | #define AES_CTRLA_AESMODE_ECB       (AES_CTRLA_AESMODE_ECB_Val     << AES_CTRLA_AESMODE_Pos)
   82 | #define AES_CTRLA_AESMODE_CBC       (AES_CTRLA_AESMODE_CBC_Val     << AES_CTRLA_AESMODE_Pos)
   83 | #define AES_CTRLA_AESMODE_OFB       (AES_CTRLA_AESMODE_OFB_Val     << AES_CTRLA_AESMODE_Pos)
```

LIMITATIONS OF THE CHECK

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Use an AEAD mode (GCM) with unique nonces.

RETEST PROCEDURE

Re-scan the source and review the call sites listed in the evidence.

NIST SP 800-131A Rev. 2 — Transitioning the Use of Cryptographic Algorithms and Key Lengths

QSI-2026-17843 · CODE-ECB-MODE

High

ECB block-cipher mode used in external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h

Potential cryptographic risk | Confidence: Medium | preliminary — requires further verification

1 usage(s) of ECB mode (first at line 184).

Affected asset	file:external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h
Data source	external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h
Detection method	Regular-expression matching of language-specific API patterns.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	38.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

ECB reveals patterns in encrypted data and provides no integrity.

REPRODUCTION STEPS

```
Open line 184 in the supplied file external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h
and inspect the quoted lines (see evidence).
```

EVIDENCE

file_excerpt · external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h · line 184 · SHA-256 b4ca0728db5ccfb4... · File 58fcbbcc35638a6a...

```
182 | typedef enum {
183 |     MBEDTLS_MODE_NONE = 0,              /**< None.                       */
184 |     MBEDTLS_MODE_ECB,                   /**< The ECB cipher mode.        */
185 |     MBEDTLS_MODE_CBC,                   /**< The CBC cipher mode.        */
186 |     MBEDTLS_MODE_CFB,                   /**< The CFB cipher mode.        */
```

LIMITATIONS OF THE CHECK

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Use an AEAD mode (GCM) with unique nonces.

RETEST PROCEDURE

Re-scan the source and review the call sites listed in the evidence.

NIST SP 800-131A Rev. 2 — Transitioning the Use of Cryptographic Algorithms and Key Lengths

QSI-2026-17844 · CHAINMAP-QV

Medium

Transaction signing for 3 chain(s) relies on elliptic-curve signatures

Long-term post-quantum risk | Confidence: High

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** True by construction. messages/btc.proto, cardano.proto and eth.proto define the supported chains, whose signature schemes are set by consensus rather than by the wallet. The wallet cannot migrate them unilaterally, so this is a readiness map rather than a defect.

The wallet's message definitions cover Bitcoin, Cardano, Ethereum and EVM chains. These chains authorize transactions with ECDSA, Ed25519, Schnorr (BIP-340), which Shor's algorithm breaks on a cryptographically relevant quantum computer. A wallet cannot move a chain to post-quantum signatures on its own: the chain has to adopt a scheme first.

Affected asset	chain-map
Data source	messages/btc.proto
Detection method	Message definitions (.proto) matched to QSI's table of chain signature schemes.
Analyzer	chain_map 1.3.0 · 2026-11-06 13:05 UTC
Priority score	56.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Once a CRQC exists, an attacker who sees an account's public key could derive the private key and move funds; how soon keys are exposed depends on each chain (address reuse, account model).

REPRODUCTION STEPS

```
Open the .proto files named in the evidence.
```

EVIDENCE

file_excerpt · messages/btc.proto · line 4 · SHA-256 9397cf36c17bba81…

```
package shiftcrypto.bitbox02;
```

file_excerpt · messages/cardano.proto · line 4 · SHA-256 9397cf36c17bba81…

```
package shiftcrypto.bitbox02;
```

file_excerpt · messages/eth.proto · line 4 · SHA-256 9397cf36c17bba81…

```
package shiftcrypto.bitbox02;
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Chains are recognized from the names of the message definitions (.proto files and packages); the schemes are the chains' documented account signatures, not an analysis of the wallet's signing code.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them. Keep the signing layer crypto-agile per chain (algorithm identifiers in messages and storage, room for larger keys and signatures) and track each chain's post-quantum proposals, so that a new scheme can be added when a chain activates it.

RETEST PROCEDURE

Re-run when a chain activates a post-quantum signature scheme or the wallet adds one.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17804 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/JuliusWiedemann.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/JuliusWiedemann.asc#line 1
Data source	releases/pubkeys/JuliusWiedemann.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)

CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/JuliusWiedemann.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/JuliusWiedemann.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 4c174a88f6da9f8d…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/JuliusWiedemann.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 4c174a88f6da9f8d…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17805 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/JunghoonLee.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/JunghoonLee.asc#line 1
Data source	releases/pubkeys/JunghoonLee.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/JunghoonLee.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/JunghoonLee.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 33671dc21911a175…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/JunghoonLee.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 33671dc21911a175…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17806 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/My1.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/My1.asc#line 1
Data source	releases/pubkeys/My1.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/My1.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/My1.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File c3884ccc2c92a15a…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17807 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/My1.asc

Long-term post-quantum risk | Confidence: High

A RSA 2048-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/My1.asc#line 1
Data source	releases/pubkeys/My1.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/My1.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/My1.asc · line 1 · SHA-256 a744425426415d58… · File c3884ccc2c92a15a…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

key · releases/pubkeys/My1.asc · line 1 · SHA-256 a744425426415d58… · File c3884ccc2c92a15a…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17808 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/TheCharlatan.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/TheCharlatan.asc#line 1
Data source	releases/pubkeys/TheCharlatan.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/TheCharlatan.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/TheCharlatan.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File c229c79d3677332d…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/TheCharlatan.asc · line 1 · SHA-256 484f2172c4dd53b2… · File c229c79d3677332d…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/TheCharlatan.asc · line 1 · SHA-256 484f2172c4dd53b2… · File c229c79d3677332d…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/TheCharlatan.asc · line 1 · SHA-256 484f2172c4dd53b2... · File c229c79d3677332d...

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17809 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/beerosagos.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/beerosagos.asc#line 1
Data source	releases/pubkeys/beerosagos.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/beerosagos.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/beerosagos.asc · line 1 · SHA-256 18fb5f0b9bd19412... · File 77547ad2b0bd39e5...

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/beerosagos.asc · line 1 · SHA-256 484f2172c4dd53b2... · File 77547ad2b0bd39e5...

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17810 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/benma.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/benma.asc#line 1
Data source	releases/pubkeys/benma.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/benma.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/benma.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File bebf447de75bcc50…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/benma.asc · line 1 · SHA-256 484f2172c4dd53b2… · File bebf447de75bcc50…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/benma.asc · line 1 · SHA-256 484f2172c4dd53b2… · File bebf447de75bcc50…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17811 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/brianddk.asc

Long-term post-quantum risk | Confidence: High

A RSA 2048-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/brianddk.asc#line 1
Data source	releases/pubkeys/brianddk.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/brianddk.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/brianddk.asc · line 1 · SHA-256 99f8a76d7955bc1c... · File fa82b4b5c2c4dd6c...

```
OpenPGP primary key: algorithm ID 1 (RSA) 2048-bit
```

key · releases/pubkeys/brianddk.asc · line 1 · SHA-256 a744425426415d58... · File fa82b4b5c2c4dd6c...

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

key · releases/pubkeys/brianddk.asc · line 1 · SHA-256 a744425426415d58... · File fa82b4b5c2c4dd6c...

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17812 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/cstenglein.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/cstenglein.asc#line 1
Data source	releases/pubkeys/cstenglein.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/cstenglein.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/cstenglein.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 1697a4111b88a52d…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/cstenglein.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 1697a4111b88a52d…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17813 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/frknakk.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/frknakk.asc#line 1
Data source	releases/pubkeys/frknakk.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/frknakk.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/frknakk.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File a3d9760761bedbb8…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/frknakk.asc · line 1 · SHA-256 484f2172c4dd53b2… · File a3d9760761bedbb8…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17814 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/guggerf.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/guggerf.asc#line 1
Data source	releases/pubkeys/guggerf.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/guggerf.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/guggerf.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 1b1ae8bf1e853163…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/guggerf.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 1b1ae8bf1e853163…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17815 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/jonathancross.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/jonathancross.asc#line 1
Data source	releases/pubkeys/jonathancross.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/jonathancross.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/jonathancross.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 4092962b5065adad…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17816 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/jonathancross.asc

Long-term post-quantum risk | Confidence: High

A RSA 2048-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/jonathancross.asc#line 1
Data source	releases/pubkeys/jonathancross.asc
Detection method	Public key parsing.

Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/jonathancross.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/jonathancross.asc · line 1 · SHA-256 a744425426415d58… · File 4092962b5065adad…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

key · releases/pubkeys/jonathancross.asc · line 1 · SHA-256 a744425426415d58… · File 4092962b5065adad…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

key · releases/pubkeys/jonathancross.asc · line 1 · SHA-256 a744425426415d58… · File 4092962b5065adad…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17817 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/nickez.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/nickez.asc#line 1
Data source	releases/pubkeys/nickez.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)

CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/nickez.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File dfeee318e1e971fa…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 484f2172c4dd53b2… · File dfeee318e1e971fa…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 484f2172c4dd53b2… · File dfeee318e1e971fa…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 484f2172c4dd53b2… · File dfeee318e1e971fa…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 484f2172c4dd53b2… · File dfeee318e1e971fa…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 484f2172c4dd53b2… · File dfeee318e1e971fa…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/nickez.asc · line 1 · SHA-256 484f2172c4dd53b2… · File dfeee318e1e971fa…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17818 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/offbyn.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/offbyn.asc#line 1
Data source	releases/pubkeys/offbyn.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/offbyn.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/offbyn.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File b6136c3272711fcd…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/offbyn.asc · line 1 · SHA-256 484f2172c4dd53b2… · File b6136c3272711fcd…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/offbyn.asc · line 1 · SHA-256 484f2172c4dd53b2… · File b6136c3272711fcd…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/offbyn.asc · line 1 · SHA-256 484f2172c4dd53b2… · File b6136c3272711fcd…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17819 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable Ed25519 key (Ed25519) in releases/pubkeys/psteiger.asc

Long-term post-quantum risk | Confidence: High

A Ed25519 Ed25519 public key was found (line 1).

Affected asset	file:releases/pubkeys/psteiger.asc#line 1
Data source	releases/pubkeys/psteiger.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/psteiger.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/psteiger.asc · line 1 · SHA-256 6dbf836a10cb5714… · File fe123e4a583a9dc4…

```
OpenPGP primary key: algorithm ID 22 (Ed25519) Ed25519
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17820 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/stadicus.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/stadicus.asc#line 1
Data source	releases/pubkeys/stadicus.asc
Detection method	Public key parsing.

Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/stadicus.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/stadicus.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 6b7a866cccd4028f…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/stadicus.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 6b7a866cccd4028f…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/stadicus.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 6b7a866cccd4028f…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/stadicus.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 6b7a866cccd4028f…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17821 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/strnbrg.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/strnbrg.asc#line 1
Data source	releases/pubkeys/strnbrg.asc

Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/strnbrg.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/strnbrg.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 3c725e0b54589026…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/strnbrg.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 3c725e0b54589026…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17822 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/sutterseba.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/sutterseba.asc#line 1
Data source	releases/pubkeys/sutterseba.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/sutterseba.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/sutterseba.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 5c279fb6619893e6…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/sutterseba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 5c279fb6619893e6…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17823 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/tomasvrba.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/tomasvrba.asc#line 1
Data source	releases/pubkeys/tomasvrba.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/tomasvrba.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File 88b2e37b7a8cccf7…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 88b2e37b7a8cccf7…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 88b2e37b7a8cccf7…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 88b2e37b7a8cccf7…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 88b2e37b7a8cccf7…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 88b2e37b7a8cccf7…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/tomasvrba.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 88b2e37b7a8cccf7…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17824 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (3072-bit) in releases/pubkeys/userofreddit44.asc

Long-term post-quantum risk | Confidence: High

A RSA 3072-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/userofreddit44.asc#line 1

Data source	releases/pubkeys/userofreddit44.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/userofreddit44.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/userofreddit44.asc · line 1 · SHA-256 315636b2f7e00661… · File e67ca6a20f1bee44…

```
OpenPGP primary key: algorithm ID 1 (RSA) 3072-bit
```

key · releases/pubkeys/userofreddit44.asc · line 1 · SHA-256 6ccbdce52e43463e… · File e67ca6a20f1bee44…

```
OpenPGP subkey: algorithm ID 1 (RSA) 3072-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17825 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/x1ddos.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/x1ddos.asc#line 1
Data source	releases/pubkeys/x1ddos.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/x1ddos.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/x1ddos.asc · line 1 · SHA-256 18fb5f0b9bd19412… · File c6cd70e5656c75ac…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · releases/pubkeys/x1ddos.asc · line 1 · SHA-256 484f2172c4dd53b2… · File c6cd70e5656c75ac…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17826 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (2048-bit) in releases/pubkeys/x1ddos.asc

Long-term post-quantum risk | Confidence: High

A RSA 2048-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/x1ddos.asc#line 1
Data source	releases/pubkeys/x1ddos.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/x1ddos.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/x1ddos.asc · line 1 · SHA-256 a744425426415d58… · File c6cd70e5656c75ac…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

key · releases/pubkeys/x1ddos.asc · line 1 · SHA-256 a744425426415d58… · File c6cd70e5656c75ac…

```
OpenPGP subkey: algorithm ID 1 (RSA) 2048-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17827 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (3072-bit) in releases/pubkeys/x1ddos.asc

Long-term post-quantum risk | Confidence: High

A RSA 3072-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/x1ddos.asc#line 1
Data source	releases/pubkeys/x1ddos.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/x1ddos.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/x1ddos.asc · line 1 · SHA-256 6ccbdce52e43463e... · File c6cd70e5656c75ac...

```
OpenPGP subkey: algorithm ID 1 (RSA) 3072-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17828 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable Ed25519 key (Ed25519) in releases/pubkeys/yaziza.asc

Long-term post-quantum risk | Confidence: High

A Ed25519 Ed25519 public key was found (line 1).

Affected asset	file:releases/pubkeys/yaziza.asc#line 1
Data source	releases/pubkeys/yaziza.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/yaziza.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/yaziza.asc · line 1 · SHA-256 6dbf836a10cb5714... · File 2108a376fad98106...

```
OpenPGP primary key: algorithm ID 22 (Ed25519) Ed25519
```

key · releases/pubkeys/yaziza.asc · line 1 · SHA-256 065d4edee7f1048b... · File 2108a376fad98106...

```
OpenPGP subkey: algorithm ID 22 (Ed25519) Ed25519
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17829 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in releases/pubkeys/yaziza.asc

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 1).

Affected asset	file:releases/pubkeys/yaziza.asc#line 1
Data source	releases/pubkeys/yaziza.asc
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 1 in the supplied file releases/pubkeys/yaziza.asc and inspect the quoted lines (see evidence).
```

EVIDENCE

key · releases/pubkeys/yaziza.asc · line 1 · SHA-256 484f2172c4dd53b2… · File 2108a376fad98106…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17838 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable ECDSA key (secp256r1) in external/cryptoauthlib/app/pkcs11/README.md

Long-term post-quantum risk | Confidence: High

A ECDSA secp256r1 public key was found (line 280).

Affected asset	file:external/cryptoauthlib/app/pkcs11/README.md#line 280
Data source	external/cryptoauthlib/app/pkcs11/README.md
Detection method	Public key parsing.
Analyzer	certificates 1.3.0 · 2026-11-06 13:05 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 280 in the supplied file external/cryptoauthlib/app/pkcs11/README.md and inspect the quoted lines (see evidence).
```

EVIDENCE

key · external/cryptoauthlib/app/pkcs11/README.md · line 280 · SHA-256 87a3decbb47c307b... · File 336cffbc3ef5787f...

```
ECDSA public key, secp256r1
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-17856 · DEP-KNOWN-ADVISORY

Medium

idna 3.4 is affected by GHSA-65pc-fj4g-8rjx (CVE-2026-45409, PYSEC-2026-215)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-65pc-fj4g-8rjx (CVE-2026-45409, PYSEC-2026-215) for PyPI package idna at version 3.4: Internationalized Domain Names in Applications (IDNA): Specially crafted inputs to idna.encode() can bypass CVE-2024-3651 fix. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/idna@3.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-65pc-fj4g-8rjx
```

EVIDENCE

tool_output · OSV.dev GHSA-65pc-fj4g-8rjx · line 22 · SHA-256 1b180b3ab9431745…


```
{"id": "GHSA-65pc-fj4g-8rjx", "aliases": ["CVE-2026-45409", "PYSEC-2026-215"], "modified":
"2026-09-10T03:50:45.700124422Z", "summary": "Internationalized Domain Names in Applications (IDNA):
Specially crafted inputs to idna.encode() can bypass CVE-2024-3651 fix", "severity": [{"type":
"CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L"}, {"type": "CVSS_V4", "score":
"CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:L/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.15).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-65pc-fj4g-8rjx

QSI-2026-17857 · DEP-KNOWN-ADVISORY

Medium

idna 3.4 is affected by GHSA-jjg7-2v4v-x38h (CVE-2024-3651, PYSEC-2024-60)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-jjg7-2v4v-x38h (CVE-2024-3651, PYSEC-2024-60) for PyPI package idna at version 3.4: Internationalized Domain Names in Applications (IDNA) vulnerable to denial of service from specially crafted inputs to idna.encode. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/idna@3.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	6.2 — CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-jjg7-2v4v-x38h
```

EVIDENCE

tool_output · OSV.dev GHSA-jjg7-2v4v-x38h · line 22 · SHA-256 b260da6fe71df346...


```
{"id": "GHSA-jjg7-2v4v-x38h", "aliases": ["CVE-2024-3651", "PYSEC-2024-60"], "modified":
"2026-09-10T03:50:12.774088619Z", "summary": "Internationalized Domain Names in Applications (IDNA)
vulnerable to denial of service from specially crafted inputs to idna.encode", "severity": [{"type":
"CVSS_V3", "score": "CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H"}, {"type": "CVSS_V4", "score":
"CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.7).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-jjg7-2v4v-x38h

QSI-2026-17858 · DEP-KNOWN-ADVISORY

Medium

jinja2 3.1.2 is affected by GHSA-cpwx-vrp4-4pq7 (CVE-2025-27516, PYSEC-2026-1471)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-cpwx-vrp4-4pq7 (CVE-2025-27516, PYSEC-2026-1471) for PyPI package jinja2 at version 3.1.2: Jinja2 vulnerable to sandbox breakout through attr filter selecting format method. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/jinja2@3.1.2

Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-cpwx-vrp4-4pq7
```

EVIDENCE

tool_output · OSV.dev GHSA-cpwx-vrp4-4pq7 · line 28 · SHA-256 e28b14906d20d0da...


```
{"id": "GHSA-cpwx-vrp4-4pq7", "aliases": ["CVE-2025-27516", "PYSEC-2026-1471"], "modified":
"2026-09-10T03:49:48.526758681Z", "summary": "Jinja2 vulnerable to sandbox breakout through attr
filter selecting format method", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:P/
PR:L/UI:P/VC:H/VI:H/VA:H/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.1.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-cpwx-vrp4-4pq7

QSI-2026-17860 · DEP-KNOWN-ADVISORY

Medium

jinja2 3.1.2 is affected by GHSA-h5c8-rqwp-cp95 (CVE-2024-22195, PYSEC-2026-1473)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-h5c8-rqwp-cp95 (CVE-2024-22195, PYSEC-2026-1473) for PyPI package jinja2 at version 3.1.2: Jinja vulnerable to HTML attribute injection when passing user input as keys to xmlattr filter. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/jinja2@3.1.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.4 — CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-h5c8-rqwp-cp95
```

EVIDENCE

tool_output · OSV.dev GHSA-h5c8-rqwp-cp95 · line 28 · SHA-256 3bcb93126b62e63f...


```
{"id": "GHSA-h5c8-rqwp-cp95", "aliases": ["CVE-2024-22195", "PYSEC-2026-1473"], "modified": "2026-09-10T03:50:00.236017440Z", "summary": "Jinja vulnerable to HTML attribute injection when passing user input as keys to xmlattr filter", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.1.3).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-h5c8-rqwp-cp95

QSI-2026-17861 · DEP-KNOWN-ADVISORY

Medium

jinja2 3.1.2 is affected by GHSA-h75v-3vvj-5mfj (CVE-2024-34064, PYSEC-2026-1474)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-h75v-3vvj-5mfj (CVE-2024-34064, PYSEC-2026-1474) for PyPI package jinja2 at version 3.1.2: Jinja vulnerable to HTML attribute injection when passing user input as keys to xmlattr filter. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/jinja2@3.1.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.4 — CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-h75v-3vvj-5mfj
```

EVIDENCE

tool_output · OSV.dev GHSA-h75v-3vvj-5mfj · line 28 · SHA-256 8d178561724d57b3...


```
{"id": "GHSA-h75v-3vvj-5mfj", "aliases": ["CVE-2024-34064", "PYSEC-2026-1474"], "modified":
"2026-09-10T03:50:13.786450101Z", "summary": "Jinja vulnerable to HTML attribute injection when
passing user input as keys to xmlattr filter", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/
AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:L/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.1.4).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-h75v-3vvj-5mfj

QSI-2026-17864 · DEP-KNOWN-ADVISORY

Medium

pygments 2.14.0 is affected by GHSA-mrwq-x4v8-fh7p (CVE-2022-40896, PYSEC-2023-117)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-mrwq-x4v8-fh7p (CVE-2022-40896, PYSEC-2023-117) for PyPI package pygments at version 2.14.0: Pygments vulnerable to ReDoS. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/pygments@2.14.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.5 — CVSS:3.1/AV:L/AC:L/PR:N/UI:R/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-mrwq-x4v8-fh7p
```

EVIDENCE

tool_output · OSV.dev GHSA-mrwq-x4v8-fh7p · line 34 · SHA-256 6da07c6d923c9925...


```
{"id": "GHSA-mrwq-x4v8-fh7p", "aliases": ["CVE-2022-40896", "PYSEC-2023-117"], "modified":
"2024-10-24T22:27:47.389299Z", "summary": "Pygments vulnerable to ReDoS", "severity": [{"type":
```

```
"CVSS_V3", "score": "CVSS:3.1/AV:L/AC:L/PR:N/UI:R/S:U/C:N/I:N/A:H"}, {"type": "CVSS_V4", "score":
"CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:P/VC:N/VI:N/VA:H/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.15.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-mrwq-x4v8-fh7p

QSI-2026-17865 · DEP-KNOWN-ADVISORY

Medium

requests 2.28.2 is affected by GHSA-9hjg-9r4m-mvj7 (CVE-2024-47081, PYSEC-2026-1872)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-9hjg-9r4m-mvj7 (CVE-2024-47081, PYSEC-2026-1872) for PyPI package requests at version 2.28.2: Requests vulnerable to .netrc credentials leak via malicious URLs. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/requests@2.28.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-9hjg-9r4m-mvj7
```

EVIDENCE

tool_output · OSV.dev GHSA-9hjg-9r4m-mvj7 · line 36 · SHA-256 54566624d8668e64…


```
{"id": "GHSA-9hjg-9r4m-mvj7", "aliases": ["CVE-2024-47081", "PYSEC-2026-1872"], "modified":
"2026-10-19T20:00:04.716114530Z", "summary": "Requests vulnerable to .netrc credentials leak via
malicious URLs", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:H/
I:N/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default

against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.32.4).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-9hjg-9r4m-mvj7

QSI-2026-17866 · DEP-KNOWN-ADVISORY

Medium

requests 2.28.2 is affected by GHSA-9wx4-h78v-vm56 (CVE-2024-35195, PYSEC-2026-1873)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-9wx4-h78v-vm56 (CVE-2024-35195, PYSEC-2026-1873) for PyPI package requests at version 2.28.2: Requests `Session` object does not verify requests after making first request with verify=False. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/requests@2.28.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.6 — CVSS:3.1/AV:L/AC:H/PR:H/UI:R/S:U/C:H/I:H/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-9wx4-h78v-vm56
```

EVIDENCE

tool_output · OSV.dev GHSA-9wx4-h78v-vm56 · line 36 · SHA-256 38975a9d04ed33a4...


```
{"id": "GHSA-9wx4-h78v-vm56", "aliases": ["CVE-2024-35195", "PYSEC-2026-1873"], "modified":
"2026-09-10T03:50:13.740879755Z", "summary": "Requests `Session` object does not verify requests
after making first request with verify=False", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/
AV:L/AC:H/PR:H/UI:R/S:U/C:H/I:H/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.32.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-9wx4-h78v-vm56

QSI-2026-17867 · DEP-KNOWN-ADVISORY

Medium

requests 2.28.2 is affected by GHSA-gc5v-m9x4-r6x2 (CVE-2026-25645, PYSEC-2026-2275)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-gc5v-m9x4-r6x2 (CVE-2026-25645, PYSEC-2026-2275) for PyPI package requests at version 2.28.2: Requests has Insecure Temp File Reuse in its extract_zipped_paths() utility function. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/requests@2.28.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	4.4 — CVSS:3.1/AV:L/AC:H/PR:L/UI:R/S:U/C:N/I:H/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-gc5v-m9x4-r6x2
```

EVIDENCE

tool_output · OSV.dev GHSA-gc5v-m9x4-r6x2 · line 36 · SHA-256 f74704bb5c636714...


```
{"id": "GHSA-gc5v-m9x4-r6x2", "aliases": ["CVE-2026-25645", "PYSEC-2026-2275"], "modified":
"2026-09-10T03:50:39.207922076Z", "summary": "Requests has Insecure Temp File Reuse in its
extract_zipped_paths() utility function", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:L/
AC:H/PR:L/UI:R/S:U/C:N/I:H/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.33.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-gc5v-m9x4-r6x2

QSI-2026-17868 · DEP-KNOWN-ADVISORY

Medium

requests 2.28.2 is affected by GHSA-j8r2-6x86-q33q (CVE-2023-32681, PYSEC-2023-74)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-j8r2-6x86-q33q (CVE-2023-32681, PYSEC-2023-74) for PyPI package requests at version 2.28.2: Unintended leak of Proxy-Authorization header in requests. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/requests@2.28.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	6.1 — CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:C/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-j8r2-6x86-q33q
```

EVIDENCE

tool_output · OSV.dev GHSA-j8r2-6x86-q33q · line 36 · SHA-256 913766f1fff73012…


```
{"id": "GHSA-j8r2-6x86-q33q", "aliases": ["CVE-2023-32681", "PYSEC-2023-74"], "modified": "2026-09-10T03:50:01.756149727Z", "summary": "Unintended leak of Proxy-Authorization header in requests", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:C/C:H/I:N/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.31.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-j8r2-6x86-q33q

QSI-2026-17870 · DEP-KNOWN-ADVISORY

Medium

urllib3 1.26.15 is affected by GHSA-34jh-p97f-mpxf (CVE-2024-37891, PYSEC-2026-1995)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-34jh-p97f-mpxf (CVE-2024-37891, PYSEC-2026-1995) for PyPI package urllib3 at version 1.26.15: urllib3's Proxy-Authorization request header isn't stripped during cross-origin redirects. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	4.4 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-34jh-p97f-mpxf
```

EVIDENCE

tool_output · OSV.dev GHSA-34jh-p97f-mpxf · line 60 · SHA-256 70e2b3135eba6c41…


```
{"id": "GHSA-34jh-p97f-mpxf", "aliases": ["CVE-2024-37891", "PYSEC-2026-1995"], "modified": "2026-09-10T03:50:14.811193644Z", "summary": "urllib3's Proxy-Authorization request header isn't stripped during cross-origin redirects", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:H/I:N/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (1.26.19, 2.2.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-34jh-p97f-mpxf

QSI-2026-17873 · DEP-KNOWN-ADVISORY

Medium

urllib3 1.26.15 is affected by GHSA-g4mx-q9vg-27p4 (CVE-2023-45803, PYSEC-2023-212)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-g4mx-q9vg-27p4 (CVE-2023-45803, PYSEC-2023-212) for PyPI package urllib3 at version 1.26.15: urllib3's request body not stripped after redirect from 303 status changes request method to GET. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev

Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	4.2 — CVSS:3.1/AV:A/AC:H/PR:H/UI:N/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-g4mx-q9vg-27p4
```

EVIDENCE

tool_output · OSV.dev GHSA-g4mx-q9vg-27p4 · line 60 · SHA-256 9068d64596de74c3...


```
{"id": "GHSA-g4mx-q9vg-27p4", "aliases": ["CVE-2023-45803", "PYSEC-2023-212"], "modified":
"2026-09-10T03:50:03.706889410Z", "summary": "urllib3's request body not stripped after redirect
from 303 status changes request method to GET", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/
AV:A/AC:H/PR:H/UI:N/S:U/C:H/I:N/A:N"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:A/AC:L/AT:P/PR:H/
UI:N/VC:H/VI:N/VA:N/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (1.26.18, 2.0.7).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-g4mx-q9vg-27p4

QSI-2026-17875 · DEP-KNOWN-ADVISORY

Medium

urllib3 1.26.15 is affected by GHSA-pq67-6m6q-mj2v (CVE-2025-50181, PYSEC-2026-1999)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-pq67-6m6q-mj2v (CVE-2025-50181, PYSEC-2026-1999) for PyPI package urllib3 at version 1.26.15: urllib3 redirects are not disabled when retries are disabled on PoolManager instantiation. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:H/PR:L/UI:N/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-pq67-6m6q-mj2v
```

EVIDENCE

tool_output · OSV.dev GHSA-pq67-6m6q-mj2v · line 60 · SHA-256 788a2cf91f9eaf45…


```
{"id": "GHSA-pq67-6m6q-mj2v", "aliases": ["CVE-2025-50181", "PYSEC-2026-1999"], "modified": "2026-09-10T03:50:25.299291456Z", "summary": "urllib3 redirects are not disabled when retries are disabled on PoolManager instantiation", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:L/UI:N/S:U/C:H/I:N/A:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.5.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-pq67-6m6q-mj2v

QSI-2026-17876 · DEP-KNOWN-ADVISORY

Medium

urllib3 1.26.15 is affected by GHSA-qccp-gfcp-xxvc (CVE-2026-44431, PYSEC-2026-141)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-qccp-gfcp-xxvc (CVE-2026-44431, PYSEC-2026-141) for PyPI package urllib3 at version 1.26.15: urllib3: Sensitive headers forwarded across origins in proxied low-level redirects. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-qccp-gfcp-xxvc
```

EVIDENCE

tool_output · OSV.dev GHSA-qccp-gfcp-xxvc · line 60 · SHA-256 c7033c35f2861e3d...


```
{"id": "GHSA-qccp-gfcp-xxvc", "aliases": ["CVE-2026-44431", "PYSEC-2026-141"], "modified":
"2026-09-10T03:50:47.272765640Z", "summary": "urllib3: Sensitive headers forwarded across origins in
proxied low-level redirects", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/
UI:N/S:U/C:L/I:N/A:N"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:H/AT:P/PR:N/UI:N/VC:H/VI:N/
VA:N/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.7.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-qccp-gfcp-xxvc

QSI-2026-17877 · DEP-KNOWN-ADVISORY

Medium

urllib3 1.26.15 is affected by GHSA-v845-jxx5-vc9f (CVE-2023-43804, PYSEC-2023-192)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-v845-jxx5-vc9f (CVE-2023-43804, PYSEC-2023-192) for PyPI package urllib3 at version 1.26.15: `Cookie` HTTP header isn't stripped on cross-origin redirects. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/urllib3@1.26.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.9 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:H/I:H/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-v845-jxx5-vc9f
```

EVIDENCE

tool_output · OSV.dev GHSA-v845-jxx5-vc9f · line 60 · SHA-256 4436cbacc6451b22...


```
{"id": "GHSA-v845-jxx5-vc9f", "aliases": ["CVE-2023-43804", "PYSEC-2023-192"], "modified":
"2026-09-10T03:49:24.753610811Z", "summary": "`Cookie` HTTP header isn't stripped on cross-origin
redirects", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:H/I:H/
```

A:N"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:H/UI:N/VC:H/VI:H/VA:N/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (1.26.17, 2.0.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-v845-jxx5-vc9f

QSI-2026-17879 · DEP-KNOWN-ADVISORY

Medium

zipp 3.15.0 is affected by GHSA-jfmj-5v4g-7637 (CVE-2024-5569, PYSEC-2026-2074)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-jfmj-5v4g-7637 (CVE-2024-5569, PYSEC-2026-2074) for PyPI package zipp at version 3.15.0: zipp Denial of Service vulnerability. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/zipp@3.15.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	6.2 — CVSS:3.0/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-jfmj-5v4g-7637
```

EVIDENCE

tool_output · OSV.dev GHSA-jfmj-5v4g-7637 · line 62 · SHA-256 27a0d57ee49a9e56…

{"id": "GHSA-jfmj-5v4g-7637", "aliases": ["CVE-2024-5569", "PYSEC-2026-2074"], "modified": "2026-09-10T03:49:18.747455573Z", "summary": "zipp Denial of Service vulnerability", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.0/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:N"}], "declared_in_manifest": true}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default

against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.19.1).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-jfmj-5v4g-7637

QSI-2026-17831 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: firmware and boot verification (root of trust) (3 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

7 reference(s) to ECDSA across 3 file(s), e.g. src/bootloader/bootloader.c:537. Role, judged from paths, function names and nearby code: firmware and boot verification (root of trust). These public keys are fixed in the bootloader or the device: with a cryptographically relevant quantum computer (CRQC) an attacker forges a firmware signature, and replacing the keys needs a bootloader update in the field.

Affected asset	source-code:ECDSA:root_of_trust
Data source	src/bootloader/bootloader.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	43.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/bootloader/bootloader.c:537 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/bootloader/bootloader.c · line 537 · SHA-256 a692ec93e05c715e... · File f168d61e94931ac4...

```
  536 |         // First is sanity check
  537 |         if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, random, sizeof(random), curve_p256) !=
0) {
  538 |             if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, hash, sizeof(hash), curve_p256) ==
0) {
```

file_excerpt · src/bootloader/bootloader.c · line 538 · SHA-256 c8910947b0750cbb... · File f168d61e94931ac4...

```
  537 |         if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, random, sizeof(random), curve_p256) !=
0) {
  538 |             if (pukcc_ecdsa_verify(_root_pubkeys[i], sig, hash, sizeof(hash), curve_p256) ==
0) {
  539 |                 valid++;
```

file_excerpt · src/bootloader/bootloader.c · line 609 · SHA-256 110ab61f5a0020b3... · File f168d61e94931ac4...

```
  608 |         // First is sanity check
  609 |         if (pukcc_ecdsa_verify(pubkey, sig, random, sizeof(random), curve_p256) != 0) {
  610 |             if (pukcc_ecdsa_verify(pubkey, sig, hash, sizeof(hash), curve_p256) == 0) {
```

file_excerpt · src/bootloader/stage0/stage1_sigcheck.c · line 165 · SHA-256 40465ea406a87913... · File 5704d66ba1ca763b...

```
  164 |         const size_t signature_offset = (size_t)key_idx * BB02_STAGE1_SIGNATURE_LEN;
  165 |         const uint8_t valid = _is_zero_u8(pukcc_ecdsa_verify(
  166 |             pubkeys[key_idx],
```

file_excerpt · src/rust/bitbox02-rust-c/src/firmware_c_api.rs · line 23 · SHA-256 0239908a79adffb8... · File 029d99f0ae93aafe...

```
   22 |     let mut hal = crate::HalImpl::new();
   23 |     util::bb02_async::block_on(bitbox02_rust::secp256k1::init(&mut hal))
   24 |         .expect("secp256k1 context initialization failed");
```

file_excerpt · src/rust/bitbox02-rust-c/src/firmware_c_api.rs · line 24 · SHA-256 d2c149057a83a51f... · File 029d99f0ae93aafe...

```
   23 |     util::bb02_async::block_on(bitbox02_rust::secp256k1::init(&mut hal))
   24 |         .expect("secp256k1 context initialization failed");
   25 |     bitbox02_rust::main_loop::main_loop(&mut hal)
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Plan a hybrid firmware signature (the classical one plus ML-DSA or SLH-DSA, both required) and the path to ship the new keys in a bootloader update.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17833 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: key wrapping and data encryption at rest (1 file(s))

Long-term post-quantum risk | Confidence: Medium

Analyst review (QSI review, 2026-10-28): confirmed by the analyst. The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

4 reference(s) to X25519 across 1 file(s), e.g. src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:15. Role, judged from paths, function names and nearby code: key wrapping and data encryption at rest. Keys or data encrypted to these public keys are stored: anyone who copies them now decrypts them once a CRQC exists (harvest now, decrypt later).

Affected asset	source-code:X25519:key_wrapping
Data source	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	43.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:15 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 15 · SHA-256 1c153d0b1adf9cd8… · File e15d530066a76934…

```
14 | use sha2::{Digest, Sha256};
15 | use x25519_dalek::{PublicKey as X25519PublicKey, StaticSecret as X25519StaticSecret};
16 | use zeroize::{Zeroize, Zeroizing};
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 19 · SHA-256 4b67afd4d285deef… · File e15d530066a76934…

```
18 | const AUTH_SEED_LABEL: &[u8] = b"bitboxsync-auth-ed25519-seed-v1";
19 | const WRAP_SEED_LABEL: &[u8] = b"bitboxsync-wrap-x25519-seed-v1";
20 |
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 60 · SHA-256 f8ee14a435cd94f2… · File e15d530066a76934…

```
59 |     let wrap_secret = [REDACTED]
60 |     let wrap_public_key = X25519PublicKey::from(&wrap_secret).to_bytes();
61 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its own assessment.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17840 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDH: key wrapping and data encryption at rest (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

23 reference(s) to ECDH across 1 file(s), e.g. external/cryptoauthlib/python/cryptoauthlib/atcab.py:974. Role, judged from paths, function names and nearby code: key wrapping and data encryption at rest. Keys or data encrypted to these public keys are stored: anyone who copies them now decrypts them once a CRQC exists (harvest now, decrypt later).

Affected asset	source-code:ECDH:key_wrapping
Data source	external/cryptoauthlib/python/cryptoauthlib/atcab.py
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	43.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open external/cryptoauthlib/python/cryptoauthlib/atcab.py:974 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · external/cryptoauthlib/python/cryptoauthlib/atcab.py · line 974 · SHA-256 87004ff616be2223… · File 0b92b5429170956e…

```
973 |     """
974 |     Base function for generating premaster secret key using ECDH.
975 |
```

file_excerpt · external/cryptoauthlib/python/cryptoauthlib/atcab.py · line 977 · SHA-256 8e5f1c528fbe7b2e… · File 0b92b5429170956e…

```
976 |     Args:
977 |         mode                Mode to be used for ECDH computation (int)
978 |         key_id              Slot of key for ECDH computation (int)
```

file_excerpt · external/cryptoauthlib/python/cryptoauthlib/atcab.py · line 978 · SHA-256 204aab89e32c1093… · File 0b92b5429170956e…

```
977 |         mode                Mode to be used for ECDH computation (int)
978 |         key_id              Slot of key for ECDH computation (int)
979 |         public_key          Public key input to ECDH calculation. X and Y
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its own assessment.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17803 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (14 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file.

> ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

50 reference(s) to ECDSA across 14 file(s), e.g. py/bitbox02/bitbox02/bitbox02/secp256k1.py:3. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	py/bitbox02/bitbox02/bitbox02/secp256k1.py
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open py/bitbox02/bitbox02/bitbox02/secp256k1.py:3 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · py/bitbox02/bitbox02/bitbox02/secp256k1.py · line 3 · SHA-256 f1b7aa46e3d0b50c… · File 6903d0adce82573c…

```
  2 |
  3 | """secp256k1 util functions"""
  4 | import hashlib
```

file_excerpt · src/rust/bitbox-secp256k1/build.rs · line 6 · SHA-256 c0e8a8076dd2e70e… · File 91983851f534bb1e…

```
  5 | fn main() {
  6 |     let secp_dir = PathBuf::from("depend/secp256k1-zkp");
  7 |     let callbacks = PathBuf::from("src/default_callbacks.c");
```

file_excerpt · src/rust/bitbox-secp256k1/build.rs · line 18 · SHA-256 25e9545e63d24ed0… · File 91983851f534bb1e…

```
 17 |     build
 18 |         .file(secp_dir.join("src/secp256k1.c"))
 19 |         .file(secp_dir.join("src/precomputed_ecmult.c"))
```

file_excerpt · src/rust/bitbox-secp256k1/build.rs · line 37 · SHA-256 47d7abfe9efe1b2b… · File 91983851f534bb1e…

```
36 |
37 |     build.compile("secp256k1");
38 | }
```

file_excerpt · src/rust/bitbox-secp256k1/src/lib.rs · line 7 · SHA-256 27351363967fb644... · File 42fc499883f91dea...

```
 6 |
 7 | pub use bitcoin::secp256k1::constants::PUBLIC_KEY_SIZE;
 8 | use bitcoin::secp256k1::ffi::CPtr;
```

file_excerpt · src/rust/bitbox-secp256k1/src/lib.rs · line 8 · SHA-256 1ae312353045cae5... · File 42fc499883f91dea...

```
 7 | pub use bitcoin::secp256k1::constants::PUBLIC_KEY_SIZE;
 8 | use bitcoin::secp256k1::ffi::CPtr;
 9 | use bitcoin::secp256k1::{All, Secp256k1};
```

file_excerpt · src/rust/bitbox-secp256k1/src/lib.rs · line 9 · SHA-256 63fd03f40121775b... · File 42fc499883f91dea...

```
 8 | use bitcoin::secp256k1::ffi::CPtr;
 9 | use bitcoin::secp256k1::{All, Secp256k1};
10 |
```

file_excerpt · src/rust/bitbox02-rust-c/src/secp256k1.rs · line 3 · SHA-256 32c3110f30785a45... · File 54c1ff4e1f9ed19a...

```
 2 |
 3 | use bitcoin::secp256k1::{
 4 |     Message, PublicKey,
```

file_excerpt · src/rust/bitbox02-rust-c/src/secp256k1.rs · line 42 · SHA-256 26f131e03c434e4b... · File 54c1ff4e1f9ed19a...

```
41 |     let Ok(signature) =
42 |         bitcoin::secp256k1::ecdsa::Signature::from_compact(signature_compact.as_ref())
43 |     else {
```

file_excerpt · src/rust/bitbox02-rust-c/src/secp256k1.rs · line 56 · SHA-256 bf55765f36e34101... · File 54c1ff4e1f9ed19a...

```
55 |     unsafe {
56 |         ffi::secp256k1_ecdsa_verify(
57 |             ffi::secp256k1_context_no_precomp,
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17832 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (5 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

11 reference(s) to X25519 across 5 file(s), e.g. src/rust/bitbox-noise/src/lib.rs:16. Role, judged from paths, function names and nearby code: channel key exchange (TLS, Noise, handshakes). Traffic protected by this key exchange can be recorded now and decrypted later; a hybrid exchange keeps the classical guarantees and adds post-quantum ones.

Affected asset	source-code:X25519:channel
Data source	src/rust/bitbox-noise/src/lib.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox-noise/src/lib.rs:16 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox-noise/src/lib.rs · line 16 · SHA-256 bec1cc899b2ee576… · File 6787c6b484f801c7…

```
   15 | pub mod testing;
   16 | mod x25519;
   17 |
```

file_excerpt · src/rust/bitbox-noise/src/lib.rs · line 19 · SHA-256 57cd8f7157563d1d… · File 6787c6b484f801c7…

```
18 | pub use noise_xx::{Error, HandshakeHash, HandshakeResult, State};
19 | pub use x25519::{X25519, genkey};
20 |
```

file_excerpt · src/rust/bitbox-noise/src/noise_xx.rs · line 8 · SHA-256 784affd9139fdfbf… · File 4ade64136c560f52…

```
 7 |
 8 | use crate::x25519::{PrivateKey, PublicKey, X25519, genkey};
 9 | use noise_rust_crypto::{ChaCha20Poly1305, Sha256, sensitive::Sensitive};
```

file_excerpt · src/rust/bitbox-noise/src/x25519.rs · line 49 · SHA-256 43d70ab747c41c47… · File 63617ea3d8acc0f0…

```
48 |     fn pubkey(k: &Self::Key) -> Self::Pubkey {
49 |         let static_secret = [REDACTED]
50 |         *x25519_dalek::PublicKey::from(&static_secret).as_bytes()
```

file_excerpt · src/rust/bitbox-noise/src/x25519.rs · line 54 · SHA-256 fae4ff917820e021… · File 63617ea3d8acc0f0…

```
53 |     fn dh(k: &Self::Key, pk: &Self::Pubkey) -> Result<Self::Output, ()> {
54 |         let k = x25519_dalek::StaticSecret::from(*k.deref());
55 |         let pk = x25519_dalek::PublicKey::from(*pk);
```

file_excerpt · src/rust/bitbox-noise/src/x25519.rs · line 55 · SHA-256 9cb9b67a0b6ccdef… · File 63617ea3d8acc0f0…

```
54 |         let k = x25519_dalek::StaticSecret::from(*k.deref());
55 |         let pk = x25519_dalek::PublicKey::from(*pk);
56 |         Ok(*k.diffie_hellman(&pk).as_bytes())
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/everest.h · line 25 · SHA-256 805839a03b00bfe9… · File ddc6c96ec9598fe1…

```
24 |
25 | #include "everest/x25519.h"
26 |
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c · line 35 · SHA-256 9ca103a64c6b7d9f… · File b4dc100b030f5568…

```
34 |
35 | #include "x25519.h"
36 |
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c · line 81 · SHA-256 fbaaec05ef452c03… · File b4dc100b030f5568…

```
80 |     base[0] = 9;
81 |     Hacl_Curve25519_crypto_scalarmult( buf, ctx->our_secret, base );
82 |
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c · line 134 · SHA-256 21b4c7046cce778f… · File b4dc100b030f5568…

```
133 |
134 |     Hacl_Curve25519_crypto_scalarmult( buf, ctx->our_secret, ctx->peer_point);
135 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols).

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17835 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (2 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

3 reference(s) to Schnorr (BIP-340) across 2 file(s), e.g. src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs:289. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Schnorr (BIP-340):transaction_signing
Data source	src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs:289 and inspect the quoted lines; the
evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs · line 289 · SHA-256 0ff082b56713f8bc… · File b8a6f3c01008c063…

```
  288 |                 let tweaked = bitcoin::key::TweakedPublicKey::dangerous_assume_tweaked(
  289 |                     bitcoin::XOnlyPublicKey::from_slice(payload).map_err(|_|
Error::Generic)?,
  290 |                 );
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 12 · SHA-256 08a764387fbe8ba1… · File 4e33062aad0ad482…

```
   11 | use bitcoin::hashes::Hash;
   12 | use bitcoin::secp256k1::{self, PublicKey, Scalar, Secp256k1, SecretKey, XOnlyPublicKey};
   13 |
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 22 · SHA-256 fd1fd79d198d0e81… · File 4e33062aad0ad482…

```
   21 |     /// Generated silent payment output public key.
   22 |     pub pubkey: XOnlyPublicKey,
   23 |     /// DLEQ proof that the host can use to verify that the pubkey has been created
correctly.
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17836 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: transaction and message signing (4 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/

> {lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

6 reference(s) to Ed25519 across 4 file(s), e.g. src/rust/bitbox02-rust/src/hww/api/cardano/address.rs:308. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Ed25519:transaction_signing
Data source	src/rust/bitbox02-rust/src/hww/api/cardano/address.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/hww/api/cardano/address.rs:308 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/hww/api/cardano/address.rs · line 308 · SHA-256 23f10be088fb5fde… · File 2edeeab72109e5d9…

```
307 | ) -> Result<[u8; ADDRESS_HASH_SIZE], ()> {
308 |     let xpub = crate::keystore::ed25519::get_xpub_twice(hal, keypath).await?;
309 |     let pubkey_bytes = xpub.pubkey_bytes();
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction.rs · line 55 · SHA-256 e86a0885d901ed74… · File 5e94e09ed5e9e28b…

```
54 | ) -> Result<ShelleyWitness, ()> {
55 |     let result = ed25519::sign(hal, keypath, tx_body_hash).await?;
56 |     Ok(ShelleyWitness {
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/cardano/xpubs.rs · line 29 · SHA-256 49b540bc531f7b21… · File ca845c4de879d93b…

```
28 |
29 |         let xpub = crate::keystore::ed25519::get_xpub_twice(hal, keypath).await?;
30 |         let mut xpub_bytes = Vec::with_capacity(64);
```

file_excerpt · src/rust/bitbox02-rust/src/keystore/ed25519.rs · line 6 · SHA-256 eaa112ccabe881af… · File 85cb066d8f6e599c…

```
 5 | use crate::hash::Sha512;
 6 | use bip32_ed25519::{ED25519_EXPANDED_SECRET_KEY_SIZE, Xprv, Xpub};
 7 |
```

file_excerpt · src/rust/bitbox02-rust/src/keystore/ed25519.rs · line 10 · SHA-256 0d1f71033a48c3c6... · File 85cb066d8f6e599c...

```
 9 | use digest::Digest;
10 | use ed25519_dalek::VerifyingKey;
11 | use ed25519_dalek::hazmat::{ExpandedSecretKey, raw_sign};
```

file_excerpt · src/rust/bitbox02-rust/src/keystore/ed25519.rs · line 11 · SHA-256 449e9e250d19db94... · File 85cb066d8f6e599c...

```
10 | use ed25519_dalek::VerifyingKey;
11 | use ed25519_dalek::hazmat::{ExpandedSecretKey, raw_sign};
12 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17802 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (10 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Role resolved for all six files, and they are three different roles rather than one. src/pukcc/pukcc.c:181 is pukcc_ecdsa_verify, the ATSAM PUKCC hardware public-key controller performing ECDSA *verification* — the bootloader's signature check, so a root-of-trust item, and verification uses public keys only, which puts it outside any timing concern. src/rust/streaming-silent-payments/{lib,hash}.rs is BIP-352 silent payments, i.e. address derivation on secp256k1. src/rust/util/src/p256.rs is the U2F/FIDO P-256 signing described in the crypto-bigint decision above. Each has a different migration deadline, which is why leaving them merged as "role not determined" would be the wrong input to a plan.

20 reference(s) to ECDSA across 10 file(s), e.g. py/bitbox02/bitbox02/bitbox02/bitbox02.py:21. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general

Data source	py/bitbox02/bitbox02/bitbox02/bitbox02.py
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open py/bitbox02/bitbox02/bitbox02/bitbox02.py:21 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · py/bitbox02/bitbox02/bitbox02/bitbox02.py · line 21 · SHA-256 c1697b58eea8e082… · File fd62ecc298b45f06…

```
 20 |
 21 | from .secp256k1 import (
 22 |     antiklepto_host_commit,
```

file_excerpt · src/pukcc/pukcc.c · line 181 · SHA-256 53085ff2be44540f… · File b5049dd725362a42…

```
180 | // Returns 0 on success
181 | uint8_t pukcc_ecdsa_verify(
182 |     const uint8_t* public_key,
```

file_excerpt · src/pukcc/pukcc.h · line 30 · SHA-256 dae3c783558d1997… · File a7650ddfd0bfe8a9…

```
 29 |
 30 | uint8_t pukcc_ecdsa_verify(
 31 |     const uint8_t* public_key,
```

file_excerpt · src/rust/bitbox02-rust-c/src/lib.rs · line 23 · SHA-256 e527a51a0f0f62d0… · File c872df632ac63a1f…

```
 22 | #[cfg(feature = "factory-setup")]
 23 | mod secp256k1;
 24 |
```

file_excerpt · src/rust/streaming-silent-payments/src/hash.rs · line 7 · SHA-256 33201f813f216ace… · File ec49885fa0902faf…

```
  6 | use bitcoin::hashes::{Hash, HashEngine, sha256t_hash_newtype};
  7 | use bitcoin::secp256k1::{PublicKey, Scalar};
  8 |
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 12 · SHA-256 08a764387fbe8ba1… · File 4e33062aad0ad482…

```
 11 | use bitcoin::hashes::Hash;
 12 | use bitcoin::secp256k1::{self, PublicKey, Scalar, Secp256k1, SecretKey, XOnlyPublicKey};
 13 |
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 43 · SHA-256 08787745f17bf569… · File 4e33062aad0ad482…

```
 42 | pub struct SilentPayment<'a> {
 43 |     secp: &'a Secp256k1<secp256k1::All>,
 44 |     network: Network,
```

file_excerpt · src/rust/streaming-silent-payments/src/lib.rs · line 124 · SHA-256 14bc8680433716b8… · File 4e33062aad0ad482…

```
123 | fn create_dleq_proof(
124 |     secp: &Secp256k1<secp256k1::All>,
125 |     a_sum: &SecretKey,
```

file_excerpt · src/rust/util/src/p256.rs · line 28 · SHA-256 925484476bdd2348… · File 7662a6309089b842…

```
 27 | ) {
 28 |     use p256::ecdsa::{Signature, SigningKey, signature::hazmat::PrehashSigner};
 29 |     let signing_key = SigningKey::from_slice(private_key.as_ref()).unwrap();
```

file_excerpt · external/cryptoauthlib/lib/mbedtls/atca_mbedtls_ecdsa.c · line 51 · SHA-256 f4cdd327029f645a… · File bc7452fc887862b6…

```
 50 |  */
 51 | int mbedtls_ecdsa_sign(mbedtls_ecp_group *grp, mbedtls_mpi *r, mbedtls_mpi *s,
 52 |                        const mbedtls_mpi *d, const unsigned char *buf, size_t blen,
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17834 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: authentication and identity (certificates, tokens, device attestation) (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later,

> answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

4 reference(s) to Ed25519 across 1 file(s), e.g. src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11. Role, judged from paths, function names and nearby code: authentication and identity (certificates, tokens, device attestation). Short-lived authentication stays safe until a CRQC exists; long-lived identities (device attestation, CA and SSH keys) need a plan earlier.

Affected asset	source-code:Ed25519:authentication
Data source	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 11 · SHA-256 3637a75aa4245184… · File e15d530066a76934…

```
  10 | use chacha20poly1305::aead::{AeadInPlace, KeyInit};
  11 | use ed25519_dalek::VerifyingKey;
  12 | use hkdf::Hkdf;
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 56 · SHA-256 211b70e7af67bb49… · File e15d530066a76934…

```
  55 |     let auth_public_key =
  56 |         VerifyingKey::from(&ed25519::expanded_secret_key_from_seed(&auth_seed)).to_bytes();
  57 |
```

file_excerpt · src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs · line 410 · SHA-256 272c2c902cbf96d8… · File e15d530066a76934…

```
 409 | fn sign_payload(keys: &IdentityKeys, payload: &[u8]) -> [u8; SIGNATURE_LEN] {
 410 |     let expanded_secret_key = [REDACTED]
 411 |     ed25519::sign_with_expanded_secret_key(&expanded_secret_key, payload).signature
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17837 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

1 reference(s) to Schnorr (BIP-340) across 1 file(s), e.g. src/rust/bitbox02-rust/src/keystore.rs:897. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Schnorr (BIP-340):general
Data source	src/rust/bitbox02-rust/src/keystore.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open src/rust/bitbox02-rust/src/keystore.rs:897 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · src/rust/bitbox02-rust/src/keystore.rs · line 897 · SHA-256 4c865a127c8bb4a8… · File b7918fdd4b52ef1b…

```
896 |         .map_err(|_| ())?;
897 |     let sig = SECP256K1.sign_schnorr_with_aux_rand(
898 |         &bitcoin::secp256k1::Message::from_digest(*msg),
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17839 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDH: role not determined (4 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

7 reference(s) to ECDH across 4 file(s), e.g. external/cryptoauthlib/lib/calib/calib_command.h:181. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDH:general
Data source	external/cryptoauthlib/lib/calib/calib_command.h
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open external/cryptoauthlib/lib/calib/calib_command.h:181 and inspect the quoted lines; the evidence
lists further files.
```

EVIDENCE

file_excerpt · external/cryptoauthlib/lib/calib/calib_command.h · line 181 · SHA-256 9b4d335853d5e714… · File daa823f84a4a6a32…

```
  180 | #define ATCA_WRITE        ((uint8_t)0x12)  //!< Write command op-code
  181 | #define ATCA_ECDH         ((uint8_t)0x43)  //!< ECDH command op-code
  182 | #define ATCA_COUNTER      ((uint8_t)0x24)  //!< Counter command op-code
```

file_excerpt · external/cryptoauthlib/lib/calib/calib_command.h · line 357 · SHA-256 55dbcbc6be030b16… · File daa823f84a4a6a32…

```
  356 | #define ECDH_MODE_COPY_OUTPUT_BUFFER        ((uint8_t)0x0C)
  357 | #define ECDH_KEY_SIZE                       ATCA_BLOCK_SIZE     //!< ECDH output data size
  358 | #define ECDH_RSP_SIZE                       ATCA_RSP_SIZE_64    //!< ECDH command packet
size
```

file_excerpt · external/cryptoauthlib/lib/calib/calib_command.h · line 358 · SHA-256 0b55dd8a591437d1… · File daa823f84a4a6a32…

```
  357 | #define ECDH_KEY_SIZE                       ATCA_BLOCK_SIZE     //!< ECDH output data size
  358 | #define ECDH_RSP_SIZE                       ATCA_RSP_SIZE_64    //!< ECDH command packet
size
  359 | /** @} */
```

file_excerpt · external/cryptoauthlib/lib/calib/calib_ecdh.c · line 159 · SHA-256 cc79ff1975ef75f3… · File c6471c07c0685ef5…

```
  158 |         {
  159 |             ATCA_TRACE(status, "ECDH Failed"); break;
  160 |         }
```

file_excerpt · external/cryptoauthlib/lib/pkcs11/pkcs11_key.c · line 953 · SHA-256 c41a3072038479a7… · File 90f5684d266788eb…

```
  952 |             /* Because of the number of ECDH options this function unfortunately has a
complex bit of logic
  953 |                to walk through to select the proper ECDH command. Normally this would be
left up to the user
  954 |                to chose */
```

file_excerpt · external/cryptoauthlib/python/setup.py · line 224 · SHA-256 2ebf64b8e13175e1… · File b79e2c2b9b52dce0…

```
  223 |         download_url=_DOWNLOAD_URL,
  224 |         keywords='Microchip ATECC508A ATECC608 ECDSA ECDH',
  225 |         project_urls=_PROJECT_URLS,
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17842 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable X25519: role not determined (6 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** The remaining rows carry roles the engine determined and review agrees with, each checked against the cited file. ECDSA at src/bootloader/bootloader.c:537 and stage0/stage1_sigcheck.c is firmware and boot verification, with the public keys fixed in the bootloader: the hardest item to migrate, because it needs new hardware or a signed transition path. X25519 at bitbox-noise/{lib,noise_xx,x25519}.rs is the Noise channel to the host — recordable now, decryptable later, answered by a hybrid exchange. X25519 at bitboxsync.rs is key wrapping at rest. ECDSA across bitbox-secp256k1 and bitbox02-rust (bip32.rs, bip39.rs, hww/api/bitcoin/signmsg.rs) is transaction and message signing on secp256k1. Ed25519 at bitbox02-rust/src/keystore/ed25519.rs is transaction signing on the Cardano path.

6 reference(s) to X25519 across 6 file(s), e.g. external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h:18. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:X25519:general
Data source	external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h:18
and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h · line 18 · SHA-256 6b26d494a6e0efb7… · File 463d56088a31cbac…

```
   17 |
   18 | void Hacl_Curve25519_crypto_scalarmult(uint8_t *mypublic, uint8_t *secret, uint8_t
*basepoint);
   19 |
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/vs2010/Hacl_Curve25519.h · line 18 · SHA-256 6b26d494a6e0efb7… · File c2e5a6ee616f8410…

```
   17 |
   18 | void Hacl_Curve25519_crypto_scalarmult(uint8_t *mypublic, uint8_t *secret, uint8_t
*basepoint);
   19 |
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/x25519.h · line 190 · SHA-256 93ca5e6bc26dc7e7… · File da28aebd8413ebfb…

```
  189 |
  190 | #endif /* x25519.h */
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/Hacl_Curve25519.c · line 724 · SHA-256 2e635634cbb35ab0… · File ff662cfae5e65d96…

```
  723 |
  724 | void Hacl_Curve25519_crypto_scalarmult(uint8_t *mypublic, uint8_t *secret, uint8_t
*basepoint)
  725 | {
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/everest.c · line 28 · SHA-256 fef5bf54076fc78f… · File 2d7f6fccc2e11f3e…

```
   27 |
   28 | #include "everest/x25519.h"
   29 | #include "everest/everest.h"
```

file_excerpt · external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/legacy/Hacl_Curve25519.c · line 769 · SHA-256 f6b2d5d0e7eb831d… · File fd769ba72bed1ca6…

```
  768 |
  769 | void Hacl_Curve25519_crypto_scalarmult(uint8_t *mypublic, uint8_t *secret, uint8_t
*basepoint)
  770 | {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17852 · DEP-KNOWN-ADVISORY

Medium

bytes 1.5.0 is affected by GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007) for crates.io package bytes at version 1.5.0: bytes has integer overflow in BytesMut::reserve. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/bytes@1.5.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-434x-w66g-qw3r
```

EVIDENCE

tool_output · OSV.dev GHSA-434x-w66g-qw3r · manifest · SHA-256 58c8425f679122ea...


```
{"id": "GHSA-434x-w66g-qw3r", "aliases": ["CVE-2026-25541", "RUSTSEC-2026-0007"], "modified":
"2026-09-10T03:50:33.995841351Z", "summary": "bytes has integer overflow in BytesMut::reserve",
"severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/
SI:N/SA:N/E:P"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (1.11.1).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-434x-w66g-qw3r

QSI-2026-17841 · CODE-INSECURE-RANDOM

Medium

Non-cryptographic RNG near security-sensitive identifiers in external/lvgl/docs/src/_static/jquery.js

Potential cryptographic risk | Confidence: Low | preliminary — requires further verification

'Math.random()' at line 2 on a line mentioning key/token/secret/password/nonce.

Affected asset	file:external/lvgl/docs/src/_static/jquery.js
Data source	external/lvgl/docs/src/_static/jquery.js
Detection method	Regular-expression matching of language-specific API patterns.
Analyzer	source_code 1.3.0 · 2026-11-06 13:05 UTC
Priority score	25.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Predictable values may allow token or key guessing.

REPRODUCTION STEPS

```
Open line 2 in the supplied file external/lvgl/docs/src/_static/jquery.js and inspect the quoted
lines (see evidence).
```

EVIDENCE

file_excerpt · external/lvgl/docs/src/_static/jquery.js · line 2 · SHA-256 22dc2b5ae897fe9b... · File ff1523fb7389539c...

```
    1 | /*! jQuery v3.6.0 | (c) OpenJS Foundation and other contributors | jquery.org/license */
    2 | !function(e,t){"use strict";"object"==typeof module&&"object"==typeof module.exports?
module.exports=e.document?t(e,!0):function(e){if(!e.document)throw new Error("jQuery requires a
window with a document");return t(e)}:t(e)}("undefined"!=typeof window?window:this,function(C,e)
{"use strict";var t=[],
```

LIMITATIONS OF THE CHECK

Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Use a CSPRNG (secrets, crypto.getRandomValues, crypto/rand).

RETEST PROCEDURE

Re-scan the source and review the call sites listed in the evidence.

NIST SP 800-90A Rev. 1 — Recommendation for Random Number Generation Using Deterministic Random Bit Generators; CWE-338 — Use of Cryptographically Weak Pseudo-Random Number Generator (PRNG)

QSI-2026-17853 · DEP-KNOWN-ADVISORY

Low

certifi 2022.12.7 is affected by GHSA-248v-346w-9cwc (CVE-2024-39689, PYSEC-2024-230)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-248v-346w-9cwc (CVE-2024-39689, PYSEC-2024-230) for PyPI package certifi at version 2022.12.7: Certifi removes GLOBALTRUST root certificate. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/certifi@2022.12.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	39.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-248v-346w-9cwc
```

EVIDENCE

tool_output · OSV.dev GHSA-248v-346w-9cwc · line 13 · SHA-256 3d74205f26ab1e3c...


```
{"id": "GHSA-248v-346w-9cwc", "aliases": ["CVE-2024-39689", "PYSEC-2024-230"], "modified":
"2026-09-10T03:50:15.994602411Z", "summary": "Certifi removes GLOBALTRUST root certificate",
"severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2024.7.4).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-248v-346w-9cwc

QSI-2026-17863 · DEP-KNOWN-ADVISORY

Low

pygments 2.14.0 is affected by GHSA-5239-wwwm-4pmq (CVE-2026-4539, PYSEC-2026-2987)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-5239-wwwm-4pmq (CVE-2026-4539, PYSEC-2026-2987) for PyPI package pygments at version 2.14.0: Pygments has Regular Expression Denial of Service (ReDoS)

due to Inefficient Regex for GUID Matching. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:PyPI/pygments@2.14.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	39.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	3.3 — CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:N/I:N/A:L

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-5239-wwwm-4pmq
```

EVIDENCE

tool_output · OSV.dev GHSA-5239-wwwm-4pmq · line 34 · SHA-256 3a3de252c0c01189…


```
{"id": "GHSA-5239-wwwm-4pmq", "aliases": ["CVE-2026-4539", "PYSEC-2026-2987"], "modified":
"2026-09-10T03:50:59.780634752Z", "summary": "Pygments has Regular Expression Denial of Service
(ReDoS) due to Inefficient Regex for GUID Matching", "severity": [{"type": "CVSS_V3", "score":
"CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:N/I:N/A:L"}, {"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/
AT:N/PR:L/UI:N/VC:N/VI:N/VA:L/SC:N/SI:N/SA:N/E:P"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (2.20.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-5239-wwwm-4pmq

QSI-2026-17848 · DEP-ADVISORY-UNRATED

Low

crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)

Confirmed technical vulnerability | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** crypto-bigint 0.5.5 (multiple timing variability issues in operations intended to be constant-time) is reachable in shipped firmware, and the path is worth stating exactly because it is narrower than it looks. src/rust/util/Cargo.toml:17 takes p256 as an *optional* dependency with features ["arithmetic", "ecdsa"], and lib.rs guards it with #[cfg(feature = "p256")]. That feature is switched on in exactly one place: src/rust/bitbox02-rust-c/Cargo.toml, inside the app-u2f feature block ("util/p256"). So p256 → elliptic-curve → crypto-bigint is compiled in whenever the U2F/FIDO app

> is built — the Multi edition — and is absent from a Bitcoin-only build. What it does there is secret-dependent: util/src/p256.rs exports rust_p256_sign, which calls SigningKey::sign_prehash with a private key over the C FFI, i.e. U2F authentication signatures on NIST P-256. The keys at stake are therefore FIDO credentials, NOT bitcoin spending keys: those go through bitbox-secp256k1 (libsecp256k1 in C), which this advisory does not touch. Stated precisely: no exploit is demonstrated, and the attacker model is physical possession plus timing measurement, which is the threat a hardware wallet exists to resist — hence medium rather than low. Fix available in crypto-bigint 0.7.6; maintainers should match the advisory's named operations against p256's scalar arithmetic.

OSV.dev reports RUSTSEC-2026-0387 for crates.io package crypto-bigint at version 0.5.5: Multiple timing variability issues in operations intended to be constant-time. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crypto-bigint@0.5.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	23.9 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0387
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0387 · manifest · SHA-256 eab63427da988e95...


```
{"id": "RUSTSEC-2026-0387", "aliases": [], "modified": "2026-10-24T03:45:02.365137418Z", "summary":
"Multiple timing variability issues in operations intended to be constant-time", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.7.6) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0387

QSI-2026-17849 · DEP-KNOWN-ADVISORY

Info

keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)

Known vulnerable component | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** keccak 0.1.4's unsoundness is in the opt-in ARMv8 assembly backend behind the asm Cargo feature, which is off by default and which no manifest in src/rust enables. Its consumers are bitbox02-rust and sha3, both using the portable implementation. Upgrading to 0.1.6 is still the cheap move during routine maintenance.

OSV.dev reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012) for crates.io package keccak at version 0.1.4: Unsoundness in opt-in ARMv8 assembly backend for `keccak`. The affected code sits behind the `asm` Cargo feature, off by default, which no audited manifest enables: the advisory does not apply unless another dependency enables it.

Affected asset	dependency:crates.io/keccak@0.1.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	35.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-3288-p39f-rqpv
```

EVIDENCE

tool_output · OSV.dev GHSA-3288-p39f-rqpv · manifest · SHA-256 638750a493b76f12…


```
{"id": "GHSA-3288-p39f-rqpv", "aliases": ["RUSTSEC-2026-0012"], "modified":
"2026-09-10T03:50:59.332505686Z", "summary": "Unsoundness in opt-in ARMv8 assembly backend for
`keccak`", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:H/AT:P/PR:N/UI:N/VC:N/VI:L/
VA:N/SC:N/SI:N/SA:N/E:U"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.1.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-3288-p39f-rqpv

QSI-2026-17850 · DEP-KNOWN-ADVISORY

Info

time 0.3.44 is affected by GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009)

Known vulnerable component | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** time 0.3.44's stack-exhaustion issue is in the RFC 2822 and RFC 3339 parsing entry points (time::Date::parse, time::OffsetDateTime::parse and friends). time is pulled in by the workspace's own util crate, which uses it for datetime formatting in util/src/datetime.rs, not for parsing attacker-supplied strings, and no audited file names the affected functions. automated severity: Medium.

OSV.dev reports GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009) for crates.io package time at version 0.3.44: time vulnerable to stack exhaustion Denial of Service attack. Some product source files did not fit the size limits, so the affected functions were not looked up.

Affected asset	dependency:crates.io/time@0.3.44
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	35.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-r6v5-fh4h-64xc
```

EVIDENCE

tool_output · OSV.dev GHSA-r6v5-fh4h-64xc · manifest · SHA-256 f9407c25f7cc6bac...


```
{"id": "GHSA-r6v5-fh4h-64xc", "aliases": ["CVE-2026-25727", "RUSTSEC-2026-0009"], "modified":
"2026-09-10T03:50:35.583241382Z", "summary": "time vulnerable to stack exhaustion Denial of Service
attack", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:H/AT:N/PR:L/UI:A/VC:N/VI:N/
VA:H/SC:N/SI:N/SA:H"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.3.47).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-r6v5-fh4h-64xc

QSI-2026-17846 · DEP-ADVISORY-UNRATED

Info

anyhow 1.0.31 is affected by RUSTSEC-2026-0190 (no severity rating)

Known vulnerable component | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** anyhow 1.0.31 (unsoundness in Error::downcast_mut) enters only through prost-derive, which is a proc-macro crate, so it runs in the code generator at build time and is never linked into the firmware image. Traced in src/rust/Cargo.lock: anyhow ← prost-derive ← prost ← bitbox-proto ← bitbox02-rust ← bitbox02-rust-c. Error::downcast_mut is not called anywhere in the audited tree. automated severity: Low.

OSV.dev reports RUSTSEC-2026-0190 for crates.io package anyhow at version 1.0.31: Unsoundness in `Error::downcast_mut()`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. Some product source files did not fit the size limits, so the affected functions were not looked up.

Affected asset	dependency:crates.io/anyhow@1.0.31
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	20.9 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0190
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0190 · manifest · SHA-256 12d4b3eeb5a7dc67…


```
{"id": "RUSTSEC-2026-0190", "aliases": [], "modified": "2026-06-29T14:15:03.650725453Z", "summary":
"Unsoundness in `Error::downcast_mut()`", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.0.103) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0190

QSI-2026-17851 · DEP-ADVISORY-UNRATED

Info

anyhow 1.0.75 is affected by RUSTSEC-2026-0190 (no severity rating)

Known vulnerable component | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** anyhow 1.0.31 (unsoundness in Error::downcast_mut) enters only through prost-derive, which is a proc-macro crate, so it runs in the code generator at build time and is never linked into the firmware image. Traced in src/rust/Cargo.lock: anyhow ← prost-derive ← prost ← bitbox-proto ← bitbox02-rust ← bitbox02-rust-c. Error::downcast_mut is not called anywhere in the audited tree. automated severity: Low.

OSV.dev reports RUSTSEC-2026-0190 for crates.io package anyhow at version 1.0.75: Unsoundness in `Error::downcast_mut()`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. Some product source files did not fit the size limits, so the affected functions were not looked up.

Affected asset	dependency:crates.io/anyhow@1.0.75
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	20.9 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0190
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0190 · manifest · SHA-256 12d4b3eeb5a7dc67…


```
{"id": "RUSTSEC-2026-0190", "aliases": [], "modified": "2026-06-29T14:15:03.650725453Z", "summary":
"Unsoundness in `Error::downcast_mut()`", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.0.103) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0190

QSI-2026-17845 · DEP-NO-PQC-LIBRARY

Info

No post-quantum capable cryptographic library detected

Insufficient data | Confidence: Medium

> **Analyst review (QSI review, 2026-10-28): confirmed by the analyst.** Accurate and the single most quotable row for a migration budget: across 40 dependency manifests the tree declares curve25519-dalek, ecdsa, ed25519-dalek, p256, secp256k1 and x25519-dalek, and nothing PQC-capable. On a Cortex-M target that is not an oversight — ML-DSA and SLH-DSA signature and key sizes are the constraint, and a no_std PQ implementation that fits the flash and RAM budget is itself the engineering problem. Worth stating as a finding precisely because the fix is not "add a crate".

Across 48 dependency manifests, QSI found crypto libraries (curve25519-dalek, ecdsa, ed25519-dalek, p256, secp256k1, x25519-dalek) but none from its PQC-capable list.

Affected asset	dependencies
Data source	dependency manifests
Detection method	Dependency name matching.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	15.0 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Readiness indicator only: PQC support may come from the platform (OpenSSL, OS, cloud KMS) rather than application dependencies.

REPRODUCTION STEPS

```
Review the dependency manifests listed in the evidence.
```

EVIDENCE

dependency · py/pyproject.toml, py/requirements.txt, src/rust/Cargo.lock, src/rust/Cargo.toml, src/rust/bins/bitbox03-boot0/Cargo.toml, src/rust/bins/bitbox03-boot1/Cargo.toml, src/rust/bins/bitbox03-factorysetup/Cargo.toml, src/rust/bins/bitbox03-firmware/Cargo.toml, src/rust/bitbox-aes/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-build/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-sys/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk/Cargo.toml, src/rust/bitbox-boot-utils/Cargo.toml, src/rust/bitbox-byt · SHA-256 bf8c520f29fbbee5...

```
Crypto libraries: curve25519-dalek, ecdsa, ed25519-dalek, p256, secp256k1, x25519-dalek
```

LIMITATIONS OF THE CHECK

Absence of evidence is not evidence of absence: system libraries and transitive dependencies may not be listed.

RECOMMENDATION

Confirm where PQC support will come from (TLS terminator, KMS/HSM, library upgrade).

RETEST PROCEDURE

Re-scan after upgrades.

FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM); FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)

QSI-2026-17847 · DEP-ADVISORY-INFORMATIONAL

Info

bare-metal 0.2.5: informational advisory RUSTSEC-2026-0110 (unmaintained)

Improvement recommendation | Confidence: Low

> **Analyst review (QSI review, 2026-10-28): not applicable to this code.** bare-metal 0.2.5 is flagged as deprecated, not vulnerable; the advisory reports no defect. Recording the decision to keep it, which is what an informational advisory asks for. It is relevant to post-quantum readiness only in the weak sense that an unmaintained crate will not gain PQ support, and this one carries no cryptography at all.

OSV.dev lists RUSTSEC-2026-0110 for crates.io package bare-metal at version 0.2.5 as informational (unmaintained): bare-metal is deprecated. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/bare-metal@0.2.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 13:05 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0110
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0110 · manifest · SHA-256 1377cbc63847011c...


```
{"id": "RUSTSEC-2026-0110", "aliases": [], "modified": "2026-04-24T09:45:08.136572Z", "summary":
"bare-metal is deprecated", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2026-0110

Evidence

ID	Type	Data source	SHA-256
QSI-2026-17854	tool_output	OSV.dev GHSA-6q48-6c7w-9gw2 (line 13)	87c1ba23d8d1ad688cb3d3331ff88fa3cb0ba134a6064a4568a2d5a
QSI-2026-17830	file_excerpt	scripts/generate_version_headers.py (line 230)	14c945f1f3952da4886e703323a55eaab298fcd6ce757a9042715de
QSI-2026-17855	tool_output	OSV.dev GHSA-xqr8-7jwr-rhp7 (line 13)	2f94715e09d8ea440753f6e548090fbc7961491b5ef65e35a87333a
QSI-2026-17859	tool_output	OSV.dev GHSA-gmj6-6f8f-6699 (line 28)	1fc2fcfcf66b7cde744640e67903559b4f93dfc0417870f1dfc401d
QSI-2026-17862	tool_output	OSV.dev GHSA-q2x7-8rv6-6q7h (line 28)	cf69831558e5cfcc55e9251c195c61130015555745da08e16087c26
QSI-2026-17869	tool_output	OSV.dev GHSA-2xpw-w6gg-jr37 (line 60)	8f549172278134db8bb6a128450758a9ebfe0537c99993930a95ab7
QSI-2026-17871	tool_output	OSV.dev GHSA-38jv-5279-wg99 (line 60)	528407f765ddc33699b8c258636cc375710304468ddc58652eaa66f
QSI-2026-17872	tool_output	OSV.dev GHSA-8988-9cw3-xx77 (line 60)	c51c0d680d885d638def44e1b95a18c000c1eb5bbeb4a65fa11f765
QSI-2026-17874	tool_output	OSV.dev GHSA-gm62-xv2j-4w53 (line 60)	b69c9a68201dc7d1cd1c1c737f9fd193a253ad102d274aa1778f180
QSI-2026-17878	tool_output	OSV.dev GHSA-vxq7-64xx-v4gw (line 60)	cbf2e0ae22b277477ba53ac789fc1c2df8d7fa5808195e02870fcf4
QSI-2026-17801	file_excerpt	external/samd51a-ds/include/component/aes.h (line 81)	92ef17166ab022a20ad6f4d44119934fae3cc5b8789a3e7c85b0fa7
QSI-2026-17843	file_excerpt	external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h (line 184)	b4ca0728db5ccfb494e9c1d28a1f570024b573214a88c0c50c5651e
QSI-2026-17844	file_excerpt	messages/btc.proto (line 4)	9397cf36c17bba81b4bbbc6db29f3dc450925e4566727b47ce57892
QSI-2026-17844	file_excerpt	messages/cardano.proto (line 4)	9397cf36c17bba81b4bbbc6db29f3dc450925e4566727b47ce57892
QSI-2026-17844	file_excerpt	messages/eth.proto (line 4)	9397cf36c17bba81b4bbbc6db29f3dc450925e4566727b47ce57892
QSI-2026-17804	key	releases/pubkeys/JuliusWiedemann.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17804	key	releases/pubkeys/JuliusWiedemann.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17805	key	releases/pubkeys/JunghoonLee.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07

QSI-2026-17805	key	releases/pubkeys/JunghoonLee.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17806	key	releases/pubkeys/My1.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17807	key	releases/pubkeys/My1.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17807	key	releases/pubkeys/My1.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17808	key	releases/pubkeys/TheCharlatan.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17808	key	releases/pubkeys/TheCharlatan.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17808	key	releases/pubkeys/TheCharlatan.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17808	key	releases/pubkeys/TheCharlatan.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17809	key	releases/pubkeys/beerosagos.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17809	key	releases/pubkeys/beerosagos.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17810	key	releases/pubkeys/benma.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17810	key	releases/pubkeys/benma.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17810	key	releases/pubkeys/benma.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17811	key	releases/pubkeys/brianddk.asc (line 1)	99f8a76d7955bc1cad612c6fe14ee442ba2c3ba5655096eda02d83c
QSI-2026-17811	key	releases/pubkeys/brianddk.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17811	key	releases/pubkeys/brianddk.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17812	key	releases/pubkeys/cstenglein.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17812	key	releases/pubkeys/cstenglein.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17813	key	releases/pubkeys/frknakk.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17813	key	releases/pubkeys/frknakk.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e

QSI-2026-17814	key	releases/pubkeys/guggerf.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17814	key	releases/pubkeys/guggerf.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17815	key	releases/pubkeys/jonathancross.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17816	key	releases/pubkeys/jonathancross.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17816	key	releases/pubkeys/jonathancross.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17816	key	releases/pubkeys/jonathancross.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17817	key	releases/pubkeys/nickez.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17818	key	releases/pubkeys/offbyn.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17818	key	releases/pubkeys/offbyn.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17818	key	releases/pubkeys/offbyn.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17818	key	releases/pubkeys/offbyn.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17819	key	releases/pubkeys/psteiger.asc (line 1)	6dbf836a10cb57142effcc3f89dd5e74340870bac059dd57e9f528b
QSI-2026-17820	key	releases/pubkeys/stadicus.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17820	key	releases/pubkeys/stadicus.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e

QSI-2026-17820	key	releases/pubkeys/ stadicus.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17820	key	releases/pubkeys/ stadicus.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17821	key	releases/pubkeys/ strnbrg.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17821	key	releases/pubkeys/ strnbrg.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17822	key	releases/pubkeys/ sutterseba.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17822	key	releases/pubkeys/ sutterseba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17823	key	releases/pubkeys/ tomasvrba.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17824	key	releases/pubkeys/ userofreddit44.asc (line 1)	315636b2f7e00661811d8d8feee53223db00c9277ae1194dcaf9d4f
QSI-2026-17824	key	releases/pubkeys/ userofreddit44.asc (line 1)	6ccbdce52e43463e8b282489a66bb4553923cec099cc342f213f584
QSI-2026-17825	key	releases/pubkeys/ x1ddos.asc (line 1)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc07
QSI-2026-17825	key	releases/pubkeys/ x1ddos.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17826	key	releases/pubkeys/ x1ddos.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17826	key	releases/pubkeys/ x1ddos.asc (line 1)	a744425426415d58dbeef862a6f1d8bcf6b00820639c8784270363d
QSI-2026-17827	key	releases/pubkeys/ x1ddos.asc (line 1)	6ccbdce52e43463e8b282489a66bb4553923cec099cc342f213f584

QSI-2026-17828	key	releases/pubkeys/yaziza.asc (line 1)	6dbf836a10cb57142effcc3f89dd5e74340870bac059dd57e9f528b
QSI-2026-17828	key	releases/pubkeys/yaziza.asc (line 1)	065d4edee7f1048b4c864ef703528e93278d15340d378d5c518a3b2
QSI-2026-17829	key	releases/pubkeys/yaziza.asc (line 1)	484f2172c4dd53b29fd23e321304b308266950550fb37c83431470e
QSI-2026-17838	key	external/cryptoauthlib/app/pkcs11/README.md (line 280)	87a3decbb47c307b59a5585db9eecff49e5997a1eb7350dbfae08a9
QSI-2026-17856	tool_output	OSV.dev GHSA-65pc-fj4g-8rjx (line 22)	1b180b3ab94317455b5be8428a3b6ed5d0f401ab156e3fc18e2d9b9
QSI-2026-17857	tool_output	OSV.dev GHSA-jjg7-2v4v-x38h (line 22)	b260da6fe71df346fc2fbff5eb48b5cd290f2fcaa442b8e591d4cd0
QSI-2026-17858	tool_output	OSV.dev GHSA-cpwx-vrp4-4pq7 (line 28)	e28b14906d20d0da1a5a90a9cd8900c0e9f3f8e617101292dbb5c6b
QSI-2026-17860	tool_output	OSV.dev GHSA-h5c8-rqwp-cp95 (line 28)	3bcb93126b62e63f1ce99f0dbbef076e72a4be50e744eb19c8c285d
QSI-2026-17861	tool_output	OSV.dev GHSA-h75v-3vvj-5mfj (line 28)	8d178561724d57b3a1b58e1376ccee532051838cf1bd2f1b097a503
QSI-2026-17864	tool_output	OSV.dev GHSA-mrwq-x4v8-fh7p (line 34)	6da07c6d923c992526545174c0d85c361cfb51e714fe54fb739d059
QSI-2026-17865	tool_output	OSV.dev GHSA-9hjg-9r4m-mvj7 (line 36)	54566624d8668e64e97057d595c41cb500bb80cf1058002a3401dbc
QSI-2026-17866	tool_output	OSV.dev GHSA-9wx4-h78v-vm56 (line 36)	38975a9d04ed33a4ee36cb1327ae63c2711a85bbd849c980a35cf0b
QSI-2026-17867	tool_output	OSV.dev GHSA-gc5v-m9x4-r6x2 (line 36)	f74704bb5c636714db99ca4783e4a2b3be4ccd7fe7421013ca903d4
QSI-2026-17868	tool_output	OSV.dev GHSA-j8r2-6x86-q33q (line 36)	913766f1fff73012784b74cfa767f1c83b6cd5ccb2339129a2df5bd
QSI-2026-17870	tool_output	OSV.dev GHSA-34jh-p97f-mpxf (line 60)	70e2b3135eba6c4126c184770467f1dd8b42cba4057694f5cae8217
QSI-2026-17873	tool_output	OSV.dev GHSA-g4mx-q9vg-27p4 (line 60)	9068d64596de74c3e90b47341a42014cea57c6cc121e804ce5fd84c
QSI-2026-17875	tool_output	OSV.dev GHSA-pq67-6m6q-mj2v (line 60)	788a2cf91f9eaf45aa98c3450fb81db5550fdc6d8ec5c3a2e0f8be8
QSI-2026-17876	tool_output	OSV.dev GHSA-qccp-gfcp-xxvc (line 60)	c7033c35f2861e3d21a7968a87a6a0bb5131d057bb7ab67da86828d
QSI-2026-17877	tool_output	OSV.dev GHSA-v845-jxx5-vc9f (line 60)	4436cbacc6451b228d4eb5436fea1a53cd8e207e7f17bde22899e13
QSI-2026-17879	tool_output	OSV.dev GHSA-jfmj-5v4g-7637 (line 62)	27a0d57ee49a9e56ef66b7abab8b3aca4800e47d4fb3cae8962ba12

QSI-2026-17831	file_excerpt	src/bootloader/bootloader.c (line 537)	a692ec93e05c715e251b70d5df1203b876ed7d82bd28e9c6b1c10a2
QSI-2026-17831	file_excerpt	src/bootloader/bootloader.c (line 538)	c8910947b0750cbb6c7959f9780e84bfbe3cd04a9d2bb09cbbf7cc1
QSI-2026-17831	file_excerpt	src/bootloader/bootloader.c (line 609)	110ab61f5a0020b335e42ad45c11bc760596590d7806f6ab45ae8af
QSI-2026-17831	file_excerpt	src/bootloader/stage0/stage1_sigcheck.c (line 165)	40465ea406a87913706627f03d6592256e6307cc6510acfb3fd097d
QSI-2026-17831	file_excerpt	src/rust/bitbox02-rust-c/src/firmware_c_api.rs (line 23)	0239908a79adffb8db0de707bb0d8d79652f9eee0c4c289c416254e
QSI-2026-17831	file_excerpt	src/rust/bitbox02-rust-c/src/firmware_c_api.rs (line 24)	d2c149057a83a51f66f258261f3e75f6e5d333f7e7cf8e4fb0e01a5
QSI-2026-17833	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 15)	1c153d0b1adf9cd8ecf03da5502f8e68db447b4ab09e5e61188fc53
QSI-2026-17833	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 19)	4b67afd4d285deeff00b1d8a4e6ef086fbb387cad74ce1646456a08
QSI-2026-17833	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 60)	f8ee14a435cd94f2c3694c380ab36b8cca17c6d237875854295e567
QSI-2026-17840	file_excerpt	external/cryptoauthlib/python/cryptoauthlib/atcab.py (line 974)	87004ff616be22235a821a5c8d5b9ead791c529f62209f2bc0cea63
QSI-2026-17840	file_excerpt	external/cryptoauthlib/python/cryptoauthlib/atcab.py (line 977)	8e5f1c528fbe7b2ee242f9afec3b53fc230249d43b130247f4590ed
QSI-2026-17840	file_excerpt	external/cryptoauthlib/python/cryptoauthlib/atcab.py (line 978)	204aab89e32c109386818d0b5149a61c7c3ce73f3684eb12b5709f3
QSI-2026-17803	file_excerpt	py/bitbox02/bitbox02/bitbox02/secp256k1.py (line 3)	f1b7aa46e3d0b50c48a6a1863e881efc3f3599b1f51734acef93ab9
QSI-2026-17803	file_excerpt	src/rust/bitbox-secp256k1/build.rs (line 6)	c0e8a8076dd2e70e550d1aa3c11d2672989702f68a84802e08636e7
QSI-2026-17803	file_excerpt	src/rust/bitbox-secp256k1/build.rs (line 18)	25e9545e63d24ed02198281c8da70ca8d5106db13c770d3cb0563ef
QSI-2026-17803	file_excerpt	src/rust/bitbox-secp256k1/build.rs (line 37)	47d7abfe9efe1b2bea40a263634d29da5de554eb838641f36186b8d
QSI-2026-17803	file_excerpt	src/rust/bitbox-secp256k1/src/lib.rs (line 7)	27351363967fb644277306e8bdedc7dfa7e9c4e0d23be03ec2a3b05
QSI-2026-17803	file_excerpt	src/rust/bitbox-secp256k1/src/lib.rs (line 8)	1ae312353045cae568e231b036e3b831f53cd99c54b61952e50f75c

QSI-2026-17803	file_excerpt	src/rust/bitbox-secp256k1/src/lib.rs (line 9)	63fd03f40121775b3740c6e1e00a3eb82337e0d734dfcffdce1ec08
QSI-2026-17803	file_excerpt	src/rust/bitbox02-rust-c/src/secp256k1.rs (line 3)	32c3110f30785a45f30291b04aaed8f855161da3b48a5d627b55bb9
QSI-2026-17803	file_excerpt	src/rust/bitbox02-rust-c/src/secp256k1.rs (line 42)	26f131e03c434e4bb99270182a8393193a714d40ef826c5534f8f5a
QSI-2026-17803	file_excerpt	src/rust/bitbox02-rust-c/src/secp256k1.rs (line 56)	bf55765f36e34101d8c06c2f514fa6a3aabb581d93221c944b40863
QSI-2026-17832	file_excerpt	src/rust/bitbox-noise/src/lib.rs (line 16)	bec1cc899b2ee57631342068081612e177c1e131b0062f84dbf4bd1
QSI-2026-17832	file_excerpt	src/rust/bitbox-noise/src/lib.rs (line 19)	57cd8f7157563d1d7dcdedce2582f83213eb268d9f974d8d8270085
QSI-2026-17832	file_excerpt	src/rust/bitbox-noise/src/noise_xx.rs (line 8)	784affd9139fdfbf69bb890deb430326b00ccb48a47de6340bd5071
QSI-2026-17832	file_excerpt	src/rust/bitbox-noise/src/x25519.rs (line 49)	43d70ab747c41c47e17b5ba00fdf0933116113165cc5310d5e74f37
QSI-2026-17832	file_excerpt	src/rust/bitbox-noise/src/x25519.rs (line 54)	fae4ff917820e0219e7ac326306ae796207dce8f7af4ff5e75fd432
QSI-2026-17832	file_excerpt	src/rust/bitbox-noise/src/x25519.rs (line 55)	9cb9b67a0b6ccdefe882b0cd12a8a8e62cc7886bd8c3fa9cc44154f
QSI-2026-17832	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/everest.h (line 25)	805839a03b00bfe936a91234a33a359e26d3cc0294734ba3ffba86e
QSI-2026-17832	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c (line 35)	9ca103a64c6b7d9f9370997f71d0b11e98331c8e1228627873d68b5
QSI-2026-17832	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c (line 81)	fbaaec05ef452c03ec3e2c812b4b680704b7c7a0f16e42a0ef90528
QSI-2026-17832	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c (line 134)	21b4c7046cce778fb5b5fd9cc64b7a419ba11a10d999477df74ff1d
QSI-2026-17835	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs (line 289)	0ff082b56713f8bc14102a8974bb75deeed2834b8d151ad27ce95d9
QSI-2026-17835	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 12)	08a764387fbe8ba1e77642c172a087172ccf897e6ed0de3209ef2dc
QSI-2026-17835	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 22)	fd1fd79d198d0e819f2a0ebb7be23d1edc0cd763ded77a8d3934110

QSI-2026-17836	file_excerpt	src/rust/bitbox02-rust/src/hww/api/cardano/address.rs (line 308)	23f10be088fb5fde1ae33ef6d2b64426c96b1a73a2cd6aa55b72f00
QSI-2026-17836	file_excerpt	src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction.rs (line 55)	e86a0885d901ed7473539a66e0f107c8eab3601fd2b8fdf6ef5fe27
QSI-2026-17836	file_excerpt	src/rust/bitbox02-rust/src/hww/api/cardano/xpubs.rs (line 29)	49b540bc531f7b21b87be208e380ff0417ca4977de15df6ba476c65
QSI-2026-17836	file_excerpt	src/rust/bitbox02-rust/src/keystore/ed25519.rs (line 6)	eaa112ccabe881af679105a7cfbae0ca7e26014770db0533cb5df6d
QSI-2026-17836	file_excerpt	src/rust/bitbox02-rust/src/keystore/ed25519.rs (line 10)	0d1f71033a48c3c6e0b8363e9a41c2f5e9ee07f08735238e9856e64
QSI-2026-17836	file_excerpt	src/rust/bitbox02-rust/src/keystore/ed25519.rs (line 11)	449e9e250d19db94ff94019815e29a47323da4b3df842285684544d
QSI-2026-17802	file_excerpt	py/bitbox02/bitbox02/bitbox02/bitbox02.py (line 21)	c1697b58eea8e082fa9f5090e083f0e57d75e65cdb346518fab79a4
QSI-2026-17802	file_excerpt	src/pukcc/pukcc.c (line 181)	53085ff2be44540f9ac0d6f54f71ca19c4e1ac12898aef6c3ecb093
QSI-2026-17802	file_excerpt	src/pukcc/pukcc.h (line 30)	dae3c783558d199701f2f4d6fa2a10c198ed29cea93653a188e1912
QSI-2026-17802	file_excerpt	src/rust/bitbox02-rust-c/src/lib.rs (line 23)	e527a51a0f0f62d0ee3a3e27b910a798f61d791ba1bc143ec92bfdc
QSI-2026-17802	file_excerpt	src/rust/streaming-silent-payments/src/hash.rs (line 7)	33201f813f216ace1faa8378e7cbee0f992739d6b30c90a1a740a8c
QSI-2026-17802	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 12)	08a764387fbe8ba1e77642c172a087172ccf897e6ed0de3209ef2dc
QSI-2026-17802	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 43)	08787745f17bf5694851af8a934a9791f92c751c329c4e20cd230db
QSI-2026-17802	file_excerpt	src/rust/streaming-silent-payments/src/lib.rs (line 124)	14bc8680433716b8fc52ba9a85b1577a9648567735eefacba61ba01
QSI-2026-17802	file_excerpt	src/rust/util/src/p256.rs (line 28)	925484476bdd234888084554a30574f13c31fd235705950f76e225d
QSI-2026-17802	file_excerpt	external/cryptoauthlib/lib/mbedtls/atca_mbedtls_ecdsa.c (line 51)	f4cdd327029f645a34c8dbe10f7c918b7d043ffe26b21bb5095c676
QSI-2026-17834	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 11)	3637a75aa4245184b2de8e6f4f098698530fa03b423eab7638b978e
QSI-2026-17834	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 56)	211b70e7af67bb49e09e40a7986b4e203dea661248df40bd30243f0

QSI-2026-17834	file_excerpt	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs (line 410)	272c2c902cbf96d8a213325ad72d5e60b2a3a926f8388d973edaf70
QSI-2026-17837	file_excerpt	src/rust/bitbox02-rust/src/keystore.rs (line 897)	4c865a127c8bb4a820585d2f29be3a734a01dc66058a4678b8e15a3
QSI-2026-17839	file_excerpt	external/cryptoauthlib/lib/calib/calib_command.h (line 181)	9b4d335853d5e7147b112d99ade016f7fe1862a3158ed862fc5d1b9
QSI-2026-17839	file_excerpt	external/cryptoauthlib/lib/calib/calib_command.h (line 357)	55dbcbc6be030b16f3c3b4c2ef5c73c86905d38266e2c344af0782e
QSI-2026-17839	file_excerpt	external/cryptoauthlib/lib/calib/calib_command.h (line 358)	0b55dd8a591437d1f5b60f510e56c4247e835c1955a33d9490500bb
QSI-2026-17839	file_excerpt	external/cryptoauthlib/lib/calib/calib_ecdh.c (line 159)	cc79ff1975ef75f3d142805474f71988336ac4bdec7e77815f61999
QSI-2026-17839	file_excerpt	external/cryptoauthlib/lib/pkcs11/pkcs11_key.c (line 953)	c41a3072038479a783a892f81f7afc850c71c7798fb752eb6480c7f
QSI-2026-17839	file_excerpt	external/cryptoauthlib/python/setup.py (line 224)	2ebf64b8e13175e127281f9b0d8e537854b0323a91c1f0f101a328a
QSI-2026-17842	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h (line 18)	6b26d494a6e0efb7718fecc1d561c12f021e02c91d67b217b25e424
QSI-2026-17842	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/vs2010/Hacl_Curve25519.h (line 18)	6b26d494a6e0efb7718fecc1d561c12f021e02c91d67b217b25e424
QSI-2026-17842	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/x25519.h (line 190)	93ca5e6bc26dc7e7424ffbcda870ae8b7506c5289cd5316dfba229a
QSI-2026-17842	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/Hacl_Curve25519.c (line 724)	2e635634cbb35ab0ee5084a79fefb67ecaa5885f8782e471bd9477f
QSI-2026-17842	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/everest.c (line 28)	fef5bf54076fc78fac6206b8a538ffa1b3f43863a516019d43b136b
QSI-2026-17842	file_excerpt	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/legacy/Hacl_Curve25519.c (line 769)	f6b2d5d0e7eb831dcff73317fa844f9e075b7b7e750fb599f86d126

QSI-2026-17852	tool_output	OSV.dev GHSA-434x-w66g-qw3r (manifest)	58c8425f679122ea78208908d7c42d7c0f6eb030e5fbaef6d65097f
QSI-2026-17841	file_excerpt	external/lvgl/docs/src/_static/jquery.js (line 2)	22dc2b5ae897fe9b7d69c804eda741771a2b85daae153267199b7d9
QSI-2026-17853	tool_output	OSV.dev GHSA-248v-346w-9cwc (line 13)	3d74205f26ab1e3c65bcb869a954fae59f9758d2bbf87e429684d71
QSI-2026-17863	tool_output	OSV.dev GHSA-5239-wwwm-4pmq (line 34)	3a3de252c0c0118999a686c303b22756537724efae2e91800700806
QSI-2026-17848	tool_output	OSV.dev RUSTSEC-2026-0387 (manifest)	eab63427da988e95becf061fd8169abcb9634f743aeeb7567dfa2ba
QSI-2026-17849	tool_output	OSV.dev GHSA-3288-p39f-rqpv (manifest)	638750a493b76f126adef65e1dd53e2ef494cf4b1d4c7edcc7db121
QSI-2026-17850	tool_output	OSV.dev GHSA-r6v5-fh4h-64xc (manifest)	f9407c25f7cc6bacd2695a5be9ea893203cf87df27320f0289d5e50
QSI-2026-17846	tool_output	OSV.dev RUSTSEC-2026-0190 (manifest)	12d4b3eeb5a7dc67549bc112051813542621c4c3bb2d3ddfb353886
QSI-2026-17851	tool_output	OSV.dev RUSTSEC-2026-0190 (manifest)	12d4b3eeb5a7dc67549bc112051813542621c4c3bb2d3ddfb353886
QSI-2026-17845	dependency	py/pyproject.toml, py/requirements.txt, src/rust/Cargo.lock, src/rust/Cargo.toml, src/rust/bins/bitbox03-boot0/Cargo.toml, src/rust/bins/bitbox03-boot1/Cargo.toml, src/rust/bins/bitbox03-factorysetup/Cargo.toml, src/rust/bins/bitbox03-firmware/Cargo.toml, src/rust/bitbox-aes/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-build/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk-sys/Cargo.toml, src/rust/bitbox-board-stm32u5a9j-dk/Cargo.toml, src/rust/bitbox-boot-utils/Cargo.toml, src/rust/bitbox-byt	bf8c520f29fbbee5a6f56af359cab24476ceec5fc694f1dd615a03d
QSI-2026-17847	tool_output	OSV.dev RUSTSEC-2026-0110 (manifest)	1377cbc63847011c54324d486e3a7bae12ccbda6c1c9a981451db35

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

- Classical only — channel key exchange (TLS, Noise, handshakes): X25519 (5 file(s))
- Classical only — key wrapping and data encryption at rest: ECDH, X25519 (2 file(s))
- Classical only — firmware and boot verification (root of trust): ECDSA (3 file(s))

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
firmware and boot verification (root of trust)	ECDSA (3)	QSI-2026-17831	Plan a hybrid firmware signature (the classical one plus ML-DSA or SLH-DSA, both required) and the path to ship the new keys in a bootloader update
key wrapping and data encryption at rest	ECDH (1), X25519 (1)	QSI-2026-17833, QSI-2026-17840	Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its own assessment
channel key exchange (TLS, Noise, handshakes)	X25519 (5)	QSI-2026-17832	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)
transaction and message signing	ECDSA (17), Ed25519 (7), Schnorr (BIP-340) (5)	QSI-2026-17803, QSI-2026-17835, QSI-2026-17836, QSI-2026-17844	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them

authentication and identity (certificates, tokens, device attestation)	Ed25519 (1)	QSI-2026-17834	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
role not determined	ECDSA (10), X25519 (6), ECDH (4), Schnorr (BIP-340) (1)	QSI-2026-17802, QSI-2026-17837, QSI-2026-17839, QSI-2026-17842	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
weak classically	8
reduced margin (Grover), approved	22
Grover: acceptable to NIST, below ECCG/ANSSI guidance	38
n/a	21
vulnerable (Shor)	113

Detailed findings: QSI-2026-17844, QSI-2026-17804, QSI-2026-17805, QSI-2026-17806, QSI-2026-17807, QSI-2026-17808, QSI-2026-17809, QSI-2026-17810, QSI-2026-17811, QSI-2026-17812, QSI-2026-17813, QSI-2026-17814, QSI-2026-17815, QSI-2026-17816, QSI-2026-17817, QSI-2026-17818, QSI-2026-17819, QSI-2026-17820, QSI-2026-17821, QSI-2026-17822, QSI-2026-17823, QSI-2026-17824, QSI-2026-17825, QSI-2026-17826, QSI-2026-17827, QSI-2026-17828, QSI-2026-17829, QSI-2026-17838, QSI-2026-17831, QSI-2026-17833, QSI-2026-17840, QSI-2026-17803, QSI-2026-17832, QSI-2026-17835, QSI-2026-17836, QSI-2026-17802, QSI-2026-17834, QSI-2026-17837, QSI-2026-17839, QSI-2026-17842

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit

Shor's algorithm — factoring Q1: quantum computation offline, classical access to the victim	breaks	RSA (41)	Breaks RSA at every key length: a longer modulus only raises the cost polynomially. Plan migration to ML-KEM (or a hybrid) for key establishment and ML-DSA / SLH-DSA for signatures; start with keys protecting data that must stay confidential for years (harvest now, decrypt later) and with long-lived signatures.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDH (6), ECDSA (36), Ed25519 (11), Schnorr (BIP-340) (4), X25519 (15)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	AES-128 (4), Blowfish (1), CMAC (3), Camellia (1), ChaCha20-Poly1305 (4), HKDF (1), HMAC (12), Keccak-256 (4), RC4 (1), RIPEMD-160 (1), SHA-1 (5), SHA-2 (31)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	Keccak-256 (4), RIPEMD-160 (1), SHA-1 (5), SHA-2 (31)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.
Simon's algorithm — superposition attacks on symmetric constructions Q2: superposition queries to the victim's keyed primitive	only with superposition queries	GCM mode (2)	No action for ordinary deployments of AES-GCM or CBC-MAC — the attack model does not apply. Record it as a model assumption; see the offline variant for Even–Mansour-type designs.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- keys — Insufficient data: releases/pubkeys/psteiger.asc line 1: OpenPGP algorithm ID 18 is not in the algorithm registry
- tls_code — Insufficient data: tools/atecc608/go.mod: go 1.19 — before Go 1.24 crypto/tls offers no X25519MLKEM768 by default; relevant if the module uses TLS
- certificates — error (not checked): external/cryptoauthlib/lib/atcacert/atcacert_pem.h line 37: certificate could not be parsed (ValueError)
- tls_key_exchange_groups — Insufficient data: external/lvgl/docs/doxygen_xml.py line 349: not in the algorithm registry, not assessed: {}
- tls_key_exchange_groups — Insufficient data: external/lvgl/docs/src/details/main-modules/indev/groups.rst line 4: not in the algorithm registry, not assessed: =====
- tls_key_exchange_groups — Insufficient data: external/lvgl/scripts/generate_lv_conf.py line 67: not in the algorithm registry, not assessed: re.search(r'([A-Z0-9_]+)\s+(.+)'
- tls_key_exchange_groups — Insufficient data: external/lvgl/scripts/update_version.py line 28: not in the algorithm registry, not assessed: re.search(self.RE_PATTERN
- osv_advisories — not checked: 3 package(s) are built from git repositories (forks or unreleased code), so registry advisories were not looked up for them: bip32-ed25519, bip39, rtt-target

Remediation plan

QSI-2026-17830 · P1 · 30 days root-cause fix

Remove certificate-verification overrides in scripts/generate_version_headers.py.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible compatibility impact on old clients.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	No verification overrides remain outside explicitly marked test code.

QSI-2026-17801 · P1 · 30 days root-cause fix

Use an AEAD mode (GCM) with unique nonces.

Rationale	ECB reveals patterns in encrypted data and provides no integrity.
Expected outcome	Re-scan the source and review the call sites listed in the evidence.
Complexity	medium
Tests before change	Open line 81 in the supplied file external/samd51a-ds/include/component/aes.h and inspect the quoted lines (see evidence).
Tests after change	Re-scan the source and review the call sites listed in the evidence.
Closure criteria	Re-scan the source and review the call sites listed in the evidence.

QSI-2026-17843 · P1 · 30 days root-cause fix

Use an AEAD mode (GCM) with unique nonces.

Rationale	ECB reveals patterns in encrypted data and provides no integrity.
Expected outcome	Re-scan the source and review the call sites listed in the evidence.
Complexity	medium
Tests before change	Open line 184 in the supplied file external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h and inspect the quoted lines (see evidence).
Tests after change	Re-scan the source and review the call sites listed in the evidence.
Closure criteria	Re-scan the source and review the call sites listed in the evidence.

QSI-2026-17854 · P2 · 30 days root-cause fix

Upgrade certifi to a fixed version (2026.10.18).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.

Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-6q48-6c7w-9gw2 (PYSEC-2026-4320).

QSI-2026-17855 · P2 · 30 days

root-cause fix

Upgrade certifi to a fixed version (2023.7.22).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-xqr8-7jwr-rhp7 (CVE-2023-37920, PYSEC-2023-135).

QSI-2026-17859 · P2 · 30 days

root-cause fix

Upgrade jinja2 to a fixed version (3.1.5).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-gmj6-6f8f-6699 (CVE-2024-56201, PYSEC-2026-1472).

QSI-2026-17862 · P2 · 30 days

root-cause fix

Upgrade jinja2 to a fixed version (3.1.5).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-q2x7-8rv6-6q7h (CVE-2024-56326, PYSEC-2026-1475).

QSI-2026-17869 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (2.6.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-2xpw-w6gg-jr37 (CVE-2025-66471, PYSEC-2026-1994).

QSI-2026-17871 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (2.6.3).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-38jv-5279-wg99 (CVE-2026-21441, PYSEC-2026-1996).

QSI-2026-17872 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (2.8.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-8988-9cw3-xx77 (CVE-2026-97687, PYSEC-2026-4175).

QSI-2026-17874 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (2.6.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low

Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-gm62-xv2j-4w53 (CVE-2025-66418, PYSEC-2026-1998).

QSI-2026-17878 · P2 · 30 days

root-cause fix

Upgrade urllib3 to a fixed version (2.8.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-vxq7-64xx-v4gw (CVE-2026-97689, PYSEC-2026-4177).

QSI-2026-17856 · P2 · 30 days

root-cause fix

Upgrade idna to a fixed version (3.15).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-65pc-fj4g-8rjx (CVE-2026-45409, PYSEC-2026-215).

QSI-2026-17857 · P2 · 30 days

root-cause fix

Upgrade idna to a fixed version (3.7).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports GHSA-jjg7-2v4v-x38h (CVE-2024-3651, PYSEC-2024-60).

QSI-2026-17858 · P2 · 30 days

root-cause fix

Upgrade jinja2 to a fixed version (3.1.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-cpwx-vrp4-4pq7 (CVE-2025-27516, PYSEC-2026-1471).

QSI-2026-17860 · P2 · 30 days

root-cause fix

Upgrade jinja2 to a fixed version (3.1.3).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-h5c8-rqwp-cp95 (CVE-2024-22195, PYSEC-2026-1473).

QSI-2026-17861 · P2 · 30 days

root-cause fix

Upgrade jinja2 to a fixed version (3.1.4).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-h75v-3vvj-5mfj (CVE-2024-34064, PYSEC-2026-1474).

QSI-2026-17864 · P2 · 30 days

root-cause fix

Upgrade pygments to a fixed version (2.15.0).

Rationale	Eliminates the root cause described in the finding.

Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-mrwq-x4v8-fh7p (CVE-2022-40896, PYSEC-2023-117).

QSI-2026-17865 · P2 · 30 days

root-cause fix

Upgrade requests to a fixed version (2.32.4).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-9hjg-9r4m-mvj7 (CVE-2024-47081, PYSEC-2026-1872).

QSI-2026-17866 · P2 · 30 days

root-cause fix

Upgrade requests to a fixed version (2.32.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-9wx4-h78v-vm56 (CVE-2024-35195, PYSEC-2026-1873).

QSI-2026-17867 · P2 · 30 days

root-cause fix

Upgrade requests to a fixed version (2.33.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.

Closure criteria	Dependency scan no longer reports GHSA-gc5v-m9x4-r6x2 (CVE-2026-25645, PYSEC-2026-2275).

QSI-2026-17868 · P2 · 30 days

root-cause fix

Upgrade requests to a fixed version (2.31.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-j8r2-6x86-q33q (CVE-2023-32681, PYSEC-2023-74).

QSI-2026-17870 · P2 · 30 days

root-cause fix

Upgrade urllib3 to a fixed version (1.26.19, 2.2.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-34jh-p97f-mpxf (CVE-2024-37891, PYSEC-2026-1995).

QSI-2026-17873 · P2 · 30 days

root-cause fix

Upgrade urllib3 to a fixed version (1.26.18, 2.0.7).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-g4mx-q9vg-27p4 (CVE-2023-45803, PYSEC-2023-212).

QSI-2026-17875 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (2.5.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-pq67-6m6q-mj2v (CVE-2025-50181, PYSEC-2026-1999).

QSI-2026-17876 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (2.7.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-qccp-gfcp-xxvc (CVE-2026-44431, PYSEC-2026-141).

QSI-2026-17877 · P2 · 30 days root-cause fix

Upgrade urllib3 to a fixed version (1.26.17, 2.0.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-v845-jxx5-vc9f (CVE-2023-43804, PYSEC-2023-192).

QSI-2026-17879 · P2 · 30 days root-cause fix

Upgrade zipp to a fixed version (3.19.1).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low

Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-jfmj-5v4g-7637 (CVE-2024-5569, PYSEC-2026-2074).

QSI-2026-17852 · P2 · 30 days

root-cause fix

Upgrade bytes to a fixed version (1.11.1).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007).

QSI-2026-17853 · P2 · 30 days

root-cause fix

Upgrade certifi to a fixed version (2024.7.4).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-248v-346w-9cwc (CVE-2024-39689, PYSEC-2024-230).

QSI-2026-17863 · P2 · 30 days

root-cause fix

Upgrade pygments to a fixed version (2.20.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports GHSA-5239-wwwm-4pmq (CVE-2026-4539, PYSEC-2026-2987).

QSI-2026-17849 · P2 · 30 days

root-cause fix

Upgrade keccak to a fixed version (0.1.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012).

QSI-2026-17850 · P2 · 30 days

root-cause fix

Upgrade time to a fixed version (0.3.47).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-r6v5-fh4h-64xc (CVE-2026-25727, RUSTSEC-2026-0009).

QSI-2026-17841 · P2 · 60 days

root-cause fix

Use a CSPRNG (secrets, crypto.getRandomValues, crypto/rand).

Rationale	Predictable values may allow token or key guessing.
Expected outcome	Re-scan the source and review the call sites listed in the evidence.
Complexity	medium
Tests before change	Open line 2 in the supplied file external/lvgl/docs/src/_static/jquery.js and inspect the quoted lines (see evidence).
Tests after change	Re-scan the source and review the call sites listed in the evidence.
Closure criteria	Re-scan the source and review the call sites listed in the evidence.

QSI-2026-17848 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.

Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0387, or the triage is recorded.

QSI-2026-17846 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0190, or the triage is recorded.

QSI-2026-17851 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0190, or the triage is recorded.

QSI-2026-17844 · P3 · 90 days

short-term risk reduction

Add ECDSA, Ed25519, Schnorr (BIP-340) usages at the wallet's transaction signing to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.

Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17804 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/JuliusWiedemann.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17805 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/JunghoonLee.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17806 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/My1.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).

Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17807 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/My1.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17808 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/TheCharlatan.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17809 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/beerosagos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.

Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17810 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/benma.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17811 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/brianddk.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17812 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/cstenglein.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17813 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/frknakk.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17814 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/guggerf.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17815 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/jonathancross.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17816 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/jonathancross.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17817 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/nickez.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17818 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/offbyn.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17819 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at file:releases/pubkeys/psteiger.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17820 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/stadicus.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17821 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/strnbrg.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17822 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/sutterseba.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17823 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/tomasvrba.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17824 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/userofreddit44.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17825 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/x1ddos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17826 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/x1ddos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17827 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/x1ddos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17828 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at file:releases/pubkeys/yaziza.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17829 · P3 · 90 days

short-term risk reduction

Add RSA usages at file:releases/pubkeys/yaziza.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17838 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at file:external/cryptoauthlib/app/pkcs11/README.md to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17831 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17833 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17840 · P3 · 90 days

short-term risk reduction

Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17803 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17832 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17835 · P3 · 90 days

short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17836 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17802 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17834 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17837 · P3 · 90 days

short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17839 · P3 · 90 days

short-term risk reduction

Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17842 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17844 · P4 · 90 days

root-cause fix

Plan migration of ECDSA, Ed25519, Schnorr (BIP-340) to ? (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ? or a hybrid scheme in use.

QSI-2026-17804 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17805 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17806 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17807 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17808 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.

Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17809 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17810 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17811 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.

Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17812 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17813 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17814 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high

Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17815 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17816 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17817 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.

Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17818 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17819 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17820 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17821 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17822 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17823 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17824 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17825 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.

Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17826 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17827 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17828 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.

Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17829 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17838 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-17831 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high

Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17833 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-17840 · P4 · 90 days

root-cause fix

Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-17803 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17832 · P4 · 90 days root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-17835 · P4 · 90 days root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-17836 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17802 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17834 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	

	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17837 · P4 · 90 days

root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-17839 · P4 · 90 days

root-cause fix

Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-17842 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-17847 · P4 · 90 days

root-cause fix

Replace bare-metal with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	bare-metal replaced, or the decision recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P1	QSI-2026-17830	Remove certificate-verification overrides in scripts/generate_version_headers.py.	root-cause fix
P1	QSI-2026-17801	Use an AEAD mode (GCM) with unique nonces.	root-cause fix
P1	QSI-2026-17843	Use an AEAD mode (GCM) with unique nonces.	root-cause fix
P2	QSI-2026-17854	Upgrade certifi to a fixed version (2026.10.18).	root-cause fix
P2	QSI-2026-17855	Upgrade certifi to a fixed version (2023.7.22).	root-cause fix
P2	QSI-2026-17859	Upgrade jinja2 to a fixed version (3.1.5).	root-cause fix
P2	QSI-2026-17862	Upgrade jinja2 to a fixed version (3.1.5).	root-cause fix
P2	QSI-2026-17869	Upgrade urllib3 to a fixed version (2.6.0).	root-cause fix
P2	QSI-2026-17871	Upgrade urllib3 to a fixed version (2.6.3).	root-cause fix
P2	QSI-2026-17872	Upgrade urllib3 to a fixed version (2.8.0).	root-cause fix
P2	QSI-2026-17874	Upgrade urllib3 to a fixed version (2.6.0).	root-cause fix
P2	QSI-2026-17878	Upgrade urllib3 to a fixed version (2.8.0).	root-cause fix
P2	QSI-2026-17856	Upgrade idna to a fixed version (3.15).	root-cause fix
P2	QSI-2026-17857	Upgrade idna to a fixed version (3.7).	root-cause fix
P2	QSI-2026-17858	Upgrade jinja2 to a fixed version (3.1.6).	root-cause fix
P2	QSI-2026-17860	Upgrade jinja2 to a fixed version (3.1.3).	root-cause fix
P2	QSI-2026-17861	Upgrade jinja2 to a fixed version (3.1.4).	root-cause fix
P2	QSI-2026-17864	Upgrade pygments to a fixed version (2.15.0).	root-cause fix
P2	QSI-2026-17865	Upgrade requests to a fixed version (2.32.4).	root-cause fix

P2	QSI-2026-17866	Upgrade requests to a fixed version (2.32.0).	root-cause fix
P2	QSI-2026-17867	Upgrade requests to a fixed version (2.33.0).	root-cause fix
P2	QSI-2026-17868	Upgrade requests to a fixed version (2.31.0).	root-cause fix
P2	QSI-2026-17870	Upgrade urllib3 to a fixed version (1.26.19, 2.2.2).	root-cause fix
P2	QSI-2026-17873	Upgrade urllib3 to a fixed version (1.26.18, 2.0.7).	root-cause fix
P2	QSI-2026-17875	Upgrade urllib3 to a fixed version (2.5.0).	root-cause fix
P2	QSI-2026-17876	Upgrade urllib3 to a fixed version (2.7.0).	root-cause fix
P2	QSI-2026-17877	Upgrade urllib3 to a fixed version (1.26.17, 2.0.6).	root-cause fix
P2	QSI-2026-17879	Upgrade zipp to a fixed version (3.19.1).	root-cause fix
P2	QSI-2026-17852	Upgrade bytes to a fixed version (1.11.1).	root-cause fix
P2	QSI-2026-17853	Upgrade certifi to a fixed version (2024.7.4).	root-cause fix
P2	QSI-2026-17863	Upgrade pygments to a fixed version (2.20.0).	root-cause fix
P2	QSI-2026-17849	Upgrade keccak to a fixed version (0.1.6).	root-cause fix
P2	QSI-2026-17850	Upgrade time to a fixed version (0.3.47).	root-cause fix

60 days

Priority	ID	Action	Type
P2	QSI-2026-17841	Use a CSPRNG (secrets, crypto.getRandomValues, crypto/rand).	root-cause fix
P3	QSI-2026-17848	Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.	root-cause fix
P3	QSI-2026-17846	Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.	root-cause fix
P3	QSI-2026-17851	Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type

P3	QSI-2026-17844	Add ECDSA, Ed25519, Schnorr (BIP-340) usages at the wallet's transaction signing to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17804	Add RSA usages at file:releases/pubkeys/JuliusWiedemann.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17805	Add RSA usages at file:releases/pubkeys/JunghoonLee.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17806	Add RSA usages at file:releases/pubkeys/My1.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17807	Add RSA usages at file:releases/pubkeys/My1.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17808	Add RSA usages at file:releases/pubkeys/TheCharlatan.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17809	Add RSA usages at file:releases/pubkeys/beerosagos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17810	Add RSA usages at file:releases/pubkeys/benma.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17811	Add RSA usages at file:releases/pubkeys/brianddk.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17812	Add RSA usages at file:releases/pubkeys/cstenglein.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17813	Add RSA usages at file:releases/pubkeys/frknakk.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17814	Add RSA usages at file:releases/pubkeys/guggerf.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17815	Add RSA usages at file:releases/pubkeys/jonathancross.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17816	Add RSA usages at file:releases/pubkeys/jonathancross.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P3	QSI-2026-17817	Add RSA usages at file:releases/pubkeys/nickez.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17818	Add RSA usages at file:releases/pubkeys/offbyn.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17819	Add Ed25519 usages at file:releases/pubkeys/psteiger.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17820	Add RSA usages at file:releases/pubkeys/stadicus.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17821	Add RSA usages at file:releases/pubkeys/strnbrg.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17822	Add RSA usages at file:releases/pubkeys/sutterseba.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17823	Add RSA usages at file:releases/pubkeys/tomasvrba.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17824	Add RSA usages at file:releases/pubkeys/userofreddit44.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17825	Add RSA usages at file:releases/pubkeys/x1ddos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17826	Add RSA usages at file:releases/pubkeys/x1ddos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17827	Add RSA usages at file:releases/pubkeys/x1ddos.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17828	Add Ed25519 usages at file:releases/pubkeys/yaziza.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17829	Add RSA usages at file:releases/pubkeys/yaziza.asc to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17838	Add ECDSA usages at file:external/cryptoauthlib/app/pkcs11/README.md to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P3	QSI-2026-17831	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17833	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17840	Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17803	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17832	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17835	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17836	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17802	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17834	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17837	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17839	Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17842	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-17844	Plan migration of ECDSA, Ed25519, Schnorr (BIP-340) to ? (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17804	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17805	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-17806	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17807	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17808	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17809	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17810	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17811	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17812	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17813	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17814	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17815	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17816	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17817	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17818	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17819	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-17820	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17821	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17822	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17823	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17824	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17825	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17826	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17827	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17828	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17829	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17838	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17831	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17833	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17840	Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-17803	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17832	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17835	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17836	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17802	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17834	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17837	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17839	Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17842	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17847	Replace bare-metal with a maintained alternative or record why it stays.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-17854	Re-run the dependency scan after upgrading.
QSI-2026-17830	Re-scan the source and review the call sites listed in the evidence.
QSI-2026-17855	Re-run the dependency scan after upgrading.
QSI-2026-17859	Re-run the dependency scan after upgrading.
QSI-2026-17862	Re-run the dependency scan after upgrading.
QSI-2026-17869	Re-run the dependency scan after upgrading.
QSI-2026-17871	Re-run the dependency scan after upgrading.
QSI-2026-17872	Re-run the dependency scan after upgrading.
QSI-2026-17874	Re-run the dependency scan after upgrading.
QSI-2026-17878	Re-run the dependency scan after upgrading.
QSI-2026-17801	Re-scan the source and review the call sites listed in the evidence.
QSI-2026-17843	Re-scan the source and review the call sites listed in the evidence.
QSI-2026-17844	Re-run when a chain activates a post-quantum signature scheme or the wallet adds one.
QSI-2026-17804	Re-audit after migration.
QSI-2026-17805	Re-audit after migration.
QSI-2026-17806	Re-audit after migration.
QSI-2026-17807	Re-audit after migration.
QSI-2026-17808	Re-audit after migration.
QSI-2026-17809	Re-audit after migration.
QSI-2026-17810	Re-audit after migration.
QSI-2026-17811	Re-audit after migration.
QSI-2026-17812	Re-audit after migration.
QSI-2026-17813	Re-audit after migration.
QSI-2026-17814	Re-audit after migration.
QSI-2026-17815	Re-audit after migration.
QSI-2026-17816	Re-audit after migration.
QSI-2026-17817	Re-audit after migration.
QSI-2026-17818	Re-audit after migration.
QSI-2026-17819	Re-audit after migration.
QSI-2026-17820	Re-audit after migration.

QSI-2026-17821	Re-audit after migration.
QSI-2026-17822	Re-audit after migration.
QSI-2026-17823	Re-audit after migration.
QSI-2026-17824	Re-audit after migration.
QSI-2026-17825	Re-audit after migration.
QSI-2026-17826	Re-audit after migration.
QSI-2026-17827	Re-audit after migration.
QSI-2026-17828	Re-audit after migration.
QSI-2026-17829	Re-audit after migration.
QSI-2026-17838	Re-audit after migration.
QSI-2026-17856	Re-run the dependency scan after upgrading.
QSI-2026-17857	Re-run the dependency scan after upgrading.
QSI-2026-17858	Re-run the dependency scan after upgrading.
QSI-2026-17860	Re-run the dependency scan after upgrading.
QSI-2026-17861	Re-run the dependency scan after upgrading.
QSI-2026-17864	Re-run the dependency scan after upgrading.
QSI-2026-17865	Re-run the dependency scan after upgrading.
QSI-2026-17866	Re-run the dependency scan after upgrading.
QSI-2026-17867	Re-run the dependency scan after upgrading.
QSI-2026-17868	Re-run the dependency scan after upgrading.
QSI-2026-17870	Re-run the dependency scan after upgrading.
QSI-2026-17873	Re-run the dependency scan after upgrading.
QSI-2026-17875	Re-run the dependency scan after upgrading.
QSI-2026-17876	Re-run the dependency scan after upgrading.
QSI-2026-17877	Re-run the dependency scan after upgrading.
QSI-2026-17879	Re-run the dependency scan after upgrading.
QSI-2026-17831	Re-run the inventory after refactoring.
QSI-2026-17833	Re-run the inventory after refactoring.
QSI-2026-17840	Re-run the inventory after refactoring.
QSI-2026-17803	Re-run the inventory after refactoring.
QSI-2026-17832	Re-run the inventory after refactoring.
QSI-2026-17835	Re-run the inventory after refactoring.
QSI-2026-17836	Re-run the inventory after refactoring.

QSI-2026-17802	Re-run the inventory after refactoring.
QSI-2026-17834	Re-run the inventory after refactoring.
QSI-2026-17837	Re-run the inventory after refactoring.
QSI-2026-17839	Re-run the inventory after refactoring.
QSI-2026-17842	Re-run the inventory after refactoring.
QSI-2026-17852	Re-run the dependency scan after upgrading.
QSI-2026-17841	Re-scan the source and review the call sites listed in the evidence.
QSI-2026-17853	Re-run the dependency scan after upgrading.
QSI-2026-17863	Re-run the dependency scan after upgrading.
QSI-2026-17848	Re-run the dependency scan after upgrading.
QSI-2026-17849	Re-run the dependency scan after upgrading.
QSI-2026-17850	Re-run the dependency scan after upgrading.
QSI-2026-17846	Re-run the dependency scan after upgrading.
QSI-2026-17851	Re-run the dependency scan after upgrading.
QSI-2026-17845	Re-scan after upgrades.
QSI-2026-17847	Re-run the dependency scan after the change.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
Blowfish		weak classically	legacy	1	1	external/optiga-trust-m/external/mbedtls/include/mbedtls/blowfish.h
RC4		weak classically	broken	1	1	external/optiga-trust-m/external/mbedtls/include/mbedtls/arc4.h
RIPEMD-160	RIPEMD-160	weak classically	legacy	1	1	src/rust/bitbox02-rust/src/hww/api/bitcoin/policies.rs:51
SHA-1	SHA-1	weak classically	deprecated	5	5	external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha1.c; external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha1.h; external/cryptoauthlib/lib/crypto/hashes/sha1_routines.c
CMAC		reduced margin (Grover), approved	acceptable	3	3	external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_cmac.c; external/cryptoauthlib/lib/openssl/atca_openssl_interface.c:442; external/optiga-trust-m/external/mbedtls/include/mbedtls/cmac.h
Camellia		reduced margin (Grover), approved	acceptable	1	1	external/optiga-trust-m/external/mbedtls/include/mbedtls/camellia.h
ChaCha20-Poly1305		reduced margin (Grover), approved	acceptable	1	1	external/optiga-trust-m/external/mbedtls/include/mbedtls/chacha20.h
ChaCha20-Poly1305	ChaCha20-Poly1305	reduced margin (Grover), approved	acceptable	3	3	py/send_message.py:1778; src/rust/bitbox-noise/src/noise_xx.rs:9; src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:549
HKDF		reduced margin (Grover), approved	acceptable	1	1	src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:12
HMAC		reduced margin (Grover), approved	acceptable	11	11	external/cryptoauthlib/lib/calib/calib_hmac.c; external/cryptoauthlib/lib/openssl/atca_openssl_interface.c:512; external/cryptoauthlib/python/cryptoauthlib/atjwt.py:95

HMAC	HMAC-SHA-256	reduced margin (Grover), approved	acceptable	1	1	src/rust/bitbox02/src/securechip/imp_fake.rs:16
SHA-2	SHA-512	reduced margin (Grover), approved	acceptable	1	1	src/rust/bitbox02-rust/src/keystore/ed25519.rs:85
AES-128	AES-128	Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	4	4	external/cryptoauthlib/lib/mbedtls/atca_mbedtls_wrap.c:100; external/cryptoauthlib/lib/openssl/atca_openssl_interface.c:83; external/optiga-trust-m/external/mbedtls/configs/config-suite-b.h:90
Keccak-256	Keccak-256	Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	4	4	src/rust/bitbox02-rust/src/hww/api/ethereum/address.rs:17; src/rust/bitbox02-rust/src/hww/api/ethereum/sighash.rs:3; src/rust/bitbox02-rust/src/hww/api/ethereum/sign_typed_msg.rs:274
SHA-2		Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	4	4	external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha2.c; external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha2.h; external/cryptoauthlib/lib/crypto/hashes/sha2_routines.h
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	26	26	external/cryptoauthlib/lib/crypto/hashes/sha2_routines.c:52; py/bitbox02/bitbox02/bitbox02/bootloader.py:268; py/bitbox02/bitbox02/bitbox02/secp256k1.py:18
CBC mode		n/a	acceptable	1	1	external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_cbc.c
CCM mode		n/a	acceptable	4	4	external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_ccm.c; external/optiga-trust-m/external/mbedtls/configs/config-ccm-psk-dtls1_2.h; external/optiga-trust-m/external/mbedtls/configs/config-ccm-psk-tls1_2.h
CTR mode		n/a	acceptable	2	2	external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_ctr.c; external/optiga-trust-m/external/mbedtls/include/mbedtls/ctr_drbg.h

Crypto library p256 (RustCrypto elliptic curves)	0.13.2	n/a	acceptable	2	2	src/rust/Cargo.lock; src/rust/util/Cargo.toml
Crypto library ecdsa (python-ecdsa)	0.16.9	n/a	acceptable	1	1	src/rust/Cargo.lock
Crypto library secp256k1 (secp256k1 bindings)	0.29.0	n/a	acceptable	1	1	src/rust/Cargo.lock
Crypto library x25519-dalek (ed25519-dalek)	2.0.0	n/a	acceptable	3	3	src/rust/Cargo.lock; src/rust/bitbox-noise/Cargo.toml; src/rust/bitbox02-rust/Cargo.toml
Crypto library ed25519-dalek (ed25519-dalek)	2.1.1	n/a	acceptable	2	2	src/rust/Cargo.lock; src/rust/bitbox02-rust/Cargo.toml
Crypto library curve25519-dalek (ed25519-dalek)	4.1.3	n/a	acceptable	1	1	src/rust/Cargo.lock
GCM mode		n/a	acceptable	2	2	external/cryptoauthlib/lib/calib/calib_aes_gcm.c; external/cryptoauthlib/lib/calib/calib_aes_gcm.h
TLS 1.1		n/a	deprecated	1	1	external/optiga-trust-m/external/mbedtls/configs/config-mini-tls1_1.h
TLS 1.2		n/a	acceptable	1	1	external/optiga-trust-m/external/mbedtls/configs/config-ccm-psk-tls1_2.h
ECDH		vulnerable (Shor)	acceptable	6	6	external/cryptoauthlib/lib/calib/calib_command.h:181; external/cryptoauthlib/lib/calib/calib_ecdh.c:159; external/cryptoauthlib/lib/mbedtls/atca_mbedtls_ecdh.c

ECDSA	secp256r1	vulnerable (Shor)	acceptable	1	1	external/cryptoauthlib/app/pkcs11/README.md (line 280)
ECDSA		vulnerable (Shor)	acceptable	13	13	external/cryptoauthlib/lib/crypto/atca_crypto_sw_ecdsa.c; external/cryptoauthlib/lib/crypto/atca_crypto_sw_ecdsa.h; external/cryptoauthlib/lib/mbedtls/atca_mbedtls_ecdsa.c:51
ECDSA	secp256k1	vulnerable (Shor)	acceptable	21	21	messages/btc.proto:4; messages/eth.proto:4; py/bitbox02/bitbox02/bitbox02/bitbox02.py:21
ECDSA	P-256	vulnerable (Shor)	acceptable	1	1	src/rust/util/src/p256.rs:28
Ed25519	BIP32-Ed25519	vulnerable (Shor)	acceptable	1	1	messages/cardano.proto:4
Ed25519	Ed25519	vulnerable (Shor)	acceptable	10	10	releases/pubkeys/psteiger.asc (line 1); releases/pubkeys/yaziza.asc (line 1); src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs:11
RSA	4096-bit	vulnerable (Shor)	acceptable	33	33	releases/pubkeys/JuliusWiedemann.asc (line 1); releases/pubkeys/JunghoonLee.asc (line 1); releases/pubkeys/My1.asc (line 1)
RSA	2048-bit	vulnerable (Shor)	acceptable	5	5	releases/pubkeys/My1.asc (line 1); releases/pubkeys/brianddk.asc (line 1); releases/pubkeys/jonathancross.asc (line 1)
RSA	3072-bit	vulnerable (Shor)	acceptable	3	3	releases/pubkeys/userofreddit44.asc (line 1); releases/pubkeys/x1ddos.asc (line 1)
Schnorr (BIP-340)	secp256k1, Taproot	vulnerable (Shor)	acceptable	1	1	messages/btc.proto:4
Schnorr (BIP-340)		vulnerable (Shor)	acceptable	3	3	src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs:289; src/rust/bitbox02-rust/src/keystore.rs:897; src/rust/streaming-silent-payments/src/lib.rs:12
X25519	X25519	vulnerable (Shor)	acceptable	14	14	external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h:18; external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/everest.h:25; external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/vs2010/Hacl_Curve25519.h:18
X25519		vulnerable (Shor)	acceptable	1	1	external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/Hacl_Curve25519_joined.c

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 cf3c23870e4b4c35feb21d303d5082fae1730b014e64646954327f37e2becf0c

C. Input files

File	Size	SHA-256
.cargo/config.toml	943	b3cb95e3866ea74ba9a0e56cb71a0bf3564d4242f1c06aa751705
external/CMSIS/Include/cmsis_armclang.h	20706	f7361b36a56909eff2b1cecd778443fd9da21eaa86878dabf5e8e
external/CMSIS/Include/cmsis_clang.h	20700	015fd70aaca6475443082154b7c844d6db58e757b2dbe3fa3bd26
external/CMSIS/Include/cmsis_compiler.h	9526	ef7a1f9cb34ac7731500ab17edb7a9516a9d29e984a23da90b9b1
external/CMSIS/Include/cmsis_gcc.h	30171	8fd513058bc3d10f1b3929c648023964dd32f375d5be39ad31a13
external/CMSIS/Include/cmsis_version.h	1936	3b9195c5373a2a4dbfd968b083b3d7f5f446f9ec2e126f75928ac
external/CMSIS/Include/core_cm4.h	125284	4dc3fd737b06d5cd228260815756c3236d49ad68a2efd57c939c4
external/CMSIS/Include/m-profile/armv7m_cachel1.h	13032	f7905d3e336feccda0647f214e791b424c5f92e788f1303e587ce
external/CMSIS/Include/m-profile/armv7m_mpu.h	11458	44294dc3e331deb8daae269793e579cc10920a9b88d133bbaf3a7
external/CMSIS/Include/m-profile/armv81m_pac.h	5832	fe2f5db52980237fe8bafe265b660ef9caac90bae4c478d34f243
external/CMSIS/Include/m-profile/armv8m_mpu.h	13978	287fcabda261890f568009423ab2beb163bdcc9af591b4ce9e28f
external/CMSIS/Include/m-profile/armv8m_pmu.h	22537	1323587da9fea03c96a685a9cf60c576b4d0532b3339501401332
external/CMSIS/Include/m-profile/cmsis_armclang_m.h	23616	b8257289970f5ede78fe52cbbbb1a0575a1654968067a9e8043a4
external/CMSIS/Include/m-profile/cmsis_clang_m.h	23632	d735406717f198ddf2636858bcc56c448d3e2bbd91b3e70e7bfb0
external/CMSIS/Include/m-profile/cmsis_gcc_m.h	20714	beee8eb228280fb29b41438e98de4afb4e5f265ddb3cabf5d4be6
external/CMSIS/Include/m-profile/cmsis_iccarm_m.h	30666	6307b35bc85f3f413e6ad1d200da6baa31851ce21d13f439ef6e6
external/CMSIS/Include/m-profile/cmsis_tiarmclang_m.h	41815	c39b3c60adbcf55327d8f01d3d1a78a5e83c09f73396a48960d9b
external/CMSIS/Include/tz_context.h	2452	2bcc2a5fea0567eff96dd3f2fd197b88feaa144bfe0df6146abd8
external/ST/Common/Inc/stm32u5xx_hal_conf.h	19574	70eba04baadf4b8dcb13c2a9c71387e004a9a828d843700832b5e
external/ST/Drivers/CMSIS/Device/ST/STM32U5xx/Include/stm32u5a9xx.h	2467242	26c12881f0c94386f5fd8ace417bf41ea05ccb7deb2141dfe4f2b
external/ST/Drivers/CMSIS/Device/ST/STM32U5xx/Include/stm32u5xx.h	10767	0f5a0e434a7b081836f69648d9fc94f58bbfd00cb76a4fae5b411
external/ST/Drivers/CMSIS/Device/ST/STM32U5xx/Include/system_stm32u5xx.h	2713	73675e7a1dfd4a6378f08f6dce3ec072d6244f70365276ef52704
external/ST/Drivers/CMSIS/Include/cachel1_armv7.h	13255	b211a8bbcfd62c9f0b3b3f1674927c9b58a0d95b02e449a0866ad
external/ST/Drivers/CMSIS/Include/cmsis_armcc.h	27999	5bc0d7ed8e36e9bcb868c4042fff350c702eca305df4162fb0038

external/ST/Drivers/CMSIS/Include/cmsis_armclang.h	48417	d5da2b48204f0d0cf1ff38ffccb527e5706cd0a091ff0a331eb41
external/ST/Drivers/CMSIS/Include/cmsis_armclang_ltm.h	56021	3e8ffb284f4f155de3ea93841cca853b41d5e812b53e52c64c3d4
external/ST/Drivers/CMSIS/Include/cmsis_compiler.h	9481	b51963d271c1571ca3463654c76aa7ea50c4eac5d0a2710df1041
external/ST/Drivers/CMSIS/Include/cmsis_gcc.h	63374	43bfd1fe69fbbc2ca70aaf7d43cc3e20f16e0c7f66e1bdef9b51a
external/ST/Drivers/CMSIS/Include/cmsis_iccarm.h	29069	2d255c8429e507e52cb603f2970bed4cc4a6cdcc3eb7fc7997385
external/ST/Drivers/CMSIS/Include/cmsis_version.h	1680	184c19fd3ee73632edf35a0b4d49cd48be75fbf49e6ccb19d9db0
external/ST/Drivers/CMSIS/Include/core_armv81mml.h	276640	a40954bccbf202bd65ee721ea688122561e005c9f9f9d18e08228
external/ST/Drivers/CMSIS/Include/core_armv8mbl.h	115004	458b1e6788079ded52adc7d06383701c865ca97b67c5e7a0a4488
external/ST/Drivers/CMSIS/Include/core_armv8mml.h	183191	5a68204fa31cda7d04d41c9ceaf3f4b1217908f59366f757ee778
external/ST/Drivers/CMSIS/Include/core_cm0.h	41643	ac13cb645f108666a0e57c93467872660d6bc860a839f840dc870
external/ST/Drivers/CMSIS/Include/core_cm0plus.h	49877	a09fa9c6b8b21d1920bce15b98aa56182b027e9278cca0b154812
external/ST/Drivers/CMSIS/Include/core_cm1.h	42626	944bf0c4f940eef30a632cd99d5dcfe8270e8a8431eb98c0f081b
external/ST/Drivers/CMSIS/Include/core_cm23.h	121556	77822e9738885394ec52fca6f617452b823a0b55e2664ee225e65
external/ST/Drivers/CMSIS/Include/core_cm3.h	109529	c9937781259a8897adf514445441c68570087d6a72a558d8a4afe
external/ST/Drivers/CMSIS/Include/core_cm33.h	189524	225c47c36230edaa314f1e7a45b38279f6ab4d58faeb132226db3
external/ST/Drivers/CMSIS/Include/core_cm35p.h	189539	206b427589981b78169c1d9b73e898e4e8de4efbd76fd0c2df7df
external/ST/Drivers/CMSIS/Include/core_cm4.h	120975	f5b63d52dd1557b15ca414cb59c264f595a7b5669db2a5878dffe
external/ST/Drivers/CMSIS/Include/core_cm55.h	317450	5707f716cf03f7e80151108ae94601fcfdf362dd2d83087604021
external/ST/Drivers/CMSIS/Include/core_cm7.h	138948	e974d0a91752487d0154457cd422c37667328e9ddb7ba4fa2fed4
external/ST/Drivers/CMSIS/Include/core_cm85.h	308328	75bc123889f0f5ca33c4d16b6cbdd580a3572663c899fadb2c8e2
external/ST/Drivers/CMSIS/Include/core_sc000.h	46562	c81377a69a4ea8ecaead42194f2ce8ab59bd4ee7bd638e423f5b1
external/ST/Drivers/CMSIS/Include/core_sc300.h	108524	73dd461c12c7d3a693840df3084fd8f85122095c6f21fa3d779b0
external/ST/Drivers/CMSIS/Include/core_starmc1.h	193055	92d8661f2947d00da91653d04794cde0593fdaafb3438842a0477
external/ST/Drivers/CMSIS/Include/mpu_armv7.h	11731	29206b52ee02290ed6f5a5415ebd4187de802cf176d9b3cb84439
external/ST/Drivers/CMSIS/Include/mpu_armv8.h	11370	2c7d22a344cedcc1e35380d1f76f0e6c55da5e3101e6a41fd8549
external/ST/Drivers/CMSIS/Include/pac_armv81.h	6062	d76f5cc0c59a59e035115f700b8e8d95b7b98ecb99a2924e885f8
external/ST/Drivers/CMSIS/Include/pmu_armv8.h	22786	cea8c105c5f41ba831fe4d9e8e39758068cd4c414383348dc2835
external/ST/Drivers/CMSIS/Include/tz_context.h	2687	45565c8cb04c653aabaf3cceddfc848b6a0de28c20d4f7e3d475a
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/Legacy/stm32_hal_legacy.h	240941	b23aaf341a5f9102a3030c1087cac535f369ce6bd84380f4f4fa1
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal.h	39344	be0aa206bbdbd4be224282ab540aa1b2545b41981b61893ca9c28

external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_cortex.h	16590	d839d550236888f60248d33a19ed31a5fe0bece6a6dde6b2dbf9e
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_def.h	7638	73a911c835b4c7f75b7d4c7e47847f9776c5b55dc265bb59c17df
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_dma.h	45368	0d4f824e3037433289958fb4d590807e04390bd707b503844286d
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_dma_ex.h	38799	5465ca6acdf8fce2f81d9efebfc131bcd9ad2a7fa7debfabd9e06
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_exti.h	15761	9fa3b01b38b1b9376e4ac9a6f4985e319c08420b67b229db919e8
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_flash.h	56604	e59f8e71dee58b4b1cb40f9e35e14fd895f05bc3c151b8909a58c
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_flash_ex.h	8978	8943fc2f847e40a811a27ccac5628b2b7ed19f40ab4b1dc30a813
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_gpio.h	16246	6d9e6c1abe6303f3e72f488c21fe2a1272a033a8309e6b8da4a3f
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_gpio_ex.h	16959	40848fe334a9252b50e7a941537312443ba0830ef0a3f2a2c2831
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_gtzc.h	26421	00906371d13f23982d72d85765aa40a8be03fd6e68e95d65938c1
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_i2c.h	39289	a6c5c538d478ec7ae8d4e8813a1086cbebcd45396b7e316eb507e
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_i2c_ex.h	13411	b3db3fb637456e6076b7c253226c8fe3f204e2c6bb96ed1e65f90
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_icache.h	9552	9c37a28d8f269113d93817fc5a7f7ed48ca59966a9726edd47ad4
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_pwr.h	49154	d1b3888b86f2c25b4d64a0ff6b47fc968eeb9dd0c68baf0cd7e76
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_pwr_ex.h	46635	dd70ff0d9345914c8eac245ae7750eb66be1bb93c408a2716054a
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_rcc.h	249034	5aad60929187c63f40511fcef8d762c914e9bd3b1295ea1e7cab1
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_hal_rcc_ex.h	122138	79e433bc3ff360f10f4e06aba39ff398d9c21199fdbed7338793b
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_bus.h	149841	a771a988182391a991c4b419c47d4e2cbf70f79f378add0c60fe9
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_cortex.h	50141	5c088a21f30fcb18195239281377da49b063c4caf0e9055498400
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_crs.h	24677	7e8889489c8ccd768d38fd3bd30aa8d7787fe7807f2630a66e945

external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_dma.h	325732	1114f18bd2fc05dd8dcf0a8e16d84fbd17101395a54083d494e9a
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_exti.h	58302	abedfbc2cc3afd1e620e0e73743dbbc7de6fda47a51d528fe0830
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_gpio.h	45676	e2915fdd634fe93aec7bfd6ffaa59c555e4c5edba91318b1f421c
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_pwr.h	135768	dd616b1bd6bdf3ea369c279498ba07bd0feffb7ae3568289404ac
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_rcc.h	243265	9d05359c0e477fee75927edfacebff6d862de97a08f08d8107ce2
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_system.h	90249	e482344b634537f1c518a531106be559a668252e9d0119ac623f4
external/ST/Drivers/STM32U5xx_HAL_Driver/Inc/stm32u5xx_ll_utils.h	14219	35a84353fe5d938a7c099e761a5c1c0e4684b72327898160cb75c
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal.c	33035	ffb55394183c29b2d99ca0e2cffee37f3d316127258a387dc5909
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_cortex.c	31975	adc1de7635bc42dd76a376614abe09c5e77733571ac376a0beaf5
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_dma.c	56742	55390442b90c03dbe0ec51487db7e15e3cc1b05a90cca161dd32c
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_dma_ex.c	173296	c172034158503449abb317e1afff86415afe036c1d5569e3f6db9
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_exti.c	23771	6e552a440b1d6333f950b74da575faa82340e1f2fc0383de55216
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_flash.c	23311	ae1f9cc749c14d5183041cad7b4ce4a12a8ea85875bb5347bd4d2
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_flash_ex.c	58686	858308fc88f1a75eed0608ea1afaa14ee27b60177506ad1f23202
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_gpio.c	33435	2d7b5f252bf1bf2ad0fff46ac4fa7050bf437b23802f212051a0a
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_gtzc.c	73331	4f4433f59e3d3e9b5207b6bf47ee4bad9225fca1cde40c1d58ec7
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_i2c.c	256810	2f1a167093470b000b8e93968c6e0f62ef0da27253e1094b69188
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_i2c_ex.c	14226	43589137818ca179eebc3e43085dddd7bfd2bc04b2fb253924fc6
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_icache.c	21837	f567b029cc3854aedda85357c300cb5df269fdfe440a24c93db90
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_pwr.c	37205	39a009767b01100b602becf7f9c166e813baf1380d5245c0e0a67

external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_pwr_ex.c	95646	2e7539e029556909828b4e9b2eede71fff772343ad8a93adee883
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_rcc.c	84672	e0314297b6d02761675651f6d46d01831e4631a4484c3962545a9
external/ST/Drivers/STM32U5xx_HAL_Driver/Src/stm32u5xx_hal_rcc_ex.c	144981	0a45772828e1039466f4652770b60aec034a464bb604e372626ea
external/ST/stm32u5a9j-dk/Inc/board.h	1939	21fc97e0ff9eb3c4a3887b6f8176a7e0843f2dbaded4a52de6cd2
external/ST/stm32u5a9j-dk/Inc/flash.h	1272	64d5076c23cc8b8bcd9171df777b5b93f8cf2c2682449a82e657d
external/ST/stm32u5a9j-dk/Src/board.c	5001	94f4ec7918cd843c5319594c5af8e1995c6ff5e6be8535faac8b0
external/ST/stm32u5a9j-dk/Src/flash.c	1327	a7ae0c1bf85177b8b9d6ec8502815161bff359133c6dffeda6890
external/ST/stm32u5a9j-dk/Src/stm32u5xx_hal_msp.c	2117	2cae850303d454a5092d3cd0a33c87f5dde8875a4e255003c4248
external/ST/stm32u5a9j-dk/Src/syscalls.c	5109	03c06c72eafb44b499814cf9234aba57dd830f345d9b8779b5d70
external/ST/stm32u5a9j-dk/Src/sysmem.c	2986	ad20f7b1fa1e7c73330727747222cf2ffa3245ebf035230ee9a15
external/ST/stm32u5a9j-dk/Src/system_stm32u5xx.c	13775	28d00434c85c76af756649577d6b2096f3810e2adb137203b5ebe
external/asf4-drivers/Config/hpl_cmcc_config.h	1312	f151793abbc058cdd3b2a4362295ad50e387cea72730c161bf47e
external/asf4-drivers/Config/hpl_dmac_config.h	238602	72c57807a82c3d1059522ebc97f84b408c7a81f7da961ddf0a759
external/asf4-drivers/Config/hpl_gclk_config.h	29582	78ae2f945d5af55b5a9f2869e16af0c56b2ed8511a89f22f15bdd
external/asf4-drivers/Config/hpl_mclk_config.h	2739	4b2a59f122a2add44ffe324f50884e8a18203782a285f32067e66
external/asf4-drivers/Config/hpl_nvmctrl_config.h	794	1cb6127249f449a6ac08e3782e102a35ab36a41fd9ce204862b1a
external/asf4-drivers/Config/hpl_osc32kctrl_config.h	4397	b58ea361bb036d4d196d22fe076df84c7c6dcaaa41d2a46a99d0a
external/asf4-drivers/Config/hpl_oscctrl_config.h	18017	5fb71a801744eea831d7bc0740b7ee90210682a0432d6506d1cf4
external/asf4-drivers/Config/hpl_port_config.h	10431	eb018815c4396485ed2398490680de939e3a7c3e99f46e364a96d
external/asf4-drivers/Config/hpl_rtc_config.h	10103	0d209d15f595ceebb32177e624adfae310886c0b366bf29b5e27a
external/asf4-drivers/Config/hpl_sdhc_config.h	611	545b79294fa88c876737a0df0a38d333ca66ecc1ab365750057aa
external/asf4-drivers/Config/hpl_sercom_config.h	12313	cb896c04f9b9617114ee79b5747c20b5af3b9c89114b0b259aa5d
external/asf4-drivers/Config/hpl_systick_config.h	480	10c45230fd2d62d3612e88cbcfff13b9b5acf86b0bccd090b0554
external/asf4-drivers/Config/hpl_trng_config.h	622	cf280b6c908d5676a537650c471b0045e69cca7f0b418416e673a
external/asf4-drivers/Config/hpl_usb_config.h	16526	e04608dd087c9a668054ac1de496f13de48dfbf75ae72fb9d6b77
external/asf4-drivers/Config/peripheral_clk_config.h	13094	af90e135d137c680b12757900de10bb2b118b9ffc9b818fc2bf94
external/asf4-drivers/Config/usbd_config.h	8575	37ffa9817a035ad26181f85a9f32f735a16e9355449e3054f6cae
external/asf4-drivers/diskio/diskio.h	6753	85811b83545e109da212b9185d89031b0318a03c62689dd829002
external/asf4-drivers/diskio/sdmmc_diskio.c	9156	2fd3791050cbe4d642d004eb38f3e27fa56a8eb567961abbe4dc0
external/asf4-drivers/hal/include/hal_atomic.h	3877	cecc60dd86d9ad9f1cb38c3846fe5f4ab0809f654209c4cf99ec1
external/asf4-drivers/hal/include/hal_cache.h	2596	9a4ef6ff55e895277b7c34839f8579dd82a5a0dae4095d341901b

external/asf4-drivers/hal/include/hal_delay.h	2300	e254bd63b1b2c6a52133de73a7cbcee3ea8f857fdad2b94707055
external/asf4-drivers/hal/include/hal_flash.h	6851	e09c82a89990ed9ab325d5b3d02d71b203d4d41fda197f3e57808
external/asf4-drivers/hal/include/hal_gpio.h	6478	8279d6f57d6d6ab08fa0a2b9791392665ba143c8e4acd1e2060f0
external/asf4-drivers/hal/include/hal_i2c_m_sync.h	8708	3c498f7a5a87277540af344d9adea5093e1163a4c367954847ead
external/asf4-drivers/hal/include/hal_init.h	2014	517751d60cb1722b851e7a7ed8ccfe66f9579c32b5b03474afbfa
external/asf4-drivers/hal/include/hal_io.h	3385	beb3ea3aae1f8aaf23f672cd65b8e3fb7dc8c44d274b533a871ff
external/asf4-drivers/hal/include/hal_mci_sync.h	7065	93066364411ef41034432d46ec232136f40a73c1e734bca687ac7
external/asf4-drivers/hal/include/hal_pac.h	2387	d8e44fdd98591ae2849e0528a92a57593be44c64029662a002057
external/asf4-drivers/hal/include/hal_rand_sync.h	4348	a00aa2ab017d7cdb89279bc763f148f1b35e2033df5ef7b9e556e
external/asf4-drivers/hal/include/hal_sha_sync.h	5651	ed3624440a83cfa7ca29abcf008499ec9f73d44e1df5fb2f41082
external/asf4-drivers/hal/include/hal_sleep.h	2189	0f63d55bb8f62876e6f3887bd02e4817f926b1fe762f8c332a9cc
external/asf4-drivers/hal/include/hal_timer.h	5944	a1947f71ec813e165d8aba175b6d095d55019e498f2fc7f522209
external/asf4-drivers/hal/include/hal_usart_async.h	10435	1a1ce2e9aab2a0d015120c0a7cc5c3290263b29217bca90448bb4
external/asf4-drivers/hal/include/hal_usb_device.h	8172	c80f51876cddd5fef1a54fe63c7aebf814eb6f7d310f393c9d423
external/asf4-drivers/hal/include/hpl_calendar.h	8961	90cb567474f4f1c89f0f9f889e95760887775d2ae85d464920f44
external/asf4-drivers/hal/include/hpl_cmcc.h	7351	a79aa8cb4c0f347be8f10ee6dda2f22ca137aecf55b2d50af57ce
external/asf4-drivers/hal/include/hpl_core.h	1647	8c308843b7a5c35e34cd81b15af54c681b2746d71b21ff765f20e
external/asf4-drivers/hal/include/hpl_delay.h	2505	01a8ddb6c143eb6e2d878e515e46e964428bd493de53352100250
external/asf4-drivers/hal/include/hpl_dma.h	4827	beae645e6616cb1364da3679b85147b6f2d44a489b1f081240e69
external/asf4-drivers/hal/include/hpl_flash.h	9031	bc78cbc7f60de621e7f4eeede70c92bd135e1a2b95d9693f0dd07
external/asf4-drivers/hal/include/hpl_gpio.h	6277	a1f35be42c644145d04a6e3ee4e53600a9dab12f087f0b97bfa52
external/asf4-drivers/hal/include/hpl_i2c_m_async.h	6304	91561cfe002c5dcd103b44188818621b4bd75a79ecf85352a4a57
external/asf4-drivers/hal/include/hpl_i2c_m_sync.h	5275	da13b738b4ec45c8392c1748a888009e84ba6cde5870b2217edb8
external/asf4-drivers/hal/include/hpl_i2c_s_async.h	5375	0e8c33fd15e668a700c9d8a5488667b656255d9733057ebeff612
external/asf4-drivers/hal/include/hpl_i2c_s_sync.h	5245	f5d0598b4829518be2e6a2c27577751f362303082313ba209b751
external/asf4-drivers/hal/include/hpl_init.h	2745	5c715f98a96cbe4df6263908b9cdfa8a928caa2a49611381d1d7b
external/asf4-drivers/hal/include/hpl_irq.h	2735	ea4ddd896ccea01cf58a8878af56a3336ca0187958e720d11562d
external/asf4-drivers/hal/include/hpl_mci_sync.h	8018	7b7af8a1df99d7f35322fe686fe4462a56d9e66792ae7416903a9
external/asf4-drivers/hal/include/hpl_missing_features.h	1438	c78fef96c98773d8daf5c7079c318aebca6d921de31afaf793a81
external/asf4-drivers/hal/include/hpl_pac.h	2693	07b3bce23bf4d1f5547fb5a4f0234835a73bb729ff3c71f05cdaf
external/asf4-drivers/hal/include/hpl_ramecc.h	2507	bd8fc42ee6508a389caab1e28a3f1161d9f00dc67fa7ee8496a2e
external/asf4-drivers/hal/include/hpl_rand_sync.h	3306	7f94483912a845967c1c54af07d78a92de68dafb3270cf593d1c2

external/asf4-drivers/hal/include/hpl_reset.h	2297	81db88674e7be881799def980457290dea1c25580b54ec48c3cce
external/asf4-drivers/hal/include/hpl_sha.h	1793	dc81e4dac4a42d1d485e50924a7f3bb9cde4a9880cf0ea5e91003
external/asf4-drivers/hal/include/hpl_sha_sync.h	3906	464bc960d8e61fd637b522e7bcd61dc76befa63c305fdcf7e45d7
external/asf4-drivers/hal/include/hpl_sleep.h	2290	480c40e7686c4e3ffa61cf3cd2024b57b37ddcbf69f28e8c42a50
external/asf4-drivers/hal/include/hpl_spi.h	4680	4161e16d4b40d5b28f313037f6d3361f422680588fb8e5849fb5c
external/asf4-drivers/hal/include/hpl_spi_async.h	4027	6fba055e16b46ad4fa8872e71a733e65b8801e3ef0abd33a4ee1e
external/asf4-drivers/hal/include/hpl_spi_m_async.h	8366	d03a0e3af98c9c9713d93b41f882aaccdd813c1625f786fe72d30
external/asf4-drivers/hal/include/hpl_spi_m_dma.h	6557	e5ae7632ac427b4cd38099ba53f4808e7a6a13a6e9568d08b1772
external/asf4-drivers/hal/include/hpl_spi_m_sync.h	5941	d84061ed2f95e757a894b8d052009003284ed6e8317793840e436
external/asf4-drivers/hal/include/hpl_spi_s_async.h	7919	d1b2b242a4ee39b70c48ceebfa1c8550e256abcec7d97f10b78a7
external/asf4-drivers/hal/include/hpl_spi_s_sync.h	7312	06400c8d53610d9176b4980c9029ad73e459c1071ddda78860fa8
external/asf4-drivers/hal/include/hpl_spi_sync.h	2020	e0142ebb39f54527a9f755cecb8396d27d61e34adeafa5d671217
external/asf4-drivers/hal/include/hpl_time_measure.h	2455	9a99acea0c1cf4e22646af6feecd68162d35e10559495eac43ed3
external/asf4-drivers/hal/include/hpl_timer.h	4299	810f1b185d4ceb2a5d414fdb5b20d40f452195fe763ccdf9362e7
external/asf4-drivers/hal/include/hpl_usart.h	2805	d82cc2e7e51d5a135b44b6fa8d066f9ac4cf710c085dc72eeebbe
external/asf4-drivers/hal/include/hpl_usart_async.h	8007	91ee6b12f089eb076773d7076eec7355736b11fce1535b3c16733
external/asf4-drivers/hal/include/hpl_usart_sync.h	7477	acbfcf477447cf7ca8a502abb686132720cac55fcf6733f218e5b
external/asf4-drivers/hal/include/hpl_usb.h	7775	3c4dc0b12b2a90ac68850b6df081211e32c5b4c2d634157fcf11a
external/asf4-drivers/hal/include/hpl_usb_device.h	11293	df2d59f55ee2d5a7ad22b938c676fac78e9d23822d203b78761f9
external/asf4-drivers/hal/include/hpl_usb_host.h	18701	950d8a8c839aad5a8d100e602366df862e0267da948490b238724
external/asf4-drivers/hal/include/hpl_user_area.h	4704	859beae1217e9691a27b49bbe53a92784303d3c78335065ecfc58
external/asf4-drivers/hal/src/hal_atomic.c	1874	db89288e13c7ce35dde4b25170607cd5ab2ae020309ccd3a998a6
external/asf4-drivers/hal/src/hal_cache.c	2062	4cc33a23d079c39d3b2ccaa43469e7f1c2f43c7ea8dddf21ab1eb
external/asf4-drivers/hal/src/hal_delay.c	2095	54c05baff4f0a9ac2a868d8b366cc74e549d0fc875a70f87cdf86
external/asf4-drivers/hal/src/hal_flash.c	8088	d7197dc8b031570a61972c00accde8d560cc8be03ea5cfb3b5861
external/asf4-drivers/hal/src/hal_gpio.c	1449	2f97bef2239628e6d6cf4bacd45f9c1c6eac04000655116cd299b
external/asf4-drivers/hal/src/hal_i2c_m_sync.c	5689	a9b5555a584768beec67504bf8ff4313e66312b95b9b4671c51e7
external/asf4-drivers/hal/src/hal_init.c	1559	95a43e91805799c5147985bd7804a2fb4c4d3779efe21fdee80ac
external/asf4-drivers/hal/src/hal_io.c	1933	ff0e7fbc899dfe0fc45c1810377b8120de5b184012dffb6596056
external/asf4-drivers/hal/src/hal_mci_sync.c	5785	1fd5a4c7892a3e16eeacde0d4e8f0a26cd770822f0e0bcf96fb88
external/asf4-drivers/hal/src/hal_pac.c	1522	ed29fde8a32a6deb3a799437722eebb2d81d0e81984091a692f90
external/asf4-drivers/hal/src/hal_rand_sync.c	3314	39ec437452732d1dec85be9ea1c923e4e428938ccefe2fc784541
external/asf4-drivers/hal/src/hal_sha_sync.c	7117	d6075d16bee673560f1d5836da06fd59a5998175d9b9e0add376b

external/asf4-drivers/hal/src/hal_sleep.c	2184	c72b49d5479d558925caf315840ea5b21e620edb7b61b1e60d244
external/asf4-drivers/hal/src/hal_timer.c	6505	0c2338021d4ea4e34b74c18a2a71cb35fefe3ac08890fa621fd7d
external/asf4-drivers/hal/src/hal_usart_async.c	11134	e2d7c2890dd90468824eaf669f130c3a539a326b83e4e93b405bd
external/asf4-drivers/hal/src/hal_usb_device.c	15543	2b9e65a8fef0c2a9dad9a245561bd259b4cee675264d933542fe8
external/asf4-drivers/hal/utils/include/compiler.h	1888	0af8238fed92a95fe7aa6378470eae23ca068c0daa575f0ba9b94
external/asf4-drivers/hal/utils/include/err_codes.h	2401	c536f0ca2a4911aad836a164e7e31f13f62e44a6b93bb24645e36
external/asf4-drivers/hal/utils/include/events.h	1684	2d1d2d1addf2c96c4d3464a3020bc178cc13cb8723dd961cf90c5
external/asf4-drivers/hal/utils/include/parts.h	1440	37e29a9a267008dc0b3d618434f1665ee52f48f6f28940590bb54
external/asf4-drivers/hal/utils/include/utils.h	48419	a7cacf3555c5077da7a20d24129747fc911072e0d1d7e49a06696
external/asf4-drivers/hal/utils/include/utils_assert.h	2909	9d2a2edc0efde0376185ee17f04f0230c808995f692e59eab9985
external/asf4-drivers/hal/utils/include/utils_decrement_macro.h	6588	6c267e0a57c8da5169df7edfd1e7eaa020015c8e6870e724e315c
external/asf4-drivers/hal/utils/include/utils_event.h	3427	c9f829e4bce355e5ca56ea63fde43aaa81917f5bf19f4def1607b
external/asf4-drivers/hal/utils/include/utils_increment_macro.h	7346	4593febf45443c8217f34ba532a703708426a4c91a92330f16ed0
external/asf4-drivers/hal/utils/include/utils_list.h	4360	38204dbfceb50c7b6b2a847dc5cfbc32c908381f9cf6e736fa298
external/asf4-drivers/hal/utils/include/utils_recursion_macro.h	2460	f8362947ebb5a334a976057d175410334bec312be9052648db80a
external/asf4-drivers/hal/utils/include/utils_repeat_macro.h	23132	59ab6f690e84be0ecc16d76e12ce66ada0ab4f5cb240d81d4753f
external/asf4-drivers/hal/utils/include/utils_ringbuffer.h	3567	bfcbec9bda89cf5f30cfaf179ebaa9097c87dec5aa93d2c76189a
external/asf4-drivers/hal/utils/src/utils_assert.c	1532	4a49855bf156bd76b0322d0c477b2e1038b6cbe2301f8c6d1fc07
external/asf4-drivers/hal/utils/src/utils_event.c	3736	70adb6e723bcbbc53585b51544ed1cdd34976ac0764e33463e371
external/asf4-drivers/hal/utils/src/utils_list.c	3494	741dc6007c2695727e97cf79419a4dfed60d67da08d225a1eecd9
external/asf4-drivers/hal/utils/src/utils_ringbuffer.c	2860	db8251f7d5497a207d6ecb878a05c363a4b5a4c648d763dcbc51f
external/asf4-drivers/hal/utils/src/utils_syscalls.c	3173	2f8cc421ba88995d6473ccbdfe71d1afd226aeccd3b2d9913b713
external/asf4-drivers/hpl/cmcc/hpl_cmcc.c	8720	34ff9cd93ef6b41de79d630b9b1026790e02ecaeab110cf6a2292
external/asf4-drivers/hpl/core/hpl_core_m4.c	4642	622d7047ed6165dc118dcd3513d833614e78761601c6d5a9d0f41
external/asf4-drivers/hpl/core/hpl_core_port.h	2084	b985ef04775ea0fe2211edc9fa43dadd3568bdf2bdf3df9b7b7da
external/asf4-drivers/hpl/core/hpl_init.c	2403	d59b6a2105c2af1623489a01eb0abcf46c2eb093fcf151fc5993b
external/asf4-drivers/hpl/dmac/hpl_dmac.c	8773	7d48048dd88e6df989344d34903a626527f90342bb972f45f2cab
external/asf4-drivers/hpl/gclk/hpl_gclk.c	14096	f5a7bd522ee72aef4f0a3ba1eaa79ececa991bba115f6be639f4e
external/asf4-drivers/hpl/gclk/hpl_gclk_base.h	2459	b52e883e479f96f64e3bae8de656b9b15c1fe37578c29fcd1db15
external/asf4-drivers/hpl/icm/hpl_icm.c	7232	bb1696c251fad634c9be4e9062b8e0201e118e3115c770c9e3a5c

external/asf4-drivers/hpl/mclk/hpl_mclk.c	1531	5fd59517a4e3ffb6fd6f06f315524610f55cb480cb9c7a366774e
external/asf4-drivers/hpl/nvmctrl/hpl_nvmctrl.c	21096	d796f22652b603026a72bf33b66fc9d6f32eaae0151b0a3667e61
external/asf4-drivers/hpl/osc32kctrl/hpl_osc32kctrl.c	2910	f5288ce4cd9195c777f7b01f62cf4bc75417426b349845fa61383
external/asf4-drivers/hpl/oscctrl/hpl_oscctrl.c	8695	091a9fa621d53af321728a68e9e7099b976aba5495b3b5c7f5d48
external/asf4-drivers/hpl/pac/hpl_pac.c	3445	2382d02cbef36a5ec801ee6e83cda9864e978d0dd8efd889ce893
external/asf4-drivers/hpl/pm/hpl_pm.c	1981	c544df613c3c1630242ce55581daff1b490bc9e7638702fd95e15
external/asf4-drivers/hpl/pm/hpl_pm_base.h	1504	8f977e4d6289ee9e75480ba74a95fe8d60b1bfbfa76d4c3cf6cf5
external/asf4-drivers/hpl/port/hpl_gpio_base.h	4977	576a9e052b58d39052b2253a560837abca18e3b2397f2911984cc
external/asf4-drivers/hpl/pukcc/CryptoLib_CRT_pb.h	2028	0fd91e883322baff1d72c13ba7bcb25597236ed540ba1d4d6afab
external/asf4-drivers/hpl/pukcc/CryptoLib_ClearFlags_pb.h	1868	a15b1f3342f48ed4ecddfd7f79169e3f26916fcf3ba7fa9fd4ce2
external/asf4-drivers/hpl/pukcc/CryptoLib_Comp_pb.h	1804	cb9e172f88863e5b282c173eeddc4f6d1bdaffec180967768d20f
external/asf4-drivers/hpl/pukcc/CryptoLib_CondCopy_pb.h	2019	ea9b5efc1311b82ef26b324cebeaccf320ee1f32d0a9fd90e3b79
external/asf4-drivers/hpl/pukcc/CryptoLib_Div_pb.h	1976	4184edb3bd9f6af69d3e7026c7b6f2c17d14792aa68714006830f
external/asf4-drivers/hpl/pukcc/CryptoLib_ExpMod_pb.h	2513	d9c290f5bbafc3cb5f2b96a0fdb356a7d65682a12c926ffa90c92
external/asf4-drivers/hpl/pukcc/CryptoLib_FastCopy_pb.h	1849	fd2f231f31983639536245896bc1f88f84667493866255fbf964f
external/asf4-drivers/hpl/pukcc/CryptoLib_Fill_pb.h	1808	c121ccdfd33908a411da05dc623262478612533ac76b22ab90397
external/asf4-drivers/hpl/pukcc/CryptoLib_Fmult_pb.h	1932	3cc1e69387580b7de35f603169d61b0c0dec0184b2f8c9acb8498
external/asf4-drivers/hpl/pukcc/CryptoLib_GCD_pb.h	1894	076dadfc71fd7e5129756d989b5b4383c3571ea6709a95865071c
external/asf4-drivers/hpl/pukcc/CryptoLib_GF2NEcDsa_pb.h	2260	9cc5677e4d9598138d10d643a05e04707eafac281a716f470b416
external/asf4-drivers/hpl/pukcc/CryptoLib_GF2NEccAdd_pb.h	1831	c6b58f88b75219297a0ce734d0d0c76613034e77a1aab57660b59
external/asf4-drivers/hpl/pukcc/CryptoLib_GF2NEccConv_pb.h	2713	56ecb702fec90eacab1578674a332bb074e187de9abca6ed73562
external/asf4-drivers/hpl/pukcc/CryptoLib_GF2NEccDbl_pb.h	1828	130748c9ed40aacffdb2c5ef8092886389ecd436e5498096cd4ed
external/asf4-drivers/hpl/pukcc/CryptoLib_GF2NEccMul_pb.h	1823	9a7118109fe720026a05a7edf6d7cebb804d2ac918cee689abe5d
external/asf4-drivers/hpl/pukcc/CryptoLib_Hardware_pb.h	1773	682017ea066799d74325a4d97ee099576a3b63c320598f74a7468
external/asf4-drivers/hpl/pukcc/CryptoLib_Headers_pb.h	9897	dd4fa7be2d08013a75fd63491fc768c945d810799de50c483adfb
external/asf4-drivers/hpl/pukcc/CryptoLib_JumpTable_pb.h	4771	ea3b237d92b0bb638a3ccbd1c6fe53abd6d251989ed2f5aab2431

external/asf4-drivers/hpl/pukcc/CryptoLib_PrimeGen_pb.h	2119	f7e274fb433a0607a1e9de86aa049250aff6eeb1c9981aa592815
external/asf4-drivers/hpl/pukcc/CryptoLib_Rc_pb.h	3451	86bb24849b50c5cd6e25922c5d52a58f71d1e0725820c8e755970
external/asf4-drivers/hpl/pukcc/CryptoLib_RedMod_pb.h	2100	928ab1a472a9a3ac673cc140c5d97b42534e2bdad681c07c7b4f3
external/asf4-drivers/hpl/pukcc/CryptoLib_Rng_pb.h	2451	87f63663441f017fb757f58819bf206ed1473d8d6a5d2869fae4c
external/asf4-drivers/hpl/pukcc/CryptoLib_SelfTest_pb.h	1763	d533743164fc7f7ae2c03f8f9faafbe270d2fe2b8c66503acf3bb
external/asf4-drivers/hpl/pukcc/CryptoLib_Services_pb.h	4862	ce1452057529d1c7417764d7c8d96237e826cf65641b2c8ea3bfd
external/asf4-drivers/hpl/pukcc/CryptoLib_Smult_pb.h	1934	4f3d50812b21022fd46c2ebc9a169c7b475b84fb188f4c48a3435
external/asf4-drivers/hpl/pukcc/CryptoLib_Square_pb.h	1945	a2e858469f16322a2e8284c8173ccb0e3f56db26c44479ae515af
external/asf4-drivers/hpl/pukcc/CryptoLib_Swap_pb.h	1805	ad42c7f5653c53a8ffc0672075e011e754c40a515ec43b59fb69d
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEcDsa_pb.h	2640	c680e1d3b575a42126f62d5cdc314c2da1c4155c3cf4d5f4de115
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEccAddSub_pb.h	1904	5d5554a09e2cefa4f73083925810265744b2a795cbbbaa3d6c0f2
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEccAdd_pb.h	1821	d1afcfc71fe2555cab049480c42375326747381900a6560eb514a
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEccConv_pb.h	2679	3659c7d6087f26e4186aa01c6233237c4895e6b269b25a43b1b9d
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEccDbl_pb.h	1812	0b58826b9ece4ab6a3d28b2c46343d8304d0382f9bd605c2e8151
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEccMul_pb.h	1808	ed6c79fc6cdac0e2d49d929edaea446b6e553a2a5244e06291114
external/asf4-drivers/hpl/pukcc/CryptoLib_ZpEccQuickDualMul_pb.h	2398	5c86ff92b258e834ae1366aac3d475f5d09e13fc52cc3964e6e22
external/asf4-drivers/hpl/pukcc/CryptoLib_cf_pb.h	1866	4caeb61c372352ad52ffd61869a3d9f51f34340e3d5a937a67f3f
external/asf4-drivers/hpl/pukcc/CryptoLib_mapping_pb.h	1764	133ce455b80dca9f2166bcc8f4981625051494510f6c1622d3725
external/asf4-drivers/hpl/pukcc/CryptoLib_typedef_pb.h	7494	25f48abc69c7c261468b4740fa92893fc9114a254e2b7f12f2f45
external/asf4-drivers/hpl/ramecc/hpl_ramecc.c	2642	f7722b527ebf2b1aebe8a138a08ac4887d98dcd3c459cc2f9037d
external/asf4-drivers/hpl/rtc/hpl_rtc.c	4852	6ac36c4bf6c189f3ac234a9008723cd52e3de6faa088fd0958e54
external/asf4-drivers/hpl/rtc/hpl_rtc_base.h	1652	093b1d7588db0e70cb41a781710000b6d77f75857272a3f7d644f
external/asf4-drivers/hpl/sdhc/hpl_sdhc.c	15744	7c23d813eeacbd1dfd2381b666481f0fb993925e89c3d5ef7a75e
external/asf4-drivers/hpl/sercom/hpl_sercom.c	84402	8403abfc554414d4978c5a5611bd1ce85bc577f0cc83e1bd0f835
external/asf4-drivers/hpl/spi/spi_lite.c	9867	0b6aaa607f2cdc4fc6a28c0eedfe6bdfd2d636affd984cd85f8b4

external/asf4-drivers/hpl/spi/spi_lite.h	3513	eaf7f6c1c2d83c6cef2effa5240a552ea2d9726d6cfcd3b732553
external/asf4-drivers/hpl/systick/hpl_systick.c	2791	c7af47de984cc39dc774ecb00e2de0bc76e095b4d282aa19f69d3
external/asf4-drivers/hpl/trng/hpl_trng.c	2881	3d443cdb79b9894f5c65ede48b8d2f5b465951c97b534f8212128
external/asf4-drivers/hpl/usb/hpl_usb.c	62288	30021e816cfb62f72a3a9400b8884c9bc036b5ad8a0cae6bd1ae8
external/asf4-drivers/hri/hri_ac_d51.h	54344	b471b3b2d554a7addcfab25b741272d7a6eee4847fd75514d43f3
external/asf4-drivers/hri/hri_adc_d51.h	119963	029b3204640648b7b4e8f5027c2be74cfc4683d733b47fd4986f4
external/asf4-drivers/hri/hri_aes_d51.h	35605	f1a54fcff49df03031d2a8965ec8dd8d8c83414dca836dc5cf341
external/asf4-drivers/hri/hri_ccl_d51.h	23922	ff293d538fc2dd597139c8bd7bc6cf4910e9184b61b241ad54418
external/asf4-drivers/hri/hri_cmcc_d51.h	10262	094d8fb7c92f9f6468188f9c74653ecab08bf22ffdf4ba844c972
external/asf4-drivers/hri/hri_d51.h	2375	fb15b07aafb9fecff237bcd8f8ab69d03d33b5e38504fb9f806cc
external/asf4-drivers/hri/hri_dac_d51.h	50209	bf13d7cd4cd315b8cd15fb48935889fc6c9138b882224e9e265ab
external/asf4-drivers/hri/hri_dmac_d51.h	219222	a637d7085631fc80361b8d94b1545d60f895dac7c946ab24a3080
external/asf4-drivers/hri/hri_dsu_d51.h	35905	607a43d4cc2b50843b4db11fd7ef946c40018a9850addbdebb0cb
external/asf4-drivers/hri/hri_eic_d51.h	55700	18e926641d37ba129a32b5bca01256d181561474a581eeaeaabec
external/asf4-drivers/hri/hri_evsys_d51.h	56659	915fcf1997fa369adad12d2b437f7725e225f2c7cf4743f535c51
external/asf4-drivers/hri/hri_freqm_d51.h	13672	95277a1ae04ec7bb104202271274947bb4130851663014e5d1620
external/asf4-drivers/hri/hri_gclk_d51.h	25859	b7b35f8b57442290cf5b502cdbf6f8e2e8a6a4376581205b02fe1
external/asf4-drivers/hri/hri_hmatrixb_d51.h	7903	c148ba15466555f5253172afcdb786ae7d5c6ebd0bbf384d07af7
external/asf4-drivers/hri/hri_i2s_d51.h	87926	ae94a47c9c1adeec79ed14db6ef4e6c9824d972ef8d4c569b302b
external/asf4-drivers/hri/hri_icm_d51.h	21744	c85a4d3a216a767ec6a877719d434b301414cca1a8ee31d9b5920
external/asf4-drivers/hri/hri_mclk_d51.h	94032	78dd43dec0e73ec0b681dd28b0fd93f62f95d85c82825d617abc3
external/asf4-drivers/hri/hri_nvmctrl_d51.h	50196	b2a36107c2fc9f38b00c27bb3f394cb7e3e32bdfbd35f566cdcf0
external/asf4-drivers/hri/hri_osc32kctrl_d51.h	39739	c78363a4e0870ac36822aacb50b84d686aafc75e6d0b5085f59fe
external/asf4-drivers/hri/hri_oscctrl_d51.h	153893	b46a7ee2e6807aef0d3a182c28c6ebd2a313450b9f0f089b11b8e
external/asf4-drivers/hri/hri_pac_d51.h	42409	de8e820ba75a58d3d592e446e2eb67e0708f4bcd9d04e2de705f6
external/asf4-drivers/hri/hri_pcc_d51.h	7868	ef4d3232ab2f3709c14b22126e4ed8e96cb1960f99ee56eabebf9
external/asf4-drivers/hri/hri_pdec_d51.h	80311	77f52d23b1a88b535f565969c4f909f72567e1e36d96fd502375f
external/asf4-drivers/hri/hri_pm_d51.h	23086	b3608d92d49d4ee84a852584207c4fe80bb9623149040d2e1b8f7
external/asf4-drivers/hri/hri_port_d51.h	85724	b1c6d6cc436fae499b35ea2aab9b049c38758ad0949793ac2447c
external/asf4-drivers/hri/hri_qspi_d51.h	60193	eb7fc6fe9fdc54781494a18329d0af13a88d7269cbe461241a0d3
external/asf4-drivers/hri/hri_ramecc_d51.h	10805	9e529997cd83333828f3b82e852869456a5c3917bcb414c23f674
external/asf4-drivers/hri/hri_rstc_d51.h	4288	43471574d299d489312c42834ba55a1a07a68ecb3941526f75320
external/asf4-drivers/hri/hri_rtc_d51.h	356515	f9c252a7ac1e0baf13931479bfa449e5f5068d8604b4eef87b313

external/asf4-drivers/hri/hri_sdhc_d51.h	223935	c3826c3beb2c31231f72525a2e0f623824ac11de0e2c3445df2be
external/asf4-drivers/hri/hri_sercom_d51.h	302258	9320a3f062807e2a5c8a1dcf3e01327429cac83434cbed6de8a77
external/asf4-drivers/hri/hri_supc_d51.h	49356	8eabcdc3980a9428fa4003ddd3ef4ebb5fcb78c24d2fec1ab4b57
external/asf4-drivers/hri/hri_tc_d51.h	93837	3fb89518295d4e77dd7e83817bc735d77c67614bb8e25856db340
external/asf4-drivers/hri/hri_tcc_d51.h	289575	890860bd2d6c1c5e99f8715032388d6e56eb3892b583d60f5251a
external/asf4-drivers/hri/hri_trng_d51.h	10663	f93ef7547da17f1c8fa1ff90170476eccd1e769af757b45acc30e
external/asf4-drivers/hri/hri_usb_d51.h	354474	e6d14e9ea14b0304f2dd44b4d8aec07e5e3a1fb6136389cc3c208
external/asf4-drivers/hri/hri_wdt_d51.h	18441	691945e6f92a429e2068f167d4aac36cf0191df6f8d937c61b061
external/asf4-drivers/qtouch/include/qtm_acq_samd51_0x000f_api.h	1077	432de7866a01b1ca978240dea47d81572651148603e6b06b47a73
external/asf4-drivers/qtouch/include/qtm_acq_same54_0x000f_api.h	13446	f309079b7e6b27130322a15c95f1c9bbe30fbc82ad86803afd95d
external/asf4-drivers/qtouch/include/qtm_binding_layer_0x0005_api.h	7811	853c7d2ec996e13ac89945c2ab138fb4dff0c4d5d12ec6ca3d406
external/asf4-drivers/qtouch/include/qtm_common_components_api.h	6189	d5f19bbba531b6c2b18655312929af4722f790de8c6a1efd3240f
external/asf4-drivers/qtouch/include/qtm_scroller_0x000b_api.h	4985	efe8d5abcfd2b79a05e908072cac908e8bd7bfdaa9d935029cbba
external/asf4-drivers/qtouch/include/qtm_touch_key_0x0002_api.h	8184	97214b7b5af0eb742c9fd00e1895e7ce1893baa2ab832faddd4f8
external/asf4-drivers/qtouch/include/touch_api_ptc.h	2364	3228eaa831b19c670ac66207b8f5965a200e1f9d50268a0358138
external/asf4-drivers/qtouch/license.h	7274	bcb02cdca42690d15e4d1385c4d1d57e5fb882e99a3437ae42bfa
external/asf4-drivers/sd_mmc/sd_mmc.c	38799	6f08817c8e76ac8597685cb64559c6d7e0d44b9d57e28d5cf9656
external/asf4-drivers/sd_mmc/sd_mmc.h	10885	4932d0baffd05f5c7dd4083ba52c11a6408d25993b8ac9002fa3f
external/asf4-drivers/sd_mmc/sd_mmc_protocol.h	44290	a55af38a659f08fec2b3a0afc308b79f9bdeea29e2c4b8048ddac
external/asf4-drivers/usb/class/hid/usb_protocol_hid.h	18586	e37898ef14bea30fcd007e9f281e8fcd6fcf1b2ea53f030f86fbd
external/asf4-drivers/usb/device/usbdc.c	24565	27b2999e673efd37499f4fe57fc758919ae04891e53c32ce6dbb7
external/asf4-drivers/usb/device/usbdc.h	7456	a74520ef0f9ffbee2c7d80afa81cbc87cd1c1c8acbd2eb36bb745
external/asf4-drivers/usb/usb_includes.h	4508	4c0ec815d94b0ced96e967da950f6ead588adf29ef674589db92f
external/asf4-drivers/usb/usb_protocol.c	4288	be475c1d0b5a42cda96e7217b4c9d0b4acd8f0f4c4f08b88ba29d
external/asf4-drivers/usb/usb_protocol.h	25218	305dffdd8d1ec030c80b13b9d6e32e0455063ea2dd8a81a6187e8
external/atca_config.h	790	5e3d48c93b6e11ae6154b6003b83bc725d477f155bb8f7016dd36
external/mbedtls_config.h	2993	bdd0ca3b7f47578557a8059b6bd1b1729b427c952fa8c550d0b1c
external/optiga_config.h	5769	e0741d8a61db528afd7660d332d83a2cd6deba38ea514ddbd3aa7
external/samd51a-ds/gcc/gcc/startup_samd51.c	45555	3acad5194aa7be5ca2273af9fc6c4913a1a4acde46e825bdf79c0

external/samd51a-ds/gcc/system_samd51.c	1695	f54a5b2febc9671ae8b633ab51111fbc8ff1cb33b00e5d25d2db2
external/samd51a-ds/include/component-version.h	1962	6a109468076f1307748994331fa3f3b28c2d43b0ad6a55dcf5a29
external/samd51a-ds/include/component/ac.h	43843	009c567c2f4155bf526a66997369eda51d288a69575ea0323d881
external/samd51a-ds/include/component/adc.h	65662	8af32bfa63c1c1f3fb7de6679e2157fd625ad910afc27a4a79ded
external/samd51a-ds/include/component/aes.h	25607	aa117523ab12a7faacc1afee879be710851774456d8bb70204e70
external/samd51a-ds/include/component/can.h	245804	093465d99290c8e0d7e62f0c504dc4d04521a059aee31af5e2d7d
external/samd51a-ds/include/component/ccl.h	16432	81f2bdb883a3ea58d4873e25ca7c627dcdd90058aee5cbfbecfec
external/samd51a-ds/include/component/cmcc.h	24851	0a286b1cd8d2e962fd021a7b0c83bdbaeec2469792558e01ecbe9
external/samd51a-ds/include/component/dac.h	39416	1e97189db313d734da6ab05463ce4b433e02cda7d9a23a25d74b9
external/samd51a-ds/include/component/dmac.h	113464	65c6e1e2886bd1d2f401dc756c2c7b35ffb29093771e68bbdf624
external/samd51a-ds/include/component/dsu.h	91387	1b135b4ffd85b30ece7f2d80a794f0c03f49d073c83343c523282
external/samd51a-ds/include/component/eic.h	37003	169eb9121f291aac485fceacb77f6c76a52de2a95e442dbb4bb1a
external/samd51a-ds/include/component/evsys.h	44353	edfa59daac85613e031fa24a997096b3aee4c2fa8a25bae044807
external/samd51a-ds/include/component/freqm.h	13845	f6ba6ea0b2618d67cf98e4728c535c1ebfb6814ccbe324db5d630
external/samd51a-ds/include/component/gclk.h	21116	d2aa2101309fb980696fe439d551742b90f44d595c3fc3e5f55a5
external/samd51a-ds/include/component/hmatrixb.h	3438	2fde77b34fff846b984641f4e6b6751483a92d704034d0e405201
external/samd51a-ds/include/component/i2s.h	61284	99b805bf3923a836ddf4d88954235b193a5882f8cc11093f0abdf
external/samd51a-ds/include/component/icm.h	41828	d176e43b49dfd865cc4b98211b5ed1b6391b6eed6e9d45189e881
external/samd51a-ds/include/component/mclk.h	34840	a15fe47dd19b47a5a1f207a940d694790c80b42856912192d6675
external/samd51a-ds/include/component/nvmctrl.h	62659	736da0e7e8223acff30ac5b4cf0730e5ff796d87a23d75ed28207
external/samd51a-ds/include/component/osc32kctrl.h	21935	16e53fff483a3cfe540afcf0d9bd032ef29d117d4029f79d6956a
external/samd51a-ds/include/component/oscctrl.h	63931	4a687faf9b1e824a8398e9fc58c4e3360809fe774672ef0557e99
external/samd51a-ds/include/component/pac.h	51510	e819ea71cb0ba59f327c0989dc5d069b001ba6884967c211dcbf7
external/samd51a-ds/include/component/pcc.h	15502	2505b23a1292a160ae1b39c91e9058cab86997b6205cfefa8b958
external/samd51a-ds/include/component/pdec.h	54513	1fffafef84c5f0dfd27cbade9ad78cb3b064e99d3e43df6eb211e
external/samd51a-ds/include/component/pm.h	16417	5d27ae66524474b40399ad0c58a53bdf8c5107e3803274a2f33e4
external/samd51a-ds/include/component/port.h	27576	3b11d12312c1f9c6e7ed42a958816a3635f2c56200d4867c48b6a
external/samd51a-ds/include/component/qspi.h	40215	a89b5dcd034703a5a6728fabc95606053bac656ee15813e9a524a
external/samd51a-ds/include/component/ramecc.h	10325	6ff9267c0a0daac15d63127a5efccbfd552fd678f7c133cd5cec8
external/samd51a-ds/include/component/rstc.h	6202	f743b5bcfe16db10ece3efcd1c09aae3c5d83cb080a01c35bcadf
external/samd51a-ds/include/component/rtc.h	175646	0d561d7619a10811d33563851f5ab892ed02ac6da4018b0c9a290
external/samd51a-ds/include/component/sdhc.h	216816	16a1123b9e17a0ae6538cee211dd18828c9386286a59806ce1501
external/samd51a-ds/include/component/sercom.h	128866	f4cd0d5ff26f42952b03617aa947e8f7fdafed7cb95be6692d573

external/samd51a-ds/include/component/supc.h	42386	3caa785dae92f395f41196fc13b8a3d858bbbc7dc36f798290349
external/samd51a-ds/include/component/tal.h	149332	0abdcc821e4fb1266879a01c5ee5fb371f0c36354d8859aaae4ac
external/samd51a-ds/include/component/tc.h	61893	07a9de0c3b9ba89790fc130f7bac59ffac1f725a455da82f14649
external/samd51a-ds/include/component/tcc.h	146064	7378baac1d2d3a07bbfa6e013f138862a3961006f3ef77c829f9f
external/samd51a-ds/include/component/trng.h	9494	a26efab320022f24f94bdfba350d7d0f0f4d1232cdfebc42ab3c0
external/samd51a-ds/include/component/usb.h	140383	4b242d6e254d3208c21ce6b729171321909a0d246303486976ce5
external/samd51a-ds/include/component/wdt.h	20549	865ba6d637efc15d3a3c301db2e6a95aa02749acfc6391300c727
external/samd51a-ds/include/instance/ac.h	4609	a23321163fd6d6b6c70456f1de1600ec2d67e9e995004030fe0fd
external/samd51a-ds/include/instance/adc0.h	6654	c4d5ea1c872bf1b0e64df85382da8f5535fdfff4b27a272c09d10
external/samd51a-ds/include/instance/adc1.h	6725	5836756078d20cee86343858b6780346e8e335276c559e3b14a69
external/samd51a-ds/include/instance/aes.h	6952	bd5066041b54e6f3c5a2f4d42a8bc09a22f14c88848f050a83151
external/samd51a-ds/include/instance/can0.h	12047	a20bcdd0720c54e2dda86e112e73787f24e0268b278c2a7435fe1
external/samd51a-ds/include/instance/can1.h	11934	1402bd26abd857d8dc718905614994e42aded797b37714718ccdf
external/samd51a-ds/include/instance/ccl.h	2675	2dd13890492e9117ee712f4df7505773fd784fe8bea17804a8401
external/samd51a-ds/include/instance/cmcc.h	3442	63786baa38876a9aba0a6492dadefe7e59bdb67d8269c245dcb82
external/samd51a-ds/include/instance/dac.h	5016	dfa8fcd9662364a70a50aa08409517408c64e6587f69e156c592c
external/samd51a-ds/include/instance/dmac.h	58807	b7f3c5719ee91f1a1b404611c809268f8e7d6d19bb1cf6cc487b2
external/samd51a-ds/include/instance/dsu.h	9032	48e2d32bd7e3b1a4a384d69f2ab4d8f38bb6219ba5e89b274d764
external/samd51a-ds/include/instance/eic.h	4449	f906244a5256693fed98637d1857c3b8f543eba244143134b2cc5
external/samd51a-ds/include/instance/evsys.h	58944	7bad44d3cdc3808191328f4024ef77ff2190cca4705344a89baf9
external/samd51a-ds/include/instance/freqm.h	3107	fa238c880c38bde570bfe9318d81caa98d49e568048acf9d500e8
external/samd51a-ds/include/instance/gclk.h	16320	d62ea0f8e6d323047a6c3c7e58578cc0820a16934e77d9c79936a
external/samd51a-ds/include/instance/hmatrix.h	9072	46a00d10a0e65ab8bd966a5eee503b8463e84c2f87e43a3273b0a
external/samd51a-ds/include/instance/i2s.h	4195	8a87922c5148e79b79017749a4eab634e0f838487712c7db1cc36
external/samd51a-ds/include/instance/icm.h	4770	63ecba2bfd70c8637430b49a8dca4eab92fb650d64c5e386706d1
external/samd51a-ds/include/instance/mclk.h	3145	2451b0943874596ea013543e581a1abad537fb7f26c584687d813
external/samd51a-ds/include/instance/nvmctrl.h	4764	bd198ff2bb16137e551a3f440a11aa39cb52fd5657e991ae31a53
external/samd51a-ds/include/instance/osc32kctrl.h	3374	f718298ac6ca3eb85246c5b2015769ec3a373413a430bfc3065f1
external/samd51a-ds/include/instance/oscctrl.h	9404	ac683d81c36a75638cadcc19a9d6410a1dc6027882ede9bd23917
external/samd51a-ds/include/instance/pac.h	4310	205c202f3b716ffc843eed3a903032ee15e52da153c9ddc5a4823
external/samd51a-ds/include/instance/pcc.h	2904	321ed94d838f9113ff76e1668761dc9357ee8fde86dde5ae49dc3
external/samd51a-ds/include/instance/pdec.h	4997	be6f0c806e40df2b7b367d21e8bb584326bbffd15f40f19f60518
external/samd51a-ds/include/instance/pm.h	3029	fb04ed4d1eab24f3a0aaacb587e4cb7420e8b238650ca169e3df9

external/samd51a-ds/include/instance/port.h	14920	51d06558014fadc43992f49226db75fef6b070386d8e30d3f8f69
external/samd51a-ds/include/instance/pukcc.h	2306	a6e95de5a4fd22c09d771abfda8a5d2646d7f43d26177c74d597a
external/samd51a-ds/include/instance/qspi.h	3989	e7dd9c1b34f841c7d015dcf7d37e684733ec8d7ae9fa006a00f28
external/samd51a-ds/include/instance/ramecc.h	2489	bfdf55fd56bcd524ace6e714da3c63f4e60baf575b242e452723d
external/samd51a-ds/include/instance/rstc.h	1759	f85d5be85fe73504f2a0f3686933095d2a68cfca19c9a86aab4c8
external/samd51a-ds/include/instance/rtc.h	12265	fc1b40545f0069b2af11073c75156057bd536645fc345ae025b19
external/samd51a-ds/include/instance/sdhc0.h	11125	58583687fba58d56f10d744b084bbc53187ba30382814cd3f6780
external/samd51a-ds/include/instance/sdhc1.h	11125	df6d2fbd0167e23761229fd4ac4a641aafec725adb72a85b2dd49
external/samd51a-ds/include/instance/sercom0.h	14324	3f65c4b6f9ae5a5b286b1f6f084800abf245475a31ad272993a95
external/samd51a-ds/include/instance/sercom1.h	14324	f84e56a416d8e3b5c41fbf6d5bd9fca08120eb6d4a7dee9e9c25a
external/samd51a-ds/include/instance/sercom2.h	14325	a49abb125e29afd3f316fe88e0708ffc71d3f480def6846588dfc
external/samd51a-ds/include/instance/sercom3.h	14325	eb17328f2f56136dbf32859d35fb6fc89af1c708314700c331e5e
external/samd51a-ds/include/instance/sercom4.h	14325	c32c05647815f02e3d3101d97e669a3e11fd156445d7dfe814270
external/samd51a-ds/include/instance/sercom5.h	14325	91e92e3f7ab720ef079a4dfde7f05ff00fe16268cbe6dc298138d
external/samd51a-ds/include/instance/sercom6.h	14325	fc8030aef0fd7f02e0d42bb452d34add603985af1b7195e8bd17e
external/samd51a-ds/include/instance/sercom7.h	14325	1c2ef9acc848885a61abc6914124308a0e5bd06f24226a48f5612
external/samd51a-ds/include/instance/supc.h	3423	651194193151b6d77ed9fe6020e2b6ad70b40efc78a2a548d01e3
external/samd51a-ds/include/instance/tal.h	51688	a80905367a6a552e05fdfa188f897c132b0a835efdd816ec4b1cd
external/samd51a-ds/include/instance/tc0.h	7651	19940f669d1346c77c5efbe19728cea80e79b3de88d0a1cd736ba
external/samd51a-ds/include/instance/tc1.h	7651	542b7b1ad008902b90cd3c59dffd0ef5640b8857f8599176eeeac
external/samd51a-ds/include/instance/tc2.h	7651	891296ffe11d9caf3ec32cc7d863c3f1c0e9ac5d30cc0654fa724
external/samd51a-ds/include/instance/tc3.h	7651	760c2e1ce3a30b703a10e4228bb8e12e227ea229d4ab81bf09d78
external/samd51a-ds/include/instance/tc4.h	7651	d8247d03d14b3cd1638dc452afce712b5b41757e168139ec2a6f1
external/samd51a-ds/include/instance/tc5.h	7651	ab7ea618eb9cf37ac9847339e4803174ea72bc015fe67876007a3
external/samd51a-ds/include/instance/tc6.h	7651	438213dcae61fcd4798bd91dcafeed5d46c76c0bb75b20a3eeb34
external/samd51a-ds/include/instance/tc7.h	7651	f2a0551d0d3ef280ee7947acba30d3537a13666d495f6000e2fc4
external/samd51a-ds/include/instance/tcc0.h	8792	26dc7f685789fa6ddb1849fc4a6f4a0c0bc4d9ce26249bfec731e
external/samd51a-ds/include/instance/tcc1.h	7888	fd67d160887c270ff92c53a64cc1e5d7b9863923d81c7b3c51953
external/samd51a-ds/include/instance/tcc2.h	7080	90bccf2d7518252739f7504000611782ab85b62ab0833bc45d10e
external/samd51a-ds/include/instance/tcc3.h	6407	3fe8fe6400967da703c0edc321704848ef67b8a4d6294f222b188
external/samd51a-ds/include/instance/tcc4.h	6407	90dc5f2fcb3949db6db1f69c721df389770292281dea1bba27d50
external/samd51a-ds/include/instance/trng.h	2275	8d2c8368a9d22e8e863d9d4c270f3992e028b8dd0a64e0a37b31b
external/samd51a-ds/include/instance/usb.h	36661	095de164cca5f9df0f939b93a5290877ea4e7dd569352b16138dc

external/samd51a-ds/include/instance/wdt.h	2654	44796f78b838ee00b7dcbc8951513e5b08f671818fbb8e2cdeadf
external/samd51a-ds/include/pio/samd51j20a.h	132487	9fe311d69d97adafc1469a6ec45fa3dfc814cf4b7d7d0865101b3
external/samd51a-ds/include/sam.h	1689	00190accf1e328471874b96548c5ec1fcaed1a871c7db5ac00cc9
external/samd51a-ds/include/samd51.h	1810	62c99f36fb052bd36ffdb72931e36de663c944d29bd9b2c932894
external/samd51a-ds/include/samd51j20a.h	73653	cbac37b093963fc1b96a1668369546ab2f419f79243adf725f4cd
external/samd51a-ds/include/system_samd51.h	1194	f7ce006d7b5078b003ed2181e30b582a8c067a9f55b556520019e
messages/antiklepto.proto	284	89caa2946283a469e5c783b9dd886d99728a40dd36b0e6156abe8
messages/backup.proto	1713	4616978a3a2b14c4523d1823c1a2b48373bcf731e5e35ff59f483
messages/backup_commands.proto	724	17a8e01da16b238061f256546900a21af5b3a5bbe150b39e5d519
messages/bitbox02_system.proto	1509	5e3ff16c8571d177180788a164d64bfb61e0e94415307a1f5f861
messages/bitboxsync.proto	1814	8d87a05e25ba6cfec7f68a878906677ed218d64f663f6a20b2472
messages/bluetooth.proto	705	d7944887f55ee5e5196e4c62ea60564ad2a94e9d47257e49d05db
messages/btc.proto	9302	6f2fe1d3b1663e2617fd9587e6ec21242d02140624c797f222db4
messages/cardano.proto	4114	8e11c3628cc98b02d5904edf3a4d21e0492bad53ad74c89009b08
messages/common.proto	870	369ca12555a7810d196055abb643594d0723c7807fd46e5cca8a4
messages/eth.proto	5601	3e2422076d196a16b59f743651cb1d6beecd9dae5a183bfc0c813
messages/hww.proto	3047	1faae60899577cf627e7c54b319b34f883aa126393d94d0cc1b84
messages/keystore.proto	1894	ed5c5d677e3c017c55d71ce274246a4cd55cc8a867b259eaa9be7
messages/mnemonic.proto	282	9cd94eff3b2bdd7a9d60981db9f487d2104119088ff7a912a8536
messages/perform_attestation.proto	436	632df1278569322d391e4648de139ddf81ca83b26b5d14a0b3f08
messages/system.proto	195	706284195da36379d9d511f54e0ecc6a910b8fa8fa8158a474595
py/bitbox02/bitbox02/__init__.py	882	d9c472a05bb8d5684e48938e061b5f9a10eb405c53853b22791c3
py/bitbox02/bitbox02/bitbox02/__init__.py	1148	cfcf44d5cca32b8142d9d7d93829f89a61f5b77eaae024476f471
py/bitbox02/bitbox02/bitbox02/bitbox02.py	60915	fd62ecc298b45f0684968cf1b77d2e8357fa01583c3be6699e398
py/bitbox02/bitbox02/bitbox02/bootloader.py	10571	25ca224abc783f35996b1ef338e852eda9ac34329dd627f266689
py/bitbox02/bitbox02/bitbox02/secp256k1.py	3700	6903d0adce82573c7e3b5bcf40ef4ab09b964b5ff859bdb408365
py/bitbox02/bitbox02/communication/__init__.py	498	0cc6de915b3383e9c0c26be6d9cdd0f8edd701517b8c94748433f
py/bitbox02/bitbox02/communication/bitbox_api_protocol.py	47007	5aaea77dde27d11e51bb17dc8f73451971d82c0cfd475e0c543fe
py/bitbox02/bitbox02/communication/communication.py	1047	bf927b525b0f3b6800b71bb56a3a0b5311378207151484 67eb721
py/bitbox02/bitbox02/communication/devices.py	4522	27e2ceab732301eaa115fb5cc655414d7f496d56a146d42f71e29
py/bitbox02/bitbox02/communication/generated/__init__.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca4959

py/bitbox02/bitbox02/communication/generated/antiklepto_pb2.py	1401	002c48b4a60d4ccffed0d11a0c9404042ee25ebfe11eafe363d0e
py/bitbox02/bitbox02/communication/generated/backup_commands_pb2.py	2102	76a031ae9de53c9cfee71697e9c30ef36e3cdf6aa5c93a0cb6181
py/bitbox02/bitbox02/communication/generated/bitbox02_system_pb2.py	3362	7b7336678c9891da9d457a05d2e4f8dea403d66b6464c0ba652c5
py/bitbox02/bitbox02/communication/generated/bitboxsync_pb2.py	4721	4d1ccbc51d2721ca65b8b0f552e0b9b823ca3f7de898e6ff7300b
py/bitbox02/bitbox02/communication/generated/bluetooth_pb2.py	2457	e1b063fbcae61d5359b0f75ae7e27239e0b1330f13b7fdba49bf6
py/bitbox02/bitbox02/communication/generated/btc_pb2.py	14672	a81dfc78da5294d567398b1deb8b958be2bedbf46723673be413d
py/bitbox02/bitbox02/communication/generated/cardano_pb2.py	7698	7438e0c6df1a52b61d83ae7b37c508d1abcb23c08d68d7c5a98b9
py/bitbox02/bitbox02/communication/generated/common_pb2.py	2044	f76860eac5b1fbfd43107106e5bbd88099c9c8064c6631b2213c8
py/bitbox02/bitbox02/communication/generated/eth_pb2.py	8782	9672d0458e47e65533e4b4937c669a8a95679ba59375ae6e30373
py/bitbox02/bitbox02/communication/generated/hww_pb2.py	6892	9ef9b913ec39d1b08486e630c41481a129488faba729f1096ffa9
py/bitbox02/bitbox02/communication/generated/keystore_pb2.py	2898	5a7b7c7d197d889cf23bf0c7f0daaf2ed95d85967cb571ede2e7c
py/bitbox02/bitbox02/communication/generated/mnemonic_pb2.py	1386	c30ee7944adbbac452288b0b7580c9da11e24cf6ba3fb4b42ca5c
py/bitbox02/bitbox02/communication/generated/perform_attestation_pb2.py	1444	4177a19aae7d7b3ffa532606295188ba87475e9650bb4af0ef559
py/bitbox02/bitbox02/communication/generated/system_pb2.py	1186	52405697ace035f84e1743af4ae4046f550c6de80a10a9ef9af3b
py/bitbox02/bitbox02/communication/u2fhid/__init__.py	159	7583c55d986b90efdb2fbbd39b022968f6e1fcd3e37c01ea29250
py/bitbox02/bitbox02/communication/u2fhid/u2fhid.py	5339	6791dd27b621d46a37729704ab182f7b4cd89d45d592d215a4a17
py/bitbox02/bitbox02/util.py	7474	d879c3f4c1bfa9a9166d8974bb5fe4e1c63bc2807100a2edf2f2d
py/bitbox02/setup.py	2374	df4b02be80f5664e130fe7329538a87a81341f796325ffa3f9396
py/load_firmware.py	12912	c2e8e0955cd508374d023dbb978c7dc45b4dd30a81f168475ed1f
py/pyproject.toml	31	55e581d5626a46609d16456b1d075222dd645f52d16bf493bf44d
py/requirements.txt	152	055756030dfe1bc5d65a42ef9e785f015c7170aee3d15a766dc93
py/send_message.py	90420	c9c1cf0d64bb467e00dada86bf99d1f0b86c0c5d1c5e709c2e150
py/u2f/__init__.py	563	42a99f304be6223c23109d3b83492abf5bb4c4b30e50794c496b0

py/u2f/bitbox02.py	3558	28f6b2e9fff944c91ff004ae53ec8b14b50518764fefe1587ae8e
py/u2f/u2f.py	11815	5e0fd84da87b2f0183ae0f3c1183ee247b909937e6ca4421008dd
releases/describe_signed_firmware.py	3419	61ff7716f98d83efff1ef364c6b173bf7deeca2662caea3c83a9f
releases/pubkeys/JuliusWiedemann.asc	3199	4c174a88f6da9f8d93faf08291aa32101e69017fa73102d353c63
releases/pubkeys/JunghoonLee.asc	3139	33671dc21911a1750070735f332cce7c84380bd1ab7cc74fd6971
releases/pubkeys/My1.asc	4618	c3884ccc2c92a15ad9be2523730182a40d523e5a3ec3a4b9abfe7
releases/pubkeys/TheCharlatan.asc	20351	c229c79d3677332d6eb406d406e2c7daef67fcad6439bb8224c4e
releases/pubkeys/beerosagos.asc	3134	77547ad2b0bd39e591192f3d4df3e57e48c6acc1cfbcf7a7a4262
releases/pubkeys/benma.asc	6238	bebf447de75bcc5015b0f9699a459fb21f61c230bc1a54c29b62e
releases/pubkeys/brianddk.asc	2513	fa82b4b5c2c4dd6c4066aa113212f89e99aa6f33c29f6c2ab5793
releases/pubkeys/cstenglein.asc	3175	1697a4111b88a52d7a7d7ccc941ed11338d9578732f05fb0b2569
releases/pubkeys/frknakk.asc	3139	a3d9760761bedbb8f2b156c0d5fa3551e5aee570659e92af6e675
releases/pubkeys/guggerf.asc	3139	1b1ae8bf1e853163d565a32344854b57b2ceadf9246384b7fdbc2
releases/pubkeys/jonathancross.asc	167853	4092962b5065adad6acb1602ef0a0d5f20f32af83d52ce3b18564
releases/pubkeys/nickez.asc	34777	dfeee318e1e971fadd704779c023b0690800b63656d4f56e0c505
releases/pubkeys/offbyn.asc	14696	b6136c3272711fcddd8ca49f9916a79a54677bbcb38a32859112c
releases/pubkeys/psteiger.asc	648	fe123e4a583a9dc4d271ffd0af7c54ec135728575a5c171dc0fdd
releases/pubkeys/stadicus.asc	6892	6b7a866cccd4028fc58c5de6f8629919f7f5062c32e58c7866ead
releases/pubkeys/strnbrg.asc	3175	3c725e0b54589026d856b7050b29c6c2c323306e2c1a826156955
releases/pubkeys/sutterseba.asc	3155	5c279fb6619893e66372f5752c954d79a586920b49367bd3b681e
releases/pubkeys/tomasvrba.asc	21018	88b2e37b7a8cccf75e40f7019a669f4699237716d38831d0c2910
releases/pubkeys/userofreddit44.asc	3772	e67ca6a20f1bee441598fe784ec7789ceed4a82e0a1c725deef34
releases/pubkeys/x1ddos.asc	7091	c6cd70e5656c75ac96db731fcbc37c6cfbb26f0d1720a0f6b8ac3
releases/pubkeys/yaziza.asc	1530	2108a376fad98106ace21d60169de8eef8cec4dbc7626b66b7276
scripts/bootloader_update.py	12651	942f2c3383a9d764a1ba19dbaf1742af5641f6965d9bffeea6a46
scripts/build_config.py	9820	6eae2fc0b63a3637b2192c0195fa2dcabff85fc7370089f40c3dc
scripts/build_product.py	4631	4d5d00b246a2428d721c05d0229b8d171dfcb5d8572c0b8cb531e
scripts/copy_st_drivers.py	11544	1fdcf9f29b9e4a51531bf8e3108e0cbe4af5f40c4482438b725b0
scripts/create_release.py	14352	ffdeafdd62d7c7b55ac3916bc170791df35c44e4b190586410cc9
scripts/create_release_assertions.py	6458	c2ac06c2099ada49cba1eabbf77d97164be0f20609741e05e03eb
scripts/generate_version_headers.py	17004	675bbd250e4d75df70f430abc30595ab8bc0ebf69592a44982a6e
scripts/graphics/convert.py	1832	9488be1ec0cc6aafd3f9d88fdaea5fb226f4e3dc628b26895cae2
scripts/image_header.py	11499	b0b7aef7a47fb3ebdf6ce8309aca81c87d78d940aae82395afe7e

scripts/probe.py	5185	1e00bfa09028e533e317890133a1a549ba9f04f7d0aab89b3ae0e
scripts/signed_firmware.py	3144	480f05f317d9cb451a3d7e2d3b5e6ce21149cf7663a0daea2b086
src/atecc/atecc.c	24952	b08f8a566a7090a9f9ca736145e9565a28add3f47c8572f3a3013
src/atecc/atecc.h	1332	fa90590d9c07db88f66b7714dd7e74603c2f5c228332228df162c
src/bootloader/boot_args.h	1954	4211ed331ec60a4cebd8a66a05bc924d06a1eaf6808d6247d96c5
src/bootloader/bootloader.c	36479	f168d61e94931ac4ebd646e1317716b95b7ceb6ad14a2642b08d4
src/bootloader/bootloader.h	589	06c21dcaacb655508bf4cf0f7fce19c7b556c589717595a45fb34
src/bootloader/bootloader_format.c	2480	a8d11e946c4155c4ad0bd9cde5c67e6bd7bb2677af3e2a14dae78
src/bootloader/bootloader_format.h	1521	201e9cc589180cef9de3abfce4b2e98139402faf6b99ebe9accf5
src/bootloader/bootloader_product.h	1455	bc0884527d99033bc323721392226a690153d3f874fd573493e02
src/bootloader/mpu_regions.c	4998	ce6c37a9c92a7ef982192a5e7e1df86a4f147fca5c9f227352766
src/bootloader/mpu_regions.h	633	2e0d365b60c5f4af7308cc1a3bf726beabda6581ad8d989ad111b
src/bootloader/stage0/stage0.c	21703	e7602bdbf3060508b511e757832adbe9a9995bc6dd13477ba481d
src/bootloader/stage0/stage0_descriptor.c	648	e466f37f5e5ab14d521c3bef5ca673a0d824a5dad24bf67b0616d
src/bootloader/stage0/stage0_descriptor.h	1483	9a5d822def8bdb79e9505ef0453bb8214024bc4667d893f4beeca
src/bootloader/stage0/stage0_flash.h	7485	923840cec8b59d2a0150b15a393a36ff4ca70547602a3769a1475
src/bootloader/stage0/stage0_runtime.c	1000	27538c86f302b07025f7be542832db5d2843e8fa6893bc5237249
src/bootloader/stage0/stage0_startup.c	1872	570ef28ad3f830c70d9713b0dbe9a2323172df00e1240964459cf
src/bootloader/stage0/stage1_sigcheck.c	9311	5704d66ba1ca763bac3bd4e998eb68c26d63ddcc74c8912267a4d
src/bootloader/stage0/stage1_sigcheck.h	333	cd78c8accb34bfcc118cdf91425314a0a79cd67d91eeb560080cb
src/bootloader/startup.c	7383	0790af69af4ff3105e7b5be5f3107d14a66043b7ca893f402be97
src/bootloader_upgrade/bootloader_upgrade.h	4938	5509b50fcadf1b81604abb566a2655e91ea46525627c4bede0ce5
src/bootloader_upgrade/firmware_installer.c	6308	fbe4a58ce6c5ee5a49b10482f7cb082c5a5ab846d7cf4dfae408f
src/bootloader_upgrade/firmware_installer.h	198	db2fe193d39c09380b6cec28f9190146e837048cf3c9f7ff5473e
src/bootloader_upgrade/firmware_installer_check.c	3641	2ffa4ba9da129ca84ec2eb826fa2233378ea07bedd12e6188c32a
src/bootloader_upgrade/firmware_installer_check.h	626	e6b04cd3dba6479e1d4a52f0d018c75bfa28ec4a531cc5641b580
src/bootloader_upgrade/stage1_pubkeys.c	1453	d0dbfb66799b31c124d1d47a5536b3d0ee40afb323563a7f21046
src/bootloader_upgrade/stage1_pubkeys.h	268	6861334872aa5690cb7d9c0f6fca8b2da3c0224ccb1f2e09c2898
src/common_main.c	2237	6a5a78f034fdb3ff90f6b29937b3f7370b75996d6f8599ae1f28d
src/common_main.h	258	00c2607686f4b979a45b494dd6465be74028d95f3758fa9cdcccd
src/compiler_util.h	156	766b232901874eff32c3045b845cdbf8b7f26bede78a78e6ea666
src/da14531/da14531.c	148	0e137eb5f9ed0c6017372cebe6a13c665f179c73d7ee74c5281f6
src/da14531/da14531.h	757	699f504277cca1dec2db35d3a729c3d2496af56bee2d715460a9c

src/da14531/da14531_binary.h	615	ab36b4d19ab2e92690987e13a9bb764070dc4fe1b37f582b13dfb
src/da14531/da14531_handler.c	10934	470474701180c1c3a831ca43c73b3323456cc31f063eac2cc536b
src/da14531/da14531_handler.h	488	c4f4950fa1e7095bf80defffee2d29224512c369d4d42a51c3b37
src/da14531/da14531_protocol.c	14092	fccf1437f23c480e2b2675366367ab4ce8d4f41ebef6584a5c729
src/da14531/da14531_protocol.h	2026	b35f9a6a828becdd2ae5b9e93f8615982afd923518df14ee269d3
src/delay.c	2214	d9b814da8a0b29b9316c7d4d9062aad566376b3f580111e44e62b
src/delay.h	527	b181e4281855a1a7ee890769b72d7884157729456e11760b93319
src/factorysetup.c	40551	f23340c080fa71056e24109f64c2404ff90acf2b6bcc516e5daa2
src/firmware.c	1346	e8c3cd856d1329684a6519d907dc516953a5da6260bb895cb8288
src/flags.h	4465	20b4d9f439ebac1783b6a6a9b58d3f44f5795a4da84148d9f7772
src/hardfault.c	1227	5d2a769cf3e77da97afcea5b416c9853b2c8db0db8bfb6c52a9de
src/hardfault.h	858	5944df9738bce65d30ad8a83310a4ba489613bc075efd653c2807
src/hww.c	8090	a01aa7f5b50ae9766c9316cdfc8a4fea04f22202b5dfeec770bbb
src/hww.h	1190	3dc95f986eab99ef6868410b64498f5af12e661ab68871c0c8c6f
src/i2c_ecc.c	2276	ed06ab53c14f7911372708f60f040238decc514e7163c2decb99e
src/i2c_ecc.h	1462	c36dd2e4b88288c1d4d5422884b2eaf273edc3973dc880ea06c30
src/memory/bitbox02_smarteeprom.c	4544	2d4869e79d0baa5614a11659bea610b382cf25f17fb5ba76683ca
src/memory/bitbox02_smarteeprom.h	891	a014fb1fc58e11fcd563a9281dd3b992c0aa75385b54be6124ae9
src/memory/memory.c	35229	10d47933c8921fb95a5c15b3e91291f273a5ace9f1ee9c98568bf
src/memory/memory.h	10717	f5f44c8b0780a106f193a012d03db97abf90230eed6b88491df81
src/memory/memory_shared.c	8175	4b5997eff681a1063e56e689a03adae0a6edfa5ccddd5eb12f966
src/memory/memory_shared.h	5977	b7c85bf18214d96b882c4e5ababf4bfe5194a8d29e3db15d9d07d
src/memory/memory_spi.c	2980	5c7ec0943599eb710ab7d2b1681fa72fa2e7148ad26f1f2d73e90
src/memory/memory_spi.h	1279	8810697f34c32b38045a4e398daf86d59353a28cb0b35a9bfe0bb
src/memory/mpu.c	1183	428c3a5c05260bea2d6c8fd7a9648904c05ef48e756ce13d54ec8
src/memory/mpu.h	3189	e3e8eeaa6c7f657e82c8afbf80ee43af9268ac058bfe93c5bbe28
src/memory/nvmctrl.c	383	51c13e773d9816fa9b3d8a4fb402ca07aec6d745072f4616c3c04
src/memory/nvmctrl.h	312	eb730b09fd1ba0120da5f956d8dfb4d2a39f5ca55f8660d8777dd
src/memory/smarteeprom.c	5033	28653e4583b809e72e8bb6104a23db25c6fa963dd12f0a4b7c9b1
src/memory/smarteeprom.h	3092	eac71c154b3af713597f18c878aa2c2d2b72b43a811df1220b467
src/memory/spi_mem.c	11033	9f2bba7727e4ee26a6e3c29b3108259feb98bce5515d8397ff9df
src/memory/spi_mem.h	3912	032202c6f683a3a269ceba9e51ad78d9b11bd12924eb29b765caf
src/optiga/optiga.c	36862	d81c6f4b45d9923458bf7b218170acfaf729a6d36876d9bea34b8

src/optiga/optiga.h	4906	d992cdf0cda34979f1a75dbca2c1307f5aaf6fded168246fe36a9
src/optiga/optiga_ops.c	5215	faeb33491aae26ecb601778ffad183100f4fd698bda52e9b1321d
src/optiga/optiga_ops.h	1694	ab2ed1d32962652a76c14fc95fd58b1ad24d273d3a692a896d2cd
src/optiga/pal/pal.c	1558	3769a1635f88870578cbd926257df76749f1b9774f78bcf82569f
src/optiga/pal/pal_gpio.c	1979	89e0f438ddc439ed69d259aa4f760fe92f41f2f5751fcd1b570b2
src/optiga/pal/pal_i2c.c	11391	8e9c342201674cdb6241e0b3fd9dfc7efa17100944b49c8660fff
src/optiga/pal/pal_ifx_i2c_config.c	2906	e4ff9782e587b6fb265086c5dace6a2ae8d2d15709d7a79dcd085
src/optiga/pal/pal_logger.c	2858	ed4810ab3dbb786ec7cd1902053f342869822b9ff765497cd256f
src/optiga/pal/pal_os_datastore.c	2763	178127d32e56421ee047b0cc0e6ee0afa935fa85c717c7da6cdf1
src/optiga/pal/pal_os_event.c	3355	245c5a1d4c696fdbb90764e6b2b5b8dd59fd531929e856f3a57b2
src/optiga/pal/pal_os_lock.c	2603	71ffab63bbf2f395b09df0a1b198a6d418473d35b06263f646398
src/optiga/pal/pal_os_memory.c	470	7b29109eb25ffa9f501f444ac0e7675a8e9cea8068a395a6b85ef
src/optiga/pal/pal_os_timer.c	2550	ecf6f183c7cb7b6c865f9e58ad392ebf6a69718c7faddceeceed0
src/pac_ext.h	654	1b829db5cbc08b8d68626be28855dc5508bd39233aadb20cd5b34
src/platform/bitbox02_pins.h	1308	7b73cfee547ae31cadde79c3c727001b7f8268b132a446d7884bb
src/platform/driver_init.c	13012	ce0ab53cd6fab718023fca835d52705f61e6b1b412d85fba33e74
src/platform/driver_init.h	2021	057862dc09feac0d0d67e191dd027d26cc0d91677b29838111b46
src/platform/platform_config.h	1732	c6a8ff515fcf1c87f79c366b42870d3ad34b09adac5b02a4d6451
src/platform/platform_init.c	2465	cc2d88ab9b7eee346288271974f1200f40ce6b33f233877990d18
src/platform/platform_init.h	359	d4c05fefa903b6446086660da71863f44bc3baa11b76f7148caf1
src/platform/sbrk.c	1187	5552fa9172238701b473d3b0bfa0df1b1906e85037596958d1230
src/pukcc/curve_p256.c	4832	f4aba16f4398bbf89102e62b826cefc8c02d3a4731a73387393d4
src/pukcc/curve_p256.h	211	c620f92a051fc37d6c4b976a72cadc61f24595dbfa1dfed6326dd
src/pukcc/pukcc.c	9798	b5049dd725362a4217e5797928ba8ad32e3c7235d029f366a411e
src/pukcc/pukcc.h	1169	a7650ddfd0bfe8a9c23dce685d1d648c412ea08b65c28f1403569
src/qtouch/qtouch.c	28561	a3ce935421670987dcc50cbb7fa710d2e507fd5f5dff108a1c53e
src/qtouch/qtouch.h	10273	ece235d4e91d9b3b459490b835463f2c1e9a621ea7e327c42cc5c
src/random.c	2290	8e70d7c104bf78ace2348c526b5e4e12625ad8bb56b0938be2bcf
src/random.h	711	8c2bca629ef2ac0214138b7c3756b271540426323fb18c0a80b1d
src/reset.c	648	817e226f032c6a0fc76fde0f4906ff0c6151f609bb4d6c38d4fc8
src/reset.h	243	7050c993ba457968af7b3411d33e22754620c9aee2a8000305488
src/rust/.cargo/config.toml	1932	a6bddaebf210d9245b6e206167b95b578e86e14a44e8ebf6c588f
src/rust/Cargo.lock	37444	26c6fdf274b15f25b33b594f5adaae1be5d54998c1ab4584a91a5

src/rust/Cargo.toml	4099	73bfb23592887b4a84691a76f5034f4c29830dbcfeff26eb17d8b
src/rust/bins/bitbox03-boot0/Cargo.toml	602	b45eb014db8ef2e7ac79a791989422bcfbae6bf8e6bd6254b37d8
src/rust/bins/bitbox03-boot0/build.rs	1188	ce807537a25ae8a54e4d07708b44fa86de7cd518bd820c4fd493c
src/rust/bins/bitbox03-boot0/src/main.rs	1411	65d8b40c8876936d6b38e3fec8aa11fdc622c8a984a0ba697abe7
src/rust/bins/bitbox03-boot1/Cargo.toml	602	dd41d2262a5b31bb5233966eb1a761d326f1ceff70144a451f406
src/rust/bins/bitbox03-boot1/build.rs	2885	56d2bc01330a026f7ecca4fbdd782a6ab33a2f99e50a75cbae2f8
src/rust/bins/bitbox03-boot1/src/main.rs	1452	5147e04dc133dc02a987c6d5fea26176da929eb8adbb89d36cd57
src/rust/bins/bitbox03-factorysetup/Cargo.toml	900	7e0ad6e8bdd8aae31eac633c9da465cbc325a4b27f587e3a6f320
src/rust/bins/bitbox03-factorysetup/build.rs	2285	d29edafa35af5298df345ada8e61b938812591fb73cf0655bb022
src/rust/bins/bitbox03-factorysetup/src/main.rs	3628	a253e1c4415b9ffd333355e274b1ec08c9cfa1adc0eb9af9cbb97
src/rust/bins/bitbox03-firmware/Cargo.toml	819	93fc8dab006fc6c5b9b5336b7ac129445d9a47a4aca353989674a
src/rust/bins/bitbox03-firmware/build.rs	3198	a7109b841bc352e2af96fd651504faa5bb14d263843df3f99a3e1
src/rust/bins/bitbox03-firmware/src/main.rs	556	e02dc06cc0918423fbc2fca5b373064fd2086c68ee174e9e7b41f
src/rust/bitbox-aes/Cargo.toml	315	f8c0eda103f0f2a76ea59a2ee7469215f09951ef490a28b9bb1a1
src/rust/bitbox-aes/src/lib.rs	189665	d1b6231d59cddb178fbde87ab0f9ab35fb8bdbdc5c1d197e8298d
src/rust/bitbox-board-stm32u5a9j-dk-build/Cargo.toml	331	f3b43b17755cd35be3bb7cdc2a5a3f635864ba7a72098d51c0898
src/rust/bitbox-board-stm32u5a9j-dk-build/src/lib.rs	2349	ef079b608145f4834765b3aea1cee1f7ab519853e1e9892f712d8
src/rust/bitbox-board-stm32u5a9j-dk-sys/Cargo.toml	363	e86eece21be38507a20b4c38642399bf2e5d404343c0ad8ac8cf3
src/rust/bitbox-board-stm32u5a9j-dk-sys/build.rs	4573	94c13a1fdaf427adcdff8c386e2a44644c58c21ea8e7f2e7261fe
src/rust/bitbox-board-stm32u5a9j-dk-sys/src/lib.rs	272	1b57158882a13471de4be80cc918e22e9628a33222e9d20fdd21b
src/rust/bitbox-board-stm32u5a9j-dk-sys/wrapper.h	141	100e3781691854b92ceb38c0eb2f61f40f7da3c1245e7b158a648
src/rust/bitbox-board-stm32u5a9j-dk/Cargo.toml	497	b41260fdc41c8a6146e9325384c72c19e3eea5bf026ca13af9e37
src/rust/bitbox-board-stm32u5a9j-dk/build.rs	2484	342a59fec95ff796491f0382609d4885c17a70bd9c4ca4350cb5b
src/rust/bitbox-board-stm32u5a9j-dk/src/lib.rs	629	98780551b6e10d5bdb56bca266d7c84c7bd5b2f1ec9cfb4d985ca
src/rust/bitbox-board-stm32u5a9j-dk/src/memory.rs	4370	59076a19e7152ab2f5a5aab0d9005a31fdfa297bb68e03f37a18b
src/rust/bitbox-boot-utils/Cargo.toml	143	25690b5da7d4452621cd5ab17a91bea7ffe634075f88dc5a97bba
src/rust/bitbox-boot-utils/src/image_header.rs	3066	0a54aa9230fd76555626b278bfac1ea7a7a280d93ac20635a1efb
src/rust/bitbox-boot-utils/src/lib.rs	22385	517532d8b9006f867adb50754bcdbf773e43afb4fc7e90abfbbf0
src/rust/bitbox-bytequeue/Cargo.toml	230	1b9256563d1f7b139b57c44e71673f17b4e6adf34023ebfddb15a
src/rust/bitbox-bytequeue/src/lib.rs	11527	a297010473f7b7f5043a2cead7181cccbc5dd5acf852059bd2cca
src/rust/bitbox-core-utils/Cargo.toml	535	ea9a88c066735da317bb3c937e841a12c93b005dc41cbff764708
src/rust/bitbox-core-utils/src/lib.rs	103	0096bd43bcfd4ce09e49a2e95943e5bf27f8f73611d08b61dcbd9

src/rust/bitbox-core-utils/src/random.rs	4073	27fc4f4a389b1010b5ce0f37de67f00751724b1a9fdf83d10e2e6
src/rust/bitbox-core-utils/src/salt.rs	1973	0a5be445ef7f2d5a2b1df43462fff5b4dd52b7a2e5cb6e5492ca6
src/rust/bitbox-da14531/Cargo.toml	401	dc7a28ec383a3eb83ecb0f364d10b1da99b32f8be098e99048869
src/rust/bitbox-da14531/src/lib.rs	7209	ff8cb7cd0e1ee5290e86ce22f3629aec737ebde2ca1ca5cf87334
src/rust/bitbox-debug/Cargo.toml	275	82ccb7a869fb1c4fcfa222bd5fa9e74389a268f45d746282a625b
src/rust/bitbox-debug/src/lib.rs	949	140a69f4d3b87b65e651e7af56f409c94b8db6324409636938c24
src/rust/bitbox-executor/Cargo.toml	267	98befdb3a6d49e8b576b0ac2f21918db7f94b0d24620878be6e3c
src/rust/bitbox-executor/src/lib.rs	7140	facd59247f5511a85d021b99e248223d4c98ca45e97a89fd2824f
src/rust/bitbox-framed-serial-link/Cargo.toml	360	0b715df72f2c74113151e0b699e79c6661e8f2091e0bb80434aab
src/rust/bitbox-framed-serial-link/src/lib.rs	17048	c97d73bd553608be30463d109f86b23b0284e973c5ffdaadef961
src/rust/bitbox-hal/Cargo.toml	252	8e639839c006a38521fe6ef232ce70588f8b88a404f68bf005352
src/rust/bitbox-hal/src/eeprom.rs	1948	200715d25813eeb9d50c237a7c8d4a0861cf7de64d38f81b6fe0c
src/rust/bitbox-hal/src/lib.rs	1913	456f9367a84f57322c65ae911c5e1aeb41c33c020343204d4b78d
src/rust/bitbox-hal/src/memory.rs	4151	ed950634d4ebb11fa50c62f323f2dcd92901986820d8c0d769146
src/rust/bitbox-hal/src/random.rs	172	3d53abdd13fc8d9c29b0534d41864d1f8fb97c9a04c38da2ad112
src/rust/bitbox-hal/src/sd.rs	588	7a3c3a204b2985238560a1951307ae884f86f5c9c76e66391b0ad
src/rust/bitbox-hal/src/securechip.rs	4226	da71595a0cca42f468cb939ff2a6fbd9c62eed395caf4d4d16425
src/rust/bitbox-hal/src/system.rs	1019	6da346170863b3ba80e5d7ab3e456fa322d94d0a36ab1e1a1769c
src/rust/bitbox-hal/src/timer.rs	159	ab7bfc9e02e33d3b86168f606ced6332ea33d39b5503fae655b11
src/rust/bitbox-hal/src/ui.rs	8042	bc9fb5c0d36d1a1e879cd4d02e116998272c37f1e537d48eb6a50
src/rust/bitbox-lvgl-sys/Cargo.toml	248	389cb4c1c28f0ccc4a422025b75061db160cc8fdbfb68b51f77fd
src/rust/bitbox-lvgl-sys/build.rs	9866	d28e575ad9baa893bce904d3b2b74b2db3fb600a1b4ee36f6a911
src/rust/bitbox-lvgl-sys/lv_conf.h	51220	5a13d194e9d7013099caf89cea5091b78ff16a22b9c9ea928f727
src/rust/bitbox-lvgl-sys/src/lib.rs	341	fcab8b13940551094a4a7a4fe477bc6186711f50e0dfa2026dfab
src/rust/bitbox-lvgl-sys/wrapper.h	338	acd8d58b7615012977b0147392b8c616a71205c3b2cbc66b9953a
src/rust/bitbox-lvgl/Cargo.toml	268	daefc6d741706e60098419b9ec4d5b2ceb5a3de23878b9bc72060
src/rust/bitbox-lvgl/src/color.rs	473	71d7bea77d1589672a954ef696f73db64bb36192eae7d0f37898c
src/rust/bitbox-lvgl/src/display.rs	5374	916b3d2e5a95aea091b5f9368115cc017ce770ed6c7f2e1fd2f00
src/rust/bitbox-lvgl/src/font.rs	1900	5f87b8d677c62d374ffe1c4d4c557abcec777306bcf8e8c877840
src/rust/bitbox-lvgl/src/indev.rs	2037	bbf58a031fdbf5b41ef0f9b6616b54a24891ed27481c92a9b41d6
src/rust/bitbox-lvgl/src/lib.rs	3486	7b69d1adbd4abb46f93f8ab9312bf857383eb1f18acf73e311621
src/rust/bitbox-lvgl/src/log.rs	985	fe230bbebb5892fa4ea4fa3d6f3fe7b0034bf21fd318a2265db03
src/rust/bitbox-lvgl/src/style.rs	3816	6db64a1617b4f2647ec3d2c6de756f7a027b773a27bd7d92f2778

src/rust/bitbox-lvgl/src/system.rs	105	ffd20d867f7d4328e68d59f135e9134ca691dd5851cd271a609b6
src/rust/bitbox-lvgl/src/tick.rs	157	ad20354b120701fbd19ef4be25425cdc219eeb71e5951ea0c0ab2
src/rust/bitbox-lvgl/src/timer.rs	130	3eabc6b60fd7223fbef597a37088779ffda143d1bc6985f3918f2
src/rust/bitbox-lvgl/src/util.rs	1312	f3f9f97063188336dd790b1aba5c0b085020096af6a57711c0401
src/rust/bitbox-lvgl/src/widgets/arc.rs	3409	0a116c3cc38d71e7fb015b929fe518a3dcbd44e7bf0d046390c0f
src/rust/bitbox-lvgl/src/widgets/bar.rs	2207	1684518d5e6da6338ef0df80538b98941221ecb55002d737b76a4
src/rust/bitbox-lvgl/src/widgets/button.rs	521	3f8a2ac68192d05fc1539694ab44e2f6f8b28c6fc72bb1d2b7574
src/rust/bitbox-lvgl/src/widgets/buttonmatrix.rs	6488	5024d703f760384f238652d387ded9200aa1baebce14e48d5c084
src/rust/bitbox-lvgl/src/widgets/canvas.rs	4115	a5b70aa02b2edb1924c0945b781853a9172f0ba1f18a66dc830d2
src/rust/bitbox-lvgl/src/widgets/class.rs	2361	efa9537454a60a2d4db805c8139483aee33a8fd71e7a1a9d3dc59
src/rust/bitbox-lvgl/src/widgets/image.rs	3280	fd27b5b532c89fea4ad00cd82e0aad8551b4b8720565d3e09a0a3
src/rust/bitbox-lvgl/src/widgets/keyboard.rs	4508	267840305dfce7dd03cee5bbd73c66b2c0acf6484fbd6937ab971
src/rust/bitbox-lvgl/src/widgets/label.rs	3497	9c01936e4c2ffea788935298186fa3165630ec5d5d7993cb1aad1
src/rust/bitbox-lvgl/src/widgets/line.rs	2822	65f099ad38111fdbd0092cea34299168ef3886c47f10a94569fd8
src/rust/bitbox-lvgl/src/widgets/mod.rs	2567	4a1f2a556ae514d9a87b742134f675d8f8fa041a1a832699c434a
src/rust/bitbox-lvgl/src/widgets/obj.rs	26478	986a3fb1a9bd5628a7428f9feb7f6d9c2b0aa2436ac7e25ca5fbb
src/rust/bitbox-lvgl/src/widgets/slider.rs	2700	7ff614b820dbb41728c77822445f8d9acabf0eb2405fdbc54c13c
src/rust/bitbox-lvgl/src/widgets/span.rs	18294	d53070663d6ebff5243cbffc2a73460492fd4d5e6465a7a099479
src/rust/bitbox-lvgl/src/widgets/spinner.rs	774	c3f69a2baf9c498ba1574b2133fb1a10fc3f66e01ca0efe6735f2
src/rust/bitbox-lvgl/src/widgets/textarea.rs	6641	c0a550e929d8ef527d767c582394858c4d0a3a9eeb1b3182a4298
src/rust/bitbox-lvgl/src/widgets/util.rs	3599	64c737a005a0aba480725702fe5539a2fbfd92ad419678a82bfd0
src/rust/bitbox-mcu-stm32u5/Cargo.toml	356	4980c20d6863a9faddd52520b0ea0f39205f5ea2baa2441d203e6
src/rust/bitbox-mcu-stm32u5/src/lib.rs	87	33f59b81e49b867519f3e09635f27a1a46f6bc4acd9d0f4aacbbc
src/rust/bitbox-noise/Cargo.toml	620	1903f4c84f747fe0de9b1af349aa437b9ca54624f36075226df23
src/rust/bitbox-noise/src/lib.rs	556	6787c6b484f801c7d4e95443f0512b7247bff8dba8f871c54b92f
src/rust/bitbox-noise/src/noise_xx.rs	10043	4ade64136c560f523c56ae0e567721938ae097ab6a192c547f589
src/rust/bitbox-noise/src/x25519.rs	2023	63617ea3d8acc0f018c1664ce5abae381d71c29cec4b9f994bd04
src/rust/bitbox-platform-host/Cargo.toml	534	e13d76508db7d5e832dbcea9d159fd868d9705976d3f96491af58
src/rust/bitbox-platform-host/src/eeprom.rs	1808	e90f8286ddc5068a56e3b59f033f32e0292148117c35f826df0ee
src/rust/bitbox-platform-host/src/lib.rs	199	aa279b040b8a9357dd020dba45c6996af8780a46caa027bd1adb8
src/rust/bitbox-platform-host/src/memory.rs	12192	702d8734b4db0d0a259d6bc831b273411179bf57fdad065bd47c6
src/rust/bitbox-platform-host/src/random.rs	2083	fde390232194f51b1530025eec812bda2d16781b2ff11b7fa72b6
src/rust/bitbox-platform-host/src/sd.rs	3060	3f40d002d85cd970f8c70473f32197c54313c5766203d5fce33f2

src/rust/bitbox-platform-host/src/securechip.rs	7297	a383fe084917e785e7794e8950397b3fb5d39628ea0a926efdcb1
src/rust/bitbox-platform-host/src/timer.rs	1617	0e643681b0e7089742a91cb2621e39fa2123f634f50e908f61407
src/rust/bitbox-platform-stm32u5-sys/Cargo.toml	345	f00953d4fdd7b0a339a4275ba3a016813153790f1440bda76edff
src/rust/bitbox-platform-stm32u5-sys/build.rs	4624	e5a03d01598e2c749dc94ca18a3d1c069e1726108c863a167648b
src/rust/bitbox-platform-stm32u5-sys/src/lib.rs	1804	7779a494a7b1f81a99cfe6da8e827ffaafa632b1e15d543bc8b55
src/rust/bitbox-platform-stm32u5-sys/wrapper.h	67	7bdf790812e8172badd61a9db85127dff2425a4654ecad202c184
src/rust/bitbox-platform-stm32u5/Cargo.toml	718	5501510cdaf7eb902f231f719b19f3fec7a542b4dc3160310cacc
src/rust/bitbox-platform-stm32u5/build.rs	597	ab8ee6746d73105d4ea8cdb73205d7b3422b288c16cd5922c2ea5
src/rust/bitbox-platform-stm32u5/src/flash.rs	3007	a128bfb727a5d77c521eadc35601cc385be833f8d0ce7089aa9bb
src/rust/bitbox-platform-stm32u5/src/inner.rs	2010	f3c9459d7051aab14209a7e53d6995a9c9c2aa500ff9b1c910f52
src/rust/bitbox-platform-stm32u5/src/lib.rs	185	2ee9aac744280d9c128b6d74eab47a33a60e8f4524aaf75cf6ae1
src/rust/bitbox-proto/Cargo.toml	529	21decdfa25e99c6645d4bfa993c898bdace001c059e9f7fa7b002
src/rust/bitbox-proto/build.rs	808	6d4f6136eadec490c81d39aae758968650da4c2442022749a034d
src/rust/bitbox-proto/src/generated/shiftcrypto.bitbox02.backups.rs	3421	45f3b31bcbddbb90b7b8b29ebb0fa4cde683b573ea309bfa41e9c
src/rust/bitbox-proto/src/generated/shiftcrypto.bitbox02.rs	86465	5b538703f43ef6f30896a79c051a3e37caf7b6c365b5493db6d53
src/rust/bitbox-proto/src/lib.rs	1372	1d36872343cc062e807a7315fead89a405251f59fb66be5980de9
src/rust/bitbox-secp256k1/Cargo.toml	484	77442775d85b6ad99b047180d03c077b5427273e767a06175c5be
src/rust/bitbox-secp256k1/build.rs	1565	91983851f534bb1e29d57360162a303f9847ba6378c5873aa4088
src/rust/bitbox-secp256k1/src/default_callbacks.c	297	da45e3b9ba84a8289455a354a0e12d934ac515a09c0c0080afd15
src/rust/bitbox-secp256k1/src/lib.rs	15733	42fc499883f91dea548d9ff0b11ad9770a60a560c74271563e66e
src/rust/bitbox-securechip-sys/Cargo.toml	259	c8ecf19222651c1788eb0599179b20ee850fff55cc34e3ab9c369
src/rust/bitbox-securechip-sys/build.rs	7537	3c435995bb9a2ac536ea75f6bc5d2857a14214b43e24d496e5413
src/rust/bitbox-securechip-sys/src/lib.rs	311	1531ac58123bde8556099e327af91ec8e5db645bd971854fa6876
src/rust/bitbox-securechip-sys/wrapper.h	123	c2f4dac6ec8ac622887b6938beb8ea123b16d32fc64c151256fc5
src/rust/bitbox-securechip/Cargo.toml	862	6f3d69d24c02306d7c1b1ebb8d0e6f53e386217cc4132f786bdf4
src/rust/bitbox-securechip/src/atecc.rs	3668	a8cbd36ebbda21ff5f525a3842e75e94fc76bf377aba97ee8c924
src/rust/bitbox-securechip/src/lib.rs	3301	9d71da45b4b72c61b0f0648a6e2756ade7604486ecb0f2d331ca1
src/rust/bitbox-securechip/src/optiga.rs	37182	dd80d44ba7dd070e3c0c1c2ef42c318cfabf7e96eaaa9817aae6e
src/rust/bitbox-securechip/src/optiga/der.rs	1945	51c9a5b5999f696830104a30fc38669b68985cc1fa15ebe1a2c87
src/rust/bitbox-securechip/src/optiga/ops.rs	20626	28f6ef019296d7087c62a4ade4089952ed70e8b5f4f1151b1f206
src/rust/bitbox-securechip/src/optiga/ops_fake.rs	15685	de3f330803a923572ae8e073ae4af8a057a0391c32a6499d62bd4
src/rust/bitbox-u2fhid/Cargo.toml	212	3e941a6766699c0e11a57e5a81779d4563e0ef4cf9b71bdc56a4b

src/rust/bitbox-u2fhid/src/lib.rs	23418	7b70b84d5579fba6237c5a176e5479c5b94502f655d63e81e0474
src/rust/bitbox-usb-report-queue/Cargo.toml	291	053b288e4a5823e95b6362dfd133f63f2bd87634db7343b95c931
src/rust/bitbox-usb-report-queue/src/lib.rs	10931	5a161e6b53648bda9c93dc66bbe58e5f1de2019925404073a2ebe
src/rust/bitbox02-rust-c/Cargo.toml	3932	2c730e8b213a5d8e3a381d0a5e4c2745e0ba2a12344883db570fd
src/rust/bitbox02-rust-c/src/async_usb.rs	2063	47a9499822950a6254d095bc1561302952fe6335fa5563cc400b7
src/rust/bitbox02-rust-c/src/c_alloc.rs	905	471866858b22495db4c369feba9db9b5af816cb01d5e4c87f0f88
src/rust/bitbox02-rust-c/src/communication_mode.rs	378	c8d296555831770fa32351f31bad19787ed3693d366c9f42f2e4c
src/rust/bitbox02-rust-c/src/firmware_c_api.rs	5436	029d99f0ae93aafe5fa6331d4ed1787be0189241340a750180529
src/rust/bitbox02-rust-c/src/lib.rs	5125	c872df632ac63a1f4735660e6ac132b661f80f0939a38c0791420
src/rust/bitbox02-rust-c/src/secp256k1.rs	6624	54c1ff4e1f9ed19a0c92a26d1f83cc8a2fea72509e0394ce4ed01
src/rust/bitbox02-rust-c/src/u2f_c_api.rs	9830	f9dd4d7daa5fcf8b4361292668bfaf314c9e817e2a6d8bdeb8f22
src/rust/bitbox02-rust-c/src/u2f_c_api_stubs.rs	781	e7f0952fdbfff2c7a2f00c76c01ad871c4ce0bcc8d783a445321b
src/rust/bitbox02-rust/Cargo.toml	4607	70708f2472b1cc14d6973c14214edd75afdc82ef08293b74fa321
src/rust/bitbox02-rust/build.rs	2374	0fdd53625ca0ef342a07b6511534ccc0413dfb7ae06c455dbc58e
src/rust/bitbox02-rust/src/async_usb.rs	16034	f24da5058d7c408f62069012f5f2463b1c76a5421a3d4e370c556
src/rust/bitbox02-rust/src/attestation.rs	2902	c20b01cb077b8baaa32a6ee878c0fd3038266ad5659829bdfbb1e
src/rust/bitbox02-rust/src/backup.rs	21345	4efaf420eec850fec73301ebd5a101c28acd12a1fbcad82d7a2cd
src/rust/bitbox02-rust/src/bip32.rs	13090	69904362ecbb2911101474f83924b9dd5cbd17e7ea748a733573e
src/rust/bitbox02-rust/src/bip39.rs	10472	3066de8005955ca3128da1bab484d55ba24d368a5ea113d36f1d8
src/rust/bitbox02-rust/src/communication_mode.rs	1970	f0301d116abedb634542e247de509b5893e48290c851a420a948c
src/rust/bitbox02-rust/src/general.rs	367	51fc5e6dda75ecf8de8e25c673afc1bb44214079f9fc848b5f1b7
src/rust/bitbox02-rust/src/hal.rs	120	18d3943a548c4b5d51e366ea87f01d61e419958ea8f91002d931e
src/rust/bitbox02-rust/src/hash.rs	1517	4b667a9836bade2671d5c320ba17c8644e245b613d4e6cee8d5f1
src/rust/bitbox02-rust/src/hww.rs	28291	2927bcad9778746803fd0da2c14bf7959bea20c137c5fd572a5e3
src/rust/bitbox02-rust/src/hww/api.rs	8874	4e88939f98f0466f8f635eac082d397f58abb032590ea8f0a29af
src/rust/bitbox02-rust/src/hww/api/backup.rs	14916	5374eff995386d703bcf4520b299484ee5673049bcb93f968f49c
src/rust/bitbox02-rust/src/hww/api/bip85.rs	4647	ca089a3a18442b292edfe900b31b8c59f600cdaa5e9cfb287c468
src/rust/bitbox02-rust/src/hww/api/bitboxsync.rs	41098	e15d530066a769341285a06cd933c9a5bb5fbffee95c9d6abc10e
src/rust/bitbox02-rust/src/hww/api/bitcoin.rs	53465	181410dc293fae8eb397eeb72e4b0ac6094efb5740c9d8de249b1
src/rust/bitbox02-rust/src/hww/api/bitcoin/bip143.rs	3095	1f374135c58bae745e88d9500fc29164ac723d522752bf67fbce0
src/rust/bitbox02-rust/src/hww/api/bitcoin/bip341.rs	5494	7217fe9028053addc3c36337b2a43958d7e56a3a4d8a92c1e630d
src/rust/bitbox02-rust/src/hww/api/bitcoin/common.rs	26882	b8a6f3c01008c0631fb04cbc9b8f389c6cf30de321502f9941101

src/rust/bitbox02-rust/src/hww/api/bitcoin/keypath.rs	18724	d5d08ddb0cf20c3ca0497a723a08d748f863118b90d170ddf1a0d
src/rust/bitbox02-rust/src/hww/api/bitcoin/multisig.rs	42734	cff6a02dc53347895b0f6c7562c316eb354eba5a81ecaf8cf111d
src/rust/bitbox02-rust/src/hww/api/bitcoin/params.rs	2291	fae19e0b1a338960fb97055efcd6bd6c1d1572962faf960644cdf
src/rust/bitbox02-rust/src/hww/api/bitcoin/policies.rs	82001	abc68ccee396717d58844920139f2d29bb75a4fabc74f97689747
src/rust/bitbox02-rust/src/hww/api/bitcoin/registration.rs	12363	338b00774cb29f0dbfe5a480dc6c944f70fa72d112e976fd4e9bb
src/rust/bitbox02-rust/src/hww/api/bitcoin/script_configs.rs	5506	17594a528cf37f0398f0deb18b16e996060fb670dae8520de29c2
src/rust/bitbox02-rust/src/hww/api/bitcoin/signmsg.rs	25339	7f2d957f821c1930f16dc8c1de5982ece4f50b3468307cdbc8925
src/rust/bitbox02-rust/src/hww/api/bitcoin/signtx.rs	138278	195b930b906cd84cefc9bb4ceecae73268d702b22b019f80b1899
src/rust/bitbox02-rust/src/hww/api/bitcoin/xpubs.rs	6970	4bc855177239b7b5cf15dd545a461bc421f39555d8f7fb3a65b67
src/rust/bitbox02-rust/src/hww/api/bluetooth.rs	14999	54fc1a9ca2d9412a3708551436f0d0b2b4c93ff2aaae3f51deee8
src/rust/bitbox02-rust/src/hww/api/cardano.rs	782	fb2b77c02fecf5d5b0927aff56ec1ee7c4d90fdbaf3e112043b66
src/rust/bitbox02-rust/src/hww/api/cardano/address.rs	25289	2edeeab72109e5d9c5244ae16cf22b7397aa658364c5f27466538
src/rust/bitbox02-rust/src/hww/api/cardano/keypath.rs	9489	669a3af625f388c46d5718aeb4cd88476c16a2629e3d12134b811
src/rust/bitbox02-rust/src/hww/api/cardano/params.rs	1354	d7afb92488f4c471889b2d5db2526630a474377dcbd903d85659d
src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction.rs	74982	5e94e09ed5e9e28ba9935be3120f5926f5ebcf6c4a275e79c2a25
src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction/cbor.rs	11409	89e1d37d2bfb99b67ab18817241be589831874e5f56a5f0075f3a
src/rust/bitbox02-rust/src/hww/api/cardano/sign_transaction/certificates.rs	5772	039d3fb80dec40bf724d7daaf3dd4ab7c9d3f1d35b159a04503ea
src/rust/bitbox02-rust/src/hww/api/cardano/xpubs.rs	5566	ca845c4de879d93b868925ace61c0471acf3041011d8bb63f9162
src/rust/bitbox02-rust/src/hww/api/change_password.rs	8093	df7e6672fc66ba2ce24c092ff0b58224b7181bf87fd978f501eea
src/rust/bitbox02-rust/src/hww/api/device_info.rs	3187	cf7700970753c5467429c85b88e4f21dd87f88aed83c0e879cfc8
src/rust/bitbox02-rust/src/hww/api/electrum.rs	3010	9cf68840f7aed0f19a71e0ed7771e423a77cf662ae54ffc196e88
src/rust/bitbox02-rust/src/hww/api/error.rs	4335	b6a13ec95227e726cf487267d3a0ed53bc9cab5b1d3860d497dee
src/rust/bitbox02-rust/src/hww/api/ethereum.rs	5189	873ca1378407cae796dfac9032da1bc6e980612f5f2846592ff2d
src/rust/bitbox02-rust/src/hww/api/ethereum/address.rs	6735	57664aa3edfb4b15fc9341ccf65cc9c7da221fd0a59b988b4a4b2
src/rust/bitbox02-rust/src/hww/api/ethereum/amount.rs	4447	78effd3b340e7034d0c5bb3332151d9fca0b944c312f2235be007
src/rust/bitbox02-rust/src/hww/api/ethereum/keypath.rs	5510	e7129e999976e13aaed04d41604ae3e7d65b5d3e8c6da7d5ece90

src/rust/bitbox02-rust/src/hww/api/ethereum/params.rs	6318	6d45acd07d99ea9e40064d668797b13f5b24947beca278a8ec67e
src/rust/bitbox02-rust/src/hww/api/ethereum/pubrequest.rs	11492	1fb38f29c8acc366e5e28ee31505eda46b33d29abca6006453cfb
src/rust/bitbox02-rust/src/hww/api/ethereum/sighash.rs	27142	cccb0d7501d1db215c00bbc9586b3d873625b63606d2a072319eb
src/rust/bitbox02-rust/src/hww/api/ethereum/sign.rs	96116	1891a4b2d603dcb45aedd9b2fc409ddaf9d27a813264145b7b9b3
src/rust/bitbox02-rust/src/hww/api/ethereum/sign_typed_msg.rs	117444	890c26321761c405f6ec0b238a67ecfa3aa3ad9d2c594f2bcbb85
src/rust/bitbox02-rust/src/hww/api/ethereum/signmsg.rs	10476	b4de40b5a63033ec24e52f0077c80a3a73a9471fdc6d10d528932
src/rust/bitbox02-rust/src/hww/api/payment_request.rs	67439	c76bd2db8e0205ee3cf3e26cb00132da08975241342a6f8893996
src/rust/bitbox02-rust/src/hww/api/reset.rs	2174	44fa3b048c424a65d24d853cd2fe6aaa805c37b6f23b2c7bc0bca
src/rust/bitbox02-rust/src/hww/api/restore.rs	9331	3afe08e63f02f6e0e547bced73824bb37800e52551da477c4cc95
src/rust/bitbox02-rust/src/hww/api/rootfingerprint.rs	1540	bec5a60dbc30d5c9fbc4b387d121c35cc82420262422c85693f79
src/rust/bitbox02-rust/src/hww/api/sdcard.rs	2540	9ac1af0cd1531ed86ab3dee9b9ca1711573ef131617453faec1a1
src/rust/bitbox02-rust/src/hww/api/set_device_name.rs	4581	b336413a7ca7043f89df8cfeb781e7c602739c44b6d17fd21811d
src/rust/bitbox02-rust/src/hww/api/set_mnemonic_passphrase_enabled.rs	2612	c1361dc350dcc62f74137349a37b5190afa02dfae0952e622569b
src/rust/bitbox02-rust/src/hww/api/set_password.rs	4915	34eb42602dc38d169e1994f8e360c960562739af09700e633fa60
src/rust/bitbox02-rust/src/hww/api/show_mnemonic.rs	7482	a5180d83d601f5774114f1cba451a2783358e952e52cbbc421a1d
src/rust/bitbox02-rust/src/hww/api/system.rs	1785	a0527f375722906e809a1410e5cbbd8404ec3d19bbd8d4d278e67
src/rust/bitbox02-rust/src/hww/api/unlock.rs	6665	29ec66437733d134968bb4e5b366d71cf0e47aeacbcaa8e5d157a
src/rust/bitbox02-rust/src/hww/noise.rs	5095	84f8975fbe603db802539bd6b6973505ce5938787617d4f53fa39
src/rust/bitbox02-rust/src/hww/transport.rs	16808	371927da8fdefc6f47d8a5cdb7518242446f2340efdcca17ca0b9
src/rust/bitbox02-rust/src/keystore.rs	83987	b7918fdd4b52ef1b088e28bf88f3442f8243d706fdf1db79c60e7
src/rust/bitbox02-rust/src/keystore/ed25519.rs	10193	85cb066d8f6e599c0f1eeafe7aa139d6c51c0bb678c4d0308173d
src/rust/bitbox02-rust/src/lib.rs	869	54ccda57e672294cdd51137f4e529824830c58457644c67d8b6d2
src/rust/bitbox02-rust/src/main_loop.rs	6571	8f8dc3d87d7a372ba046e88c3e844c1ca14b8ff5340d715af4122
src/rust/bitbox02-rust/src/reset.rs	4440	fc08d0007d657de5b8f162034006b1eefd42822948c09ad247aac
src/rust/bitbox02-rust/src/secp256k1.rs	287	4da9a5cfc6fee6086c57c29d11367c6473c6f2bc7e7ec31c593ef
src/rust/bitbox02-rust/src/u2f.rs	6712	3d6bffb5411151496b3962d2f9c5ee559fc86e060f86791f8b2ca
src/rust/bitbox02-rust/src/version.rs	166	49bbc061dde4aa1a081e6e79d3e0a65ef4067cadfeb636c0b02a4
src/rust/bitbox02-rust/src/workflow.rs	296	885ca432cb372563ae134d554254bdbd8b914a880c12c5e15517d

src/rust/bitbox02-rust/src/workflow/confirm.rs	1449	84c4285756710f56d7ccabd75246644e49ae186c2d1d3aae6a457
src/rust/bitbox02-rust/src/workflow/mnemonic.rs	34778	2257e6aceeca61f3939defe9bb98fa092f5ce6bcd4a424b84aa29
src/rust/bitbox02-rust/src/workflow/pairing.rs	1690	345b1b3d5c9873b12eff6f03f40d279bdd27dd077daeddab88199
src/rust/bitbox02-rust/src/workflow/password.rs	6278	f0a995ecf8584cd5c58cb37f20686d39e9654c5cfa35289d267b3
src/rust/bitbox02-rust/src/workflow/transaction.rs	3554	6451402352fb8ff628bba81fabd478aec289f32385735b8b68a13
src/rust/bitbox02-rust/src/workflow/unlock.rs	17747	065962fe57d9cb6640869677d8f5489bc0fc188a060f746dd83b5
src/rust/bitbox02-rust/src/workflow/verify_message.rs	13559	4a2c0f3cea125caeae7d0371c7629ae08b03f3cb6d76f183cd6d1
src/rust/bitbox02-rust/src/xpubcache.rs	11790	fcc3f7f8340292a2d682a108772c490f6d41827c31f331a0edd32
src/rust/bitbox02-sys/Cargo.toml	390	4492ef51f8a188413a4ee87a0dc30f58c2cd5a28f890a63148055
src/rust/bitbox02-sys/build.rs	21680	6370b2397f94e9e9531ae43ecb9dfba9e819e1d7f0dae4c23a3e8
src/rust/bitbox02-sys/src/lib.rs	892	8ed5a6852000ff1e71639e70b0d8243624213c16d85a0f7b64aff
src/rust/bitbox02-sys/wrapper.h	2101	ce7ee138f934a422cd5a49f23498dab908a330bbc1ebba755c99b
src/rust/bitbox02/Cargo.toml	1669	b70d719688aa910b578befca33864e88ee5c4011f2c03e68cb22e
src/rust/bitbox02/src/da14531_handler.rs	584	a4ef27ce8aa7d67aed9ec004d45235b0ebbb9cf9f6cc3f27b4f29
src/rust/bitbox02/src/da14531_protocol.rs	880	63a96c8132ae1e3c14081bd8cef8b1f4617c4ea1ef586091cd8dd
src/rust/bitbox02/src/event.rs	248	bf922d48b6110e575cf44bc25bbdafe56949fd54ef922730cf2b0
src/rust/bitbox02/src/hal.rs	2058	d6915485b75c894ba22b5a62c8b90d521f53157ce6c2d217d4bea
src/rust/bitbox02/src/hal/eeprom.rs	942	8154b22c49dfc6aba4a2c3b9940a0691738d899e4fe1f086c35a9
src/rust/bitbox02/src/hal/memory.rs	13523	473863fe6af3fb418ab72f97c99e8312d305a39517412a675e08e
src/rust/bitbox02/src/hal/random.rs	783	2e1db38e7270a448d0ec60d8edfaec3156bdca102d6c3daa14974
src/rust/bitbox02/src/hal/sd.rs	1306	7ab3174a13fd91733d38d08b24bd2ccafbcf1243dfecaa37d9352
src/rust/bitbox02/src/hal/securechip.rs	11172	1cd7cce092509b4025fea0317afe9aa7ef2e59bcf9958b3c20d0d
src/rust/bitbox02/src/hal/system.rs	1477	6b1b1d5546d581e3ffc1fe9d7da92e52d312e0078cdee9db506dc
src/rust/bitbox02/src/hal/timer.rs	2636	85c0bed052cc43976f4c401ffdf5d83bfe1d7d15b2b84a6c2aa0d
src/rust/bitbox02/src/hal/ui.rs	14681	8e7d3ad2359f392ae852b68da7d6723f831adabedaf5883e9d00f
src/rust/bitbox02/src/hid_hww.rs	321	724870b9fac503e4e30a9dcfa7f3bc066419776ff26fdc2ca94f8
src/rust/bitbox02/src/hid_u2f.rs	321	8643cda0f6b2d05d969ce9173354761b2264745d3c4b65277a5e8
src/rust/bitbox02/src/hww.rs	101	d4cbb84bb30eae55a5d2a57c3b4cd3d1581c465305f0a7aa8228c
src/rust/bitbox02/src/lib.rs	3431	b6e93585807f0e85bf91e713932d289f18759f3e9f90871f3e1e2
src/rust/bitbox02/src/memory.rs	29086	d3ad0bad7493c2002c330e45c3fde403fb95d37db43f9cd3f9c7e
src/rust/bitbox02/src/platform.rs	387	d2ecae56b160e9760f16ced2e9a17174b270fe43e0d7b714358bf
src/rust/bitbox02/src/random.rs	1425	edcc95719949a75adde6a34205951544cd8960c8f8742cf82d7e1

src/rust/bitbox02/src/screen.rs	1206	6b5d0d4eab685a82c934dea778c963e411c91aabd8a62f943931d
src/rust/bitbox02/src/screen_saver.rs	686	f08978cd4f9adc7285150e7fc806a61df24fda669027b9137356b
src/rust/bitbox02/src/sd.rs	3633	332eed271f3a1b655dd3bca4a6e4de60a011e46d9da11fdabfc70
src/rust/bitbox02/src/securechip.rs	259	74f1549267b89c45abeb33faa7b537897aad0c8fc1257ce160dc5
src/rust/bitbox02/src/securechip/imp.rs	6338	83020355fa265b9e4e37c87c4f36c5909b62d628162b1982f065c
src/rust/bitbox02/src/securechip/imp_fake.rs	3211	330cd9da55af9c84a08d8f350e84e2fdc8c1c73ee96ea77c91681
src/rust/bitbox02/src/spi_mem.rs	1334	f62ac7882ba064c5e08ef0f6e6d6fd466bfa68adda78f14a34c55
src/rust/bitbox02/src/u2f.rs	198	1a80974c4b199c6418d90887e5895b2e89ea249936bfd652315fe
src/rust/bitbox02/src/u2f_packet.rs	463	1de6b32c0488616e6343e5ceeb4131d1621a35824139984fb1510
src/rust/bitbox02/src/uart.rs	822	12c23b2e2d988d49d67a3efb9647408309d40fa0e95b1f4054090
src/rust/bitbox02/src/ui.rs	632	29e754c46c1185d2fefa2261446a4d6cdf656ff2f2675233ea373
src/rust/bitbox02/src/ui/types.rs	3770	09f8ea3ffa8ca24ade9793ffd205a1eda3ac3216bb291daf72fc3
src/rust/bitbox02/src/ui/ugui.rs	72	c3c17298da8d70199030dfc0c054cd953f08460fb924f74ad9dab
src/rust/bitbox02/src/ui/ui.rs	32463	df7dbba10ee96a9ded04a6cf9891b49bb239d8374911e637073d3
src/rust/bitbox02/src/ui/ui_stub.rs	2885	44a7c3b57ab698f176f1a6dd661d2b4fdd8e44481e70f3b52da95
src/rust/bitbox02/src/usb.rs	153	965aefb4815cdccdf10f5a2898efd9506e6fff5de3ca33e5ce923
src/rust/bitbox02/src/usb_packet.rs	912	b54e3d1f804137366c10d343e0b69b0433384ec1ff156e15d108f
src/rust/bitbox02/src/usb_processing.rs	1240	60de0f4120276030ec4689de0b3e2df8e83779f34627eeee87533
src/rust/bitbox03/Cargo.toml	400	6927165100e35a7e5b2e199208a4c7db809060ea479f375649961
src/rust/bitbox03/src/eeprom.rs	792	ddfb8813943a33930450b01e2bb1616811ee5ba688a51a5cec5f3
src/rust/bitbox03/src/io.rs	61	2953777db481a5a05c64b72d4f3a9ba953aa2f72a9b38c2ffc270
src/rust/bitbox03/src/io/touchscreen.rs	1866	ae775069eb670b7d1f0e62545590e920d12b10a817b7e031a9829
src/rust/bitbox03/src/lib.rs	2578	38cd67b71d70f90fee279114ad5d1398c695eaa859950b85d5a02
src/rust/bitbox03/src/memory.rs	3894	4245c2ea7d8b4506f821543c2b4f92c9c03b7da8c6575f7d6ac34
src/rust/bitbox03/src/random.rs	259	19138960ca09317cf42a71bd818c0171ddd703c0db9c1ff6a0019
src/rust/bitbox03/src/sd.rs	727	6d7d16479a0cdd1f9fbd270ded12f5d8231208e48299c62bf1086
src/rust/bitbox03/src/securechip.rs	1946	42c32de5b793c2c21ef8cd188192fb821c778cb6b47e5c9d37ecc
src/rust/bitbox03/src/system.rs	446	d748f74fd561901eeec5b1e43bfc50a30d3515b57090f468f7860
src/rust/bitbox03/src/timer.rs	239	71b861386ab450012cb83ff87a1a5799dd4ebfb234b3f7781a65d
src/rust/bitbox03/src/ui.rs	14682	1d353457089c94e7747b248670e64a5203f4407e0aee997d325d5
src/rust/bitbox03/src/ui/choice.rs	3868	161d493f0600ce9c57d8ecdeafde6066ebb5ecd02e94c0e201c8d
src/rust/bitbox03/src/ui/confirm.rs	6097	38396a0ad70cbc306c39df4293d497916bcffde5674b5918c7f7b
src/rust/bitbox03/src/ui/demo.rs	3421	1d9858404809ee7f4b0d9bcbba9b76fa7360386db1064c7784a6b

src/rust/bitbox03/src/ui/enter_string.rs	98270	3742ca93ba5aeed6c2a0a5d1a1a4d0f0797c8b0ee01e5217072e1
src/rust/bitbox03/src/ui/keyboard.rs	33770	8e3bf28083cf243c0c9f45f7c30675ba0655c488ef01480bf8452
src/rust/bitbox03/src/ui/keypad.rs	5537	67c0c3ed07ff6dfbd20609f8f95854356e7bf107bcd00217a3fbf
src/rust/bitbox03/src/ui/menu.rs	4895	cf75ab296292809f1b3c571e2e47024268d50aa889cf81dac52a2
src/rust/bitbox03/src/ui/nav_button.rs	8533	695ae86711d9a7bb9b5e4dbbcb96f988de3200878dae044a86b7c
src/rust/bitbox03/src/ui/recovery_words.rs	21855	8693f42ae62696f45f923dfadb51c55349385f2ff23484a158fb8
src/rust/bitbox03/src/ui/slide_to_confirm.rs	33146	f2cc8a66ddfef7be8e0d521e23104cdfb7eeba3ff1129c99adb13
src/rust/bitbox03/src/ui/status.rs	2746	28cfc602ea285c799d834462b8013ee85bd14e11a0eb94f647321
src/rust/erc20_params/Cargo.toml	309	9e7f84d4c08350c9fd01151b5e00bc58ae558fb6b612678ffc0b5
src/rust/erc20_params/build.rs	5606	036ca79ff51cde8f3d847464408401 85f617b63d2068367058c56
src/rust/erc20_params/src/lib.rs	7510	2692c60e866fd7ef4fbc5c3691b101e43f50255e99c3fe8d877fd
src/rust/fatfs-sys/Cargo.toml	253	8e99c5b88d0e76f8c315afb2a1b7cd57da52bcc40e673b717b7d0
src/rust/fatfs-sys/build.rs	590	19030cba4386dfedb0e812698684f5b7f225079e2c9e3218fe448
src/rust/fatfs-sys/depend/fatfs/documents/res/app1.c	1135	17f99c12e7017ceaf25734165745b5b43b734fdd6fdabf5c8c6a0
src/rust/fatfs-sys/depend/fatfs/documents/res/app2.c	2236	11c9bf944aca5e55c7642225e1a59f5998a3dc43a302b6f1c5310
src/rust/fatfs-sys/depend/fatfs/documents/res/app3.c	4133	e5b1c71593750cda3cb11ca638f75d55e23578515fcebcc7607cb
src/rust/fatfs-sys/depend/fatfs/documents/res/app4.c	10567	11ff9994c90bd50a04754f6aae344cd2863b2de142d401ec86f81
src/rust/fatfs-sys/depend/fatfs/documents/res/app5.c	1411	ed551e157dc26a5902e59beba3dabe3052d1cacdb806ccfcefa76
src/rust/fatfs-sys/depend/fatfs/documents/res/app6.c	1974	209a05020537c1a63cff1d5a4c48a8510ac05fddce400d2e7f440
src/rust/fatfs-sys/depend/fatfs/source/diskio.c	5385	435567bf00b4607a803339ec31d9402ea12536cbf65bb65ef9301
src/rust/fatfs-sys/depend/fatfs/source/diskio.h	2733	1c2b29f53e061c55d43b2e412c3a61320b2621bac7561a05e2755
src/rust/fatfs-sys/depend/fatfs/source/ff.c	266970	58ec7816a45870331b2a9c0eb977db19d1f0d95d7416a1482df07
src/rust/fatfs-sys/depend/fatfs/source/ff.h	17365	42920ae4d6080bfc2a72d92a9969b5e94eeaff070e5e71d90da9a
src/rust/fatfs-sys/depend/fatfs/source/ffconf.h	12714	6dc2ff2adf09b990a5b91eb253543061bd77ed498f92f4887837b
src/rust/fatfs-sys/depend/fatfs/source/ffsystem.c	6172	4ef367d61c6060d2bedeac57985ec293cf29d33faa98d15fbd78c
src/rust/fatfs-sys/depend/fatfs/source/ffunicode.c	1974995	3f1632c1f1e64b0b683b0d7e51341b9870728f398604869dd8211
src/rust/fatfs-sys/src/lib.rs	11	bdf6c7cf1fc01f69cc1d4e52948d1f246fdd310e6323d5dd1efd2
src/rust/rust-toolchain.toml	178	adc23c7a875dbd0c12bae71d2e35f74ef87baf17d0660dd4ec245
src/rust/streaming-silent-payments/Cargo.toml	404	b7a6dba7cfde137b16a86d3d65157dc965d70835aa6f6ee6cea66
src/rust/streaming-silent-payments/src/hash.rs	1776	ec49885fa0902fafb0b1bbd6ac6e4444aee0a3edb622adbd887f2
src/rust/streaming-silent-payments/src/lib.rs	14461	4e33062aad0ad48284857ac80a9b626078474c94ab7901ba3f2c1
src/rust/util/Cargo.toml	878	543d8a138679badb32951a85c9c276ce05fd7ba1ca05952148f1f
src/rust/util/src/ascii.rs	2089	fde025eaad4c2af6ae324939a7fdf659c09af403649596354181d

src/rust/util/src/bb02_async.rs	1686	6ed61692ba8092d58add3b157ed5eb9cb1fa81dd448f53cf5b56b
src/rust/util/src/bip32.rs	1412	457688519fc8b6a097ccabd27c189b1f6ed218437d8b478e02945
src/rust/util/src/bytes.rs	7737	9e7f5104142dc0fd492bcaf693ebbb2c4bc98a91f19cf3c929b41
src/rust/util/src/cell.rs	1286	4fb9bdf81aa82bac48d10dd8ba8b2443e2fd5d523ef6b6275cc09
src/rust/util/src/datetime.rs	3468	741a7d3fdce76a0a30d332b29f73f237691783190c06f9c3922b0
src/rust/util/src/decimal.rs	4784	01a1b5c401e594edfbbe05a5261deaf57841f8a63c5f762238f07
src/rust/util/src/futures.rs	60	8fdafb7122f20d1b5ce988e860031c57512db4fcc1d0a86d01e53
src/rust/util/src/futures/completion.rs	3750	339f12b2cffad430257aee544aba322fdcdd803a23b52dc443480
src/rust/util/src/lib.rs	2166	dbba9c609740021556b8cde831364aa22ae8415addf76f481941e
src/rust/util/src/log.rs	4249	d9b518ed3e5292c284bf90958e551320c39b18c1bf88f79094ded
src/rust/util/src/name.rs	1558	998da04d404cfd53209df659581886544d34ae1a4c314a0c04408
src/rust/util/src/p256.rs	3298	7662a6309089b842bfba84fb5a187419ef8588678035d9f9766bc
src/rust/util/src/sha2.rs	11704	ea2cb924dd5d2189531c80450997fba14624ba4244fc2063ded4b
src/rust/util/src/strings.rs	9623	bb60b4f1763c9444ff1675b49c130c87ed3394b6ef479589abb16
src/screen.c	3476	cfe33ab69a62ae78bf8bd1a52d007ecbe0a2d8799f1917ce5052f
src/screen.h	943	35e89291cfb88bbc08bcd4ab8b913e406808ac84fc5a112743ad6
src/sd.c	10969	8dcf171b7d515317a15e17a90a6fb1073bb4175e9eb3561e415aa
src/sd.h	1898	c91c4fd60a64f13e33871264bc51668534a51576535f4b39378f2
src/sd_mmc/sd_mmc_ext.c	698	79fb100d033d6068fc812405f98f56ebfdc9536fc350fbc80bec1
src/sd_mmc/sd_mmc_ext.h	206	6b4c4b41838ce53d4723c1566344eb30dfa8292e3ff033e33dbc6
src/sd_mmc/sd_mmc_start.c	430	ffd9c7aa3251780e2374147433d3bad4e01b670ad9b0e6eb0c0c5
src/sd_mmc/sd_mmc_start.h	51	ec31f931047363fac9aee9778d3afc0b523683a9d5b21701c424f
src/securechip/securechip.h	2253	553fd604572b88af6d037102db994a94ffbf5bab836db0783d439
src/system.c	1093	b6ff1e7ab0ec14f265f906eaae439977df28d9b9bfe7835aee62a
src/system.h	262	b2188c4cdc59fd46318d22edd41cf2ddfbe25023c3226490c7b27
src/touch/gestures.c	7780	15d289b74f8d1afd120c857425dc1b5614c3aac3bd767564056ca
src/touch/gestures.h	1036	f1431c6bc572c1ff0bb639b596d83bfbda72e30440550db2f7af5
src/u2f.c	32289	5f6bfa1e7cc4b94edf3c23fcfe699c74c484856841debe934de3f
src/u2f.h	1076	85ee65d73887ef958af72a5495e5134c5619b2db53a1817379c7e
src/u2f/u2f_app.c	2770	0cdd0a8722f62744d8f89ef5d303b007836f4fb5ab5c700866391
src/u2f/u2f_app.h	1429	4922af347b06b58cf350f854f0e3d824d34c4d6ebbd5252ec37e7
src/u2f/u2f_packet.c	5265	b255472c0baaf4fa28be35ceaf7f367bfac3c144800080815b0ac
src/u2f/u2f_packet.h	1048	62fec15e5a9779ba29e3876e6979f65a3e9918c7758a467c30990

src/uart.c	3229	8a2f1cc8191b42d45b03d0712e73a238b6c1de7bd126d8325cc1b
src/uart.h	545	e43783cf909e54c395b4d4f7b9f8059ec4ee8c99a080ed4acdc9f
src/ui/component.h	1821	a07473e0ffc5c839fba5d583fa2a5d543facc81064f087f04b3d7
src/ui/components/button.c	5022	cc451912b4ccd09fed913d03fd0369ad53acaffaf8778e24b9c02
src/ui/components/button.h	1399	3a1736c1bd6fc90a784a73bf31d7afa16714dbc5c16996049f9aa
src/ui/components/confirm.c	4677	ff7fc3bff893ce08e68a27c1a73c3d21a061d03b574d7557dff7e
src/ui/components/confirm.h	1636	e8bed0d30352f14a1301317974040fd9a631ffd1193fed5e1e255
src/ui/components/confirm_gesture.c	4737	b74d570fd5e75f68404265f705fe36fdfd0b611122732ec21c458
src/ui/components/confirm_gesture.h	493	447f87d9695c6941ad76066593024fd4128ccf286db7ea105b9be
src/ui/components/confirm_swap.c	3118	e54acf522bfd2eea84a76be00993c645a92995312eb3b385fa97f
src/ui/components/confirm_swap.h	686	7a4a9ffc3cd075256c6135bf9436a36e4b0e492eced8187af5d9a
src/ui/components/confirm_transaction.c	4813	810d55e494198ee188e15aa696d2354c6ef6041fa79181d385dc7
src/ui/components/confirm_transaction.h	1200	61f867ffb8af1557dc717e5b667f5f52ed7d4ce2255ec8f2ed591
src/ui/components/empty.c	679	cd1c8fc5e607e841c9d92235e2ebcfba6c8cbcd6425cb39060f2d
src/ui/components/empty.h	225	6c17f7e1f19f22a91275f7c6093def976bd7de8c997ac927ffd7f
src/ui/components/icon_button.c	5054	ecb4e2b2970c787c8db7deed590b4633ffa6f100c7b86bae5d752
src/ui/components/icon_button.h	616	4ee6d9866e86b07ca04465d3164b8773d83bedc6752cb856ed25e
src/ui/components/image.c	2682	9000e883ffc7ddc70e34bc49bf145bc31eb3af716ec6ef9b79e42
src/ui/components/image.h	753	f749e5a75baf96d686d30acccfb1b1c4d739353799d340ced306c
src/ui/components/info_centered.c	1662	be0d954ae5733ed1ea1eb5fe759236704a98e246c45f9b0677190
src/ui/components/info_centered.h	550	2cc0857e51f6966a3225bb5796e8e1295f8d5bd92154a16244d54
src/ui/components/keyboard_switch.c	5153	4ee1d5a0b16f35b8e986e3d16cb50086340db1c5a665c4fbf79f2
src/ui/components/keyboard_switch.h	958	78786a829330eaf101021614d0f9aa94d76319965e16ef37e1653
src/ui/components/knight_rider.c	1836	e227df5a99bfb0fb90159492462438e4871b089d96f676ae0cef2
src/ui/components/knight_rider.h	425	5b38b3b1f8a035e4446c84eda0a9a305cb80b80f1d7ef7bc89208
src/ui/components/label.c	8871	208edee08788d00a3ccdde38fcb505a6279b0a90c7cefd00bafcb
src/ui/components/label.h	3426	f135077b533da03cc075498e897570a3f0161bef23aa48864becb
src/ui/components/left_arrow.c	3444	09d2d637d63f8a5c6c61ec0c6276d7f2f6fa97b23d58a6582af3f
src/ui/components/left_arrow.h	511	3196849704e2eba89ba6c5ea85f0ea13127dd1081445818ec9fa4
src/ui/components/lockscreen.c	3456	199c5bd8d11f4aece24ca0ec834764cf799cb2db3175fee5bb89e
src/ui/components/lockscreen.h	223	900e7f56572d6f06caec37d075aa969818a165ccaa7c5bc4b9173
src/ui/components/menu.c	9273	21ebdaac62d659f4a8019c82ab870e764626fa348923a3231e59d
src/ui/components/menu.h	1345	706c6ff3f2345513b3c10cff0806c5b85b27cb4a35c087d80716f

src/ui/components/orientation_arrows.c	6207	71686192af175d7e295191e46879b26be6b44c59ed696329cc153
src/ui/components/orientation_arrows.h	530	1bf76607ee6050cffed3efe2faa9a9e8ad81190411f803ee0dac2
src/ui/components/progress.c	1648	947e90bc3a27026e979ce9f48f5ed71c30d9e62e11449c27ef72d
src/ui/components/progress.h	402	8232044dd9aef38e46af994c30c99b30303d61f2d6c8ec442d3ec
src/ui/components/right_arrow.c	3650	83a10b58ca690de826c330750f2e24a09eebe43a5390a30423607
src/ui/components/right_arrow.h	515	8ba6ea528f44d08b61d363e81d5f65d2caca802205d89c8b18df1
src/ui/components/screensaver.c	3122	474414d341c7f942ca9435d3eef090a1e4d0cc6bd507eacd20fb4
src/ui/components/screensaver.h	375	7ec6bd40c73af2d8f8fb5cf196568acfed60b2979463df6cc9c59
src/ui/components/sdcard.c	2748	8c319b47b9d4c4041cafdf47c7094e2ff5daadc6b7d522e977ebd
src/ui/components/sdcard.h	389	2287aa5eced8006a9e9b407923b89841de7cc505a78bb84ec5372
src/ui/components/status.c	1672	56d47212b55dd91f6cc887b1a1d877ebc8a8f6d3a4f778aa37ee5
src/ui/components/status.h	507	afa759656dc103c185a053bf5ec2643fc5a7f3df5464ab170938b
src/ui/components/trinary_choice.c	3257	4b7f85ae36693ca5a89323187fb542474277fed672670351e5ea0
src/ui/components/trinary_choice.h	905	86a614340e27105664d95ae72c1579823cdacad01250dcb35aa24
src/ui/components/trinary_input_char.c	14903	bc419ea37ed53cfb0c960c2aecde3c9b040ad7aa5e07081116e8f
src/ui/components/trinary_input_char.h	1663	ed246da425389a34d86d366639ce4a8ca5a1b8b93458de5b4a6f3
src/ui/components/trinary_input_string.c	18762	c565e0b6e2e4f32072a611c257b76f29d20efb27aea23a38c2667
src/ui/components/trinary_input_string.h	2123	d7c341feef38c0723581490bcc465a243460a5085792d83baad8f
src/ui/components/ui_images.c	14893	38788e6b993403db59eb8b9675c05c5b5f82ef12f9d059491b98d
src/ui/components/ui_images.h	1814	9d0d48dc2a867a42c1c1f8935d6964906f18c509b5183d9ef5f18
src/ui/components/unlock_animation.c	10949	8cfedb0276426a8b6ad17cc94382d747285b74e95c820cb0b8a1f
src/ui/components/unlock_animation.h	269	031cb5d1a4a989b037b9c91785b7128f7e4a02621c4bb72973eb8
src/ui/components/waiting.c	2901	876058da24e77fa58c9f4799480b7def7ffcd96bc3b80d5705fc7
src/ui/components/waiting.h	611	0c497151feb5ed87172f218a79524b725ba06c6b8ad2dac99e912
src/ui/event.h	563	ceb3e26bfa9d5441f571eb846273ecade740f6b1353311ce55e99
src/ui/event_handler.c	1520	7e183e8ce31c33254a45ec1a02e2fd7d3fb91e69bb1cef7e5e190
src/ui/event_handler.h	267	43ec16c77d45de807db6cd8a50463e1fc7bc4a79443493973eacc
src/ui/fonts/inter_bold_32.c	220230	a1b38b3f0de200baeff70ca8d4342fd1e003c319b455b3bd8797a
src/ui/fonts/inter_bold_48.c	477264	8f262bd976dddec9c9fc69fb6cbedad692ef7fb05ebc10219aecb
src/ui/fonts/inter_medium_20.c	109924	6d2067839ae61ff2ceb6c4b85da0025bc8baffe5462486e46bf55
src/ui/fonts/inter_medium_32.c	218261	9445298cfc9ce4141e95c867e65aaf87a5cd2a5c6d6ae71ffae4b
src/ui/fonts/inter_regular_24.c	130640	0f898381eda14ad4bf04eb3d1eca09bae336e85b0c35565ca06a7
src/ui/fonts/inter_regular_32.c	195082	14680f1a8e90933a0d6d84e0e8760548cc3a456392c5703a3dc9d

src/ui/fonts/inter_regular_48.c	416631	fa6169cb2efdf02dd8ba3dc4642001d1e5efcc24ede2872390200
src/ui/fonts/lvgl.h	391	11c17c628828fa80cf4d76e36e738c309ee0e5bd18344dd817562
src/ui/fonts/lvgl_font_compat.c	3435	67a8045562eb177d9798515529acd8c3927cd7045953c9960e8b7
src/ui/fonts/monogram_16.c	16168	a0e5970175aaeabbac7299cadbb545920af1f31d076539c44d0ff
src/ui/fonts/monogram_16.h	158	9c1f589005022ab443d2e0320d282e4aa4c5799b3d255fab19a8d
src/ui/fonts/monogram_16_bootloader.c	4840	5de1f46de5c48efdb1ee97ddde36b48ab68c1dd84ed87d92795c4
src/ui/fonts/monogram_16_bootloader.h	191	be9c3f187d51c4c9d06de25ef871450fec51b85ee964da350ec7b
src/ui/fonts/password_12.c	1405	3e9db0e36887259c555fe1ba6393ef9efa4f66c8f7bcb9daca4d1
src/ui/fonts/password_12.h	198	c52f920e705997665f8d6a6a14cf4071f8953dae7f047ee8781bf
src/ui/fonts/password_9.c	1388	b089160d8076d8b1b0fec785e8ad9e17870aaaebd543a44ce4059
src/ui/fonts/password_9.h	154	90f8f69a18d7dc70b40620af8974468d44b61f899f6db6139ba61
src/ui/fonts/regular_11.c	32527	7b364b0fc185ca3f17222133fea1ce2ea7551077cdabdeb1e9a40
src/ui/fonts/regular_11.h	154	0a98c23bac34f471d61acf4fa92a2c837e07806433ca95aa131eb
src/ui/fonts/regular_12.c	33621	061359ab0984d07b83e698f7cec0082d786731378f288b383fd87
src/ui/fonts/regular_12.h	154	c284235a85a0a20cbe2546d488d6588be37e6df1b84e6d780d293
src/ui/fonts/regular_9.c	31381	7aaab51be31de788729c9fe46ec4024079fe1bbf7fb5f389c064a
src/ui/fonts/regular_9.h	150	98f7215d5be8c70c3cc3aa8e97184a5f391201367c8b73d7a3ccc
src/ui/fonts/regular_9_bootloader.c	15876	5453ccee1557b14014c4b4a7ab57004ab4b814d98f454bd04756a
src/ui/fonts/regular_9_bootloader.h	183	e1d321850785b1cf43e041776fb84f961f042c34baf1809870e1a
src/ui/fonts/regular_fonts.h	197	8a8e025aa12cff224593361e93ac0b80822f17945e297ba5271c0
src/ui/graphics/graphics.c	700	04b7f519cfb183a24fe900af85577a43332d093b5602cab775604
src/ui/graphics/graphics.h	496	a4ce7700e842555679eb3e4413370cefa48cd2a27861e272f5e0c
src/ui/oled/oled.c	4379	ed7a0ba91c16163d3b5b8cd5dbba05d155c6cc8f2a46d81277532
src/ui/oled/oled.h	2770	11bc1ea771b107fc399ab5830fc6a542d1b85f8f41dad91abe2d8
src/ui/oled/oled_writer.c	1290	1e49807ebed8391f58f225bc0e3ddbc5bc47a8be969e00ffb18f8
src/ui/oled/oled_writer.h	436	484da60c1c4ace43f60680ab6067265cb3201962065648c3475b8
src/ui/oled/sh1107.c	4742	4cdbdd49fb69f9a9ae925988c4bbfc477e61fd884492347a09375
src/ui/oled/sh1107.h	424	6f11af7cd69de4fccf04a5ca441b9d39c7b4c40f06bc9937ec172
src/ui/oled/ssd1312.c	4806	c229caef4b0b9f99872bdf2b6015d38a180a85f9aa248521c4902
src/ui/oled/ssd1312.h	412	7e80fa18a1ff73103d7420dc2f8e028b1746d879739f49dbcc264
src/ui/screen_process.c	2920	c797a0a0dd7f8a7519bcc53fcefb344d5b5fb78baf2691d3c40e6
src/ui/screen_process.h	1324	4925324ae565ea56f3b245de7e65211ba042e42d3ca1f08f9959f
src/ui/screen_saver.c	1605	e5d01795a1d6c770a03cd378030fb5badf2394bd018ed5b839dd4

src/ui/screen_saver.h	841	6d9596bf239ce5933c80ceaaf9ea6f1e8072a6abb18ced3780aa2
src/ui/screen_stack.c	2027	197e7ea15e96db7fd131ea244c960a3d4832517ebd3f0a3d79d3a
src/ui/screen_stack.h	541	15906232878e5958c681a35ed055844d98b3dd1aac15978929c87
src/ui/ugui/ugui.c	24444	220bfa84f9ce74facbe56797bb218cb57d22c4bf009f083cf8a3d
src/ui/ugui/ugui.h	5912	d6faaeecf2aa46263906f5ac5a062b10e74583c75a783dd0f3ebe
src/ui/ui_util.c	8629	9de60147f2cb3098e19bfff7bca0b9d88c5aa3a48e1f004ea14f5
src/ui/ui_util.h	5096	c27d10d4029d2bb1093a8a9c119157e89d60f49a3d3b9c78c7f89
src/usb/class/hid/hid.c	11109	bcd4ce1b19d1124edb7e01ef42f03e770287cf5d1657bc6d20775
src/usb/class/hid/hid.h	4109	577de80520fb227d0c59d54cbee8b7e2a433255b9d57b63e02de8
src/usb/class/hid/hww/hid_hww.c	4754	6386782800379810d5b871a51d656181425832582fe7186513a0d
src/usb/class/hid/hww/hid_hww.h	1709	c55cc67135f87e9364a8b27a1cbc8c9c1c1c532c1f483357194be
src/usb/class/hid/u2f/hid_u2f.c	5262	ce8506310c076ae0cb615c00b78afab8d9958f9cbc5de15cfc7e7
src/usb/class/hid/u2f/hid_u2f.h	1623	bcec43c5c70c6eefbbdc8d1e6005860562c9af74c9bda31084b76
src/usb/class/usb_desc.h	7046	5065da9afd04d585483cb4bb706a6ecfd53e3504c5a96c757959f
src/usb/class/usb_desc_bitbox02plus.h	5016	4538dd98aad944ad1a79ef1004234c299475cb6fe215ad3f4bc62
src/usb/class/usb_desc_common.h	6509	f81e45537c09af04b33ff8a5705458630940f4ab89805dad0b68b
src/usb/class/usb_size.h	166	f9ec0deac44a546890d03fd364d6f7c30337cf3af0bf10b8dfd54
src/usb/class/usb_u2f_desc.h	3758	6d7dd70956c39337cceb388e23a971840d5d4dd1e6aac0907fb21
src/usb/u2f/u2f.h	3133	d76b09a8fb63da8ecf6b6490b165c20005247b9b1a0c391070c60
src/usb/u2f/u2f_hid.h	2982	66976af1f1a767ae9c7e4632d530cfbe0e7849e4acdeb2c13d020
src/usb/u2f/u2f_keys.h	2313	a162f73d950ca857744d864a07e27d90dd9f78c0be9ddc86114a2
src/usb/usb.c	3090	0d427e3fb7ab2dbba2b65367e70ae6d2017d9c8ff3191109467c9
src/usb/usb.h	275	f40df9cad0283326621b13dae6dd0027f43acb95ce640f28c7e69
src/usb/usb_frame.c	5540	92d3fe3c443ca9989db2f96c976a931517c41394342bd98f314d1
src/usb/usb_frame.h	4220	b4e0030db88b0911f7ee8a8ffa623a6bd470b0eb543eb7b4422ce
src/usb/usb_packet.c	3034	a08a57316ff2a59ced1b13b2518bfc12ba116b19feede5d5e147e
src/usb/usb_packet.h	865	7d2d02d5d55c31764bd99c5585730cbbf2baf6442fcd63ccae826
src/usb/usb_processing.c	13698	e62fb4705df80476c55490f83bd3889603916eca52be3a6cd3a80
src/usb/usb_processing.h	2167	b13ab589a15524572845e28b92af5bcd1757e9bad4fa92d7e74d5
src/util.c	3410	c89f4e11e0a50684fd7986a85a7d84749b53f7870b6e89c7d8832
src/util.h	5262	8bfcd192ef4b5709bdc647646e9e595c2aefd6dbcd94fa79a84c1
tools/atecc608/config/config.go	11409	184f8b713ca7771f8f16299f960f301884ee2794e40a7db275484
tools/atecc608/go.mod	26	f4755a635cb61efa0b2312266e47f195381c2c2ff373ca9f1a193

tools/atecc608/main.go	14826	39a1a4d3f2edbcd638d83e20f8b5c858c432a749e2a538711b3f4
tools/prost-build-proto/Cargo.lock	10683	7082227fd2d30d56308df664b7832f0d586d68eb22710492096d6
tools/prost-build-proto/Cargo.toml	334	1f669b2998c5ed56cef2f50a1c27ecaa047877d58c4d16d4d134d
tools/prost-build-proto/src/main.rs	437	b3f4fa1151df049a8db26afe78f28128843d6dd27f02c665f0a61
tools/ttf2lvgl/ttf2lvgl.c	41257	14e0cae684c38939e0b962515b027de2999b25f16cb5137b3ba32
external/cryptoauthlib/app/api_206a/api_206a.c	15307	d5ce1f2d962566807980f8c02fba4975a854dd93453a0acfa95e0
external/cryptoauthlib/app/api_206a/api_206a.h	2891	becd7c920e878939d3c172ef89000633b0f66dd5614465f8f0c3e
external/cryptoauthlib/app/ip_protection/ symmetric_authentication.c	4967	94ef44e7695fe4dee92bcd78acdb062c0ed4765e7083097339207
external/cryptoauthlib/app/ip_protection/ symmetric_authentication.h	1648	c49d4fe6f78bea0aa00d03bca44c6257c69cc2b6174890ae2201e
external/cryptoauthlib/app/pkcs11/README.md	9751	336cffbc3ef5787fdcc41289bcab29a3668cc03d10e2bf936c726
external/cryptoauthlib/app/pkcs11/ example_cert_chain.c	11711	4de8a105d778e914baeaa5cb60304a48a658e3cc1eba43c016286
external/cryptoauthlib/app/pkcs11/ example_cert_chain.h	291	f5a21d013a2bb13d7722707f24dc270becbf9bf65a635f99e148b
external/cryptoauthlib/app/pkcs11/ example_pkcs11_config.c	6435	dcad1f6bc70e99f032de41746fbb3aa95947ef1619bcce470941e
external/cryptoauthlib/app/pkcs11/ trust_pkcs11_config.c	5959	a3a6f251fb95901129df19e70a07875c04cee0bcb7fe6a57c1286
external/cryptoauthlib/app/secure_boot/ io_protection_key.h	1570	c23acbe676d767d275ab00252cbb6c857aa7bd2c2c1de16db5b71
external/cryptoauthlib/app/secure_boot/ secure_boot.c	9473	e76cfe687fdd89da8287eeb746fa26e628bcbb6512a8fb3a4f219
external/cryptoauthlib/app/secure_boot/ secure_boot.h	2898	d2a8870794496b5b92ff057fc88b9fc0f898bd99c43f511d0da8e
external/cryptoauthlib/app/secure_boot/ secure_boot_memory.h	2336	c1b98745643ddb1b6d6fd7516b2e97a41aac6d737b8168d239baf
external/cryptoauthlib/app/tng/ tflxtls_cert_def_4_device.c	7675	174bd2615d7d94f5235a0c28f30127ffe095aef3843b78956f3b3
external/cryptoauthlib/app/tng/ tflxtls_cert_def_4_device.h	1555	cead3ee8d90f74f240feb6bfabb4324c01e91b9cd00da6991e93d
external/cryptoauthlib/app/tng/tng_atca.c	3727	98decda85d261c15f156fa2159d4e6c6f956222be004d5e87dafb
external/cryptoauthlib/app/tng/tng_atca.h	2599	5886a2065b7aee185e214b89306c6ca75f7567a543bea1eec64b7
external/cryptoauthlib/app/tng/tng_atcacert_client.c	7115	acad9769b2fe1283a641c49d6871bb4069012be07e9681e10e5bf
external/cryptoauthlib/app/tng/tng_atcacert_client.h	5987	cf89eec191bb7f31fab46b389c91390e25c1d6087ebcce8dd8da7
external/cryptoauthlib/app/tng/tng_root_cert.c	4624	6e5af7e514a2b82df638ecbe8e3149e4b7fa5c9272e58fda33f39
external/cryptoauthlib/app/tng/tng_root_cert.h	1614	3d5c093d4fca74f84bd42f044e8fac18c7af893a792353cbe2cba

external/cryptoauthlib/app/tng/tnglora_cert_def_1_signer.c	3680	38a976b4983c2c8dffddb28d70b39cbb1e67e43f7ade08026bfa3
external/cryptoauthlib/app/tng/tnglora_cert_def_1_signer.h	1560	d5ac56e11f96e7884224f22a62808c3ffbf1ca38da45682f013d0
external/cryptoauthlib/app/tng/tnglora_cert_def_2_device.c	3814	410d547617400f45670eda995e18035a99141eddb32be660e8b47
external/cryptoauthlib/app/tng/tnglora_cert_def_2_device.h	1561	700f180e789e32522f43096a9f1d967aee8ace1af061bb8226699
external/cryptoauthlib/app/tng/tnglora_cert_def_4_device.c	8508	d51973a657797423363cf872a6c928ac1013ca030e803c8e57a72
external/cryptoauthlib/app/tng/tnglora_cert_def_4_device.h	1608	f20b178d3be797085c20df335fa2d50b4fea2c31b1e3fa758cfba
external/cryptoauthlib/app/tng/tngtls_cert_def_1_signer.c	7196	42dc9ed786c7ac8597200b948558df9fe1b6226671ca5841fc7aa
external/cryptoauthlib/app/tng/tngtls_cert_def_1_signer.h	1604	b8a2b9cb4996ef4a2073b9ec974b8288cd75179a1df32a5306e83
external/cryptoauthlib/app/tng/tngtls_cert_def_2_device.c	7855	2712f973bc26a1065558e373b6a3aacf61473bd419f38b15e3996
external/cryptoauthlib/app/tng/tngtls_cert_def_2_device.h	1650	54b76930db788172a154b071d7f1030ed1753c46e983d442926f3
external/cryptoauthlib/app/tng/tngtls_cert_def_3_device.c	8458	ff07b415bcf72ff5ce20f27f3ad8f5450d1ddacdb291a6cf00e38
external/cryptoauthlib/app/tng/tngtls_cert_def_3_device.h	1602	615687631e3abc316c67635db8299d59dfeda6428c1e0b364c4ec
external/cryptoauthlib/harmony/config/cryptoauthlib.py	15305	2fa0310a4270809b4dd6f137a6a258e30dd2a61b2239222c638db
external/cryptoauthlib/harmony/config/device_common.py	1532	359798fab180049617b73d7a00edfbab2360b82ed1c5cb93165b7
external/cryptoauthlib/harmony/config/device_instance.py	9213	3feac52fe247d01a57e97481410c2d1b127510d857b30b158f45c
external/cryptoauthlib/harmony/config/pkcs11.py	5674	a6c657c04cd273b3492deaca43b3db58a4f6f99fe8cd90ed0d805
external/cryptoauthlib/harmony/config/tng.py	4447	2a3e5cc106cb09b7251713ebe8c95e167750890f2deed01ccf1e7
external/cryptoauthlib/harmony/module.py	3670	13e2cce6acd1fc65f94bffa2ef34ccb05c798b680bfa3adc71dec
external/cryptoauthlib/harmony/src/cryptoauthlib_main.h	745	e04e243dbc3d235affbca4a1fae7891a6a70d12985dfeecb9f760
external/cryptoauthlib/harmony/src/hal_i2c_harmony.h	2082	28ab8149f62a76ef256163daf02bf3b123e22e9e09538f5c6a156
external/cryptoauthlib/lib/atca_basic.c	126331	ad66baae8500efb167badb7512b93e4130b607cc6b07d55249dda
external/cryptoauthlib/lib/atca_basic.h	37087	97be5cfbe62f4841539573ad75792ae8aaf04fc8d4cba14b02046
external/cryptoauthlib/lib/atca_bool.h	1586	926074b48e3db661fe4d5652a76baee68b91bc18efb4d77b4c072

external/cryptoauthlib/lib/atca_cfgs.c	5536	5be8cb927756553b5b5d2e705eaa0c5d5c92e671472c17c106025
external/cryptoauthlib/lib/atca_cfgs.h	2529	4c1b73724196b5ae946010b1cd397fa3f16ed43d45e3b51cda97f
external/cryptoauthlib/lib/atca_command.c	3454	052196da2fe9ac0f7ad441baebb18c6bc669db1afc9ee1844c7df
external/cryptoauthlib/lib/atca_command.h	2566	3e1e6181d2b3189674ba09d7756ba8e6d7b9da9c54129f1734523
external/cryptoauthlib/lib/atca_compiler.h	8939	d6eeeb082d65cd1001c42cea28a00bfc9ec56d4f59c689525e22b
external/cryptoauthlib/lib/atca_debug.c	1643	797bc4a42065b9889215b0e1dbee3978d7186ee5432c3a0e4446c
external/cryptoauthlib/lib/atca_debug.h	246	9fe972773847b28f97ee17c0704bdc2de7d10af4bdc0f7ca3b133
external/cryptoauthlib/lib/atca_device.c	4882	34afd9aa03e931c1d1fd00bbadd767d0cfeff25e05d89f2691911
external/cryptoauthlib/lib/atca_device.h	13269	d48ef66f48efb3e0c5f7750e5a9f97602422590f99dfec5b9599a
external/cryptoauthlib/lib/atca_devtypes.h	1747	cd0082011dddf700281801090e969d589c848a802b807752a256a
external/cryptoauthlib/lib/atca_helpers.c	25755	4cb89568e6e1fd8742e7ad9acf0d24e92602db0cdddfb1af2012e
external/cryptoauthlib/lib/atca_helpers.h	3416	d5232f199ee469ae317938193582b392a0b3dba5457b278d9b619
external/cryptoauthlib/lib/atca_iface.c	9159	b3e55a5e5f8b82691ee51640d1169d86f3af682b308bd7fc4fa9d
external/cryptoauthlib/lib/atca_iface.h	7399	10260b76b9bbef7c8dc0fff866b136b019ed8c39a3ce61ff9d078
external/cryptoauthlib/lib/atca_status.h	5366	45641f02fb165c14152e3ea14b41a9c34094d559947b07c119ad3
external/cryptoauthlib/lib/atca_utils_sizes.c	3418	bd958ae085c2c5751a4e0b736b0a57c5de29f5a247fab60f88862
external/cryptoauthlib/lib/atca_version.h	1545	ba246e92a539d86d1ba5d827da852b30cc017f84345dbb4ca6a91
external/cryptoauthlib/lib/atcacert/atcacert.h	3032	852db1817d64561bdbbe2906056d766a38c2bd9a1847e7e2220ac
external/cryptoauthlib/lib/atcacert/atcacert_client.c	13525	01378b673dccca786a0bf4e4c6d2cd1b9f8e68b05e158cb64268c
external/cryptoauthlib/lib/atcacert/atcacert_client.h	8213	2ebd2bff70eb2b865e734b69bd02dc8080662dc54a22b23dea737
external/cryptoauthlib/lib/atcacert/atcacert_date.c	32659	911508f55380dd08a5f4f0dce248931e18ee10254c0471263be30
external/cryptoauthlib/lib/atcacert/atcacert_date.h	8397	e63b01a374519138f01db2cc5879e357ed9582d5ff1336afc51a0
external/cryptoauthlib/lib/atcacert/atcacert_def.c	57772	e3d470d3bbc5efe5e2bb2dc53a10980172fa0dd4ec1a5da080a15
external/cryptoauthlib/lib/atcacert/atcacert_def.h	41649	eefff3e9ea019b12aca38ce057f5b5e4584a3629f74d36650922a
external/cryptoauthlib/lib/atcacert/atcacert_der.c	16350	c13f286af27adb864023ed2d19c6d8e138bb2f45a0c601a1c1c24
external/cryptoauthlib/lib/atcacert/atcacert_der.h	7558	afaebdd08edc55db6465795aefe56ebbdaef8d0350924d8f70ef8
external/cryptoauthlib/lib/atcacert/atcacert_host_hw.c	3114	46648cf1d28767eb64a522a3d460a054df305287953d1221b11b4
external/cryptoauthlib/lib/atcacert/atcacert_host_hw.h	4450	3554a583c4d5cb10264dda55a459bf475a6a529b591c79bc11bd2
external/cryptoauthlib/lib/atcacert/atcacert_host_sw.c	2924	6706cdc3e8eda7475acbe110c1434feed6a733c8c59daabdfb4e6
external/cryptoauthlib/lib/atcacert/atcacert_host_sw.h	4336	e4cd37f9f6223473b2f2c384962cb3564212e9ff95e89019b04b3
external/cryptoauthlib/lib/atcacert/atcacert_pem.c	5824	8775d02cee595578f602085bb3e8d2f9d4a2f5992963038357b05
external/cryptoauthlib/lib/atcacert/atcacert_pem.h	5600	3a8840e28ca222153aafdb946359c11109ba1a0ceeb7b6649d5c6
external/cryptoauthlib/lib/calib/calib_aes.c	6075	b0ee8c5ee772efbbb3e605b0ee3e809ae02d8056f06535ffefcd3

external/cryptoauthlib/lib/calib/calib_aes_gcm.c	21139	a41e3eafc70d079d17d0ccc620bc25f609d659d7245f9bd60770f
external/cryptoauthlib/lib/calib/calib_aes_gcm.h	3926	c0add73cff66de9b61d8770d6ee732323fe7df634594364ebac00
external/cryptoauthlib/lib/calib/calib_basic.c	9400	a069cc571c623a21a5b0183ec00365c61cb188297f88a03f6fd64
external/cryptoauthlib/lib/calib/calib_basic.h	12173	cab07399a08f1f920d85aacda4453cecc1f74073a9aa8524340aa
external/cryptoauthlib/lib/calib/calib_checkmac.c	3718	a8641e57fe4254055139476a623553feeb524aae6ba819c87f118
external/cryptoauthlib/lib/calib/calib_command.c	20906	f1a6400f05ed6c11a851e976f53a8898826df6406199b21dd50df
external/cryptoauthlib/lib/calib/calib_command.h	53202	daa823f84a4a6a32cce2cf7d42bfd1d97fabb5d46df2c3228d0de
external/cryptoauthlib/lib/calib/calib_counter.c	4562	c1d01ea200bc8e15ef95eabd5d4e0af203f47a8fe0b377902bc33
external/cryptoauthlib/lib/calib/calib_derivekey.c	3191	594fe02c9af7db294e311b93e55e0fcc9a8936a31bfae1ca3cc94
external/cryptoauthlib/lib/calib/calib_ecdh.c	10853	c6471c07c0685ef5da7b6889d27f75f58a7f5c3f16e29c57819e3
external/cryptoauthlib/lib/calib/calib_execution.c	12160	ea99cb59f721f6aa0485bc541660f1e5795f726840c4f3ac18ff4
external/cryptoauthlib/lib/calib/calib_execution.h	2273	a98dc9774080ed54d4c261c40c66a4f9cc75886b55df46484baf6
external/cryptoauthlib/lib/calib/calib_gendig.c	3856	47ef2824db5c16ef34467382f384536083bc3a21a3198807b061f
external/cryptoauthlib/lib/calib/calib_genkey.c	5605	5b86a68cc2260b003d5bfe19e5ee8278f18765e3696f9c7ed451f
external/cryptoauthlib/lib/calib/calib_hmac.c	3552	c6c089eed42405869075e0345d87e07dec0d99ce47568b00c37ce
external/cryptoauthlib/lib/calib/calib_info.c	5163	8fbe4234e1459aab7ab630fb7c0bbd87175e5df62018fbfc6cfef
external/cryptoauthlib/lib/calib/calib_kdf.c	5520	34e91586920a2bb029c27766a94a57ab399ac6dadaaab1d72eba1
external/cryptoauthlib/lib/calib/calib_lock.c	5715	3d48b48914e2a6cad1b8a1a68ea869de0730f32c22e40760f8fca
external/cryptoauthlib/lib/calib/calib_mac.c	3667	806ec85793d069373840e227ee614b99d696cbc33c34a61cd2a71
external/cryptoauthlib/lib/calib/calib_nonce.c	8412	ecde35178d24ef5bb46bec55ef99db9fcd259381271810cb174a4
external/cryptoauthlib/lib/calib/calib_privwrite.c	8240	1d2ad034a4490d282424c969e5836f4d213f1b2d2012f394dd95b
external/cryptoauthlib/lib/calib/calib_random.c	3060	91d2e658769c5878b180cf12fcc7cf7be6e3c43afa29e4eeed1d9
external/cryptoauthlib/lib/calib/calib_read.c	25781	293b19883f692fe37e9eec6e329e82ba7caa10a476d6a710889f6
external/cryptoauthlib/lib/calib/calib_secureboot.c	8352	b5cda1a38ee600a6d3c1498c861a2a0180ee1eb235b0dba6d9c32
external/cryptoauthlib/lib/calib/calib_selftest.c	4052	1dad33906c11210bc6a35aae483112b60c92664449e8077d7c392
external/cryptoauthlib/lib/calib/calib_sha.c	20407	65d38dce718560bc3eab0a7528bb4349212deb5219c152ad4d805
external/cryptoauthlib/lib/calib/calib_sign.c	6997	b5eeba1bb7ce5ac077f3eb1f3d4001f12d3fe99ea4aa6596ff9a6
external/cryptoauthlib/lib/calib/calib_updateextra.c	3144	9a712cf53d09ccb4ac17c7c655a85054575519e15560aee14b67f
external/cryptoauthlib/lib/calib/calib_verify.c	20657	6ccbf9461762e51d163fd4b69fc3bb678d12d27b503378a327578
external/cryptoauthlib/lib/calib/calib_write.c	20897	de896466e15a5739b55ca4b66c658ae9e6ffc192e0e90164cb2b3
external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes.h	4230	d8a1af7bb9ee93f918bcc8f87c610fc29e31033abd51344d98e96
external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_cbc.c	5937	6d574d733ec2a449c0ba752a5feb3722ae6491833189fba8506f2

external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_cbcmac.c	5692	3b60e0686502863af9d557646d09e68b93b3b607c0d3ffddee079
external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_ccm.c	21098	1e2f566947cebe288aa5280c12404c3f8836ed64e2bd8422355fe
external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_cmac.c	8072	fe3ffbf587b5c2291eba649534ef689b85d9cee18fa1a45082e7a
external/cryptoauthlib/lib/crypto/atca_crypto_hw_aes_ctr.c	11347	40cdc322529bde9af6e018a92f4fc1eabeb2e2895cee6bee8743e
external/cryptoauthlib/lib/crypto/atca_crypto_sw.h	6515	d172810d2e7532b940cf8fc0b096ecc7631286ea3d370183438aa
external/cryptoauthlib/lib/crypto/atca_crypto_sw_ecdsa.c	2080	fa2c8da252a940e6bfbfd53cc1952d0fb96146f96c6c5d051b853
external/cryptoauthlib/lib/crypto/atca_crypto_sw_ecdsa.h	2126	a0f3b3dd9c260a216adc49e4f13a0edc9e2891e67496a674dd666
external/cryptoauthlib/lib/crypto/atca_crypto_sw_rand.c	1672	54353489786253ab575083702ba1b31564d2af17b719288ce82ab
external/cryptoauthlib/lib/crypto/atca_crypto_sw_rand.h	1669	6a0f009d909b9dd465b6dcbadec774c07ff90f9081b3c44c6bdac
external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha1.c	3317	cd088f7167eab6bef18f31ea83d9a356276814006707d836dde8f
external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha1.h	1965	36b6ba8f892bd697b2a97ddcdfe1626c9b84fc7820ca537d871d4
external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha2.c	7728	147701ce979daf7e95fed5e4599d71f2e8689ec0fe1f73da2cd8d
external/cryptoauthlib/lib/crypto/atca_crypto_sw_sha2.h	2492	53d8fec4e82f0921736b3295e8d0451707d5d77c74952a89c1cd8
external/cryptoauthlib/lib/crypto/hashes/sha1_routines.c	7838	6a62989a2ce78d332733511b3d05d52c5b219aab7f86fbfc0b109
external/cryptoauthlib/lib/crypto/hashes/sha1_routines.h	2364	8e8463f36541cd7bd3b1bf3353b9851b327e830bcb6af7aa08a7f
external/cryptoauthlib/lib/crypto/hashes/sha2_routines.c	9945	08c3849a3f4d1b985056e267c7eb2c147489b4c1b61239cc58355
external/cryptoauthlib/lib/crypto/hashes/sha2_routines.h	2160	1132df96ae80fbb6da122a13fd36e912014c77462efe02ed78229
external/cryptoauthlib/lib/cryptoauthlib.h	4072	136c985bee9e975bdb7b2145ef362ffb3aba0712772753b2a54d3
external/cryptoauthlib/lib/hal/atca_hal.c	8930	e7456b5445e3e6df2d7cb487942ac68517c5425a776f20e7f5d26
external/cryptoauthlib/lib/hal/atca_hal.h	10037	174121242701462c013e1041e689f87938245c75736ea65a928ad
external/cryptoauthlib/lib/hal/atca_start_config.h	186	b29a69137b3e32126b54d5e7077e9f9e1ccbd24fdacacf7197257
external/cryptoauthlib/lib/hal/atca_start_iface.h	185	a4d3739b7689019106ba8a9d479832da72af89b62b9d6ede4c1f4
external/cryptoauthlib/lib/hal/hal_all_platforms_kit_hidapi.c	9504	dfeb4bd23627caafdc9ad7f3f3bc5ef23e3c8c4627f53f6deab4f

external/cryptoauthlib/lib/hal/hal_esp32_i2c.c	14787	d73c4884df15729de1089ae8bf8722b94fc2569b305803b8462c2
external/cryptoauthlib/lib/hal/hal_esp32_timer.c	886	3ab492f1a9b6f69ae6afa65244984e5d5e89170a6aa7e108045fb
external/cryptoauthlib/lib/hal/hal_freertos.c	3344	62b51adb663528c24c980ed647a7708cb7792a25cc0f1e04c2eb3
external/cryptoauthlib/lib/hal/hal_i2c_harmony.c	17184	b177a6046b83763f4930907397ca5b85f77eac912967c07941bad
external/cryptoauthlib/lib/hal/hal_i2c_start.c	14731	2f97a83b88296088939446176b88eb763aee2ce4b4356c2d99c8b
external/cryptoauthlib/lib/hal/hal_i2c_start.h	1983	42b500a589fe060b1ac1994a8d42453d60802785049b222e72116
external/cryptoauthlib/lib/hal/hal_kit_bridge.c	12159	31877777bc183bb0a51ca3e62bfc3162999d2c05fe8be556a4621
external/cryptoauthlib/lib/hal/hal_kit_bridge.h	2273	5d72a272a62ccd5072df5ad89242c496bb1de2e20895caf5242b1
external/cryptoauthlib/lib/hal/hal_linux.c	7741	6bda8a1997786ef309ec64be421edc9022f1e48ea31dbe4484035
external/cryptoauthlib/lib/hal/hal_linux_i2c_userspace.c	12381	45f0f446a83dbb8fe068cfa31e01a42b53309c3f4eb61c156740d
external/cryptoauthlib/lib/hal/hal_linux_i2c_userspace.h	1764	92f271c142f48a5c19f81c88083f1eeaa9cc8d6fc200641dc76e8
external/cryptoauthlib/lib/hal/hal_linux_spi_userspace.c	9828	7ddfedf9487031496afeb17f24475dd59c12c12d86138b880ace5
external/cryptoauthlib/lib/hal/hal_linux_spi_userspace.h	1790	0307866a7b5de9e3a38e9b1b22119a852cdc32143461b6d441db7
external/cryptoauthlib/lib/hal/hal_sam0_i2c_asf.c	14048	29a1696595af9c62435dd8bb1d7f076f8660c9bbe98d676c9fcda
external/cryptoauthlib/lib/hal/hal_sam0_i2c_asf.h	1755	52780a0a5eebea2dc7f8937614e19e32084a20077a007181f801e
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external/cryptoauthlib/lib/hal/hal_swi_uart.c	17208	ed7474bba42e374d06d1626fdf363313e841cefec98cddad922b9
external/cryptoauthlib/lib/hal/hal_swi_uart.h	2305	f953ec28677bef82c4428328c78e18fdca9455c0deb62f856ec78
external/cryptoauthlib/lib/hal/hal_timer_start.c	2408	9cafec72d0291552aa3a86c619d10641961c066263156f3f8cc30
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external/cryptoauthlib/lib/hal/hal_uc3_i2c_asf.h	2108	8ef0df9c3ff24b735bd500863cf12f8ee361d0437e8449cbe1249
external/cryptoauthlib/lib/hal/hal_uc3_timer_asf.c	2490	fdc3693c67a7c11dbdde051554b745c9d4f98fc58a904e966f950
external/cryptoauthlib/lib/hal/hal_windows.c	4420	d0ba1a98dd1d985dc5b0cbb94b5c72867443853db8608f90b55e4
external/cryptoauthlib/lib/hal/kit_phy.h	2013	b1c095ca9d45279fca11db28b230908829f84cf99fad115f57c69
external/cryptoauthlib/lib/hal/kit_protocol.c	19206	01461be8ea78c36e5f3a03132b4c6a837f12891fdaaff8aca13bf
external/cryptoauthlib/lib/hal/kit_protocol.h	2530	40ce866c994e12f982f93c659bf35001baa3db35616bd0b7b27d9
external/cryptoauthlib/lib/hal/swi_uart_samd21_asf.c	8578	58a4b311dae88fca6d0a8ab9d7a55b27120885229b3cc5a8a56e7

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external/cryptoauthlib/lib/hal/swi_uart_start.c	7289	94bbca7da6cdd17f48e6bb9d77400e6b8fc07d1529e8887e8603c
external/cryptoauthlib/lib/hal/swi_uart_start.h	2928	fb931c1211ea9da17384961ed2d7f2623a77f7be40ade0a408102
external/cryptoauthlib/lib/host/atca_host.c	51312	192be7ba2b6e97f849d2e4bcb3c312b34bd8741599166890ab543
external/cryptoauthlib/lib/host/atca_host.h	21629	24128a71c25ff78e44febbe6c686cc49e77798aac12b05a0dd8da
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external/cryptoauthlib/lib/jwt/atca_jwt.h	2339	337cc8eee0e4bd9c21d40f4029fc9b9758861e033610e49d45e9c
external/cryptoauthlib/lib/mbedtls/atca_mbedtls_ecdh.c	4676	7b02184224a8fd6ac79b51f40f2718d73e00cf0ca367b356d635c
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external/cryptoauthlib/lib/mbedtls/atca_mbedtls_wrap.c	32586	e5352695b155b49d60ba7266be6cefbe8455ff3484126fcf35ec3
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external/cryptoauthlib/lib/pkcs11/pkcs11_info.c	2848	28176eb34f062b13f02dcd11aa3541be93f6e3550c6129ffa384e
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external/cryptoauthlib/lib/pkcs11/pkcs11_init.h	2218	cb8cfe7c01ea22c60dc4637dcbade4aacce5da93931d13777a0ec
external/cryptoauthlib/lib/pkcs11/pkcs11_key.c	44050	90f5684d266788ebfb04feb7b887139cb38f3ced88464bc9e49da

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external/cryptoauthlib/lib/pkcs11/pkcs11_os.c	2567	c7cfca5c9e72d02aaeacb251d6d2373c026d11dfc619f3d1d93a9
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external/cryptoauthlib/lib/pkcs11/pkcs11_token.c	19567	41211cb282065f42b40bb3e8c70490dc13253912a4fe4762a9a94
external/cryptoauthlib/lib/pkcs11/pkcs11_token.h	2357	dc1727ad53dfaeacd475f89e490473feafceb776ca7456958a242
external/cryptoauthlib/lib/pkcs11/pkcs11_util.c	2592	384d4db087605cc3f04043a0fd4af05505c4afc232e652efb101a
external/cryptoauthlib/lib/pkcs11/pkcs11_util.h	1672	c0c2de6af3612688fa7f232b5c74b47eafe61fa6a29e3db047b64
external/cryptoauthlib/lib/pkcs11/pkcs11f.h	27386	69ae90423046bc74d6bc4fa81c7a93dd9daee35b988dfca88a5b8
external/cryptoauthlib/lib/pkcs11/pkcs11t.h	72149	f30b9254acb8b1592f62a0f101432b381ac530716ba48ad7fb2c2
external/cryptoauthlib/lib/wolfssl/atca_wolfssl_interface.c	10099	093a1f68d4aff61717680b1065f9ea1a9465a33478fa4decd951c
external/cryptoauthlib/python/cryptoauthlib/__init__.py	1352	40fb2a9e9f2a9835ee72854060ea00f78012da0d63155ab90527d
external/cryptoauthlib/python/cryptoauthlib/atcab.py	114570	0b92b5429170956ee3b25c3dcecc05f822715a44b5f6d1b48e06e
external/cryptoauthlib/python/cryptoauthlib/atcacert.py	27170	17bca2af987ea156db0f0a46e8d7069915efa3e7f8f9f37ab74a6
external/cryptoauthlib/python/cryptoauthlib/atcaenum.py	1004	d3317de13e3ea2916df82ca8676f8dbbc9e750607d1689f07f790
external/cryptoauthlib/python/cryptoauthlib/atjwt.py	4082	b28b739ecce2f085ad0c234005f0553a7e0b8c86259f4b1621956
external/cryptoauthlib/python/cryptoauthlib/device.py	8243	21d6d279d890bdc5c1ce2d49e2d6dbab84f2be9a97cd928710eef
external/cryptoauthlib/python/cryptoauthlib/exceptions.py	4679	be356526d4db995abe6ea2b5add82931d0df1359c4adeb60ad0e5
external/cryptoauthlib/python/cryptoauthlib/iface.py	5610	acd167417a54c6184772d45c3563804f1de0b72f612fa5465a80e
external/cryptoauthlib/python/cryptoauthlib/library.py	8521	86289d795b378c255c85d77fd2dea9977e1462cb6ed4868cd83fa

external/cryptoauthlib/python/cryptoauthlib/sha206_api.py	11289	070b0dbafa75c9c7ab4966d1be314111e8e6b432b2a0c1b652235
external/cryptoauthlib/python/cryptoauthlib/status.py	6743	41344e72bb7f16616e7bed1c61598f0dc4c673835c12b82e5e4ce
external/cryptoauthlib/python/cryptoauthlib/tng.py	10283	bddcf0eab7817600b12bac69f986212176299fb9d1cf52dc90a2f
external/cryptoauthlib/python/setup.py	8396	b79e2c2b9b52dce089ce97767fa1c444a872329a0a6b3ac52942a
external/cryptoauthlib/third_party/hidapi/hidapi/hidapi.h	13870	113efe421189dfefd6b7946f1605b1eaabca70d79deb87e58d5ca
external/cryptoauthlib/third_party/hidapi/libusb/hid.c	40667	c1c9a663237a5360ee0da2087bf4c4cd655fd4d9c2b8db464ff53
external/cryptoauthlib/third_party/hidapi/linux/hid.c	19857	b712f3a6ab4005e3bc9ab67155b078a6c958f8f09f42cc69680df
external/cryptoauthlib/third_party/hidapi/mac/hid.c	29609	24618c9712cbfcf5394c84378cb31de97404ae0e41bff953a7026
external/cryptoauthlib/third_party/hidapi/windows/hid.c	25457	cb314715fa194ce86b888a6dcc7ac1e9e596e4e6991352fa78825
external/cryptoauthlib/third_party/unity/unity.c	65366	d7a7255e0fc01418a412d01780aa69c10c64fe03eebfce0e8e2be
external/cryptoauthlib/third_party/unity/unity.h	89027	7c8da363b67a7e37cd1052baf2b7043a702c36772d34c81992666
external/cryptoauthlib/third_party/unity/unity_fixture.c	8392	090875bd5b4d55966fdd6f9d984aa9883924a395d1a56a78ccbf9
external/cryptoauthlib/third_party/unity/unity_fixture.h	3220	73854da3d615a041ab114be64bb98debfa8b186ac9741682f3de4
external/cryptoauthlib/third_party/unity/unity_fixture_internals.h	1476	adf665b853438b5e35ffc72e6d0f1bd404a09188fbd4fc7c64c2e
external/cryptoauthlib/third_party/unity/unity_internals.h	86767	c97938da35602ea0759fdbe2bbae9464fc374190584df4e217cb4
external/cryptoauthlib/third_party/unity/unity_memory.c	4865	7bc202de8e950466f713b2c3918beb824a10d33cd7df43058dd46
external/cryptoauthlib/third_party/unity/unity_memory.h	1980	85ea1b5fa55bd59f03901fad4272a938a33696d9f0821d085ba6f
external/cryptoauthlib/third_party/unity/unity_to_junit.py	5939	0743f26b5ae67af204981293337f9cc9a8cb46c90aaa3f46f31fd
external/embedded-swd/dap.c	12150	fa5957cf285a88068a40cea0f7ccd82185d5f62e840ffa8d8ec29
external/embedded-swd/dap.h	2529	c9caf8a0d8c293d6d4ae8413680061c407b6de0c188f1c942ba85
external/embedded-swd/dap_config.h	4040	c884d6e8418634980324b354e558cc8708063902c695fd7bdffcf
external/embedded-swd/dap_target.c	3219	33bed9cd1c968b17a50c5eab41c51b908eaab8501bbd3b1dc1d10
external/embedded-swd/main.c	5566	5b525ade80cda064547da403bab33088091069d557fadb951f15b
external/embedded-swd/raspberry-pi/dap.c	10537	1af13298edba706a77639fcda65dee669b7571773e5a21c16a3c6
external/embedded-swd/raspberry-pi/dap.h	2572	fccdb45752ef643a8a6a26e6240abe4f55ca4538c627592590c42
external/embedded-swd/raspberry-pi/dap_config.h	3981	d83687a623aa46793496bf37054f297c821bc1e1d9844d3ac7932
external/embedded-swd/raspberry-pi/dap_target.c	5095	3d17629e6fb9250628ef47e04b49c690c5b2ca60739e9b69a3984
external/embedded-swd/raspberry-pi/hal_gpio.h	7121	6248f2a07feca6ff5fe6d8005a44de40192217f5a0b7242f248bc

external/embedded-swd/raspberry-pi/main.c	6122	a388df9fa00862fbcdc50e2cc4cda54bea5a93460a5543b7203a3
external/embedded-swd/swd_hal_gpio.h	5149	db63582c8a163d01a7a5b23543ec2854ffc224e85e11d85e3d696
external/lvgl/docs/add_translation.py	1209	4f7f3ea87cae36fbb2e51bb149be6fb46c9f50bad5ae94b1679ca
external/lvgl/docs/announce.py	3836	d498243f64e94469de0f5114250cb98b3d9d45eedd41585471a97
external/lvgl/docs/api_doc_builder.py	35433	01460953b7e211e20950de5c3629f93f7fc8e0dc09d58518d1144
external/lvgl/docs/build.py	32744	f4731d969f05e2b5f1334136e095c131ad1823bff08da687d9d36
external/lvgl/docs/config_builder.py	2107	de13b0e12b73a739b211de964460bf5b9a05ad7b6f759b6743356
external/lvgl/docs/doxygen_config.py	17842	3fd06d11c47f894e0de5e56ca2b6f9240f3777211af077c6ff9a7
external/lvgl/docs/doxygen_xml.py	60891	b4c91f0e0c3a6eb920df70e36e440e8223194592ed2badcb09e77
external/lvgl/docs/example_list.py	3762	a943490a0bbeb7b06513cdf200eb178a2c1708ae071892897ca7b
external/lvgl/docs/requirements.txt	331	6bf9b19c3da97d1d1036ecad16f9cfb18b72220514d13ad78b0cb
external/lvgl/docs/src/_ext/link_roles.py	2146	fbc140462fecce7862f04edceed96a9e6e59265ef2bcaaa617cc7
external/lvgl/docs/src/_ext/lv_example.py	5651	02f1180eb189d48b44497ed6b819ad388414576552d40c6ee35b3
external/lvgl/docs/src/_static/jquery.js	89501	ff1523fb7389539c84c65aba19260648793bb4f5e29329d2ee880
external/lvgl/docs/src/_static/js/custom.js	13577	3234d7e597bf5c95c59122aa89849afab6d6a365c2c5795f0fc91
external/lvgl/docs/src/_static/js/include_html.js	999	67002468f527c257a6bc2550bc8c792fe097b5df05f709f9c7fa7
external/lvgl/docs/src/conf.py	38876	f7ff853b3f5566b3d017db44533e2112d514e0869ceff0896c8fa
external/lvgl/docs/src/details/main-modules/indev/groups.rst	3883	b9765d511013b2ff991168636c450efecea5268180421221256cb
external/lvgl/docs/src/lvgl_version.py	1695	93c4f7ea0beb6b44d4d77e18bfe8d60f80dbf1c79362d636b5fdf
external/lvgl/env_support/cmsis-pack/lv_conf_cmsis.h	47770	4e6120b2b5148a24b104ad356be872cc9360145577a077e1024fd
external/lvgl/env_support/pikascript/pika_lv_point_t.c	376	85d246f40965bafbbcb48bf36c544bd85ff82b94cfbb4340c112b
external/lvgl/env_support/pikascript/pika_lv_timer_t.c	1395	4237a13e337846c3c8690d27098b65cc0390732da614394d28cda
external/lvgl/env_support/pikascript/pika_lv_wegit.c	12866	56941aed303d49db2a722ab443125ed3c1d10ac8f8180c2f3cf61
external/lvgl/env_support/pikascript/pika_lvgl.c	8617	1fde6d4e5d8c6cd930a55b6266242b77a159e93ad9b77426de1dd
external/lvgl/env_support/pikascript/pika_lvgl_indev_t.c	459	f21c09c63b5475ba768d7b25d5d34b0c6c0da557c50d00907d444
external/lvgl/env_support/pikascript/pika_lvgl_lv_event.c	634	25ac679fa36c0393379d128c7985be6310c4781694791717fb0b1
external/lvgl/env_support/pikascript/pika_lvgl_lv_obj.c	3756	a9788269f563d2b228fc7d97e3518f77d682a47b97069f3757c5a
external/lvgl/env_support/pikascript/pika_lvgl_lv_style_t.c	2283	5f2c4adabae75e3045a518ed0bdfd49d4beb8afa6f586d3bef157
external/lvgl/env_support/rt-thread/lv_rt_thread_conf.h	1646	8cad8d615b627e27a7d4ab176fedb686d24fccdb54b27409314fb
external/lvgl/env_support/rt-thread/lv_rt_thread_port.c	2369	098718c060699ff59c339f2beb269830c1d3c9a41045877a55d16

external/lvgl/env_support/rt-thread/squareline/lv_ui_entry.c	327	d9652420881c304d9ec81667af9aa7a7608263c5733e39e8c35e1
external/lvgl/env_support/rt-thread/squareline/ui/lvgl/lvgl.h	323	c946643df67fa07b5b170d7bb04321671ef839347e3fbb4efa257
external/lvgl/libs/nema_gfx/include/build_version.h	56	8e2782c46b5d13eface3d33395cffea72b5c46ac4c8a4ff236f4f
external/lvgl/libs/nema_gfx/include/nema_blender.h	8202	5785a5bd1ba1567c35f305f755ca6cc90c50b46cfd989d0e3b141
external/lvgl/libs/nema_gfx/include/nema_cmdlist.h	6196	0a2076f182dceb99c15b7780be7a0cccb10ead829de0b8632df4a
external/lvgl/libs/nema_gfx/include/nema_core.h	1780	74312d6770b3a8becb5b7dcbd1f4bc729a2d8c0129b9ef6a5018a
external/lvgl/libs/nema_gfx/include/nema_easing.h	10260	a499ed663b981e36b2de1fbd896ef9a7177e6a4fa178b2a0576b9
external/lvgl/libs/nema_gfx/include/nema_error.h	3402	42934847c93dbef5b2177a63d6d929790c8a384af0a4c3dca49b7
external/lvgl/libs/nema_gfx/include/nema_font.h	7084	7a1a0c63b09abc26e4dce43ebf567d30b4dfb97707353982e110f
external/lvgl/libs/nema_gfx/include/nema_graphics.h	37346	6a8d24edd082e50684518528a7aa40fb69bcc15709ee3902526a2
external/lvgl/libs/nema_gfx/include/nema_hal.h	5556	752ec40cd2f581c191ab531a88cc8c8064f9948a131375f8c3c79
external/lvgl/libs/nema_gfx/include/nema_interpolators.h	5484	11755946b9f00e63819a0cc43a569334e4b8d1d834f15e2e9368d
external/lvgl/libs/nema_gfx/include/nema_math.h	7231	8d105b7e03ed0f619b50619e9d88838fb9737f98909865baa3a75
external/lvgl/libs/nema_gfx/include/nema_matrix3x3.h	6688	9d37d476ff4bdc96e06fdeedcf7ed435220538fa0219f970d9547
external/lvgl/libs/nema_gfx/include/nema_matrix4x4.h	8405	495ef5ca0c19bb2ab826a1aae15865923052d267407f927563050
external/lvgl/libs/nema_gfx/include/nema_provisional.h	5286	ad6cdd8afb24c6b4ee6aca896fafab4e125e73ab34cdfdfcb7b50
external/lvgl/libs/nema_gfx/include/nema_raster.h	3624	f07b4d8da69424f9630ac4a275d6f2cc9a2c4dd8318f92dbc5ba5
external/lvgl/libs/nema_gfx/include/nema_sys_defs.h	1740	793904b437eb61c8fb32a18a29e2aeccc5b3c66595bfd186745f0
external/lvgl/libs/nema_gfx/include/nema_transitions.h	8600	c725eff5efa9a7e0fbf60992cbb161248ecfaf7bb4b091f6425da
external/lvgl/libs/nema_gfx/include/nema_utils.h	2033	1bcfa011a73eff0e979ed4353b720c5f5e020776ffabbccbb340a
external/lvgl/libs/nema_gfx/include/nema_version.h	2399	a526b8831aa2157247975786aa2fda5a773171528a559f8f7984c
external/lvgl/libs/nema_gfx/include/nema_vg.h	10078	0e56276dc34882de944369d018f0085b943156a3b0133131afe29
external/lvgl/libs/nema_gfx/include/nema_vg_context.h	10407	685b22a4e235568047f942a4977aa8b913b27d948a3a7b0b98fbc
external/lvgl/libs/nema_gfx/include/nema_vg_font.h	10504	89e9ae0f91364b44e33def3e9b3b16a543513871085ebcb28af54
external/lvgl/libs/nema_gfx/include/nema_vg_paint.h	8458	e0ca2a336b6a61cb462c8b6da1afda41ddb6acddfd812fd0caa13
external/lvgl/libs/nema_gfx/include/nema_vg_path.h	6810	ec6a4eab89c120c14cbd64b40b3565747fd8ad2831f8f46a111c7
external/lvgl/libs/nema_gfx/include/nema_vg_tsvg.h	2047	fe43fcca248e5bbd78859de9471a0d86c0c7a0f1c65f5cea89f1c
external/lvgl/libs/nema_gfx/include/nema_vg_version.h	2430	713c8158b78339c7cf26d9da163239119d6f06830ef1ec52e1726
external/lvgl/libs/nema_gfx/include/tsi_malloc.h	2380	326c15b478813266fd73391bcae4003e05b7ea02db150f69a0987
external/lvgl/lv_conf_template.h	51370	ac25beccf7ca971a52b761a4a20d68a37c65ad54fc52cb11e29e7

external/lvgl/lv_version.h	248	9cf9add2c92c0f70b6a764766605a4f199d3885e5d3ed4e9dac32
external/lvgl/lvgl.h	5942	9f421571d2fbeaee457e24f8aaa8a94bdc6e31e4202bf5e6a6223
external/lvgl/lvgl_private.h	4639	89c13853ed6f348f4d4c3e8737f50a5ea75dba79e569d667e2e93
external/lvgl/scripts/LVGLImage.py	49858	cf02850de3b4b8ae89f8ae5a52c4d682cc6efac4d4e9d61d94681
external/lvgl/scripts/built_in_font/built_in_font_gen.py	2465	e06f0e0feef35ea7343ce0b99e956f0827f3509ed65c59ce0150a
external/lvgl/scripts/built_in_font/generate_all.py	11561	179af7094ac7ebf75c0af9f89f25eae129860fd58e9f196b3ad79
external/lvgl/scripts/code-format.py	1943	ce8739ccdf825976d3154d5d45e9d2c3caa2c476c328c1c0a0a22
external/lvgl/scripts/filetohex.py	343	a8ab114103add8516f973982e0ad21258b04564c49467fd0d1bfb
external/lvgl/scripts/font_license_verify.py	3877	66ec4c66eac7af95e5e76d79c5a06a45f4cf1b49e179f6bf9eb43
external/lvgl/scripts/gdb/gdbinit.py	315	399b35fcfb4167b03cde418fae1936ee1527f4551408279886d98
external/lvgl/scripts/gdb/lvglgdb/__init__.py	269	2d686809021771dad9ade8ee285fb1f4b42cd210ff899954657a8
external/lvgl/scripts/gdb/lvglgdb/cmds/__init__.py	478	4cc6b5ad9a60a6007785fa30f1ed201f8ff90ef698c6fcd2fe005
external/lvgl/scripts/gdb/lvglgdb/cmds/core/__init__.py	58	0ebecc840b368a9816d77c8fa5e981c6ce23f1af1f7702c3c70b6
external/lvgl/scripts/gdb/lvglgdb/cmds/core/lv_obj.py	1796	ea9aeec2c7c46312d8be3690c6ecdaf2ac4005c5ba71c85a61a38
external/lvgl/scripts/gdb/lvglgdb/cmds/debugger.py	2082	39bc0893cb38c9f3fff8d7128db6b61065484ff8da59af61cb472
external/lvgl/scripts/gdb/lvglgdb/cmds/display/__init__.py	76	ab4dfed12aadf9cb10e44489e2ed6c4cccbf93fc90efa98d6a6c3
external/lvgl/scripts/gdb/lvglgdb/cmds/display/lv_display.py	1579	4dd3ac679127c4ac30eb6cd860880767f2283b4450a0d24373aba
external/lvgl/scripts/gdb/lvglgdb/cmds/draw/__init__.py	69	c12317334c5e4574899777012a2d5f0dc668d8946007a9fc2a493
external/lvgl/scripts/gdb/lvglgdb/cmds/draw/lv_draw.py	1681	d8743313be0decf6aaceda2dae0e83046fa0ce218ab0412eb1c17
external/lvgl/scripts/gdb/lvglgdb/cmds/misc/__init__.py	64	261955d7b21b56dedb5fb66ddfd5858dab1c0d93b29319f945485
external/lvgl/scripts/gdb/lvglgdb/cmds/misc/lv_style.py	1035	8d8901b2543109f1c8f960707b129796a203874948015df0143f0
external/lvgl/scripts/gdb/lvglgdb/lvgl/__init__.py	294	2c69acdac1dcd135b60a45bd71fc044230e1ac0834cfa03d01d76
external/lvgl/scripts/gdb/lvglgdb/lvgl/core/__init__.py	146	8c5ab80798228e16b76da4793c5d3a880407d354fc8f8fe683492
external/lvgl/scripts/gdb/lvglgdb/lvgl/core/lv_global.py	2854	18501d5634fe88f6618336903a7e0c4b774440cf9d505d46e6e04
external/lvgl/scripts/gdb/lvglgdb/lvgl/core/lv_obj.py	1843	d388f5c9c2d2fcd77e0d3a5478316ab7b046a3e09145cceb73d47
external/lvgl/scripts/gdb/lvglgdb/lvgl/display/__init__.py	66	48ae83d03ead74722706cf46eb5c26e2ecf64c43fb4431e84ce6e
external/lvgl/scripts/gdb/lvglgdb/lvgl/display/lv_display.py	1277	2d1e5c6dec9ad913ed347594e2df80450a12de62eb466cc9f398d
external/lvgl/scripts/gdb/lvglgdb/lvgl/draw/__init__.py	67	84c5710ab20061963d9a1d7587b05b72d498e93709a6cdd710e28

external/lvgl/scripts/gdb/lvglgdb/lvgl/draw/lv_draw.py	0	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca4959
external/lvgl/scripts/gdb/lvglgdb/lvgl/draw/lv_draw_buf.py	6962	4e9bbf8017264c1ab71710e79777563af54bb85bd2e597c625063
external/lvgl/scripts/gdb/lvglgdb/lvgl/misc/__init__.py	116	f97667f7e5c3abd728cd10fc3ed2ea4f4bcdca175b2f74642da80
external/lvgl/scripts/gdb/lvglgdb/lvgl/misc/lv_ll.py	1473	f8fcc1bd245b87ca41aab46a1c593360b685dc7399f62f4b20896
external/lvgl/scripts/gdb/lvglgdb/lvgl/misc/lv_style.py	151	b8b065122fdba27d0956e13a67c75455cbc957884d175d4d1526f
external/lvgl/scripts/gdb/lvglgdb/value.py	1012	892533149356cc3ed86f937d829029e0f03424b7d53637560997d
external/lvgl/scripts/gen_json/create_fake_lib_c.py	19229	8cfb968793fca2875fdfc2f41aabdef98fb79042299906b82fd3b
external/lvgl/scripts/gen_json/gen_json.py	10581	e67f8b5da262792b31788a5bf630ce71fd57dd6f882fbfab3280a
external/lvgl/scripts/gen_json/get_sdl2.py	1744	2423a34037dc9aa56e4ebf48a14cfb65199af9b52621746ecc167
external/lvgl/scripts/gen_json/pycparser_monkeypatch.py	47201	9119656fa48834288ce7a79b954f62575288ae2e1fad53affeaa4
external/lvgl/scripts/gen_json/requirements.txt	315	0b1bfaaeb328ee5660f6e28e1e02bc12f10adf409d039f44f6f40
external/lvgl/scripts/generate_cmake_variables.py	4664	1d4670abf9b97ad30e2a05de1e3b1f3565311ff19779ca7068521
external/lvgl/scripts/generate_lv_conf.py	4263	6ff8548398f9a5ffc52b76c2cbcdfb4259e762828fe59cbd3dddf
external/lvgl/scripts/image_viewer.py	802	d8e0a0cc296912bd29c1a355af27d489ea3e9e49921d6d51e7a1e
external/lvgl/scripts/jpg_to_sjpg.py	3582	fc2d077a4445d6bd8d4aaf2b1f5329af3c388c063614580f6e08a
external/lvgl/scripts/kconfig.py	12588	87f0c9f386f1e4904b4a153c8dfd5980c58c98422e508edb9af24
external/lvgl/scripts/kconfig_verify.py	1816	45ca26e71b63ef7d8d9eff6607a59dbbdf8723b12f5ab0aa8f4d6
external/lvgl/scripts/lv_conf_internal_gen.py	9318	13a3c4a22ddcd9b8be6442cc63573083149762231c2f27f299b67
external/lvgl/scripts/lv_templ_check.py	5220	d6a483e378ddbf1da553b97a4ac6da25f5afc27b900d879f6379a
external/lvgl/scripts/perf/filter_docker_benchmark_logs.py	4862	60797656e60f8432f5dd052995b6503e9c1456e735c4b125ec99f
external/lvgl/scripts/perf/serialize_results.py	4312	5f10b30a1845addf747d6b8a0065810db89fcf8dea608139d752c
external/lvgl/scripts/preprocess_lv_conf_internal.py	5676	b48746adc8b9690f67de9f8b4f92d6d6a13f88c1ab0736f40db4d
external/lvgl/scripts/properties.py	7087	2378003c388907776544cd409bc4f124dbc70a01b3b21b0e2cdbc
external/lvgl/scripts/release_branch_updater.py	13951	ee2fb8168740c20d8cbd8484d928c9b9637439e107ac28d3e08c5
external/lvgl/scripts/style_api_gen.py	38625	899725f6565c374383d65f81ec6aa4148271976f2d49528cb3334
external/lvgl/scripts/trace_filter.py	1348	b92d69e103b6c76c7b8684f7e5c3d8c655c21aed80330b5965aca
external/lvgl/scripts/update_version.py	5796	68aff60460b00a4edeb3bbdd113d80a47e6ab1c1c7f21bfa6b0a3
external/lvgl/src/core/lv_global.h	7062	3101940e6d5930dda9f9a4e1a97b2cc2a6248637752e4559313b7
external/lvgl/src/core/lv_group.c	14144	9e41401ac5dd23fd359f21bdf740aee0d4fea16275568bd42383d
external/lvgl/src/core/lv_group.h	6793	1187cb25d3c22f0d0e04b74e516a954f8581a146d3c3eeb3c7b3a
external/lvgl/src/core/lv_group_private.h	2026	efe339d1c37140179383094 10d659a4ad1fe5f9fcb733d9d995aa

external/lvgl/src/core/lv_obj.c	39822	6d47497cf60fb6065d07a90ebe454d1ba10c50d21142b2544f9b2
external/lvgl/src/core/lv_obj.h	21806	f1ea113fe1ba4429f490ef825d182933b577a4d79382fd6026926
external/lvgl/src/core/lv_obj_class.c	5888	5c33b982c91c13cd7a2baa26b5bb523af51f9c81ff18cd83c8082
external/lvgl/src/core/lv_obj_class.h	1789	51448cfa46f8b0b3031bf3552a031e5c07ce0db5caf4a5f65ffc7
external/lvgl/src/core/lv_obj_class_private.h	1914	ee176e8c0abc4dc395dae7ff1cf0e46f7de04123a39810f7f093b
external/lvgl/src/core/lv_obj_draw.c	16521	1afcb881e6e76eaa8b578087c72d7a1fa24bd62f6fb36a7f0e279
external/lvgl/src/core/lv_obj_draw.h	4981	46727969cf4c1b993aaa09cef54cc07daeecd7c17d7cf4df2e323
external/lvgl/src/core/lv_obj_draw_private.h	867	6f6076a40e5e733aab92b5ac705c54b7f3e6b13a0b8d5a3daaf04
external/lvgl/src/core/lv_obj_event.c	12927	da6526213686972b24922053926905f5a8808db0e5e01fa59c28a
external/lvgl/src/core/lv_obj_event.h	7258	8c8bc14c50686689ef4deee0dd0a035f228c2b60f4940f5672537
external/lvgl/src/core/lv_obj_event_private.h	1661	00ebd83bbb3cabc0a6b9d1e8cd459d2eeba968d12ba86bd88e547
external/lvgl/src/core/lv_obj_id_builtin.c	2690	616756187ce046dc9b19ddd6654d564b77c5771b62357e54e044c
external/lvgl/src/core/lv_obj_pos.c	41976	0ed16a104c71395e1165540bf24ee7b1658f6666ebf9019ad86a0
external/lvgl/src/core/lv_obj_pos.h	19536	4f3e2b847c5cc6aa2965b0c59e77d13a98c1ba11638b2ec2d722f
external/lvgl/src/core/lv_obj_private.h	2571	8542b1490e64eb0e74591395ac0c913a56d19ec641dd1e487dd64
external/lvgl/src/core/lv_obj_property.c	11236	0ceb9072315c72eb508006d8f68ffcac1833bc8d0cfdb6da81ef1
external/lvgl/src/core/lv_obj_property.h	9208	eae9e5e513a16a105d23963967e20c1213b1941dd109dc78c0006
external/lvgl/src/core/lv_obj_scroll.c	28233	16bf27aec253dfc711e3056c7ea09067c50dbb19afba92b536b09
external/lvgl/src/core/lv_obj_scroll.h	11006	334b0ae0003d46c1984ea4a214bf35bfa011ae4cde46d3c616427
external/lvgl/src/core/lv_obj_scroll_private.h	1061	621f784b8783cafaf7c4f9b31cbf488c0506479aa0e3496e031fe
external/lvgl/src/core/lv_obj_style.c	44128	1bb880078f512c3f914be452755b11b45c0d299fa83a81b5a7ef3
external/lvgl/src/core/lv_obj_style.h	16144	3674ef17b8324b4f86b57185a8f49743c869888ff327cbb849997
external/lvgl/src/core/lv_obj_style_gen.c	28941	8db2135e3494b13e0ad92fcbdee17899069ba5fa1603daefe317d
external/lvgl/src/core/lv_obj_style_gen.h	40131	a3f682bde495e18cdd00831c320e540461116d8c4c092b1103ea1
external/lvgl/src/core/lv_obj_style_private.h	2029	bd8f841f501e1708105bf27472fcc1e2ea594dcc1dae7d36f0666
external/lvgl/src/core/lv_obj_tree.c	24638	ae99d2dd726d9e35594fcced3e6b2242db93d27d1ec25a22bf5b2
external/lvgl/src/core/lv_obj_tree.h	11529	f9f97cb31b5c23881867078fbb56bd74846a0d085fbc9b2fc1593
external/lvgl/src/core/lv_refr.c	52497	7b3854c6d51995b85d23cc55207a3102206fbe3b5110e130e0e95
external/lvgl/src/core/lv_refr.h	1660	a42e51c6c1f50187e96ff74033b59ef7e27fdd0680ffec06eb18d
external/lvgl/src/core/lv_refr_private.h	1776	cb2a164ccab1b2888b665b979ac04055d660f061505cab73cb6b4
external/lvgl/src/display/lv_display.c	40448	178f673b4b72ea3b54f4a3d09897c815a893a4fc5f2572d8a007c
external/lvgl/src/display/lv_display.h	27397	3a5d42f1fdfce16e0a80a23e71ac85e4218820bf731c60d996882
external/lvgl/src/display/lv_display_private.h	5741	8d52dda563b07a25fe85ba506bfe74bb036bb28118ef2e6323e8e

external/lvgl/src/draw/convert/helium/lv_draw_buf_convert_helium.c	3768	26b44a4896eb19113c4e404472b455ba776d345fc6783a915ee2d
external/lvgl/src/draw/convert/helium/lv_draw_buf_convert_helium.h	1388	2549a1126579992c29137fc6bf5ee62e1e7a91a745fac92e148b2
external/lvgl/src/draw/convert/lv_draw_buf_convert.c	4300	fb946aa5069a5604f4efae7fab9b4d082b2a982a5da089cbe17ee
external/lvgl/src/draw/convert/lv_draw_buf_convert.h	649	c04acd44fb3bd38ac8dacc589cf1bb673ccaffe79fad0a6b0e710
external/lvgl/src/draw/convert/neon/lv_draw_buf_convert_neon.c	3672	b5ed283aef2a677bb9cbd266e809922e83c8936255f441496abfa
external/lvgl/src/draw/convert/neon/lv_draw_buf_convert_neon.h	1365	954859a165ff94b681f265bbfc166adb4ce389f8fde9ccf4e1106
external/lvgl/src/draw/dma2d/lv_draw_dma2d.c	12583	7c4e3d7293af2ea9294beaaacf769fa414092f469d4d19c69096c
external/lvgl/src/draw/dma2d/lv_draw_dma2d.h	785	7ba993a2007d5b9d9da637a8a5b61544c0c67944bdffe7e17c699
external/lvgl/src/draw/dma2d/lv_draw_dma2d_fill.c	3830	5ace4bfcd3edb7c60beaf7f8f30552f65063bc92a7e8e78ec1f6f
external/lvgl/src/draw/dma2d/lv_draw_dma2d_img.c	9915	76bb1b84d840ec7d8b58186859f45a3166b8b331a5ec4f9cc641a
external/lvgl/src/draw/dma2d/lv_draw_dma2d_private.h	4211	11ecb03e114b5d1ff50223a59aed9da345caaedebc60d9a4d6f2f
external/lvgl/src/draw/espressif/ppa/lv_draw_ppa.c	7850	0fd18ae3f2c51d98619b21ab345752382aa12905fee519a21f637
external/lvgl/src/draw/espressif/ppa/lv_draw_ppa.h	1082	12eef580b0b8d64e3affad72a6ffb31ef87969b290b5a5ca5b097
external/lvgl/src/draw/espressif/ppa/lv_draw_ppa_buf.c	1372	365ca9961a8910f0b13e0b4f0fb39b08be2fc0f4fd7b9c6419e4f
external/lvgl/src/draw/espressif/ppa/lv_draw_ppa_fill.c	1333	584eeb197d7dd3477df356359dd81c410b0aef11132e1e5c26686
external/lvgl/src/draw/espressif/ppa/lv_draw_ppa_img.c	2514	3cbed4ccb45697d12a6cffc35d75feab8601302854c3e32ef338e
external/lvgl/src/draw/espressif/ppa/lv_draw_ppa_private.h	4719	e529f5c343d5f94b64b2cfed83f729490e2fbe005f6fd26705870
external/lvgl/src/draw/eve/lv_draw_eve.c	4084	c9d256820f3efeb0733fed9c759ed21dd0f8013546e518c99d40e
external/lvgl/src/draw/eve/lv_draw_eve.h	917	6fbfeb4fe2a13b61634a0c5888891d4015c4407b1dc05205a27cf
external/lvgl/src/draw/eve/lv_draw_eve_arc.c	10566	098945c35a840f15f0d6cff3bc9f4f40188d120d3bb723b9511e2
external/lvgl/src/draw/eve/lv_draw_eve_fill.c	3574	853a06ea4e89abaa55a29e2cfff9677dcc5d14842f16a563f79a5
external/lvgl/src/draw/eve/lv_draw_eve_image.c	9075	3431e9a6ffb50df7a1eec7bfc847c1f9297f8e0271f1c877f506f
external/lvgl/src/draw/eve/lv_draw_eve_letter.c	6942	1d6bb9f977b8f9ce1b3207fa4d7890c2e3a3a3e264370650da61e
external/lvgl/src/draw/eve/lv_draw_eve_line.c	1631	0beecb6c6ba7a0c10ec0bea46c68f832bf03502e511405094ee37
external/lvgl/src/draw/eve/lv_draw_eve_private.h	3249	ca72d04c60ea70d9d19d8a3d83dc3c387cf78df26dba56e5fd4ce
external/lvgl/src/draw/eve/lv_draw_eve_ram_g.c	5916	b8da47a84075d348b0130029c07f7008a3ca0d480b3988e396c87
external/lvgl/src/draw/eve/lv_draw_eve_ram_g.h	823	ced2ec42b040c7a48316145db90e06c6a3464094f1e92561d4614
external/lvgl/src/draw/eve/lv_draw_eve_target.h	2817	e3f199452fffb98edbdea94cc4f29ae1f7825daf7d9e63abe342e

external/lvgl/src/draw/eve/lv_draw_eve_triangle.c	3334	aa4578bdbcd64b8facc891e6e22371789ea385adac51a2dc922f7
external/lvgl/src/draw/eve/lv_eve.c	6657	1e584761376c4585ac569b35f34171a14257246c8a2abda5502ba
external/lvgl/src/draw/eve/lv_eve.h	4787	2444ad1b0fe0a5ff9f1b959628498594077c169fe260ebc5b16ce
external/lvgl/src/draw/lv_draw.c	21952	c39fb2bcf23683ec0e84dbf042d79a6410959e83116da5dcd17d5
external/lvgl/src/draw/lv_draw.h	11851	ce774bd8650db80704ac6497857464be4da2f8488443d3c914a3c
external/lvgl/src/draw/lv_draw_3d.c	1447	5d87908d026ab2e8cc736da11b92a61d2bed4403ad6b6400fc7ac
external/lvgl/src/draw/lv_draw_3d.h	1417	6bee682f396e70fb2d80dfc9b3ac139581a4fdde43cd9b5bb846c
external/lvgl/src/draw/lv_draw_arc.c	6661	1d82fa5144fc675285b7eae4e76142cd597d3c9fe5303593ae5e7
external/lvgl/src/draw/lv_draw_arc.h	2997	92f196095341ff18c7577ece941d0156e6c2a721bd26941803ab3
external/lvgl/src/draw/lv_draw_buf.c	21548	21ad8877fc3bd99bdcea9aa77383055fabb728cc39ef7c17f67d1
external/lvgl/src/draw/lv_draw_buf.h	14726	933ab3b7b298480bbff21ae2fe7221df786d10fb29f2ac54e6443
external/lvgl/src/draw/lv_draw_buf_private.h	1067	01cbffca7f45dd65ffe329658491156ee5c1d43621ad9615e2d45
external/lvgl/src/draw/lv_draw_image.c	13581	3373652248781e086e6eeaaf012985de304367cd09741a5c17e94
external/lvgl/src/draw/lv_draw_image.h	5920	1bb2dc5db4b6d6d0948d31a7881545ff9f36c4d8ddb62a07d3086
external/lvgl/src/draw/lv_draw_image_private.h	2466	cd21588b7c5c20a25e3cf3cf7640a5a8b962f3a93d4d012682eb5
external/lvgl/src/draw/lv_draw_label.c	23656	70149624a5515f39b46e0a50155b487620dfa6b20225fa5a15118
external/lvgl/src/draw/lv_draw_label.h	8112	37b57eeae472aad4e4bbdb3386ddd5baa7f9abbec1f2520b449a2
external/lvgl/src/draw/lv_draw_label_private.h	2031	2fc8b82558c99931ffb6f333e09e7c5eb8d2f3597905a2704b583
external/lvgl/src/draw/lv_draw_line.c	1836	1985f06a7de6955d95303f3f10036934ee031d4e9e4766c4fee59
external/lvgl/src/draw/lv_draw_line.h	2272	9301c854dc2b5de50c2a140cbec45f7dbd86a1b9da198d860765e
external/lvgl/src/draw/lv_draw_mask.c	2160	311d4973746c83016adf97f445274736b6bac4a8c49541a9f6389
external/lvgl/src/draw/lv_draw_mask.h	1281	d09a5b02620c8ad2d2ddfdcd5c6df1dbfdc2ced66841fa55f9bb6
external/lvgl/src/draw/lv_draw_mask_private.h	964	0fae20393279d2bb329991c2a297dc912aa11c16881092c228d67
external/lvgl/src/draw/lv_draw_private.h	6592	e2b9f51c56b4b4728558257e504786e212548faffcc3e9cd56a50
external/lvgl/src/draw/lv_draw_rect.c	11959	8df5eeae23da8587bd1abca7c3aeb7c427c747be0c4bfa831ab4a
external/lvgl/src/draw/lv_draw_rect.h	6633	095f632dd4ae843336db0821b247b5d95df2667eea15d8712e8ca
external/lvgl/src/draw/lv_draw_rect_private.h	584	e7c2b4e419878a865c065cfaf00825c4c00d551657fb5d09b04f3
external/lvgl/src/draw/lv_draw_triangle.c	2065	98715f7c24bf532702d36d6cc342b201aa6070b84d0c5b44dd5fe
external/lvgl/src/draw/lv_draw_triangle.h	1767	0da7fa16c0b675feb103e059b0d58bd53116992a131d47d43ef7c
external/lvgl/src/draw/lv_draw_triangle_private.h	671	c920e5df472e2a018ef1da0b5a293d2f119b3c72e9017cdff44f8
external/lvgl/src/draw/lv_draw_vector.c	29698	a27ed7f84dc6a483147348655673fec01a35889a5279ec69b5a3b
external/lvgl/src/draw/lv_draw_vector.h	22541	98b30ffa70fe02f2b7bebe7e808b99f34c0bf2d59c0aa208cf96b
external/lvgl/src/draw/lv_draw_vector_private.h	3493	e8d3604efeba91926491cf296349605f80273bfedaf9a8a1eefce

external/lvgl/src/draw/lv_image_decoder.c	14622	8d001a4e4a024b9a3d247abd1fc7a31d1fc09f14d8a02807d5955
external/lvgl/src/draw/lv_image_decoder.h	8498	cf8cb23198e97741d52771974ea7276fd2c79b59c425a020ed4da
external/lvgl/src/draw/lv_image_decoder_private.h	3926	e3c20296ecf46c9741f1884aa89822889020dc35973695f864cf9
external/lvgl/src/draw/lv_image_dsc.h	4564	f6c5b2290a39dd8906305a5d3340a20feafbd3ce99bd458bb0ab1
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx.c	13949	8a3f71ccd93e7f4cb397c3383971edaf4a2ce93a72c0060893864
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx.h	4043	f3ae2b3783e49b832cb8f4d9202a77d8fd371f8a46d96a20746c0
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_arc.c	3760	4d83f93dbd578159c8e181bb3abbfae81cff39aa9919c31c261a3
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_border.c	8501	9986854a223fce80112170d67e832dc1290a01d917b0c69827d42
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_fill.c	7609	8005fcfa6bd5fafa1ce4a647a538f0378a7f0992428e3dbc3e040
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_img.c	9118	bf4f5579334e5b58e9213d46d4fccab95556a1109c9b2976cd781
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_label.c	33003	1ad6f9664e3cde141509a8e466b0e28c0cf6d464d084850c48432
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_layer.c	2123	b4c37d4af2dbfc864c0b903d681c98b9f29c9c26b310d338c0f89
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_line.c	3801	e082813e86c083cebc5d5dc189976e7921316588fae9f34deb43c
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_stm32_hal.c	5103	73cfa4bda569c8f6d71931fc55d53a113ae2d75eba8af0cd34ba8
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_triangle.c	6818	50580187911ab6a6045c4b6018e9db271ffefb1d3a085e3695ec6
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_utils.c	4538	b1f7036456c1a8fa7a706a33634225183f6d897fe392a0c438b27
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_utils.h	3645	8d612cc41b5c8358a44e23abb9ad160c446ad4e55024a00a8d509
external/lvgl/src/draw/nema_gfx/lv_draw_nema_gfx_vector.c	6101	0640d8ae6132c14d636b4a1781c037daa17964e0df37cf08ab06c
external/lvgl/src/draw/nema_gfx/lv_nema_gfx_path.c	5036	998c95499a1f0161a51e80e52985adfc87db8ffd76d2f28a9dcf7
external/lvgl/src/draw/nema_gfx/lv_nema_gfx_path.h	2951	e8085d601c42c45a4293c6c699d2c0a3e0eb594abef303f0853b7
external/lvgl/src/draw/nxp/g2d/lv_draw_buf_g2d.c	2038	08652d22f1a267949f8c604262fddfc6e20786fd996c67efbeed7
external/lvgl/src/draw/nxp/g2d/lv_draw_g2d.c	9266	afdc785d77336b03ab273f7fc8a19f28232c023d95ed2a76828a9
external/lvgl/src/draw/nxp/g2d/lv_draw_g2d.h	1172	23d2b49f0260b54312663e8ec09667f7170da1a724659b2ba9f10
external/lvgl/src/draw/nxp/g2d/lv_draw_g2d_fill.c	5360	982971531a8be210e2f6d640435cc30c3321bdad2ff77c97d79b6
external/lvgl/src/draw/nxp/g2d/lv_draw_g2d_img.c	10189	ead930179547172c88c336c8b0520f55b0c0b2c03b8b64e6f03b1
external/lvgl/src/draw/nxp/g2d/lv_g2d_buf_map.c	7480	5557e697ec6f9e53a9e50d6b9ca722f72d898eb3787a9173aa817

external/lvgl/src/draw/nxp/g2d/lv_g2d_buf_map.h	1376	eeb9efdf194b8d2649c8434aa0899b475a2489883027a77405e36
external/lvgl/src/draw/nxp/g2d/lv_g2d_utils.c	4376	9c9cb49ad993ac40ea28f582e368d696ef3493c42bf94c6ea92d8
external/lvgl/src/draw/nxp/g2d/lv_g2d_utils.h	1821	327e2a8610a407d612adfa3ffa2441e0f503f3bd49bfa873dc6f5
external/lvgl/src/draw/nxp/pxp/lv_draw_buf_pxp.c	2988	26a23284b6cef33c081baa32c09a641aa4ab5e21bea419acc8630
external/lvgl/src/draw/nxp/pxp/lv_draw_pxp.c	14308	1082a2cc04a32195f7709b245e4835e201781bc0685f532b83bef
external/lvgl/src/draw/nxp/pxp/lv_draw_pxp.h	1667	59c57bc35e618145b7d90e1520fa35476d69be71956c77e3de924
external/lvgl/src/draw/nxp/pxp/lv_draw_pxp_fill.c	4537	38e29397af24e8726c8a97b3becbddad6f15818be2f51902a691a
external/lvgl/src/draw/nxp/pxp/lv_draw_pxp_img.c	13109	9e5ecd83552171a84cdc75a615484e8720cebdf59b8216e9ebf27
external/lvgl/src/draw/nxp/pxp/lv_draw_pxp_layer.c	4226	52747b09c518d0885e7ae4fa21104a796feffb70a8f1867aef5ca
external/lvgl/src/draw/nxp/pxp/lv_pxp_cfg.c	1756	3e6896af0328593ba3a8fac91906c0f044072ddcf9d1fd49f57d9
external/lvgl/src/draw/nxp/pxp/lv_pxp_cfg.h	1955	683f6ba650d4a8f019343089220c2c7960fe7b7c49147e2dd3f02
external/lvgl/src/draw/nxp/pxp/lv_pxp_osa.c	3478	7e4f45345847abfbe6f41b08879bebb5eb78349e517d1fabde843
external/lvgl/src/draw/nxp/pxp/lv_pxp_osa.h	983	498117629c6f40c98a99ac003f80030ac706d6c8da1f87b3e8fb6
external/lvgl/src/draw/nxp/pxp/lv_pxp_utils.c	3794	626787a5ba5ad10201d7fd0113cf082c1bfda7c86a8c32794fc3c
external/lvgl/src/draw/nxp/pxp/lv_pxp_utils.h	1819	d3df2791035e240abfe184521162e18a36c9383bcb62b84932026
external/lvgl/src/draw/opengles/lv_draw_opengles.c	24674	ad7ab5c62d0277475e590fdf2fb0cd4058df2d29a6aaffe2e41a4
external/lvgl/src/draw/opengles/lv_draw_opengles.h	707	545e95d96ac46ddd3d96373f5cb8cabc7a9551b0bc231660e4b47
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d.c	19946	f053e39b176f7b698c11a700fb7a36ceb521888994ac9003adf66
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d.h	3839	c1e25cee536f373059b345f0482c301722b20e6012e15d6c70324
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_arc.c	6955	6799666a7884b0dee2d6fb8b787885a1d0a505f96f425d4e4adc8
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_border.c	15271	5d4139208574ea7fc7ce6624c5745553f847d9226d06394d85698
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_fill.c	11912	2a8bcfae5090dbf8570d1f2855da6b91e6f856db676ead663b48b
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_image.c	11682	983eb6edcf154eb109ea94466cc39eb9f5ee39a512713267c033d
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_label.c	7343	7e55461f97ee73489e32751e3621e1d40ce2f62a0fc113b215458
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_line.c	2225	2848032d838c5d201329bec08955592b5d151fccddcf2ed0deb54
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_mask_rectangle.c	1455	e6c405281319801c26efc11d9d1b2a831c8bc3a44fa7972e0d148
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_triangle.c	4507	c0df1dcfb31a018084c6ebc1ebf766cef3d74f4178683cd99f8db

external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_utils.c	3663	703b15dad391091739692b1534b2cdde5b2044221121fe44e7f5a
external/lvgl/src/draw/renesas/dave2d/lv_draw_dave2d_utils.h	843	7df826a57f751c26f11655b8248344f6359655ab21dac3586b15f
external/lvgl/src/draw/sdl/lv_draw_sdl.c	18896	3cbc348ddcb33512f433fa04db3408e236415a11cde88c6bad107
external/lvgl/src/draw/sdl/lv_draw_sdl.h	2268	11c0190a36fdaf75bec381a01c7b946593cd0e48e65e467ec3156
external/lvgl/src/draw/sw/arm2d/lv_draw_sw_arm2d.h	26909	62aed206e4d11943395cef613b66a67b5088ffa1a7eb01439c5dc
external/lvgl/src/draw/sw/arm2d/lv_draw_sw_helium.h	1106	4f035faa6d0959f853f2f53049e1ee32ec8fceb1cc637b4691f50
external/lvgl/src/draw/sw/blend/arm2d/lv_blend_arm2d.h	39784	37ca545d7e8095ef8df5d01e21d805270c080f07ce9b916caa258
external/lvgl/src/draw/sw/blend/helium/lv_blend_helium.h	47947	2d2f8b8569c7b985e40614bae6c4d90847ff12c6b055397b73213
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend.c	9678	61631e4f9b4e9117fe17387a22da2f11428952255da46869b7e22
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend.h	1363	0865a61b9cc65c54e1b5ce029465f0955a6c77e6126ca38d9f28b
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_private.h	2372	5e889bf2ba481a13f2b20003761a11709f734dbd47ce0de922f80
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_al88.c	41160	bf746eae222b1a5162916cd4240973ed0b31f4d4bc1117acfbc77
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_al88.h	849	9eff75b0e87916ff2c0c3407a3fc8a3e6309e5be0e5d7485b24ea
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_argb8888.c	57757	c3d7dbb731076fe4a906a97611e77ad5419893efaec067a898606
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_argb8888.h	873	dbf68d652e1abc0a8cf7c15ef3d178ae642e35e8c33ad51019246
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_argb8888_premultiplied.c	37235	40de554dfcfca2081769a861960abbcc35dd89582c64f1a3f9a0e
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_argb8888_premultiplied.h	956	538e58c98bcde31fcff3b5ff2217922a869910f88f6e96fc3c40b
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_i1.c	54018	2209204b678672bd09924418205bb78e1960f17aee0df696bd506
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_i1.h	837	c0dc5caf960277d2a562675b1f43e08d73cf4f0aaf17958a4b6f0
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_l8.c	40219	c4c26a11d1e3018dff6f9be34a57ba2979cf1f2913ad9e4b7cbf1
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_l8.h	837	a4734fbf57818420c94986f38cdda2d51dd4b635c2debdb2fae79
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_rgb565.c	68705	b25dfda8103b8c5844b06d705fafc341533bcf82f7b87c069f6d5
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_rgb565.h	861	c4567efcf56ceb2e1743a0c5174ab84f9ef28371cbce01b39b5cd

external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_rgb565_swapped.c	75389	42ac068b7b79cf40ad0ea9006976ef51eca125251c0bed3f04804
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_rgb565_swapped.h	909	9d666a65971eee9c7f53b378b1c60c6b2416c01a3dcf056c104b4
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_rgb888.c	48857	6d070bbff6d038fd795d25fcaeede12e080bc7d0440e75063c314
external/lvgl/src/draw/sw/blend/lv_draw_sw_blend_to_rgb888.h	1039	dc54eee5def983eceb520fa2f1f5852fa1d3af1b40d55d474166a
external/lvgl/src/draw/sw/blend/neon/lv_blend_neon.h	856	d2425f67137d2c5cf78e2259bff78aeae006c0806116a47c4df3a
external/lvgl/src/draw/sw/blend/neon/lv_draw_sw_blend_neon_to_rgb565.c	100019	ddb05548df7076f5dca4fc4a565926d003938643262a424060a5d
external/lvgl/src/draw/sw/blend/neon/lv_draw_sw_blend_neon_to_rgb565.h	8189	17f6c440d7514f796501fc3e698afaae42972bfa9696fe0c45734
external/lvgl/src/draw/sw/blend/neon/lv_draw_sw_blend_neon_to_rgb888.c	75484	2474db675a391795172ae856a3649062f894c9f72dfb3428d3732
external/lvgl/src/draw/sw/blend/neon/lv_draw_sw_blend_neon_to_rgb888.h	8036	4e40c15062e01172772a9e1ffdb1a553e30077e9589e21287c0be
external/lvgl/src/draw/sw/lv_draw_sw.c	13944	815767c3484098bd5d505968d4fb24d579adbf8fa9a76b05429e7
external/lvgl/src/draw/sw/lv_draw_sw.h	6298	dcbb10b5cb1135c20d65eb6732637e40f76b42e4d13dd29f61d6a
external/lvgl/src/draw/sw/lv_draw_sw_arc.c	11289	a1b0eb4671e553590848813d8b1993b1dea3544eb96a6987edd2e
external/lvgl/src/draw/sw/lv_draw_sw_border.c	11417	9d9b8083284c459edcd4f09d7d98f878250b01c670366cc503cab
external/lvgl/src/draw/sw/lv_draw_sw_box_shadow.c	27811	7fd22e75af363ac148f9e10e37a14233cc3056dd30ba907dbcc98
external/lvgl/src/draw/sw/lv_draw_sw_fill.c	12974	fb44ea37597348ad96ea1bc89c8361d6b5be034aa6d7409207f67
external/lvgl/src/draw/sw/lv_draw_sw_grad.c	19929	b5089de0fb5e3ee4720b1bc8300adf2076d0af6aa1e997fadfaf6
external/lvgl/src/draw/sw/lv_draw_sw_grad.h	5034	984308dbe083609ae6e6b2d7411ffa9c2050674feebe3420a6119
external/lvgl/src/draw/sw/lv_draw_sw_img.c	41944	aaf1f0d1605fec25ff996591710bd6b6ae6961ea010bed3b16b81
external/lvgl/src/draw/sw/lv_draw_sw_letter.c	16240	1a816965ccb153dab44136c3c8d6241e4f128a9d121056240bdc5
external/lvgl/src/draw/sw/lv_draw_sw_line.c	13595	40ca00c3fad2850a4f375553ae1e62ae9c52b18f99dd9efa32d37
external/lvgl/src/draw/sw/lv_draw_sw_mask.c	43448	79840ecf88dc2e1ac5f9b33cf33144078a93cbfb66f721f5be0f9
external/lvgl/src/draw/sw/lv_draw_sw_mask.h	6967	672b4d9981bdbfe8d006fdaf6034534cd6883f20c9299081b6493
external/lvgl/src/draw/sw/lv_draw_sw_mask_private.h	3863	6e5e74fd4ab4d7ac6e6e0ff6956786893d7bd5a4dbfa4361385d2
external/lvgl/src/draw/sw/lv_draw_sw_mask_rect.c	3887	e1f3c696daef7677a0928bdfea687f65d25312b88ef8e44db3f6a
external/lvgl/src/draw/sw/lv_draw_sw_private.h	1355	4596522585731ad498b83dc759f03653e1848e68f1c0c81848dd3
external/lvgl/src/draw/sw/lv_draw_sw_transform.c	38613	7563df59d359659efcbf70973a388c8b26eb6137b2ee6d78a44b7
external/lvgl/src/draw/sw/lv_draw_sw_triangle.c	7118	1d5d3ce1e18efb912d4d6f47b46ba394462028051cf3e795d5f7c
external/lvgl/src/draw/sw/lv_draw_sw_utils.c	20110	9fcad9796d421f99a88ceae4c498d9e23042c82e809a67df90370

external/lvgl/src/draw/sw/lv_draw_sw_utils.h	3957	b9371ce464c848fffb7233dd61ee296c26cb3417f46938df2fdb4
external/lvgl/src/draw/sw/lv_draw_sw_vector.c	17270	64d26f57983373f3dde3d90225798328115952f4618147f37fdf3
external/lvgl/src/draw/vg_lite/lv_draw_buf_vg_lite.c	1313	598b9b388f3e7f9200f3d648a6f5f0cfe8f03d1e7b3b216550000
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite.c	9360	a35a4c0f82c8a5e7ba26c616ab50040ac38a7a2778285b7f5fea1
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite.h	2723	0bc611ff2059d4f9036526ae081f342914b18d59e3fc3e5b43070
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_arc.c	6656	66ec3c4eb6f48169307c8ba0f3f02fa6174e33bc9ede856c7ad79
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_border.c	13793	9e333e0a3cb87f2a1fb4ed48ff6961a6cd594e3429c47bd700554
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_box_shadow.c	2685	023787977bdf32bd9860609a7906dd591815342b5e988431ce6fe
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_fill.c	3046	53f9d280672233d60ced1760c77987ce9960301ca754ed9864ec5
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_img.c	8069	51978c364d80ca0aba277bc69089946cfa611d06ee1d7db6ae818
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_label.c	20162	aaae9ad2265224f10766ac469c51df7a752e0c3045d5a04319e41
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_layer.c	1776	d443f3cc2b11eba46bca90b2e9df14efa807781fe5375697bd783
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_line.c	6082	2f20b0232fa4372dc365221881135e98f1c8d1ef448fbeba2278c
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_mask_rect.c	4920	90806e37fb80b5a1912bfe5a092334865a9ee8d927b2c585ff9d8
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_triangle.c	2705	b22092f6f5fb2384fd54acd462b6d1d3f6663ab66a7cb6b8b9aff
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_type.h	1494	cf6a3e619677498a760f9406552298c6172a33aa69102e87cda4e
external/lvgl/src/draw/vg_lite/lv_draw_vg_lite_vector.c	16979	575ecb436bf90152b61418485ba9e7b805981179d39cc53699732
external/lvgl/src/draw/vg_lite/lv_vg_lite_decoder.c	11494	eae1b3b636e2d56bee4d4816c85e10259fd4759277ca628727c5d
external/lvgl/src/draw/vg_lite/lv_vg_lite_decoder.h	716	9bea602776b03689fe5b5f59b934a61fee52293b9d8cd25020850
external/lvgl/src/draw/vg_lite/lv_vg_lite_grad.c	29769	fe108d9d5c2c9aa2af18ccc357b762314a70cc8608ea6c9ae47dc
external/lvgl/src/draw/vg_lite/lv_vg_lite_grad.h	2841	b3000d010ba4087098fd1043a6dfe097ed83a202f4d304e0ced1d
external/lvgl/src/draw/vg_lite/lv_vg_lite_math.c	1034	6ec7978735aaec7605d3642a2117e7061986cb0d40703c983d553
external/lvgl/src/draw/vg_lite/lv_vg_lite_math.h	1447	db649b02817074778294f90a23cbde87f95a46cd7d1826e4f519a
external/lvgl/src/draw/vg_lite/lv_vg_lite_path.c	20843	5b6d972466183730974c8ba7181dd90ff31591ee23957b8252393
external/lvgl/src/draw/vg_lite/lv_vg_lite_path.h	4398	2ed6cc0924a7792089aa7efdbb6393b894fb438dc8f55cea244b8
external/lvgl/src/draw/vg_lite/lv_vg_lite_pending.c	2977	53ed7fc2bf687d24331e0472f7998d0a4da490c0dd55089bcb71a
external/lvgl/src/draw/vg_lite/lv_vg_lite_pending.h	2119	dd98bcf394a2c0448486dc087667abb3eaeae03f482e600d9766d
external/lvgl/src/draw/vg_lite/lv_vg_lite_stroke.c	12148	9edda809acff250de8bfbe6655606cf7884e08e479adad8461d0d
external/lvgl/src/draw/vg_lite/lv_vg_lite_stroke.h	1916	878f6248cc0da84cf946d8b75701c270f239295cbb5d2e1a42e0b
external/lvgl/src/draw/vg_lite/lv_vg_lite_utils.c	46419	03b1c354e6d2d39bb5ef4d07f449160f9c3cf3f0b5b446e708ac8
external/lvgl/src/draw/vg_lite/lv_vg_lite_utils.h	11165	106452077e428d48049aa4beba84a53506808f5f7ef727cdfdebf

external/lvgl/src/drivers/display/drm/lv_linux_drm.c	30389	578a80b9f422f4410a86af59a31a1845dd5d2c2bb1180f29d1403
external/lvgl/src/drivers/display/drm/lv_linux_drm.h	4058	c45d281d6fbfad07b2826cce92aa631a3bdb84a88a204e4fe8691
external/lvgl/src/drivers/display/drm/lv_linux_drm_common.c	2689	5ab61fbc6fccff4aaaaa7c1e8dd4f7c0507513d8d04fa4904c0f5
external/lvgl/src/drivers/display/drm/lv_linux_drm_egl.c	23441	f230226e365addac8a03cc8159f5841c4c79c2bbca80ee2cbe79b
external/lvgl/src/drivers/display/drm/lv_linux_drm_egl_private.h	1519	a97b9d62b8517715238cb596da6688b06eb4356b10aa921be94f8
external/lvgl/src/drivers/display/fb/lv_linux_fbdev.c	12414	f4a258cceaf8d88ee9aebe57ae075efc50913d8246820baf7b941
external/lvgl/src/drivers/display/fb/lv_linux_fbdev.h	978	597c23b9fce5bcbed8a043585872c36adeac86038815b91dd7e31
external/lvgl/src/drivers/display/ft81x/lv_ft81x.c	16207	fa590caf11cdde2eeca8bdd18f117ebe001ace8cda4a11fae97e5
external/lvgl/src/drivers/display/ft81x/lv_ft81x.h	3002	274a32a0104cfe5b9bd06831cc4b08b7228b5429d033cfc6bfec4
external/lvgl/src/drivers/display/ft81x/lv_ft81x_defines.h	32280	90a7f28424d96e201f51ce9f5c0f8b3bc7d96603fff33d48053ca
external/lvgl/src/drivers/display/ili9341/lv_ili9341.c	3858	1843fc84c50c6cd135864752a42b3ad39fcb979ed38a2346e5f4a
external/lvgl/src/drivers/display/ili9341/lv_ili9341.h	2463	d64e6b253ea17a7445df7a18813d7e278320bc23f468924cadd81
external/lvgl/src/drivers/display/lcd/lv_lcd_generic_mipi.c	11626	e6db705a7467cb8e76cb1e619d9ac121ce31c09c4d3d8786b6e80
external/lvgl/src/drivers/display/lcd/lv_lcd_generic_mipi.h	13743	4c3e8bff229771dfce6d46ec8e4a1ad867af8ee474c0983f748df
external/lvgl/src/drivers/display/lovyan_gfx/lv_lgfx_user.hpp	1103	a63a0a92f428cd4702811c0b38653866fe9a72363425c1086914a
external/lvgl/src/drivers/display/lovyan_gfx/lv_lovyan_gfx.cpp	4192	73260a0231fa5c9dc63588ac9220c8cbb5af4e234e12f6bf98489
external/lvgl/src/drivers/display/lovyan_gfx/lv_lovyan_gfx.h	824	ca95498b3430316e70019afc467cb97c778d4431b40a8f8f6a996
external/lvgl/src/drivers/display/nv3007/lv_nv3007.c	4620	35b006ebed76755f7896a9b5249b1ebd495f49c9b870aa2343898
external/lvgl/src/drivers/display/nv3007/lv_nv3007.h	2446	8067399de0e506b922912e1119cf9d07412cb82771b839e588d5f
external/lvgl/src/drivers/display/nxp_elcdif/lv_nxp_elcdif.c	6120	9c3910fcc81947f63c8d82fa63784ea399e0e2ae0d5f9e6c73e7c
external/lvgl/src/drivers/display/nxp_elcdif/lv_nxp_elcdif.h	2716	3c5c8efdd091fc7ea65b6c8c5c8f9ad9d4ff9ed4770c5c356f449
external/lvgl/src/drivers/display/renesas_glcdc/lv_renesas_glcdc.c	10581	20c3f2fbce8543f537158346e5f667156bdd7ddfaee27d8b9a164
external/lvgl/src/drivers/display/renesas_glcdc/lv_renesas_glcdc.h	1175	f011bdb819609fa8b15aba55c145eccb1cebe9fe2298c6b83b52c
external/lvgl/src/drivers/display/st7735/lv_st7735.c	2815	3e72a08a0cc201252fa932e89bf09f8453b76ab2c62ff82853de6
external/lvgl/src/drivers/display/st7735/lv_st7735.h	2445	50d5e8756c0bbe7e50164a6aa83ec391cc004203513741b50e80b

external/lvgl/src/drivers/display/st7789/lv_st7789.c	3405	fba5a64365089202cc802abf5e5fe267d2670810d1531be3e7414
external/lvgl/src/drivers/display/st7789/lv_st7789.h	2446	895d19a03d674ed131f4ac7a3ed169809fd4851cd70154da0fb9e
external/lvgl/src/drivers/display/st7796/lv_st7796.c	4173	1205a543091051367c5f53d1bdbf9feb819d11f95128602278380
external/lvgl/src/drivers/display/st7796/lv_st7796.h	2445	ed7354b1587ae2b9a01a58603fe09b81fc51d3ed2d6ec00f4c937
external/lvgl/src/drivers/display/st_ltdc/lv_st_ltdc.c	9956	b29d16172ed77aeedadbbda422546541fc65fecbe84dfdf80d164
external/lvgl/src/drivers/display/st_ltdc/lv_st_ltdc.h	1767	12802eb8fcfc1a8819ce92341607a09f138bdf362bbc20ef621fd
external/lvgl/src/drivers/display/tft_espi/lv_tft_espi.cpp	3127	a4a80f734a5162f03cc2e0800535f5e29e77455547532fbb67586
external/lvgl/src/drivers/display/tft_espi/lv_tft_espi.h	725	fa191eb8b82497016e734c88b0433bc48a94ea9919da7b616402a
external/lvgl/src/drivers/draw/eve/lv_draw_eve_display.c	9131	55c673d5755b3c726ae666b46a7f419ca0a93ffb64362b1cea70c
external/lvgl/src/drivers/draw/eve/lv_draw_eve_display.h	5233	511a0c79d9958a506ddf4d306e15a7186fea9b5c46596244c4909
external/lvgl/src/drivers/draw/eve/lv_draw_eve_display_defines.h	55974	c42b2fe7c2d0b738330af46b5ccc5c99f5cdee9531e687ab1b3f9
external/lvgl/src/drivers/evdev/lv_evdev.c	23497	8d552ad90054450a36594d8a983a8a39e10e70e9cd65059d44709
external/lvgl/src/drivers/evdev/lv_evdev.h	3169	46d2db37aa3d0ab398df9c4de6a374f23a7e4d5825aaa9270f39f
external/lvgl/src/drivers/evdev/lv_evdev_private.h	636	8809d95f2ff47cb8e69cd181d201ce45ff12c336f00684a84f0ae
external/lvgl/src/drivers/libinput/lv_libinput.c	22681	a100da0d6933788839db59a63f5ed608fe82eddead2ac6dc5676d
external/lvgl/src/drivers/libinput/lv_libinput.h	2855	35417de00e0b20dd18a394c136b01f02562b930b81b89483302f3
external/lvgl/src/drivers/libinput/lv_libinput_private.h	1798	a390580971a6eb78df2aae3b0503007e2ce437cec3f4c78dc6cb0
external/lvgl/src/drivers/libinput/lv_xkb.c	4281	7028055834da6aa234ab1578875b7155a0344e42e0140cbbf6730
external/lvgl/src/drivers/libinput/lv_xkb.h	1335	7ffc7ecc19e3808171685ea7d7c9e8721bf120f7f89c83b097718
external/lvgl/src/drivers/libinput/lv_xkb_private.h	814	5addb372832853b95e9275117ddae8b9d480b55852c3d59d76caa
external/lvgl/src/drivers/lv_drivers.h	1976	a62f6cc8e1b85453b003880180643b017608e31af9d1d0f894698
external/lvgl/src/drivers/nuttx/lv_nuttx_cache.c	2948	3bce5b0dfd7dc86915f63286836973bde044e48f933da79b94eaf
external/lvgl/src/drivers/nuttx/lv_nuttx_cache.h	601	f4c649e574422b470c156103805512db23a38c433f3509f13f877
external/lvgl/src/drivers/nuttx/lv_nuttx_entry.c	7237	b9ac4529a667b01f6ce6fd947846cdd8852e806f471a90483560d
external/lvgl/src/drivers/nuttx/lv_nuttx_entry.h	3258	502b299ccc2a9e39882f27595d5db3ed31042bc6f696bd324a94c
external/lvgl/src/drivers/nuttx/lv_nuttx_fbdev.c	14163	10125f69afa288bf2dc6098589852cdef707229a783693203c333
external/lvgl/src/drivers/nuttx/lv_nuttx_fbdev.h	961	548617e25ae1c45417b4dda90698c0f39be1f9056d34672a5cd23
external/lvgl/src/drivers/nuttx/lv_nuttx_image_cache.c	6488	8589d4a5db3fca9e4324ad0187a2917c96956db0ee7a9e0bccc64
external/lvgl/src/drivers/nuttx/lv_nuttx_image_cache.h	729	d7c1b5e6a079afc7f67df996a96fd8d97eef92a9c901b9405afbc
external/lvgl/src/drivers/nuttx/lv_nuttx_lcd.c	6428	128f17043200456b37e20072dfbbfb9ed431ecbcf88610de7bcb7
external/lvgl/src/drivers/nuttx/lv_nuttx_lcd.h	725	2b5b27d1dce048517247b981fa39bdcd58601e8f3eef4d031ccd5

external/lvgl/src/drivers/nuttx/lv_nuttx_libuv.c	9328	83dccc9731828b17768a638acb6f00bc059770bf72444fa8f1546
external/lvgl/src/drivers/nuttx/lv_nuttx_libuv.h	1166	1cbc89bff67c97fa70897c03a2964517f25e8f9c80eb29ef37b04
external/lvgl/src/drivers/nuttx/lv_nuttx_mouse.c	5004	5dc96193b49ae340821d42c0630cd13b8acafb57d22009dcb9647
external/lvgl/src/drivers/nuttx/lv_nuttx_mouse.h	900	c0187ecbf603f9be5c957a82c345b693fa4dab21f0b789b77a99b
external/lvgl/src/drivers/nuttx/lv_nuttx_profiler.c	2783	9424f2ace74632be50d366a9267f5a7a6c5d887f06360e1b4cd26
external/lvgl/src/drivers/nuttx/lv_nuttx_profiler.h	670	617861f393e090eab96bc52fa801ce27f22f31fb52bfa91a65435
external/lvgl/src/drivers/nuttx/lv_nuttx_touchscreen.c	6612	ed4357f0cafad4dc91f9d2bd242d0e7a34ab9f5ec411be9e22639
external/lvgl/src/drivers/nuttx/lv_nuttx_touchscreen.h	942	6cb7ccca66b96c6f6f8928ca9c8bd6462201c5dbf8e1d20d60680
external/lvgl/src/drivers/opengles/assets/lv_opengles_shader.c	8768	c69798030f537370b941aa365bac6ebfccee3ab8a148b5cc2c16e
external/lvgl/src/drivers/opengles/assets/lv_opengles_shader.h	775	b44ba66d8ec357d6beb48926b4bc425610470616b59130d237bc6
external/lvgl/src/drivers/opengles/glad/include/EGL/eglplatform.h	5317	3e9962fb42f7ba918c6ce41b3006465c88d4773107f8865c47989
external/lvgl/src/drivers/opengles/glad/include/KHR/khrplatform.h	11211	e5152137e667069b624d2555bb3ccf18ceb4110e0b6c3809a590c
external/lvgl/src/drivers/opengles/glad/include/glad/egl.h	29149	ad8f200ee64e3e466a8fd4635a3a774e53f4c88e8a4f1f2316507
external/lvgl/src/drivers/opengles/glad/include/glad/gles2.h	52985	899f85d3da31315e17c514c475bb3b645765d6c8d1e8714c74442
external/lvgl/src/drivers/opengles/glad/src/egl.c	17947	aad5ba6461d42961f44099ef843982eb9ca647d138258bec0fe38
external/lvgl/src/drivers/opengles/glad/src/gles2.c	29734	ad4af201d75ee1bc412f6f33a1a07ae46f1b5b8dfc9865ba2e0f7
external/lvgl/src/drivers/opengles/lv_opengles_debug.c	1064	c8b30cd6f46e085c8feea011b412535a17b651eb4e904552bd6c1
external/lvgl/src/drivers/opengles/lv_opengles_debug.h	948	5da13c9a57c223045651f701d048c14849a2d1fe78ad03c487705
external/lvgl/src/drivers/opengles/lv_opengles_driver.c	21813	b38f8686ea80a11cde2fab37a185f03eba8a84ecd1b8a886e1b94
external/lvgl/src/drivers/opengles/lv_opengles_driver.h	2856	2d97a7466f90f3d0ca3dacc15142f8ac06b14d098694b8e260463
external/lvgl/src/drivers/opengles/lv_opengles_egl.c	17366	355370d4c21d37bed7cb7f1d3fab7b4e8aa03d6ceeaa3d88d901c
external/lvgl/src/drivers/opengles/lv_opengles_egl.h	1189	4442c18d56fb4be597f2a18d7d6fa0a406ff3afd944a1391fd602
external/lvgl/src/drivers/opengles/lv_opengles_egl_private.h	2404	effeb9557cfafb13e4e82665986328bce8669328c3a758ec5bae9
external/lvgl/src/drivers/opengles/lv_opengles_glfw.c	22608	108edca42175da074aa4befb79141b884e1bf80b07b136c9864cf
external/lvgl/src/drivers/opengles/lv_opengles_glfw.h	2686	650bdb9a1756effe450e61ca87274c50e516bf123b1bf97730a53
external/lvgl/src/drivers/opengles/lv_opengles_private.h	2330	16ece7520705eb67c2a6889a6a4170e30ee2cdc7b742a2dd8640e

external/lvgl/src/drivers/opengles/lv_opengles_texture.c	8630	3e24a78e3edd9f2090e8446c9acec86757b7471ee2598f4ebe5c4
external/lvgl/src/drivers/opengles/lv_opengles_texture.h	2019	c53c603cac4fe5d586a058a3bf8d953100fad6b07d75c6c25840b
external/lvgl/src/drivers/opengles/lv_opengles_texture_private.h	1147	a99e002e3cb636573383c906e15aa0063ba14fa475e4336af28d0
external/lvgl/src/drivers/opengles/lv_opengles_window.h	3105	4e754aa023140cfe72dffc36967df37b58a00faa91db03979298e
external/lvgl/src/drivers/opengles/opengl_shader/lv_opengl_shader_internal.h	3911	71bc2f5d7b47592f762e6d4739cd9d16babc57a0188d5d570cc85
external/lvgl/src/drivers/opengles/opengl_shader/lv_opengl_shader_manager.c	21813	8c70e7393b904e16edbf3436461f5f605af202f3011fe65b9975a
external/lvgl/src/drivers/opengles/opengl_shader/lv_opengl_shader_program.c	3309	8a660df9bb3b587db178505941256ed6278d466c76811833a9de5
external/lvgl/src/drivers/qnx/lv_qnx.c	15688	c56a8d45841ab43202ad2d25e4138fa69fcdd4b51baebb2d3d9cd
external/lvgl/src/drivers/qnx/lv_qnx.h	2104	f6c1bcefce8be2e657197f1a36983d823b90db23839b96f54c9ed
external/lvgl/src/drivers/sdl/lv_sdl_keyboard.c	5981	8506876cdafb7841c897f3368e77d9a62171acc5a2f8e13f9537c
external/lvgl/src/drivers/sdl/lv_sdl_keyboard.h	718	53fa5ef7779bca17c50de2cb9badd964bf9146fda373655f53fce
external/lvgl/src/drivers/sdl/lv_sdl_mouse.c	6167	640cc17d3e0ad42300d428f0babe6c6a8ef592147298bab462eae
external/lvgl/src/drivers/sdl/lv_sdl_mouse.h	635	a6bb44241ba8de6768c6d59e3afa87787c4a2c53b84332ce47caa
external/lvgl/src/drivers/sdl/lv_sdl_mousewheel.c	3711	af4ee511aa4ed7261bd8718ec6b3405648174ccd048800f3f690a
external/lvgl/src/drivers/sdl/lv_sdl_mousewheel.h	720	90dcee8eb66a8ea9f8c3bdaba04c068889857efef1963f6b30b3c
external/lvgl/src/drivers/sdl/lv_sdl_private.h	847	be020917737a650ddd16505414b9026adafe4dea383ca106ebb07
external/lvgl/src/drivers/sdl/lv_sdl_window.c	17903	77336ce61c14defdd81bc122be81258ec1796ab1cc73558af7f69
external/lvgl/src/drivers/sdl/lv_sdl_window.h	1599	22d07e4ed15992a11f39b56bd4a5f0bd510e8b86aabffed842d30
external/lvgl/src/drivers/uefi/lv_uefi.h	4029	ae55b5c18afdb934e5fbdb4f57894e2a629a05f9006844d13de26
external/lvgl/src/drivers/uefi/lv_uefi_context.c	1961	99ff738925e23a907ea30a5813d5d4eaa596e48f712d48ff3a881
external/lvgl/src/drivers/uefi/lv_uefi_context.h	1458	c03d161b4d017e1a5ed095a20e801c35f397c5d48c17c47b44f5f
external/lvgl/src/drivers/uefi/lv_uefi_display.c	8253	675ec8cd4c18dbd5e0102dd1426805bcb0fb7a80ac9de1ad38cc0
external/lvgl/src/drivers/uefi/lv_uefi_display.h	1253	5bea6ad59c2c76aa6b15b5fe575ba78ce53abd1067d12ca62e16d
external/lvgl/src/drivers/uefi/lv_uefi_edk2.h	585	b2e8bba8db7952e9da2c9d37a413a93f6eea2b8c7544e96bcd27d
external/lvgl/src/drivers/uefi/lv_uefi_gnu_efi.h	187	cfdcbcf9b3d4647923cbd99d995cd4c58690a7d0d7c60662e801b
external/lvgl/src/drivers/uefi/lv_uefi_indev.h	3250	b22bfc4e8b44dcc94a329ddd4c792a1ca492c15590e6412a6be51
external/lvgl/src/drivers/uefi/lv_uefi_indev_keyboard.c	11112	db9a8124a7b8cd2fe89a65eb2214ee1b66fa9f32ebfb2d7aa9184
external/lvgl/src/drivers/uefi/lv_uefi_indev_pointer.c	9752	20793168b305d82655a41ef2ae5e77738ffbc0459fd2a4f8b3be5
external/lvgl/src/drivers/uefi/lv_uefi_indev_touch.c	10263	c45b234a33cbda277a65df633a2a3e9ba480fcb9bb6923655b078

external/lvgl/src/drivers/uefi/lv_uefi_private.c	6328	643ad6c9d8c872c39d51f6de66a29fd30580d48e87efc2ca77521
external/lvgl/src/drivers/uefi/lv_uefi_private.h	2774	1a2fdabcd6aab35d1e2e5b19b8f1a9955d35018a80f517cff5ae2
external/lvgl/src/drivers/uefi/lv_uefi_std_wrapper.h	3297	0d94750752e94f6b6aee5b40d7c270549cf50538ba5eff7d17573
external/lvgl/src/drivers/wayland/lv_wayland.c	17896	ee8391509bbbc386cafdb30b9906bf653d063440942337fe7bb22
external/lvgl/src/drivers/wayland/lv_wayland.h	1086	c46975afae0bb65cf6d70a74fe9591412254f7e253878e9d7b121
external/lvgl/src/drivers/wayland/lv_wayland_private.h	12423	fc6c35a43c7e171069b6033dab42f3de7bd4d44ae320502e863b4
external/lvgl/src/drivers/wayland/lv_wayland_smm.c	18155	4d9b808d6ac6a24ef9e4eae934c50bc81c6d82b6aa3a972431a99
external/lvgl/src/drivers/wayland/lv_wayland_smm.h	2397	7a0312b54eed81e77855ccf1f9305427bc300511c62ba7cb8026d
external/lvgl/src/drivers/wayland/lv_wl_cache.c	3705	bb40a30337bbefef4cf5b94ed750faf23a1ec356f572c2a5f2d7f
external/lvgl/src/drivers/wayland/lv_wl_dmabuf.c	25480	344cf05068d5e1deef15acebbe448c94c41487461788b339c3a98
external/lvgl/src/drivers/wayland/lv_wl_keyboard.c	8423	3843c94a3380f15ecb5f6b29a13e1bfae907a4b05f724b4e4d31c
external/lvgl/src/drivers/wayland/lv_wl_keyboard.h	924	546b3152f1624e4a1065af0dc2f82af2bee00237d0ad5a2182fd1
external/lvgl/src/drivers/wayland/lv_wl_pointer.c	7125	f869c71cd4b2bf8ab5cae37e8b6a9a3b31f940af4b68a3c4f238d
external/lvgl/src/drivers/wayland/lv_wl_pointer.h	982	6909f1ad66c6449caa9ba28b1365bbc98c0653ff3348bcb5afbf9
external/lvgl/src/drivers/wayland/lv_wl_pointer_axis.c	1614	4a994f1379373c354618936bd59cd336172b770ca85e803ce2c3c
external/lvgl/src/drivers/wayland/lv_wl_pointer_axis.h	1011	847981e6f3bfb6c784f58f8ea0da5ae855f53efcf7820d22594df
external/lvgl/src/drivers/wayland/lv_wl_seat.c	2674	bcd2e440dba8e5ac30c7853b0113e6a150f7cd08c940568c1f27b
external/lvgl/src/drivers/wayland/lv_wl_shm.c	15976	7c3cebf9188a8fd64784649cfe59ddb3a4d167eb770a5523aa4ef
external/lvgl/src/drivers/wayland/lv_wl_touch.c	8616	e1a7eaa6e6ad07a5d32258ee3455d9ada20b0cb9e4d53404697b8
external/lvgl/src/drivers/wayland/lv_wl_touch.h	920	382ee96fc2e0f0bacdf0d81f2a0b4191cc585828c5b5f86b559ec
external/lvgl/src/drivers/wayland/lv_wl_window.c	14867	c8ea19b4ad0b9a993b4f5ff8ff14e3ac723c084ae1e52364f2696
external/lvgl/src/drivers/wayland/lv_wl_window.h	2498	4fb171c14b33b270f38dcc1eff5e66e33f5481c6c45411fbad370
external/lvgl/src/drivers/wayland/lv_wl_window_decorations.c	13372	61f6b2f5679af7fe53f169dd207bae4b6ad37512c4272230a7255
external/lvgl/src/drivers/wayland/lv_wl_xdg_shell.c	16240	17543f3ec374fdbb95223f0320d5f9e0dd3ea1c950df6725c5110
external/lvgl/src/drivers/windows/lv_windows_context.c	24748	fcc680da7d61a8b9be18c74f0db41637b63d6bbb574a5c2bd01c5
external/lvgl/src/drivers/windows/lv_windows_context.h	3378	c7d0c06b516cc27951c5baeaedc4d754bc944f3745031f2d1b52f
external/lvgl/src/drivers/windows/lv_windows_display.c	4469	39266b93622cb84824a6bc3faa45afaa5c8b76a75cb9adae59a9c
external/lvgl/src/drivers/windows/lv_windows_display.h	4082	17bc028841450212864fc32d4a43e6dd3006183b6f1d9bd252f44
external/lvgl/src/drivers/windows/lv_windows_input.c	28018	83a8b18c7fe8af8025a612639d4d7af0fbcdb56d431fa2409ab78
external/lvgl/src/drivers/windows/lv_windows_input.h	2212	32eda79c6b62e5107ea7033678993985d256b871589a4781254e7

external/lvgl/src/drivers/windows/lv_windows_input_private.h	1108	2756efae19fb95a24537c49c69ac68ce240403230aeed8f4d452b
external/lvgl/src/drivers/x11/lv_x11.h	2378	ee421aac355a9fe8504cecd8adaaa4719741c3861b79a7e7bb48b
external/lvgl/src/drivers/x11/lv_x11_display.c	14647	ad352b905d1eca667a0ad4caba4fb8ee914caa26eb2d2e15c7013
external/lvgl/src/drivers/x11/lv_x11_input.c	11665	c9f7ce3a8ce0a32f69dd20c4cd562d69d06c5ff2e071949bfad3f
external/lvgl/src/font/lv_binfont_loader.c	22180	8ecd5e70445313e4b810d81db5b860765b43e85f29e173b73391b
external/lvgl/src/font/lv_binfont_loader.h	1692	ae529c7706a7a53d9652b0380c9d54558ba2f50e51cc75c43c12a
external/lvgl/src/font/lv_font.c	4958	64e6b5b180a2986a1f1b82a2884a89114165604be0588ebff7b74
external/lvgl/src/font/lv_font.h	11377	f8628984c4f5f94a6ce6af1184f1e3e41bff567377b366619cccb
external/lvgl/src/font/lv_font_dejavu_16_persian_hebrew.c	287976	8599edbcdd1d2facbade0f07d09c6ab8bc82e8944bafc3668cd72
external/lvgl/src/font/lv_font_fmt_txt.c	22317	b462bdd8430e37dea15e69d4173b75f12b57b4e2f13e857d432f5
external/lvgl/src/font/lv_font_fmt_txt.h	8342	06cc5c66d872d7349c3449b2625a99d45e161ea88ffd75f2d9d4c
external/lvgl/src/font/lv_font_fmt_txt_private.h	921	509384e80e9a679c82cc533b895594e02d1da9461faca3b89a908
external/lvgl/src/font/lv_font_montserrat_10.c	63143	afc14fefad31c0fe072fd6d3846c89df1277b77713d458e8b3e4c
external/lvgl/src/font/lv_font_montserrat_12.c	76198	5be9c2a8878c694e8abd55643352023a7071e5bac474423337ed2
external/lvgl/src/font/lv_font_montserrat_14.c	89473	a050906a9d05b51d4678b8beecf41bdb353d161729c34fe9c9308
external/lvgl/src/font/lv_font_montserrat_14_aligned.c	183182	9b576bd9452ff8ce78a1f9f2d6421a2a9386606f4ff681e8501fd
external/lvgl/src/font/lv_font_montserrat_16.c	103154	bfe04c1059a2cb310c8c7a69e72d52e720f743e26065fd0ddc456
external/lvgl/src/font/lv_font_montserrat_18.c	122375	cc3b22b9d8c7689d4250e319136d26f4808ed59da1ca753f06305
external/lvgl/src/font/lv_font_montserrat_20.c	139125	d8b67ea4c76ae2c71bf8203521de9f74fe17ac01fd6eb7fff4918
external/lvgl/src/font/lv_font_montserrat_22.c	161216	43c3c6d4b39cb1884dbacec21c4cfee820466e39523acf082f912
external/lvgl/src/font/lv_font_montserrat_24.c	181037	530fb723e9f225b8d203f359fdb70ec59ef2e1a043a6bc6ebaad0
external/lvgl/src/font/lv_font_montserrat_26.c	206668	6a0ab0686efc751cc41c0df15e77cff137c4f515d1674b65943d7
external/lvgl/src/font/lv_font_montserrat_28.c	233057	779d47f0eb18c2563fb8bffe50dd9f797fcebced2abfe7edb9c96
external/lvgl/src/font/lv_font_montserrat_28_compressed.c	145127	6e104239c45fb0ddbcfb1fabd8125c1cfd33ea582c85588a7fb98
external/lvgl/src/font/lv_font_montserrat_30.c	261203	d9dfc1bf226beb682eef29b3f0d72f69d36395df5258b2834a13d
external/lvgl/src/font/lv_font_montserrat_32.c	284811	0f0b77e61b63d712b728e523d9139dcee810794055773d8699d97
external/lvgl/src/font/lv_font_montserrat_34.c	323533	fce37859b82016ace4d1482319d31a227824b81779e3696befb80
external/lvgl/src/font/lv_font_montserrat_36.c	354598	2388d7a652aa52a479c6410bd0ebdf31e2e6d0f8be58f61ecf5b4
external/lvgl/src/font/lv_font_montserrat_38.c	390343	823f2450ddca033e38addd2a08dd9cf2f01e840ab1a0a91744439
external/lvgl/src/font/lv_font_montserrat_40.c	431710	0e13032310141859ec8c337358488d85af560e96d0dc18c81138e
external/lvgl/src/font/lv_font_montserrat_42.c	472438	b4fe50b53c59078d46db3315afee59559e629ef6fbb52c6cc8001

external/lvgl/src/font/lv_font_montserrat_44.c	511317	bb5b7ef84e4878ce37c64f3149afa956ee05b5ef284a89ceda33f
external/lvgl/src/font/lv_font_montserrat_46.c	557657	75e6da97e36ef8d7c3c17d6e0b7b4f880265ef408bcb279521bf6
external/lvgl/src/font/lv_font_montserrat_48.c	591315	e68fa0e706d2a18d66dfbe536d9fab83068145159b40091fd261e
external/lvgl/src/font/lv_font_montserrat_8.c	52949	334eb715262c7d7df7b91a610a29afcc359f1a3c206b388cbf529
external/lvgl/src/font/lv_font_source_han_sans_sc_14_cjk.c	1020007	b7491cbd00936487ae1cf08c914ca202e64c0604d3b7c0c0aec56
external/lvgl/src/font/lv_font_source_han_sans_sc_16_cjk.c	1220369	4caaa2aff00bb59d1155227c838ced106ab4d61717afcaa66328c
external/lvgl/src/font/lv_font_unscii_16.c	24392	e794d86db88db9a1a9216ac6a564dece6f78e71eca6915f122bca
external/lvgl/src/font/lv_font_unscii_8.c	16540	fc22dae14096a1a6f9365dbcd646c62216cc693cb89c77e22d66a
external/lvgl/src/font/lv_symbol_def.h	9423	67884ce07db66697d54f518d04e34dd7d521f7235b7196ed5f828
external/lvgl/src/indev/lv_indev.c	64893	94d9d28d51946127d897f5128496fdc899c5e2ffbcfb09db35a63
external/lvgl/src/indev/lv_indev.h	14804	314b10ca15b42378b9ce520cac4d6f156e56327ec95534b6db646
external/lvgl/src/indev/lv_indev_gesture.c	32938	d075b5b878eda78d9c518e2915cc061db0e99aac4f2b9e5dcdc5c
external/lvgl/src/indev/lv_indev_gesture.h	9910	49fb7df618c84affb253f480b8a076efcf1fa03850d3357adfae2
external/lvgl/src/indev/lv_indev_gesture_private.h	3030	1fff50b148f7024fdf06e7762ed867c9730aff1dcaa6ec214ca9c
external/lvgl/src/indev/lv_indev_private.h	4853	ddf3812102d6b4d00f8c6770a4cf14911b469ca568cf0d1640d16
external/lvgl/src/indev/lv_indev_scroll.c	31063	8502fb1b70de2708e54b2f373a1efa8eafc948e37d74ba436737c
external/lvgl/src/indev/lv_indev_scroll.h	1513	278f8447c6dc5823a19ee63b8e1459454c091e31b7e22231f0edf
external/lvgl/src/layouts/flex/lv_flex.c	21643	2b5ac5ca32dc015dd3d760aa3a2aa33e50b7f148c46f09d9daa73
external/lvgl/src/layouts/flex/lv_flex.h	3018	e20e9d294f2f91f47dae2a1d176b0f00deb17c40fe2b39df8b4e1
external/lvgl/src/layouts/grid/lv_grid.c	22539	2d9131ad6ad481756dcef1bbecdd8f326b19ae3203a3447fcc2e8
external/lvgl/src/layouts/grid/lv_grid.h	2462	d8676b7913973640068eeb387c651264017f6fb0465ca85418575
external/lvgl/src/layouts/lv_layout.c	1705	d083c1b1f92085aad955a8a1f292c57e00c37ff39ef4a68b8af71
external/lvgl/src/layouts/lv_layout.h	1133	6414fb1124d73dd912bc9120916a6fdfb6c49c0cd1d0012382a35
external/lvgl/src/layouts/lv_layout_private.h	903	a77ddd34114a69c2d4c725ffc7a03c08a25a09086b862742dcc71
external/lvgl/src/libs/FT800-FT813/EVE.h	53412	bd4991bc28dde8f1153d4bc1ed324dd70d56e94f5007556652a7e
external/lvgl/src/libs/FT800-FT813/EVE_commands.c	134988	bf57833c09abd0509e0e138dbc2b4a1e22c67f35d693316784e1f
external/lvgl/src/libs/FT800-FT813/EVE_commands.h	18573	f875ed379fa26a001921b805fd41e28dbed0bf107a7012a19c6e2
external/lvgl/src/libs/FT800-FT813/EVE_config.h	1335	dd8004dc7cf5cdcb4dbb29585c0a415b7127484ea3c5db931ed64
external/lvgl/src/libs/FT800-FT813/EVE_supplemental.c	5405	48490d7271723a65a4de0122296da6a840e3e69d3e83ee37ed2d5
external/lvgl/src/libs/FT800-FT813/EVE_suppplemental.h	1786	2db47c4599bf7dafb56fda062813bd0406ce5419ec50aa8526d7b

external/lvgl/src/libs/FT800-FT813/EVE_target.h	3235	c42d6aafbde5a56fe7687332f0f6450c9e614b1caceac0cea1d83
external/lvgl/src/libs/barcode/code128.c	18957	c7ce6a8685ea560b5896f27f2a6764110639b03d968ae710dc4f6
external/lvgl/src/libs/barcode/code128.h	1926	cbbc24883857d1191321863b0dc55bba797d524992c41a1390c47
external/lvgl/src/libs/barcode/lv_barcode.c	8905	a13c086d4fe4ed12398143d059321ac322b21ce2513429ac5b62a
external/lvgl/src/libs/barcode/lv_barcode.h	3442	578bb41368cbeaeb9423d940c7643556367371a58cb57fd0238a5
external/lvgl/src/libs/barcode/lv_barcode_private.h	898	6dff8ae531c7945711a71d4e77e9b4aa4867eee691d9ed95f38b0
external/lvgl/src/libs/bin_decoder/lv_bin_decoder.c	42258	39416e4f018263a5f231cd7065fef4a849c15e5b9b56b33eadfe0
external/lvgl/src/libs/bin_decoder/lv_bin_decoder.h	1990	60588913df80866205d836c5592fc9f427c800a4cccf65683f6e7
external/lvgl/src/libs/bmp/lv_bmp.c	7780	52f839a5b1a4ed59947572266237f43427b3fe86601fabcb4e23b
external/lvgl/src/libs/bmp/lv_bmp.h	627	f33aa945052d97abc55b810b41d3adcdf59ede566fdcaac7784fb
external/lvgl/src/libs/expat/ascii.h	3760	ffdf2fb7f4bacbbdbcd5846981e609b6a542c8bd33c6bda646284
external/lvgl/src/libs/expat/asciitab.h	3598	840e5f0e715572c377be35e365d7363cae3568de2d22d56c5c9ea
external/lvgl/src/libs/expat/expat.h	44148	5cd870fd3aa0931ba249e12ececb8ef8358e75f00008c8be39702
external/lvgl/src/libs/expat/expat_config.h	4332	c2dac34784ad0ec553e1a6df74782fd164446e39d17436b06f141
external/lvgl/src/libs/expat/expat_external.h	6107	592a84df04a167b2bf87b46cd7fa201e02975e618312775fa50d1
external/lvgl/src/libs/expat/iasciitab.h	3668	1ac587935add708e8a0a6aaccdfb2935c08e562d21f179c5d9552
external/lvgl/src/libs/expat/internal.h	6421	795df2a1fd5310f91abfc55bf3e58bd0686779a2faf2a66ec1bd9
external/lvgl/src/libs/expat/latin1tab.h	3651	1e077d96b08ac8ef7d068e5501038c4c1f017f76aabb3df8e245a
external/lvgl/src/libs/expat/nametab.h	9122	95b7eba4ea83e78b777e38c5bdb3b3c12e6ad1dea2b3c6364271b
external/lvgl/src/libs/expat/siphash.h	13092	8f5fd5fe98e3da3732ed79605c32f60df7de79ba14cf68065f80c
external/lvgl/src/libs/expat/utf8tab.h	3600	1051bdd95827cc5f1da02853a13ea982e186f9cc38cc97866de36
external/lvgl/src/libs/expat/winconfig.h	2116	fa36cfe1bfba4e3a6ce48b18948fb2948c2ebe0f1892cefcc72a8
external/lvgl/src/libs/expat/xmlparse.c	287077	68ad04fe38876258a1b742a9294f3fed80a7cc56961fd2fe43860
external/lvgl/src/libs/expat/xmlrole.c	35553	c0dc74774865d8e4b10dbfb7dcc39fa801c1b98bd214ad9b8cc2f
external/lvgl/src/libs/expat/xmlrole.h	4757	c7082469eacbad69a1e59486639993e68900a66f3bb632675b122
external/lvgl/src/libs/expat/xmltok.c	54424	efa3e7c677ce15a7187dd5eb40a2f88f4c4e7c13993f5d5047237
external/lvgl/src/libs/expat/xmltok.h	12972	84ea927045a031d00f2080692f87ae2bc5d2921b8c809d5feb912
external/lvgl/src/libs/expat/xmltok_impl.c	53798	5f94aa327e58905de8cf1fbb76816c4cff0004773307d72b8e238
external/lvgl/src/libs/expat/xmltok_impl.h	3508	0a3d8b35138c2ecacbc451f24c36a2b45872074e96b6f6769e302
external/lvgl/src/libs/expat/xmltok_ns.c	4718	23fa4f71b5896564741d8ac331f753e53a4da2244fa8190a4da9a
external/lvgl/src/libs/ffmpeg/lv_ffmpeg.c	30329	0e355c2ad815d27e7691b654056985f7418bfebdfe8b0b8ea2386
external/lvgl/src/libs/ffmpeg/lv_ffmpeg.h	2252	4cee83fb5de1893fc00c7be5b5f0b2ad3677700b8007c03b07da5
external/lvgl/src/libs/ffmpeg/lv_ffmpeg_private.h	837	87d46278faf0d56ccee57428e1f4addb0e0f56a55c9fcd4e4b891

external/lvgl/src/libs/freetype/ftmodule.h	1533	c8d5ec90afe9db74bde516e03f122b1e236adfd8d5b23feb321a1
external/lvgl/src/libs/freetype/ftoption.h	38678	63aae93dcdcc40c402e52d6f0f3dac548c6e906ebc13312975f61
external/lvgl/src/libs/freetype/lv_freetype.c	14849	60cca5c319906e2be30ee42873a0b4bc9c2683379e02b9a43f929
external/lvgl/src/libs/freetype/lv_freetype.h	3968	e7e7df0a56311f1b4b238a012817efa14e56e0030bbd5995e5a5b
external/lvgl/src/libs/freetype/lv_freetype_glyph.c	8329	51da7c19c32f856e5f3271c17db1519804fb6200e6b3d96ba0be8
external/lvgl/src/libs/freetype/lv_freetype_image.c	6994	4f2bd8e157a0ae2970be3a478bb80b36f86d63bb805da74f81c9a
external/lvgl/src/libs/freetype/lv_freetype_outline.c	14465	3f5bf86d89f8d68bfb5d9d157190a446f68b92f284f4d8a508936
external/lvgl/src/libs/freetype/lv_freetype_private.h	4260	5188cdfd9f1a7d9a230a47b7d8125017771793e348c9b64cb2dd2
external/lvgl/src/libs/freetype/lv_ftsystem.c	7138	a153a7f684956da3cb0fc31607d9cbb8b23ccaca2c3586a4aa6ca
external/lvgl/src/libs/frogfs/include/frogfs/frogfs.h	7628	1536945f54d175415b39d2c750beb07dd4f3063828c829390ff34
external/lvgl/src/libs/frogfs/include/frogfs/frogfs_types.h	110	3f4fc3ebc763c3bb40df3a971f5584a71d8c5a94a584f8c434c10
external/lvgl/src/libs/frogfs/src/decomp_raw.c	1934	9f1755b0b65832e73445c76fa0020845206d62151cc0d56271c46
external/lvgl/src/libs/frogfs/src/frogfs.c	10758	00878fbbce108ba6f728b3f0627c741783a800ed91999e02c2726
external/lvgl/src/libs/frogfs/src/frogfs_format.h	2174	aa20c60989450b712a0ae1c7df94f7f75a5b5434767d7d435e591
external/lvgl/src/libs/frogfs/src/frogfs_priv.h	2076	1d813cc2932c575d145e202a16385190d0e5a0e7f9af230d723b1
external/lvgl/src/libs/fsdrv/lv_fs_arduino_esp_littlefs.cpp	6375	f4388838820db298b20fc0d4567608f9b16c4c6e0bab17dd9348c
external/lvgl/src/libs/fsdrv/lv_fs_arduino_sd.cpp	5910	d3e114fbb956dc1bcd0471081f9c78aa790f6d3b1467c6cec6e0a
external/lvgl/src/libs/fsdrv/lv_fs_cbfs.c	192	d5c0b97cee156e0097a22635397a79978146eb6e11e61a69b29cc
external/lvgl/src/libs/fsdrv/lv_fs_fatfs.c	9244	cedb11239c009ca4f3e555b0412fa10ecdb975c2f6acad1348af3
external/lvgl/src/libs/fsdrv/lv_fs_frogfs.c	8651	1bd50de532a6cad4bef9c1537a2cd322be24ed0abd9d52c7cb01e
external/lvgl/src/libs/fsdrv/lv_fs_littlefs.c	8865	dc603f5fe9d89acf83991668827051083cc292d321a6487292645
external/lvgl/src/libs/fsdrv/lv_fs_memfs.c	6673	663f9e6dd26fccc62f10de636de52d54e4ad4424cf08430ab6bc0
external/lvgl/src/libs/fsdrv/lv_fs_posix.c	11328	3e9747910b815393ff7b3a6ed98d4b7ca8d8ab22cb9bc1f1f1c81
external/lvgl/src/libs/fsdrv/lv_fs_stdio.c	10773	7a5c2a5052fbde6da0d7426a3b905735e378ee4d0b2804132f72d
external/lvgl/src/libs/fsdrv/lv_fs_uefi.c	17496	cceea4b6934e7b3502fb94af27f836cea76398422733a5f7c056a
external/lvgl/src/libs/fsdrv/lv_fs_win32.c	14376	a1a86edc1f5def81b28c3cd9caffc5ebe1eaad4caba6daca0f174
external/lvgl/src/libs/fsdrv/lv_fsdrv.h	2132	85267a9054143c5c3ad2b2d9534ac07cbf71881da79abf777cc69
external/lvgl/src/libs/gif/AnimatedGIF/src/AnimatedGIF.h	9543	16cfb600821d2a7e8762fa064e363633158f86929321696f1e8e9
external/lvgl/src/libs/gif/AnimatedGIF/src/gif.c	62992	b0c66e3559f4b3c3416c0cdbcbcd6af99deb455ed90a5b89b09f1
external/lvgl/src/libs/gif/lv_gif.c	8078	89db4162c3d4bd2681550ee0b8a358907247e311cc42a31eb4e02
external/lvgl/src/libs/gif/lv_gif.h	2455	d7bd5193b6c9ff515c18782f59fc5d1aab1ced01202af32537685

external/lvgl/src/libs/gltf/fastgltf/lv_fastgltf.hpp	4358	9ee6254bfd5869598b68955f7cbaee0eff417ab9e8de1e57173b5
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_bind.cpp	5277	0708b759f2937b8a9e52ac03fb85adeb1a8019821ca8c87817039
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_bind.h	3847	d1041b1fd0ffa148087bc351b91c5210fc1c2383ec95f6c752316
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data.cpp	7306	738af4f9dc9a262765d0027c9df5e507239e0d730d5d3dd52f0f5
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_animations.cpp	9866	e9e70115083da7afad289966d58ae345f7c92802162b6f6839353
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_cache.cpp	3146	09203fc2c5ef8a66508c612a01c93ec14fca77f36909ae3e90036
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_injest.cpp	38075	af217b63abab12a0e5aa7ea66823a11d512d6e663a7531fb5770f
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_internal.h	7411	40aa3ac0482df6e26d5d2732f814c2b8c2bd20d9b8f5fd1516a77
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_internal.hpp	14842	a41d361aafcaa6d16813f1bc24b5b12f1c0005f364260a7388aca
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_mesh.cpp	972	2d603c6d8a33e2818e402bf8d936cecb0d39c18fd352224a27a32
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_node.cpp	2507	ec8d5afd076591afba7ba182fe487377fc185763fdbfd13a56070
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_primitive.cpp	3784	a6bc7efa733e4e3119b701e8df32b8789fcae53a1d438571f62e9
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_shader.cpp	1456	e84278585c75612a2b782790a16b58b2d9535b1c0209d04323d02
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_skin.cpp	1398	41cb7f147d5e15662d8f5417e0fca32e34f5f1f609409826022c6
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_data_texture.cpp	2063	2b20dcaf815ce2aa1ffb97b84a28d1b2311644d697b8081468d24
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_model.h	4586	3e5f350939dca4facde29b419dd06d72605a65ec156783c491a35
external/lvgl/src/libs/gltf/gltf_data/lv_gltf_uniform_locations.cpp	9603	4ed3971f0ed840161430a0fc5b89ca7730c0740c759d0faf62aa5
external/lvgl/src/libs/gltf/gltf_view/assets/chromatic.c	557855	2bd0f0553b2a96be5d0ae54ab9ccfb9119dba0e87b9c0c47a70d7
external/lvgl/src/libs/gltf/gltf_view/assets/lv_gltf_view_shader.c	118827	b7a5094ab552f184dc3275d9b8e1fc5f76acdd2176ce03e9dc8bc
external/lvgl/src/libs/gltf/gltf_view/assets/lv_gltf_view_shader.h	780	38a98d33a2fba8d0a354bcb931502f2678f7ba37c0554a4a5c142
external/lvgl/src/libs/gltf/gltf_view/ibl/lv_gltf_ibl_sampler.c	25041	52339f357a11111ecba219bd433cb63ef1a5b9d4b80d026a8733b
external/lvgl/src/libs/gltf/gltf_view/ibl/lv_gltf_ibl_sampler.h	1964	8ddfdcc743f8185a40e6bbd462b49fef11e449dcedc79ff98b1f2
external/lvgl/src/libs/gltf/gltf_view/lv_gltf.h	9625	6e3bcb8bd3aad0fc6ff958a719a8b9cec12482792c6265ee46135

external/lvgl/src/libs/gltf/gltf_view/lv_gltf_view.cpp	21537	df3f99796774ed479e41795f4f262a9c36c8297fee4fb2b6bc556
external/lvgl/src/libs/gltf/gltf_view/lv_gltf_view_internal.h	3782	e91e0accc449dbe2220d43716ee1142012e99f5835296c90e3195
external/lvgl/src/libs/gltf/gltf_view/lv_gltf_view_render.cpp	62146	1c69e645f754faf3f2e682c9891c85322d56db8b0e497b6ad9c54
external/lvgl/src/libs/gltf/gltf_view/lv_gltf_view_shader.cpp	17453	31b03114051e74d7ba377c15fcb5428bf75927ac05c717f127225
external/lvgl/src/libs/gltf/math/lv_gltf_math.cpp	2025	871294c14d234ec44db0fc08dad80835ccac867ee3e68121308f8
external/lvgl/src/libs/gltf/math/lv_gltf_math.hpp	2399	838f98e47f3f428b8d0a4016fa8661383072816a65a7f9051e4d7
external/lvgl/src/libs/gltf/stb_image/stb_image.h	303477	d81d95287ae188e472fe66ddcad8635041e8f29d51a799cc77c65
external/lvgl/src/libs/gstreamer/lv_gstreamer.c	20893	3f65c28ff411cf0ba0f0855f2af83afda1398da617ed05e4539a6
external/lvgl/src/libs/gstreamer/lv_gstreamer.h	5789	3e8f33004b0731521ca947edd3d3a1642ef3466b020959424269a
external/lvgl/src/libs/gstreamer/lv_gstreamer_internal.h	1329	ac63d83ff7c18b145aaaa52e43c3689c0965d25204efa9a2038f7
external/lvgl/src/libs/libjpeg_turbo/lv_libjpeg_turbo.c	20209	37ae6afdbbe1b84009b5353ab08a6ca5f3b387e49c0ae500159c5
external/lvgl/src/libs/libjpeg_turbo/lv_libjpeg_turbo.h	770	dda5ec8e9b10055a313d0755c26d85f405e3c16937ef16b06542e
external/lvgl/src/libs/libpng/lv_libpng.c	9807	f500b381fb5c6877c6fc3fe9cc7057d6d8e1270e834ac4918f92a
external/lvgl/src/libs/libpng/lv_libpng.h	707	f70a2a12727f3ef27521527bf4fef42b4e1526bf772932d86bbd6
external/lvgl/src/libs/lodepng/lodepng.c	328422	d8356d63aaf018f50eefea3153c37c969cada4490761736afb3a5
external/lvgl/src/libs/lodepng/lodepng.h	104022	9c5db08ee60e791b0d74e1a1d2f5e4fe9310d5eace26471970685
external/lvgl/src/libs/lodepng/lv_lodepng.c	8826	845fd3b2529cf2040850e1c74b7bd7f1f30a7c0856530a0c19345
external/lvgl/src/libs/lodepng/lv_lodepng.h	715	6595eb11cdc6ae9f22dbee944e19aebca9660e600cd20116da692
external/lvgl/src/libs/lz4/lz4.c	118248	dde3a9c6bbef76d82040fc1baa66debe539472cbd7c532108aa44
external/lvgl/src/libs/lz4/lz4.h	46282	7797c082c6b10de6aa2ef0d0671c205c4067db848612fd39fabc5
external/lvgl/src/libs/qrcode/lv_qrcode.c	8097	34677f81ee2af29e8e3df7a301aeeb9068c7b64b57fa46c633653
external/lvgl/src/libs/qrcode/lv_qrcode.h	2403	42ef852423757d5c90d449fe8ac74ba9b313480a5cc3ce8e2dbe3
external/lvgl/src/libs/qrcode/lv_qrcode_private.h	817	9e2fd66f3c30b9c85536a86f1e37beeaeeefe979e56a0e4f2312b
external/lvgl/src/libs/qrcode/qrcodegen.c	45011	53158b9b2dce1ff635e0279ddae975b07fa8d1c57d4ec2445abe1
external/lvgl/src/libs/qrcode/qrcodegen.h	14758	2acb64b7c97e947726bab18f7fd1273a26f4435f90484ecd5cc9d
external/lvgl/src/libs/rle/lv_rle.c	2839	78a35fa1f874f09d3cff77a4d8be3d0a4001dd2d083f064bc46f3
external/lvgl/src/libs/rle/lv_rle.h	771	27ec8477769453a02191aebcc7e7dac2bd594cccf3824ae56ff22
external/lvgl/src/libs/rlottie/lv_rlottie.c	7400	d87a78f044b321cb0c643930850538a4936ef2ade43bbd5045eae
external/lvgl/src/libs/rlottie/lv_rlottie.h	1346	f26327195f2b8187641ba5b8a1295fb5702ec76343d6886ff8568
external/lvgl/src/libs/rlottie/lv_rlottie_private.h	1106	b986fff3bf79804ba9a63dacb5a1af4e92cb09a1d7c1a62f36efc

external/lvgl/src/libs/svg/lv_svg.c	3130	43cf51298d750f76cbd3ad237d8017d650a2e0bd02c294b90463a
external/lvgl/src/libs/svg/lv_svg.h	8021	1f40e9111068dca437f4509bf54223f15cd3cb0992991fce337bc
external/lvgl/src/libs/svg/lv_svg_decoder.c	11780	1a792ea775d3ee51f5435a5e43f3c7a32e3be5015514bc7812c0b
external/lvgl/src/libs/svg/lv_svg_decoder.h	772	a5cdddb04bac4ebad9670a676d85fbc0d1dccf5cea7b2db0571c7
external/lvgl/src/libs/svg/lv_svg_parser.c	80306	30aad4a3108dd3928ddb7d8e7a3335dcdc4031a6780b7036c35cb
external/lvgl/src/libs/svg/lv_svg_parser.h	1870	5f968f8027809be567f2f1e9dc9102eb3d68d35fccaee2c3dd42b
external/lvgl/src/libs/svg/lv_svg_render.c	87409	dcf991ed266e47de977076210c0c2ba9e1fddd0fe04bbc6adc933
external/lvgl/src/libs/svg/lv_svg_render.h	3848	6d2038840a7e1364718d5904627cfe96ce426f048ed20a88d653a
external/lvgl/src/libs/svg/lv_svg_token.c	15638	1eca40e7a44167d4729048a13e076a4267830134f8eed79c1dce8
external/lvgl/src/libs/svg/lv_svg_token.h	1581	51fb0383f2a1b7911101f8aa870191c817e8d3ea3dd6b7af7ad7d
external/lvgl/src/libs/thorvg/config.h	448	d539475746af7f9a73cdc8231a923d065df59cb3728baceddf886
external/lvgl/src/libs/thorvg/rapidjson/allocators.h	22744	beec8171ae738ed7b9e7cba42eaca945d285e5da6117c06bfe119
external/lvgl/src/libs/thorvg/rapidjson/cursorstreamwrapper.h	2260	9e619d9a57c2fdf51e89e422668d6fe3065cf227a66261ac2519f
external/lvgl/src/libs/thorvg/rapidjson/document.h	133685	92995df5ea1a67400fe3dd649ac3a5fdf97ddef17005f4f933b24
external/lvgl/src/libs/thorvg/rapidjson/encodedstream.h	10641	9f966d81e21540888470fe0901b2ba9026b0b662dd29f76f741ba
external/lvgl/src/libs/thorvg/rapidjson/encodings.h	29240	766732e0d0f4e08c7667371b3e9e2f172b0d7c35f6501651bc9c3
external/lvgl/src/libs/thorvg/rapidjson/error/en.h	13025	3cc7a00f9252b1183f3a108a9c7ceed59e408cd7b5a04f4c202a6
external/lvgl/src/libs/thorvg/rapidjson/error/error.h	13353	648743428973469227f6b419f80a979809c33bb8d66c409c8f377
external/lvgl/src/libs/thorvg/rapidjson/filereadstream.h	2974	6427374b5f814840d47f7646b5eefcc3286323922713924d0c2ae
external/lvgl/src/libs/thorvg/rapidjson/filewritestream.h	3119	eab610840c5d225ad29aa6bb1885c6cd612be8555fd00125bd218
external/lvgl/src/libs/thorvg/rapidjson/fwd.h	4007	05043f27ea8a2cc773671d90ca80d4d5c2024dfdffcc85b3fe67f
external/lvgl/src/libs/thorvg/rapidjson/internal/biginteger.h	9252	76fc3ac1eeb1308228abd834cb01c8b297318668bcffdb5df404c
external/lvgl/src/libs/thorvg/rapidjson/internal/clzll.h	2045	4ca3e63eaae4c67b049e8a6cbb532182e1d9c9023b7d992c556fc
external/lvgl/src/libs/thorvg/rapidjson/internal/diyfp.h	11580	b1c63c28dec8f82f7fdc2a39193d7a09a4c30343c3f008f060adc
external/lvgl/src/libs/thorvg/rapidjson/internal/dtoa.h	8469	839551a35826b5cf2a5289cee282714882ed4ba97b30e9c9f1e2f
external/lvgl/src/libs/thorvg/rapidjson/internal/ieee754.h	2969	3eab1c7d0b052c81bb66fbce3186b3a67851fbd2bc2bbf2b635dc
external/lvgl/src/libs/thorvg/rapidjson/internal/itoa.h	10110	a0f965f33b11ce112b7bf2fbdeaf5338fca174e33f9f47608ccdd
external/lvgl/src/libs/thorvg/rapidjson/internal/meta.h	6616	fbc1e3fa746a68a60bf236a922e19ecdf9572b814164f198e8103
external/lvgl/src/libs/thorvg/rapidjson/internal/pow10.h	3567	b8bf6bd27304d9d38f4cda49f249de0d3b97e4226e8dcdd04f75c

external/lvgl/src/libs/thorvg/rapidjson/internal/regex.h	26102	e7f64e62a005f74661a1d63a09672d870d39914bc4a2f165b4165
external/lvgl/src/libs/thorvg/rapidjson/internal/stack.h	7157	8b7efceeef0a0721ac98965ab6bef6c8507b0d0fcd205e35e0eec
external/lvgl/src/libs/thorvg/rapidjson/internal/strfunc.h	2721	6ee8658c852b9c076e1893425d718cdca4089948b7c6b91e3653c
external/lvgl/src/libs/thorvg/rapidjson/internal/strtod.h	9032	0638724ffa4198da3678dba1457cb4abfe36b48d8be78132822af
external/lvgl/src/libs/thorvg/rapidjson/internal/swap.h	1398	72acfabebd33f44869e7724603e57744da33337998b32443b66ea
external/lvgl/src/libs/thorvg/rapidjson/istreamwrapper.h	4051	e11984b1ab67d74d87dfcd478bc1059e67720ab2d0de6e8fb57c5
external/lvgl/src/libs/thorvg/rapidjson/memorybuffer.h	2534	87d8c876554e50698a3e9f3cac49896d4b1f15eb814e279d01b2b
external/lvgl/src/libs/thorvg/rapidjson/memorystream.h	2642	9bf7c6bd1a657f48ba53c386f6dd116c1d5d8471d04c92216680b
external/lvgl/src/libs/thorvg/rapidjson/msinttypes/inttypes.h	8361	f95f632862138ad569a6e3adc431d867e6fb5ee5f8cf23e8cd1e9
external/lvgl/src/libs/thorvg/rapidjson/msinttypes/stdint.h	9375	c3f7d184f187b82c617134170bf9a02c3233413cb3bfc0ee4aa36
external/lvgl/src/libs/thorvg/rapidjson/ostreamwrapper.h	2303	1f0c6cdebf9a4e31c53e59091c411e359e6e2c504790eafe55680
external/lvgl/src/libs/thorvg/rapidjson/pointer.h	63123	113861bc59a7a273e22299305a16cb3600c0eaa5dd1d7554a3021
external/lvgl/src/libs/thorvg/rapidjson/prettywriter.h	10504	27ae70723851da686c9a829f8404fa2e18e787eab9dfee16fe433
external/lvgl/src/libs/thorvg/rapidjson/rapidjson.h	25613	a5d7f757ee6ecbd8834fdf532096a5762deb568a16d8b0284ea2b
external/lvgl/src/libs/thorvg/rapidjson/reader.h	94332	214ba7447b544217dd005020010fdcf5241ac7866607b94704bfe
external/lvgl/src/libs/thorvg/rapidjson/schema.h	146749	fb800528abcbe0fc44f343d111267b3383edf85fb0a328d3b671d
external/lvgl/src/libs/thorvg/rapidjson/stream.h	6732	a3708a941a130a5679d2302368169852e5c4ca5f54758ea0e94ca
external/lvgl/src/libs/thorvg/rapidjson/stringbuffer.h	3968	5f01edff23eb9f98997a766e12873e23afa205fdd20a8cf2bebfc
external/lvgl/src/libs/thorvg/rapidjson/uri.h	19752	53d200ad6eae23399fab31687c50511b3a14cd32c0a36900151f3
external/lvgl/src/libs/thorvg/rapidjson/writer.h	26817	8b27f983418c95d36c24d484c3271129980f0db91a50193fac206
external/lvgl/src/libs/thorvg/thorvg.h	86121	dc8b6c756ca7478ae5e1164806c15b934d7fce88fc7df38c79738
external/lvgl/src/libs/thorvg/thorvg_capi.h	103534	bcd1ea248bf5741bd47fa398bfb957b9363147167f3a754dcc55f
external/lvgl/src/libs/thorvg/thorvg_lottie.h	2819	b01f57cb6924bf0372aba3b88bd3136ba70a50631a23eeea372d1
external/lvgl/src/libs/thorvg/tvgAccessor.cpp	3450	20bb66760a7527a6f5b817734332b9f864004b0f867fd00e40729
external/lvgl/src/libs/thorvg/tvgAnimation.cpp	3822	7dc4b0e0cbf09805b210b837450a0d0b1ca1350704e8719507124
external/lvgl/src/libs/thorvg/tvgAnimation.h	1655	d821663d63749dd59b05e0db38df5be2b2fc4ee09619b06c6f6bd
external/lvgl/src/libs/thorvg/tvgArray.h	4495	1e2a12674188c91449cf5c29cb101d6c532a6be38542cec859904
external/lvgl/src/libs/thorvg/tvgBinaryDesc.h	4652	4bf27031f659f6b4b5e5c6541c91909f0ae3c76f078ee577feaeb
external/lvgl/src/libs/thorvg/tvgCanvas.cpp	2679	dcb0c08007fc51c70b2a5d599112ec890cfa9fc3b8d692c47a0ab

external/lvgl/src/libs/thorvg/tvgCanvas.h	4791	0506c718bf3b9d2cc1bd9b7c89c093445f53e86b398b455865d78
external/lvgl/src/libs/thorvg/tvgCapi.cpp	31533	b09a98b36c20c249969fc438459722e06cbc08f85e3227429f7e0
external/lvgl/src/libs/thorvg/tvgCommon.h	3332	b11e439af5c6e9f271233838f79d05e6737957b9ea0899a826be1
external/lvgl/src/libs/thorvg/tvgCompressor.cpp	16779	fdda523535dd439cf6a77c3d07617284b5748869ded64901eaaa5
external/lvgl/src/libs/thorvg/tvgCompressor.h	1784	032d5b75a7a462d6e1ba6c1621fd6ab05bfa5cdffa1521e8fe51c
external/lvgl/src/libs/thorvg/tvgFill.cpp	5983	6cc53f7873993fca46b9a35c0db18baea8cc0dc7f150fc9ee6803
external/lvgl/src/libs/thorvg/tvgFill.h	3086	0f46a71ac80432069a19cf765001981c38080c071a9399dfb39ac
external/lvgl/src/libs/thorvg/tvgFrameModule.h	2200	7a7472c9c8785db5b4c0fb35e12887cafd8d657b3127cf8efe051
external/lvgl/src/libs/thorvg/tvgGlCanvas.cpp	3130	e51a00346f19e5fc867cb59dde1618756e9d2dcfe3c53091000a2
external/lvgl/src/libs/thorvg/tvgInitializer.cpp	5303	c7c15aee550dc8335f8fcb82f878c0af536771247709de4154069
external/lvgl/src/libs/thorvg/tvgInlist.h	3016	ccf3172650aa2d63f5504f9c3b9270276ce9e8ee4cf4da3fd6934
external/lvgl/src/libs/thorvg/tvgIteratorAccessor.h	1576	db767a17cfdef2319d65ed0c0a59c51db7aef40b6f37ed81ff814
external/lvgl/src/libs/thorvg/tvgLoadModule.h	3366	8fc4fddf59a69e5e2ed3eef709aadf30d909125290ef639f640b8
external/lvgl/src/libs/thorvg/tvgLoader.cpp	12712	9643c50315a0b8506fa7c9d1dffa35c03a17d649c126f06167f7c
external/lvgl/src/libs/thorvg/tvgLoader.h	1929	217e76a4c56506d0544f22df1697a1541985e91ca08815192d81d
external/lvgl/src/libs/thorvg/tvgLock.h	2000	0674f716557079ee1c8e291a275a2f931534deaf32c5ee639aa1a
external/lvgl/src/libs/thorvg/tvgLottieAnimation.cpp	3148	1de32d5a682fc93764bc1eaaf5a4f21dc9e39aef2628610185f3f
external/lvgl/src/libs/thorvg/tvgLottieBuilder.cpp	55376	ce7f5bdb1c56b8b7775e850a9f8a53134a1fbf95caf74478c8d39
external/lvgl/src/libs/thorvg/tvgLottieBuilder.h	6125	e4da96de4cb4d75cd9d8414e8fa091511f824d0374ac75a446aa8
external/lvgl/src/libs/thorvg/tvgLottieCommon.h	2625	50461b091c5232763eccaade09fe3031f02e8a232a667f8c73e2e
external/lvgl/src/libs/thorvg/tvgLottieExpressions.cpp	47947	f1c9ed921af36723b37f92fc735195914ed2563b9e84745e351d8
external/lvgl/src/libs/thorvg/tvgLottieExpressions.h	6562	4787a484c7c019d75eb2bf8b1b089a36c66d8f0206f77d06947a8
external/lvgl/src/libs/thorvg/tvgLottieInterpolator.cpp	5356	dabf651e523b942def00b10665781d7fe468a9011734c87c12b9f
external/lvgl/src/libs/thorvg/tvgLottieInterpolator.h	1843	763f0fa122b01f321ec6a0b3fa3464ed648900b97ec3ba8abc24d
external/lvgl/src/libs/thorvg/tvgLottieLoader.cpp	10039	a2205417a99159929d948617626f81309d4c8245390274dc8c9b5
external/lvgl/src/libs/thorvg/tvgLottieLoader.h	2967	35bdf6f88caacaa2a832c13b50efc8954ea8f423ee6298456f271
external/lvgl/src/libs/thorvg/tvgLottieModel.cpp	16905	570ac58d20a322a52f03b32e84d66ed1ea7c9a49ac972b578d669
external/lvgl/src/libs/thorvg/tvgLottieModel.h	22235	933df7149486e97889e8ad09ee44a8f06165af0968036e151f544
external/lvgl/src/libs/thorvg/tvgLottieModifier.cpp	15849	e39ef19e0ff242c409c17a1c09a32ce7c44d6ed275b65348bcd36
external/lvgl/src/libs/thorvg/tvgLottieModifier.h	3284	071587d6a2b0ebaefba575e442bc139bf56c468c8d8f082c0a7fc
external/lvgl/src/libs/thorvg/tvgLottieParser.cpp	47104	b514a38427623477d3d65fc2d3ebc1d30c24c435eb62700822d3b
external/lvgl/src/libs/thorvg/tvgLottieParser.h	5050	ff8dc6f949c6cf510ca6e67f556fe02942654af76a71887f4ef25

external/lvgl/src/libs/thorvg/tvgLottieParserHandler.cpp	6450	68b6cd4070b83a01ffb6e6effa9a45be3ce40188fc7d7f8720c09
external/lvgl/src/libs/thorvg/tvgLottieParserHandler.h	5217	58b90961e2ab3c7ab86c8602b5b3597ac9dfe84e70c724fd2d8e8
external/lvgl/src/libs/thorvg/tvgLottieProperty.h	25046	7b112d5526410e199fe97fda8deca9e1cc286a5b5bb1f2602d7f1
external/lvgl/src/libs/thorvg/tvgLottieRenderPooler.h	2052	d38bbccdfe62c00d1b754a59095db0cc566721cf757fa56f99969
external/lvgl/src/libs/thorvg/tvgMath.cpp	11055	d7a6297f6be4de2e14da2f50243badccf3ff1ff2336cfff06cce9
external/lvgl/src/libs/thorvg/tvgMath.h	7507	2862fc814ddf890644c792c0aa6c0582a6bd8f51d58e74832fc80
external/lvgl/src/libs/thorvg/tvgPaint.cpp	15670	5fe8fe698636a4aed439d2c3184739283c39e603cb572d962c48d
external/lvgl/src/libs/thorvg/tvgPaint.h	6054	10754172acdf8bde978e6549b9237cef6964ec019ce50f9087cf3
external/lvgl/src/libs/thorvg/tvgPicture.cpp	6109	696a14aa2b7ca8d51a229eecaf948a7f274f586964f0721dedcf6
external/lvgl/src/libs/thorvg/tvgPicture.h	6013	d16599837916dab26457f3a6ffe1b8ef5c80634cac6acb535c8c3
external/lvgl/src/libs/thorvg/tvgRawLoader.cpp	2758	4d7c42aa94e4e8d4161f76d83ca5a0b91bbfd24c7c9a03c966152
external/lvgl/src/libs/thorvg/tvgRawLoader.h	1570	41f9c37fdd3a41ec04e43f5452fc0a1fe3040795611276ea89201
external/lvgl/src/libs/thorvg/tvgRender.cpp	2567	832d2ce7ea01bcc7d514a7e2d5b53b9c8912ce28d855e6919de36
external/lvgl/src/libs/thorvg/tvgRender.h	12125	874267a7e5d4c0273a02ec8811333bbd7468747df34eb6678fc89
external/lvgl/src/libs/thorvg/tvgSaveModule.h	1668	52928fe8c921eb4d17d59348c394c01b634e807adf16b33cc05ed
external/lvgl/src/libs/thorvg/tvgSaver.cpp	5403	ccd3924404d6a37dfa7b8708d17e05e2737714920e9421f290229
external/lvgl/src/libs/thorvg/tvgScene.cpp	3356	a2056d58518bda25238c98af68a402fb6f9db2c3353864c8815b8
external/lvgl/src/libs/thorvg/tvgScene.h	7712	ed43c44978864bb328a7f71cadd2ca4bc3ce22e44b290802b6b7e
external/lvgl/src/libs/thorvg/tvgShape.cpp	10750	8a37b73dfb50b5a399fc0955d04b3a92f0eceef092cda9febb3d1
external/lvgl/src/libs/thorvg/tvgShape.h	12434	72155f4a3640ad52da34e5832c5162e601ec3f7c9757847168acf
external/lvgl/src/libs/thorvg/tvgStr.cpp	7261	87d9bcf1471140568ba660a317518ff4a843b6b21ac3b415eb432
external/lvgl/src/libs/thorvg/tvgStr.h	1661	1978f13a077bb9b8c0db59fd4daf7d9bc49ebdc8fcf6dfe0c3513
external/lvgl/src/libs/thorvg/tvgSvgCssStyle.cpp	12645	43d168cb860e3ec22824ff7a68e5e4f3a43d13dc50280d2fbc055
external/lvgl/src/libs/thorvg/tvgSvgCssStyle.h	1719	8b2f71217c9d5ab67ebadab065ffa269b2d1efc5fb106b803793f
external/lvgl/src/libs/thorvg/tvgSvgLoader.cpp	138351	0131b47d7545e9fcc3693c0e3941b3932f3ac001233bd14ba76ff
external/lvgl/src/libs/thorvg/tvgSvgLoader.h	2298	b14b28ede797f3d39c510cccac1133ef82c027906497b0fccdb47
external/lvgl/src/libs/thorvg/tvgSvgLoaderCommon.h	10682	4307b541a80af9e5d32e816faa9cab00b7ccaf79e1e85fdc970b1
external/lvgl/src/libs/thorvg/tvgSvgPath.cpp	17766	0e1f3a5c9124457e6e832d0000ed02e5dbb2c5d70a8d6db1e9f69
external/lvgl/src/libs/thorvg/tvgSvgPath.h	1407	82d3d33c6a0cdc4a5f8a6ad7b38937608a147ebb171543c50f74e
external/lvgl/src/libs/thorvg/tvgSvgSceneBuilder.cpp	36622	3d6027f4c0e769ea9b80fb5b8d63de04f3d3b71bcf95e61bad0ca
external/lvgl/src/libs/thorvg/tvgSvgSceneBuilder.h	1560	3caed8c5dcf2b91204cfe52aa1b0be9ae4305a544e8e8df3b7307
external/lvgl/src/libs/thorvg/tvgSvgUtil.cpp	2628	fbb5e747121684eadd2e8129f6596779e453d3205e758b54a28be

external/lvgl/src/libs/thorvg/tvgSvgUtil.h	1405	77749b1016c3e3d6fa81c034fe7b60ce54fec5f15147ca2c5d434
external/lvgl/src/libs/thorvg/tvgSwCanvas.cpp	3962	fd90edcdace547b6deca32f04f00e8ad88f2dd11b84f8528c458e
external/lvgl/src/libs/thorvg/tvgSwCommon.h	20273	662df4788c7f0ccbe63e4688d2e98260daa203b8b95463632efab
external/lvgl/src/libs/thorvg/tvgSwFill.cpp	31658	eb03a7cef708ff4c518323db0800b35b102dad7cf5a821a038d61
external/lvgl/src/libs/thorvg/tvgSwImage.cpp	4290	86405ea251d00072fa91ce6a984b0292b675cc1326e8adc54735b
external/lvgl/src/libs/thorvg/tvgSwMath.cpp	9042	4a40115be3e1d3fbb7e2c5540dec2b6111d6577e843d5cd7c4cd2
external/lvgl/src/libs/thorvg/tvgSwMemPool.cpp	4221	85b9a4ae8356eed9891ae5ddd802ad51df5f58a0121002410d9be
external/lvgl/src/libs/thorvg/tvgSwPostEffect.cpp	7348	45e5a1682fa57f5c64e5b2331b9d72c44b5609c63b4250e2652a2
external/lvgl/src/libs/thorvg/tvgSwRaster.cpp	71214	17ff3ec3e855193e8b99e65efebbafedae300a1c2da47b16d0f6b
external/lvgl/src/libs/thorvg/tvgSwRasterAvx.h	8865	ee0978343463edc33a365f698d71a0377e1e9c2288831a8542493
external/lvgl/src/libs/thorvg/tvgSwRasterC.h	6166	70ef91c9508f1e807445399a833c120245f82ee86bf03518f85a0
external/lvgl/src/libs/thorvg/tvgSwRasterNeon.h	6733	f56d712270a0cfce3aaf194b6deaad597d6d9167152b7528a6d58
external/lvgl/src/libs/thorvg/tvgSwRasterTexmap.h	39642	bb97202102161e8e4ae9357d18be491b5d937087ae0a194d12256
external/lvgl/src/libs/thorvg/tvgSwRenderer.cpp	22607	61f0d5a60e89b202be8d575a89da154ecc998af88d3efaa41c848
external/lvgl/src/libs/thorvg/tvgSwRenderer.h	3701	25afe63e4e4017d186d3d25927b1b68e1ae1089582b80240b4c2e
external/lvgl/src/libs/thorvg/tvgSwRle.cpp	32280	1a9690a1a4844ce9b80a9498c54a809db7eb554588bb13f0aca9f
external/lvgl/src/libs/thorvg/tvgSwShape.cpp	21083	916559b8ee01407841810f50ae72bb3cc5da06d817e9ea9d674f9
external/lvgl/src/libs/thorvg/tvgSwStroke.cpp	28271	55ef4b306ef7db69435dee427f2b3aab7b196ea91a5c52b4f1a49
external/lvgl/src/libs/thorvg/tvgTaskScheduler.cpp	5897	8ca1f330f1a2d88c3499bbb2a55d40cdc7c2344dc1abff7e3b091
external/lvgl/src/libs/thorvg/tvgTaskScheduler.h	2918	d8ef0c805cd1a972573b5b89c7dbe242442a4cb6790465f15bc28
external/lvgl/src/libs/thorvg/tvgText.cpp	3249	ec8553d3790fccb9aa39d98c67e765387fd2a47ef82c73e2e74f1
external/lvgl/src/libs/thorvg/tvgText.h	5284	5259fba506b348e93d5136adb034352af0d961aee6862bc07acbc
external/lvgl/src/libs/thorvg/tvgWgCanvas.cpp	3291	a90ef1245442f861ba0baef2bfa53d197938f5a0932a67ac11fae
external/lvgl/src/libs/thorvg/tvgXmlParser.cpp	18538	766eeefd0730869b582f1c8a2579586eaa51aa2f4109192aa2891
external/lvgl/src/libs/thorvg/tvgXmlParser.h	2989	741daffc46056927f4d63daeb131bd23a7a13e51b5e9e602b77b9
external/lvgl/src/libs/tiny_ttf/lv_tiny_ttf.c	24232	ba5b184fa1e9a7826c4ad84eae868bac95cf27c0657802197fdc8
external/lvgl/src/libs/tiny_ttf/lv_tiny_ttf.h	3048	b74742be3f172e9f5826791ca02d5fdad2b7a5423e17c356c09e8
external/lvgl/src/libs/tiny_ttf/stb_rect_pack.h	21136	a77b5e58de68d892c0c880ff62f6de7a7a669503cf04fa6cd0af7
external/lvgl/src/libs/tiny_ttf/stb_truetype_htcw.h	228103	bf989936a0e69b53b455e583b583d0bffb6feaaa72b8c2a05af4b
external/lvgl/src/libs/tjpgd/lv_tjpgd.c	9608	e86658188441209ca9512c1025080e18dc7c4556a5a2e825ecf58
external/lvgl/src/libs/tjpgd/lv_tjpgd.h	595	602b5d249967fe2a489c04eed922f35d3389fc9188a2d2839d5c2
external/lvgl/src/libs/tjpgd/tjpgd.c	48709	869f12918ddd795c97ac36c281683fc3f18c81b9e0279c6219f8b
external/lvgl/src/libs/tjpgd/tjpgd.h	4056	7f768047e0b1947cd1eb55e29890137645042bfdb03fdcb56634a

external/lvgl/src/libs/tjpgd/tjpgdcnf.h	865	ca1d5bc5fd41f3800238c56ea4322d5c0824df4e5fce5eb47ef99
external/lvgl/src/libs/vg_lite_driver/VGLite/Series/gc255/0x40A/vg_lite_options.h	4831	67ee639e90fba4fcc9feb3c8a6ebe6a082ff12358e74b6e70ad65
external/lvgl/src/libs/vg_lite_driver/VGLite/Series/gc355/0x0_1215/vg_lite_options.h	5621	bc15d52c0a3565d6ac273d457c814898914274c25b5d6b77755c8
external/lvgl/src/libs/vg_lite_driver/VGLite/Series/gc355/0x0_1216/vg_lite_options.h	5621	98661466f2adb3ad1a1b16c188646beb9326f36cc6c218f9f0aa7
external/lvgl/src/libs/vg_lite_driver/VGLite/Series/gc555/0x423/vg_lite_options.h	4908	99a3f53c25cf1c98d5965d8fb7349ac34c7af5a9decb76ee3578e
external/lvgl/src/libs/vg_lite_driver/VGLite/Series/gc555/0x423_ECO/vg_lite_options.h	4901	4fa0c4e37a5bee7c168c50921bb95ec01a2dca6102384a828b981
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite.c	246765	245267ce41526aa7452c8b17851c89c2c87b3e0dc40e37f2f7690
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite_context.h	14042	64139ce95bd81c3a8e567cd1f0362d5c69a5556a499fb2abd73d7
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite_image.c	68825	646f96defc01df2ac023e7c902dd133b66907fe6e5e443142ea5e
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite_matrix.c	4999	3b6cbde78015e7c2c2fe3b996d792b483ceb16ce3117bdde38a90
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite_options.h	1813	d6c23a205d4b54f1ae122c944887126c2d8094a99af3100a75fbe
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite_path.c	222940	af08fc671e7baea596b739870223ee5ca877b2173684ce7c5f39c
external/lvgl/src/libs/vg_lite_driver/VGLite/vg_lite_stroke.c	189186	95257167e183aead7b0cb3f8d75b8b6309ff50cf1c2f4bb5a14f0
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_debug.h	2508	6682e2d4f26c927280062c62252355df15bf8bf2a199827bb0e54
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_hal.h	11755	9ec83334f5a6743b89d9e9ca380dab7584baeed30200eb801dfea
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_hw.h	3881	ce106316f44e43875e4521c3ff44e010fd2a77ecb3ee234ff80e8
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_kernel.c	47635	75ed772288fda564830824d3c69cb05b173821e787dccad756500
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_kernel.h	15987	cf0896b5db723112cd8f640cd44a72af299ef864234bc02c3dc2a
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_option.h	4612	1276d8cdaba15a74dc6eff5ff0433d6a2c4653031f39cc5892fd0
external/lvgl/src/libs/vg_lite_driver/VGLiteKernel/vg_lite_type.h	2796	00e2bff15f1fea0b52ed87d41914636ac60680c48b245c1c58f54
external/lvgl/src/libs/vg_lite_driver/inc/vg_lite.h	72778	0215eb21202ca3d9e90d357e3ceec9843bf449b848619901f744d
external/lvgl/src/libs/vg_lite_driver/lv_vg_lite_hal/lv_vg_lite_hal.c	16309	6fcfc96c0f00ccd096c87bbb4c96b54647a3556e6672e797268c7

external/lvgl/src/libs/vg_lite_driver/lv_vg_lite_hal/vg_lite_os.c	236	61b5263ff7fbd585a7c44d9b83dfe98c30d8cdb51303e25b5cd6b
external/lvgl/src/libs/vg_lite_driver/lv_vg_lite_hal/vg_lite_platform.h	1220	e50e57ab2a556889cabd8eb1a627883d5577686028a967ba08203
external/lvgl/src/lv_api_map_v8.h	14790	8b6659fbe747dee915425404cdf2a4e1f9e883e6927f28752a8c2
external/lvgl/src/lv_api_map_v9_0.h	1718	d31e9ac2965968328225aba92614d349fca2c9fa0d378ea3ea766
external/lvgl/src/lv_api_map_v9_1.h	4151	ec57f2c537290de6df954bfc8cf31b9f2a28d8cd83137eaebc3b3
external/lvgl/src/lv_api_map_v9_2.h	1720	63216986c429213255b37d857cfc6ca162aef71dd87ac92041860
external/lvgl/src/lv_api_map_v9_3.h	5094	e6739cccd0c255aa460478544c9bff7d6cde89ce0a4cd9be7836f
external/lvgl/src/lv_conf_internal.h	154141	300c93bd1db6543af213c2a614e4f09ecd1742c945d18b7bc6b20
external/lvgl/src/lv_conf_kconfig.h	11822	503d1deda84867c1b55ef19929b82205de40160a7ea8f45afa01b
external/lvgl/src/lv_init.c	11407	15738eb7eb04342eb28c1651c9d79df442087bacde3fdda1acc36
external/lvgl/src/lv_init.h	837	48acde7f63dc2aaed4bcd4e6156f64171e3432cebbf0b4fa0d6c9
external/lvgl/src/lvgl.h	630	3c739c642fbe68e57e1efa52ca197a54250c5ece0c865950d21f8
external/lvgl/src/lvgl_private.h	640	9fed0e26d5d4e13af98fd317ffff0b2d3ec43b62b8717e764c264
external/lvgl/src/misc/cache/class/lv_cache_class.h	268	e1efd6ce9bb28c42279dc069374d814339e252f6ebbd30adc2300
external/lvgl/src/misc/cache/class/lv_cache_lru_ll.c	12193	ce2477d2585f3efc16e103ecad507c4d1c8bc0b9a7e138fd87c07
external/lvgl/src/misc/cache/class/lv_cache_lru_ll.h	813	869ada3fc4002572b11dbb471e70e535f49eb296c15415078876a
external/lvgl/src/misc/cache/class/lv_cache_lru_rb.c	16289	a24cfa05f34a7de472714de952cf4184ea9a48af57597521fb9f8
external/lvgl/src/misc/cache/class/lv_cache_lru_rb.h	813	d82926b1230444c4f5e96a248238cc742b37380db17c0f8636f90
external/lvgl/src/misc/cache/class/lv_cache_sc_da.c	16560	de59aba04e8dfff491f4550568e754331719b53bcc0cfb10b2fb5
external/lvgl/src/misc/cache/class/lv_cache_sc_da.h	720	d6abb87999ac79d248b1be091e41896311c7a45a4cdb8dbd3be80
external/lvgl/src/misc/cache/instance/lv_cache_instance.h	256	fab16d7f0d51d7acfa6593af23b03a1ee470a4d979d9454bc049a
external/lvgl/src/misc/cache/instance/lv_image_cache.c	5664	d3ceb1cca351a75feace1b09c1905386d2845cbb9e85b41e9a066
external/lvgl/src/misc/cache/instance/lv_image_cache.h	1745	d2287b0d4cfbfeb3d422df0db32b9c95754bd33ec8042b84f56e7
external/lvgl/src/misc/cache/instance/lv_image_header_cache.c	5445	b46115a0baa0803c71cfbf455f3a54040438d6cb21c0c2557da29
external/lvgl/src/misc/cache/instance/lv_image_header_cache.h	1965	74a1eed9a6636c6c53f57a8f3e0bcf0cfcc8bd500b781bdbafd97
external/lvgl/src/misc/cache/lv_cache.c	9593	d45a3df0cf144cd56c70aeee10f1817f55480e1a6cb22ce43eeaf
external/lvgl/src/misc/cache/lv_cache.h	11294	ee21712d894191713297045ced7fdb2ac56445968c92352709ace
external/lvgl/src/misc/cache/lv_cache_entry.c	4292	96d2aad16cb52229084c02e54c4e7ef3f4bd829655d93faeeb892
external/lvgl/src/misc/cache/lv_cache_entry.h	3373	4318c88bc4001cdfd9dd828ce9316a92e1df14f3ef9c429f86d84

external/lvgl/src/misc/cache/lv_cache_entry_private.h	1893	3c747101cfb03ca87d659d75d3e243dbe4af52ea2db4bcb4cf563
external/lvgl/src/misc/cache/lv_cache_private.h	7589	6a4e2bad87c3d44a670002e9aefdc8df6260f3b56101d1ba43f96
external/lvgl/src/misc/lv_anim.c	23840	2682a1fd6fa908737e64d20dacaf348060000f624193257a39ce7
external/lvgl/src/misc/lv_anim.h	22375	909e1d5ed39a896c69e847bb99556b2e532c86f6f4c111a6ae49a
external/lvgl/src/misc/lv_anim_private.h	1014	f0f7e852cd7f462813b4a7112febdc576a9bb84666de53dc8dd42
external/lvgl/src/misc/lv_anim_timeline.c	9474	61e235ed14b86a45399b9094cc8d298a72991d04aeb8f2b857e15
external/lvgl/src/misc/lv_anim_timeline.h	5345	dc3dd6b5ab0a6691387328fa9a2042fccea345eaee4ec66e2d24e
external/lvgl/src/misc/lv_anim_timeline_private.h	1206	8eb0b9e887e5d4bd551ade2a0bba17e61398c1227f1f3061db293
external/lvgl/src/misc/lv_area.c	16791	f97a87781f9c36bf2ee0288401526d9027e3f7fbc0d54ca7a563d
external/lvgl/src/misc/lv_area.h	7411	5f92c841954d11a1bb27311730a174ed02922d1601a6a4ba63cf8
external/lvgl/src/misc/lv_area_private.h	3431	567579a0ed9b9ace5228d6517b493a3fe7d189ecc2c1c16b22940
external/lvgl/src/misc/lv_array.c	5471	c194246f1038177e437e576fa8cb8007d293566efcdaefebc3812
external/lvgl/src/misc/lv_array.h	7027	81c2fcfe0fb65de1aa73f2ee64ce7a86999339a52a8a953aa5b9f
external/lvgl/src/misc/lv_assert.h	2586	e2981f5118b8188088ab5b5344c8d39e37d554c5f59427d94976f
external/lvgl/src/misc/lv_async.c	2464	c1efac14a51c20d24625cf8b9283abae2990156e892c30ec3fc96
external/lvgl/src/misc/lv_async.h	1342	43d380375967f57c436ff68a81a10584343ffdd37886cc579ba96
external/lvgl/src/misc/lv_bidi.c	21559	b1efa85dedca496a49b89ceab624dc1e4be743f1af0514b4f2376
external/lvgl/src/misc/lv_bidi.h	2241	f947c2791343e2c86b70c961c7fc296b3bc02d6fc371a27d338bd
external/lvgl/src/misc/lv_bidi_private.h	3390	8a2385f09dee20f8ff2731c32ffc810c7e0615e81db0aa3889358
external/lvgl/src/misc/lv_circle_buf.c	6992	07bb1a0e02a6505ca8ad6c594c30af3cf20895625f5488d6a6f17
external/lvgl/src/misc/lv_circle_buf.h	5572	1194e0ab32d01cd24e4c4395fb8419fdad9bee8ed4003f9692a9e
external/lvgl/src/misc/lv_color.c	10422	c7b03a0e103c48a97969b9d657b76a5e4d6f743ac316e42cd0442
external/lvgl/src/misc/lv_color.h	16313	94fd3ebccdd9a7069e6f2fced6c289d7fcffe2420d25ab947b81e
external/lvgl/src/misc/lv_color_op.c	3021	c59308f21cea56668f5a1b54d599feee72b97526697e7aefd9976
external/lvgl/src/misc/lv_color_op.h	2657	3d7f7ea009ca1b1a7771a41d06869a6add81e3e29733d2ceb3922
external/lvgl/src/misc/lv_color_op_private.h	579	8368a2c9fb90d933881bbfeaacb99cbeb56342a1c5874dc6e61e6
external/lvgl/src/misc/lv_event.c	12208	d1c4bb5f304cc10b5671c0c523c88518a0418dcfda7d13447f6e6
external/lvgl/src/misc/lv_event.h	11441	e8799b9043844f2c0148f962143a02dcc3af6c598d60cffafe097
external/lvgl/src/misc/lv_event_private.h	1589	1cda17926012d39e5529b8e61c4f98ac14ae57b1f97b116818759
external/lvgl/src/misc/lv_fs.c	23360	594bce8c830614d5032ee40d102ed75e96a905893972009b286d8
external/lvgl/src/misc/lv_fs.h	11577	9962b6b3764fb881a895b8bf9524d9dbe5abc75390fb96ede4020
external/lvgl/src/misc/lv_fs_private.h	815	584db57e43d707b69148f95bc42cb7c9745af661a42313dba2f59
external/lvgl/src/misc/lv_grad.c	3180	3e0e139f8ca9dbfbc060f44161b8d0738ee6925b08d0948b431f2

external/lvgl/src/misc/lv_grad.h	7334	245b64abd976aa2169fe3073a9420be4c4c8127105d6100acf507
external/lvgl/src/misc/lv_iter.c	5337	c645dd99be4f4247be8d9f9a4caefea257ce3458db1a952de415b
external/lvgl/src/misc/lv_iter.h	3938	2e60c4319cedacac3476df8e82f0863484ff1459ac4af4037be5d
external/lvgl/src/misc/lv_ll.c	9341	2ca526b6f8880425b0d0c38b471c8b81ebbcd2dfd1f0b369b4655
external/lvgl/src/misc/lv_ll.h	4203	7a020a8d646c37f5760ec228c49955da3f731865e63bf6af4d83f
external/lvgl/src/misc/lv_log.c	3518	86e97097989d274adbda9af1536acf4d2d0c1b5a6d44b0f18f197
external/lvgl/src/misc/lv_log.h	4623	87409bbd86419e606dccc62a4555c20d67850fdc5d1e92a69852a
external/lvgl/src/misc/lv_lru.c	9387	0a73d5367879f63f05f33eff773ccd9fab14297ffbbee319b47c2
external/lvgl/src/misc/lv_lru.h	1822	d7d76fb6a0936c31aa040989d692df62986d8f3d5bb02efbcda6f
external/lvgl/src/misc/lv_math.c	14598	58e309b8dd6a450684814e043b9b1724ebe2cac1058de1597679e
external/lvgl/src/misc/lv_math.h	5459	c93ab3593e02af43048bfd266872f26a19756d3c16978fadc6684
external/lvgl/src/misc/lv_matrix.c	6988	00108f9e3f298ebfe3d1c9f8c29647cfd6ab1c6960d56bd6414db
external/lvgl/src/misc/lv_matrix.h	3575	e5636e653c9416586765a74e97a968d9429026b87bafd15b60c06
external/lvgl/src/misc/lv_palette.c	8267	87e038f4b160d274318663ab3214aade8eaadbba094095416fca7
external/lvgl/src/misc/lv_palette.h	1315	57f6853a9a1f9b014d4e75d96858c2fee09e31878b99d7fd8f2ad
external/lvgl/src/misc/lv_profiler.h	5462	ae1d658d3a4dac133642720b486022b5196aa685354d9b9d53f2d
external/lvgl/src/misc/lv_profiler_builtin.c	7417	23d32ef8bbd72f8ac5db9483946f32753c010e7f14cfd71ccaa20
external/lvgl/src/misc/lv_profiler_builtin.h	2127	83f5b32e9214fd2b42a4012e1953a4f54f7f87f2b0f3a162e2b99
external/lvgl/src/misc/lv_profiler_builtin_posix.c	2746	7d5de1faf1c5f786ca484c0340a25993eedf8b48dff86e2e99a7f
external/lvgl/src/misc/lv_profiler_builtin_private.h	1570	8235644274254cdb3fbd1c8ff4eb7100f9aced31562ad1abde514
external/lvgl/src/misc/lv_rb.c	13646	376ed3247c31f42e7841fdc9a253f4a3c8eb895da5c11a8c553b9
external/lvgl/src/misc/lv_rb.h	1500	b758d58da2016f02fd8503a747a362a0037e5adb11752caa86b9c
external/lvgl/src/misc/lv_rb_private.h	814	69915d754585ada469c9893bbdb5df89305b3da9e5cd211ca5144
external/lvgl/src/misc/lv_style.c	19723	5407b0d05c928d7be0b2b989e3b4765163d7083b1ed9047b19c41
external/lvgl/src/misc/lv_style.h	21904	0c5d713bbf38bd8d02d39634a2a7381ab2198c831cb74ed712afd
external/lvgl/src/misc/lv_style_gen.c	23231	ef357db4a7c8fe4410df855171456c75d16e3f0e86685ea4a28d1
external/lvgl/src/misc/lv_style_gen.h	24421	f91315f21669235972e64efd1f973c32be59dd74d8e3d8f8e53a0
external/lvgl/src/misc/lv_style_private.h	564	3cb6afad2988cbcc320f33fcf536cbac490772a2ca71295c75b08
external/lvgl/src/misc/lv_templ.c	717	40d1b3dab81753f66ebca0b6d0b3065bc92d317c3c54e1809628d
external/lvgl/src/misc/lv_templ.h	509	7cc6e2a787852952c13d4d42b1e9b3967bcac6c27341390a58cf4
external/lvgl/src/misc/lv_text.c	30304	37453be77fd8fc841e747de58e1d8610fa926eca97dc6a5327f00
external/lvgl/src/misc/lv_text.h	2265	750449ca510f8c7d4b2c286fb29c2e4af648c6d0f0be0e1a99491
external/lvgl/src/misc/lv_text_ap.c	10094	cf3d20b8995203508d71eb871f5f2383ec34818e6f737797b698b

external/lvgl/src/misc/lv_text_ap.h	942	a355b36465a9e2dd21234c376886d262b1e56ca8663fdc4fbf283
external/lvgl/src/misc/lv_text_private.h	10560	093c1a850fe0ff5e3e71de2ab1db0aa8c50af961b7e7d6de015d9
external/lvgl/src/misc/lv_timer.c	10363	bcec8183d48745433cb744bb1ded8a6a1aff15d47d34664bc2207
external/lvgl/src/misc/lv_timer.h	5626	87157dafe2d3ee9873950bd88243b9941d4ed0915d4daabb43955
external/lvgl/src/misc/lv_timer_private.h	1611	1a1d568decbea4d6765c9ebf5b15e01cb45a23cc2aeb9dc89108d
external/lvgl/src/misc/lv_tree.c	4968	242feeedba53d5f883155a9506c5de58ac11b27296c68b66c01b6
external/lvgl/src/misc/lv_tree.h	3061	b7bab62a7b0bc2bf4adca28688d61ab73ab42f11b33e79df24f1c
external/lvgl/src/misc/lv_types.h	11427	137469b8ddfb40890982d18ee5f787c6f2c9e695e77c7bfeb9c7c
external/lvgl/src/misc/lv_utils.c	2272	3bf2782c33509353e48169e2b769283e8fab5bf4508d9500e7755
external/lvgl/src/misc/lv_utils.h	2254	85a27f6fff5bd2f686636f3f96f1675165701b1e6f762fc07a8e7
external/lvgl/src/osal/lv_cmsis_rtos2.c	4891	bf57eba6877405164ae3be3871b190e090b1375cb1eb0636f9c3c
external/lvgl/src/osal/lv_cmsis_rtos2.h	870	6fb1ef9a8fb9b5ce63c251ccb66b8f22d81375fc6264f8ba23dd3
external/lvgl/src/osal/lv_freertos.c	17555	524e27835ec9dd9e916e9ce67e39f026e83d11ae994d929d08b91
external/lvgl/src/osal/lv_freertos.h	2570	58fd3aaaf404170659c16cce87a510b2f14f0ee3b2d630d4fce6e
external/lvgl/src/osal/lv_linux.c	6692	05639308fad6f06a323506da514d357f450f76626ade2188d72bc
external/lvgl/src/osal/lv_linux.h	1045	d1f3d049bd68593971da951925413e7bf598af58f2062491172d1
external/lvgl/src/osal/lv_mqx.c	4041	1d4adb5fb4bc2a7fcc10ec2d0305a682ba0ae15f774f16cfbda49
external/lvgl/src/osal/lv_mqx.h	733	53f52015930e67b88807b5408432f2caf5a770db9601580e1e760
external/lvgl/src/osal/lv_os.c	1268	644bdea60d965b610efb2efc77c59c351991b8cc49ce4aac90900
external/lvgl/src/osal/lv_os.h	1541	1523589a733911ef70cc2322c692ab72cc725ef503529147595a7
external/lvgl/src/osal/lv_os_none.c	824	fce07fb098f6fcdc4af2c271aaeeefacbbe25b7644cbd74336a8a
external/lvgl/src/osal/lv_os_none.h	660	74dddc25a66bfa264eb6076996c046f8c90f4cffaf793ddcbc744
external/lvgl/src/osal/lv_os_private.h	6981	e6a069dd95c7bd620c103b4debcf3fc7c43e4cbe4e5de4d1b026d
external/lvgl/src/osal/lv_pthread.c	4159	8200ffa89ca9e72c9a1ec2a27ad4d0a5a9510aa9e4e79755cf1d7
external/lvgl/src/osal/lv_pthread.h	918	dc67af93628bb5a7548ade85252963e8f96c8084da4d5579e00aa
external/lvgl/src/osal/lv_rtthread.c	4308	0e47d2f7f6e0dee5a1889287e932f825555f84def68a24eca9687
external/lvgl/src/osal/lv_rtthread.h	803	91a9691e84901b3806237dae07273d795fbedc7baf61a7c7bdad9
external/lvgl/src/osal/lv_sdl2.c	4085	dc22aa730d09e9d6b28278aabb65ebdd512fa0f50221b8957c951
external/lvgl/src/osal/lv_sdl2.h	848	b4438bedf5cbef3c670239216f7d17161095f5b9ecc30a340291f
external/lvgl/src/osal/lv_windows.c	5168	e453c309d1cd5d7ca15ca8c19a981321c0f09f9bfa3a01f0d8b2f
external/lvgl/src/osal/lv_windows.h	800	83fbfdfb6c89ac3a743a819242a6e9406a62944f2e346908bb537
external/lvgl/src/others/file_explorer/lv_file_explorer.c	27380	80ffe04c1974669c23c5956c63ac8f979b7f0832a2a5b3123faaa
external/lvgl/src/others/file_explorer/lv_file_explorer.h	4201	88f779f2ab44abbb2a2ce35eb0ff37e13293a55c372210156632b

external/lvgl/src/others/file_explorer/lv_file_explorer_private.h	1631	ee2eec609ef20c702ffcd89ae52a88d7b234680127e57eb0c228d
external/lvgl/src/others/font_manager/lv_font_manager.c	19174	a5151f16c6b443503434e2a9652b37f83dbc380c76c3d7bea6400
external/lvgl/src/others/font_manager/lv_font_manager.h	3581	98fd33dfcedafde66cf5c3cb736a19936dca99405a59a16e87fad
external/lvgl/src/others/font_manager/lv_font_manager_recycle.c	4994	4c3058e4a8c578322b6f7885ab96c1f47329385a3a0dc611ed752
external/lvgl/src/others/font_manager/lv_font_manager_recycle.h	1934	70032d2d7cfd008faa9b125258f9becbd34f639b69a31247b74f2
external/lvgl/src/others/fragment/lv_fragment.c	4356	ac125bd0061620bd0b65114a5912f746105e498c9bbd6b5388e3f
external/lvgl/src/others/fragment/lv_fragment.h	8351	9c3c12ff88dbd08a62498057a388561487842681fb6e0b4d1f167
external/lvgl/src/others/fragment/lv_fragment_manager.c	8453	36edd107275c67c6b1d87d53eebadf68faeb8db6beb75b3f9b67e
external/lvgl/src/others/fragment/lv_fragment_private.h	1385	f9412421aad95a3fbe95ea9c719b1a1fdb123fafc15fdcb91e3a3
external/lvgl/src/others/gridnav/lv_gridnav.c	14610	302f684fc8cc8d6c6c30c0e9203794f097a10adeb32f2fc4ed32c
external/lvgl/src/others/gridnav/lv_gridnav.h	2574	65fb4ced703e4eb0359a6dc249fc5c61c27c708a522d0e3fc0fb6
external/lvgl/src/others/ime/lv_ime_pinyin.c	40355	7c69ed3ab9a23b0eb4ffff23f3475242cd990c69d065be27b9c91
external/lvgl/src/others/ime/lv_ime_pinyin.h	2792	c2a95e087a4dfa4c1bc3fa8db193fe3556bfa296b5bb938a86414
external/lvgl/src/others/ime/lv_ime_pinyin_private.h	1789	3ff2434a93cd32b0745da9974864af706acbe0c79f7da31c8163e
external/lvgl/src/others/imgfont/lv_imgfont.c	3002	b78cbf7b92746f3e9fce1ab6bc5411c628e8a6702964e8afe32ac
external/lvgl/src/others/imgfont/lv_imgfont.h	1444	134619f5278016d7406191816d506f97e3c9a3eb76abd6c48ca69
external/lvgl/src/others/monkey/lv_monkey.c	4803	553a89b1eea0a65d7cdc150a829273e4cb03cd1d7c98ed0c392b9
external/lvgl/src/others/monkey/lv_monkey.h	2422	4b40ab13480ae56fa55442341ee26266f0425dc77cc52466195d5
external/lvgl/src/others/monkey/lv_monkey_private.h	626	f2b18e5230d65ee0e4c1c9edf9936e1dcc24326ad83263778e7f9
external/lvgl/src/others/observer/lv_observer.c	33981	f2dda22b8c18796b49b73b60f2354969826bdb4c2cffb79d40035
external/lvgl/src/others/observer/lv_observer.h	24263	70395d0d596f81f91612ae4a967b1d13ad3a77f7cd66ca2d73b70
external/lvgl/src/others/observer/lv_observer_private.h	1863	ed1c33fcd98c2093a2d244b475b605ab3973df74cd7a56428f61a
external/lvgl/src/others/snapshot/lv_snapshot.c	6553	4e22d544380f72c0db8dc6c55fe25dfcbe26ab8b2c1c6f9761bfb
external/lvgl/src/others/snapshot/lv_snapshot.h	3265	267d83666b874e795f46928f87980e2a354ca21b54d2978644e9a
external/lvgl/src/others/sysmon/lv_sysmon.c	13894	2558b74ecef2fad07d79f790185bd5bdcffb3255d8436292901ae
external/lvgl/src/others/sysmon/lv_sysmon.h	2583	8fb01d8343f671f12b481dc16291cab3416d8db432afaf3eeea1e
external/lvgl/src/others/sysmon/lv_sysmon_private.h	2081	e7cc2404a6d1461f2bc86463441ebc7166caf0d94128abffa8dc0
external/lvgl/src/others/translation/lv_translation.c	9352	78ea3080c589a6d3c3aca740c956047602719c4d8fe84d035d625

external/lvgl/src/others/translation/lv_translation.h	4868	531ffe9bd555bee7b35bd7d35dd8a6aa58efff71a6951d2770f08
external/lvgl/src/others/translation/lv_translation_private.h	1132	6fbe61c001a6b4b13c7719658d5fa251224e2d6752767744a82ea
external/lvgl/src/others/vg_lite_tvg/vg_lite.h	69659	1a7bbe4f52957197646ee72db3d9d94acff123e720c2de9473b31
external/lvgl/src/others/vg_lite_tvg/vg_lite_matrix.c	3952	58d313b99765f1f460323610fc7a12c0104f473bd42d723bd2403
external/lvgl/src/others/vg_lite_tvg/vg_lite_tvg.cpp	101625	ff43b8c947d157f2a127cfc989d6e74816d7a357c2387a21812cd
external/lvgl/src/others/xml/lv_xml.c	35696	638486b538a70956de840b08a47bf8a2ac13b344a6130ebefa91e
external/lvgl/src/others/xml/lv_xml.h	4571	9f13145d7192c40125d11ae3d674c1cd46c53cf848b2897b07b9b
external/lvgl/src/others/xml/lv_xml_base_types.c	24588	419fc993d598f20f622f4a1a2d8ac24fe66262e09c3fdeb441c47
external/lvgl/src/others/xml/lv_xml_base_types.h	5210	6670fddddae3604136d537c1a99fdadcdd81d5b94c2ec735fa7d8
external/lvgl/src/others/xml/lv_xml_component.c	36609	f4e1af79f3d296cd2cefb65d0c491810acefc6dd391e2e4dfbb64
external/lvgl/src/others/xml/lv_xml_component.h	2324	ea2a2667acbd0f5395073f15fc09b4e7ffbe2f4ab3083e1eb7f40
external/lvgl/src/others/xml/lv_xml_component_private.h	1947	347c75b4f3d4421792333663b264c9bdd87fdeb51e7488750e493
external/lvgl/src/others/xml/lv_xml_load.c	10119	17b3e761cf894c8a4f1858ef914e7dadb792524b74873c3123a53
external/lvgl/src/others/xml/lv_xml_load.h	1930	83f15962f5cd38895c2ecaf196aac273f852a8323963ddc6e5e9e
external/lvgl/src/others/xml/lv_xml_load_private.h	679	9228dd097e2d64c96044ce9c062434535281d087bd295ebd0dc67
external/lvgl/src/others/xml/lv_xml_parser.c	3428	670548b6f14b04d1e7ba43d4656bb0194f87ebc5c80ae0443e9a2
external/lvgl/src/others/xml/lv_xml_parser.h	1991	ffb143c282be817c381cc09f0f4c1eca2d6aab7f01b34f4b6ce03
external/lvgl/src/others/xml/lv_xml_private.h	1454	56e3da1d485a627f712ced3072c4277923826946698b4c77e8d2e
external/lvgl/src/others/xml/lv_xml_style.c	16539	173349148203f5b3a3d4c34be6efb3c80da6e81a93e84283d2a9a
external/lvgl/src/others/xml/lv_xml_style.h	2085	913a9574d8aee3a823456d2877b1d54e837a35dc7c20bca6d7022
external/lvgl/src/others/xml/lv_xml_translation.c	5233	1a5f601a2527022330d1588c0a64dd49f88621e390023d314c180
external/lvgl/src/others/xml/lv_xml_translation.h	1037	68a230ef54a3965db037ae8ecad04dffa9c258ab776e12e96aa92
external/lvgl/src/others/xml/lv_xml_update.c	3227	c3285a86dedf65898a6dfc4b471ea99ac31dfb3fb1b1b0fba9406
external/lvgl/src/others/xml/lv_xml_update.h	1007	19c921bc1da120189782bfc8d6328b1d52fe1b356203e0696d998
external/lvgl/src/others/xml/lv_xml_utils.c	6759	68474a9f7393010dfe5654d867b0ed2b0c228cb2f15049ec81c25
external/lvgl/src/others/xml/lv_xml_utils.h	2143	64de4da79c701e4aa90aa43ba501e8e6e3d021cf3f9524a25f8b7
external/lvgl/src/others/xml/lv_xml_widget.c	2688	262a9c06ee5480956408d181823538541664cf1c56735ca546993
external/lvgl/src/others/xml/lv_xml_widget.h	2664	5b43e423a2e6589705c1442ae743a8e260bee30d35b8b4cd11841
external/lvgl/src/others/xml/parsers/lv_xml_arc_parser.c	2914	2b4601232835e63bcba60472f3da8d6ae7ad000a802d5d5993bc1
external/lvgl/src/others/xml/parsers/lv_xml_arc_parser.h	712	746bab9adb668054711195feffd5b2e7115e93cb5f53b532027f3

external/lvgl/src/others/xml/parsers/lv_xml_bar_parser.c	3556	d3f19fd8d5c0b6457e61a73cc3adba0629f26ba141837a40617e9
external/lvgl/src/others/xml/parsers/lv_xml_bar_parser.h	712	c32248230e0c882e1cdb63808a78f8f2d41cb376c1e6948c8863f
external/lvgl/src/others/xml/parsers/lv_xml_button_parser.c	1132	6e92a92a57462b556b73d597d78cd7060448c2816d1ae12faccbf
external/lvgl/src/others/xml/parsers/lv_xml_button_parser.h	734	d3e3fefc677e5a1004e00560dd9a033f47a72b1c54b5c57aead09
external/lvgl/src/others/xml/parsers/lv_xml_buttonmatrix_parser.c	6616	9826dad5b1c2623a762b188cef9b92ef7a81d8f0f70cb659cf628
external/lvgl/src/others/xml/parsers/lv_xml_buttonmatrix_parser.h	776	21cac6dc02e1a1e0aeca715f765ec27ec8b03fd7afdea7f7afd9c
external/lvgl/src/others/xml/parsers/lv_xml_calendar_parser.c	2753	3994df3d3c5f4de058bdc72e29d644659c2dd6543548f52b8f924
external/lvgl/src/others/xml/parsers/lv_xml_calendar_parser.h	1111	8f36da0c03896acec0ea9ae3ad4dbf15340685a80b9f37dc6cbef
external/lvgl/src/others/xml/parsers/lv_xml_canvas_parser.c	1115	060956284027fd5e164a14800bd1dbbc29160129063a7d37215a8
external/lvgl/src/others/xml/parsers/lv_xml_canvas_parser.h	717	acf0e5863d2ad25edb01e5facec0f56036567e7176f1cc97b40bf
external/lvgl/src/others/xml/parsers/lv_xml_chart_parser.c	6497	dc992ba2bedae2ec9b22478202354686f2bf85ac5aa9244a5e06b
external/lvgl/src/others/xml/parsers/lv_xml_chart_parser.h	1234	4007a0ce34b5ef2d2905465f0f9f780b17a411f0a238faf22e38a
external/lvgl/src/others/xml/parsers/lv_xml_checkbox_parser.c	1366	08e8dd64ab869e32546300919b2ba0adaf48d9f633a8ca3bb64c5
external/lvgl/src/others/xml/parsers/lv_xml_checkbox_parser.h	728	6106e4d102cdbd4fa9e019b2097167845cd5a601e18a30db20a5b
external/lvgl/src/others/xml/parsers/lv_xml_dropdown_parser.c	2381	266cdf46bbce5bcdfdf1ebf84f3e8a2626f2be1ffe71bfedb0659
external/lvgl/src/others/xml/parsers/lv_xml_dropdown_parser.h	919	d3b9d1d4fe9d04ee474fec98f1f173d6eaff1ba5f0bd2b0d458ee
external/lvgl/src/others/xml/parsers/lv_xml_image_parser.c	3392	b1e97d1b8567e710acba67860b61b8ab9ef79b0e7964349f1c968
external/lvgl/src/others/xml/parsers/lv_xml_image_parser.h	775	e99ebdcce89fd5a637503ac64ab6886a8ad8092474eaaa6fd6d73
external/lvgl/src/others/xml/parsers/lv_xml_keyboard_parser.c	2524	907d36dccd1306548c527e431302cc2a1d3135592d21f5735fe8d
external/lvgl/src/others/xml/parsers/lv_xml_keyboard_parser.h	748	e461872ac793348f2680492cb2c98820ef1079c55ae2edcdbd82c

external/lvgl/src/others/xml/parsers/lv_xml_label_parser.c	2820	d487b34f2e2a8abe5d7ead7c4920610c82d07be935c55c93cac8b
external/lvgl/src/others/xml/parsers/lv_xml_label_parser.h	727	41160d0a2b65368356e14a0e5b25f3cff592c4ee451dffa394222
external/lvgl/src/others/xml/parsers/lv_xml_obj_parser.c	45589	e07fe3f0317abfa569aae6cbd1866d1209165561c2b186272ad78
external/lvgl/src/others/xml/parsers/lv_xml_obj_parser.h	3060	e49e1d2f9c66c4dd96f1c095b468b8b114196a5eb31b87c5cf0f2
external/lvgl/src/others/xml/parsers/lv_xml_qrcode_parser.c	1764	651e063dbb32b748a515dc5774ccbb2afe4f13c803ad53e49abc2
external/lvgl/src/others/xml/parsers/lv_xml_qrcode_parser.h	736	b50aedff5997d7d2e0151a0159637094a338333fe20976b6c1791
external/lvgl/src/others/xml/parsers/lv_xml_roller_parser.c	2762	591db8e3a88bedbd622e5e754ddf7a28f2fb5a25e130c56782806
external/lvgl/src/others/xml/parsers/lv_xml_roller_parser.h	733	dac7485a5071fb5afcd7a7e1c34e921ce59d4805057fa1f344fe3
external/lvgl/src/others/xml/parsers/lv_xml_scale_parser.c	5343	e99150ecc891151e1e4ae84de9685b37c545670902374acbf6e95
external/lvgl/src/others/xml/parsers/lv_xml_scale_parser.h	898	3cb35c87cb9f7967df523f55c7c91abf949e52eacccbd8b1f6fe8
external/lvgl/src/others/xml/parsers/lv_xml_slider_parser.c	3634	f76062dd01459681826cad65a7dd27f9ebedd8212b9277978dccb
external/lvgl/src/others/xml/parsers/lv_xml_slider_parser.h	733	a22820d86c1eec078ce06018394f2de70511ac412b5e3668e12d5
external/lvgl/src/others/xml/parsers/lv_xml_spangroup_parser.c	3539	56e2ee37264bf05d2a473b416903275f8402c0e3a4c81fff3ffd3
external/lvgl/src/others/xml/parsers/lv_xml_spangroup_parser.h	923	ce1e357c8140946aa0066c38711640886bde56645c97aa9437c33
external/lvgl/src/others/xml/parsers/lv_xml_spinbox_parser.c	2331	4b3c7701fc56a094a0c7dfa10134c343500f102f0538a51f7fe83
external/lvgl/src/others/xml/parsers/lv_xml_spinbox_parser.h	740	8bc90a095a5b8a57da1d4a2869d442fd711996c15e5387ce82aa7
external/lvgl/src/others/xml/parsers/lv_xml_switch_parser.c	1911	e77bec91a3bef1b2516b25759dda0ca2ccd0d8987c5be799db6b6
external/lvgl/src/others/xml/parsers/lv_xml_switch_parser.h	716	c0c2efe53d8cc9ad3c453ac944beed0270a50a005170760c9656b
external/lvgl/src/others/xml/parsers/lv_xml_table_parser.c	4050	3bbb241433f6289aebf417d242bf0e12a47bada70cffc19a6b04a
external/lvgl/src/others/xml/parsers/lv_xml_table_parser.h	1063	935d546c7006fb4d98ae522574ca98b522521de03b2b52360edd2

external/lvgl/src/others/xml/parsers/lv_xml_tabview_parser.c	3295	7dd6d99c0c2e41a803a7aa803bfb79c31addf49425b73a95f278c
external/lvgl/src/others/xml/parsers/lv_xml_tabview_parser.h	1269	dda379f1da9de438a76ed75bf225cb2755cb7e7b442e5bf499687
external/lvgl/src/others/xml/parsers/lv_xml_textarea_parser.c	2020	927052390280e70d68608baf60e74cd83b9ff3bae81aaeaa8a590
external/lvgl/src/others/xml/parsers/lv_xml_textarea_parser.h	748	ea90fbc60bd608104f9f5c7850413766ad3f68b995694372180be
external/lvgl/src/stdlib/builtin/lv_mem_core_builtin.c	6431	a186e9a3ccb47906f960b8b50cf9a38727e2d4c6f2cab0a048bc2
external/lvgl/src/stdlib/builtin/lv_sprintf_builtin.c	30272	4efe0afd79c703eaef6f8629194aa79831480a5e7acfa9ecbc1c1
external/lvgl/src/stdlib/builtin/lv_string_builtin.c	6596	025a30ff53e5d8b6a44a2ca8f1247f2f39f0b1ebb486ebdb101ed
external/lvgl/src/stdlib/builtin/lv_tlsf.c	38016	fb67f7034a5fb9f4176fa2d0c7c31937a60c955e0dd5351dee241
external/lvgl/src/stdlib/builtin/lv_tlsf.h	3742	d133a30d9dfdd2cec4a6e26d3124aa9d079d397d276df338fd836
external/lvgl/src/stdlib/builtin/lv_tlsf_private.h	862	11d7dc0fed370a35c8d65bcb685bc32dd82ec0efa6f65fefceb60
external/lvgl/src/stdlib/clib/lv_mem_core_clib.c	1498	f626a9ea8cf60dee9ef8ecc0d8032286ab7ee420e1d7605129c27
external/lvgl/src/stdlib/clib/lv_sprintf_clib.c	1082	b5053f9c6b8f8398eb280fe64a034aa153fdfdd83b7795356bf4b
external/lvgl/src/stdlib/clib/lv_string_clib.c	2884	76cac93eb37945ecfb835ba8c9e21f58d6adefa26cabed6d63aef
external/lvgl/src/stdlib/lv_mem.c	4379	00d8226e661600d059aecad01f28fadb2bc469cd7cd87a84c7a06
external/lvgl/src/stdlib/lv_mem.h	4272	93adf94a575981657bd15176488e833aa18065acc42604dc72a61
external/lvgl/src/stdlib/lv_mem_private.h	554	9201a978bed0a0409a8c08f31080a4a56f6189ea2e8336b134517
external/lvgl/src/stdlib/lv_sprintf.h	1410	43f6f5359097c346c78e09eb3827a0e5bbe62260af57150122e21
external/lvgl/src/stdlib/lv_string.h	7741	668a5be6caffb6cf21964a47d9dffd18b3d937c30080d8af5099e
external/lvgl/src/stdlib/micropython/lv_mem_core_micropython.c	1802	e4099b52b4f4ffe22caf1eb7f6a7b892e2ad3dcf8852ac421fb5f
external/lvgl/src/stdlib/rtthread/lv_mem_core_rtthread.c	1637	da0fe8d34f416ca1d1f44de85e5910f0dd455812a9b9ac0ae5d9e
external/lvgl/src/stdlib/rtthread/lv_sprintf_rtthread.c	1225	d7599be3b9416ef5ae2bfcfa1db51560bca20c8d9ea1b8b9f34db
external/lvgl/src/stdlib/rtthread/lv_string_rtthread.c	3251	544daafe2e237bc70fb803d83ed9b8288278af46375c1f5101b9f
external/lvgl/src/stdlib/uefi/lv_mem_core_uefi.c	2327	e26069f90a8135cb07442ea75f8f4cfcdcf61be09f63d96d5564b
external/lvgl/src/themes/default/lv_theme_default.c	54412	413abb720f698ec622411b84cdaea7d2b245aba763e901cc61f50
external/lvgl/src/themes/default/lv_theme_default.h	1448	2f569db7bcca41560cef96b28dd20274412d6784dd73810378c46
external/lvgl/src/themes/lv_theme.c	3005	4b8f12f433517cb02e211621b7cae73146f94cc05004572b86577
external/lvgl/src/themes/lv_theme.h	2555	0897d702dddfcbe00a84b216474ba4cd7d99f531e9dae7560c3d7
external/lvgl/src/themes/lv_theme_private.h	1012	be35765d5d3c23b9c06bd2a6ba17f8edc7263eab33bacf2437ffe
external/lvgl/src/themes/mono/lv_theme_mono.c	21032	1d0e8f548ec9b9f266523191f215f114a4611ed527420dac81b68

external/lvgl/src/themes/mono/lv_theme_mono.h	1209	14b152f62b6d98815d47f4a953ab35c30cfa646c4524a29d2e4c3
external/lvgl/src/themes/simple/lv_theme_simple.c	13953	33887b4dc97e020f1f6d69a6fa4100c0d7c1a1ab48ea161dd2c65
external/lvgl/src/themes/simple/lv_theme_simple.h	1172	20e1c612ba2f2d1383c936b9d544da4d29e25aa45085de520a79f
external/lvgl/src/tick/lv_tick.c	2548	3a72269e2c36f37119c9ce0baba438f1f8b88fa62199e9ef43851
external/lvgl/src/tick/lv_tick.h	2409	1e77b5ba509c0f8aa24b7f6418d00084cb033facb1a47ff926c3e
external/lvgl/src/tick/lv_tick_private.h	725	a4250ae24958892696061c06e52b46bb64047044309de369759d1
external/lvgl/src/widgets/3dtexture/lv_3dtexture.c	3753	4f3fe1d1a5a60af87fdc4b374e0a5ed4a854f068c623ee92db66c
external/lvgl/src/widgets/3dtexture/lv_3dtexture.h	1815	8d714c108fb9bade8c7d708272279501de3e65f25c13ce8280aba
external/lvgl/src/widgets/3dtexture/lv_3dtexture_private.h	803	ce6b78b3eedf2a6a317abb615c465eb9b19e8950236ff01025b15
external/lvgl/src/widgets/animimage/lv_animimage.c	7146	60c52250234aad96eaa4ddc3678318ac86baad646273a68c5db30
external/lvgl/src/widgets/animimage/lv_animimage.h	5188	e9ebeb5adcafb34920339dd42014502a68a49e6d2eeb83b68fd41
external/lvgl/src/widgets/animimage/lv_animimage_private.h	892	a2b07314afd9beba8a6a3bffb5978cfa700873903ae3571820500
external/lvgl/src/widgets/arc/lv_arc.c	36673	ed6347f5ff868930ef45d245a98023e8e3ae17ea6f9fa273ad409
external/lvgl/src/widgets/arc/lv_arc.h	8111	c7a8609222fa28694a1646b55c5fba90f202a9589bdb3f8da3f8f
external/lvgl/src/widgets/arc/lv_arc_private.h	1637	1eb31bc2b518360a0f6247f3d9cc66afc720ad9644d7e10e7dde9
external/lvgl/src/widgets/arclabel/lv_arclabel.c	19709	3c9ab9a2d8abf2ec6f855d4499a8aa1170815ef8f72fbc2163554
external/lvgl/src/widgets/arclabel/lv_arclabel.h	7455	282031fe85d46abc3ca6f2ff80b14b71fef305189872415c18cbe
external/lvgl/src/widgets/arclabel/lv_arclabel_private.h	2212	ab9ea6aadc652d192ceed23f4cce9998dbea39c9b674a2cdf5f31
external/lvgl/src/widgets/bar/lv_bar.c	24258	ac24457e63f831a7fdab81e44ee669591073a09fff0ca35f16f1c
external/lvgl/src/widgets/bar/lv_bar.h	4731	fb739ad5392370b762532c0a61df1c3ef7a0ff77ced9085849e38
external/lvgl/src/widgets/bar/lv_bar_private.h	1411	6c55351866f55b2401680065ed1bb517286fb2bc4f4963fc4b163
external/lvgl/src/widgets/button/lv_button.c	1594	792b2e7d2d440b9b099ef5c9dab62c2e6fd62bf018f30825f2417
external/lvgl/src/widgets/button/lv_button.h	849	a5373650a63c4818004036d109c72886d1067acd77901cc5be675
external/lvgl/src/widgets/button/lv_button_private.h	777	3c58c69ee43b7c119eb5026a63789ee7905586cb58fbfc6d3746e
external/lvgl/src/widgets/buttonmatrix/lv_buttonmatrix.c	39637	899105e43c1098368b562f6bc8e88ea2db16d4ad25bca69f993e8
external/lvgl/src/widgets/buttonmatrix/lv_buttonmatrix.h	10309	2a58ddd955def471bcf46309a46f14143edcf6c407e65d84e8375
external/lvgl/src/widgets/buttonmatrix/lv_buttonmatrix_private.h	1534	217c197a0def005b7eefb9447a4856ae8137754a8d208d518d5ae
external/lvgl/src/widgets/calendar/lv_calendar.c	17049	77931dc4509fd2e64d6b569802261524927 98ae48f23af6e616c7
external/lvgl/src/widgets/calendar/lv_calendar.h	5516	1f102a7a1a167e39d4682fc5a6f8bafa2c09b28f9b6caf810c112

external/lvgl/src/widgets/calendar/lv_calendar_chinese.c	11127	3c5adca5a2ea8184cd6c0399844907ce7bec850ed7ac4b97b517b
external/lvgl/src/widgets/calendar/lv_calendar_chinese.h	1596	d6819a00581c5b5c72d15024a42b0a5109949045d127de3aa2210
external/lvgl/src/widgets/calendar/lv_calendar_header_arrow.c	4644	5a42b9eab6151336fad0c045c4f5ce7273cdc4b86e50e481db2d0
external/lvgl/src/widgets/calendar/lv_calendar_header_arrow.h	1014	ed322fb42ac911ccbc63f7f1e37a2ae2c5f8465cbd2685e51ceb4
external/lvgl/src/widgets/calendar/lv_calendar_header_dropdown.c	6233	4892e5e58cb97eabc4bec838398b85caf329f0fc7b02a88d1a1ef
external/lvgl/src/widgets/calendar/lv_calendar_header_dropdown.h	1629	c39ffbacd33237f45ac3c53a3786991e9be5a7ee4177f1ae59954
external/lvgl/src/widgets/calendar/lv_calendar_private.h	1707	8780ecad6b85c35f734a2a20da5b8311c84d62d5c2ac33219f74f
external/lvgl/src/widgets/canvas/lv_canvas.c	12315	f5e65f05b3ccbcbe2e35920a924c1e61b55e696cdc6e3f86b2c97
external/lvgl/src/widgets/canvas/lv_canvas.h	5971	2b547c53a7d3ea96963b225ebb10bda3779477d90a5a44cd8819f
external/lvgl/src/widgets/canvas/lv_canvas_private.h	791	f6aa387198c498914abad0ac058e025f9842d479b5aabd7880929
external/lvgl/src/widgets/chart/lv_chart.c	55633	5ab1050c6981a037f915295f5f9a2f536a8679b2a78ae127f437d
external/lvgl/src/widgets/chart/lv_chart.h	15420	b0e2d896e06fb3ed7919aa96dc27c34894714a942a3f06ecfdc82
external/lvgl/src/widgets/chart/lv_chart_private.h	1852	6e53ca0b9241edb3c15d9e470b700f5cf4cbfd88f950bc7aae023
external/lvgl/src/widgets/checkbox/lv_checkbox.c	9269	c2a2c97b5012d542e783f1246d3125b515a0cee5c0a8d9fce941a
external/lvgl/src/widgets/checkbox/lv_checkbox.h	1762	5448695c34a149458f04d10cb55a1f00a529c602da9cf3e2c8b13
external/lvgl/src/widgets/checkbox/lv_checkbox_private.h	838	eb48a09475723e95b84995c1828571c700bc350235f901f66d58a
external/lvgl/src/widgets/dropdown/lv_dropdown.c	42213	3b9d793dbe4ac37c7879d2b1b93e8df12db4602d7d6ff6a37712d
external/lvgl/src/widgets/dropdown/lv_dropdown.h	8473	9724afc5ee20d6311487fafda59f001bfc57039d615c46eaefb53
external/lvgl/src/widgets/dropdown/lv_dropdown_private.h	1753	c664192b033e10c3b1ee6916d71f2dbfe07d50309955944adcc08
external/lvgl/src/widgets/image/lv_image.c	33426	b8fa1ff90d3199c417d4b3e1ae474ce0100d2b7df265c8fca83c7
external/lvgl/src/widgets/image/lv_image.h	11944	4674449acf594ee03e1f9ba7c24795e86fa31e63c7682292ef774
external/lvgl/src/widgets/image/lv_image_private.h	1776	eacf3dde3339bf0b0150eaee07f5bcdb2099eba28daefebf1dddc
external/lvgl/src/widgets/imagebutton/lv_imagebutton.c	11722	4451d5dd7b1348860782f2dbe164d99503076d4e6626d2b513a8a
external/lvgl/src/widgets/imagebutton/lv_imagebutton.h	3663	64d1bbebd3a490a8e9c57bc825059c8379a21b88fbac2efaa89c1
external/lvgl/src/widgets/imagebutton/lv_imagebutton_private.h	1208	b42f109ef7dfc5201bd62232a0463ef181743a318ce9aac35aee7
external/lvgl/src/widgets/keyboard/lv_keyboard.c	18283	ebeaf92da138676a72c705db67d9f2b56f6658e646047fadb2d9b

external/lvgl/src/widgets/keyboard/lv_keyboard.h	5281	077da9d353ec8275059a6ed4643c781df8298ee351556a9e376aa
external/lvgl/src/widgets/keyboard/lv_keyboard_private.h	971	3dca68ab692f6570df240e0f9706fabb2d3850a313061f167ad73
external/lvgl/src/widgets/label/lv_label.c	50177	82dc5fe50869a85312b1cefd37017293a024221ab4c71ddf06b43
external/lvgl/src/widgets/label/lv_label.h	10099	720121d2292b43350bc1e986e7f7ece38e5d6be8042cff1c45e36
external/lvgl/src/widgets/label/lv_label_private.h	1783	b0d96fbbd7d216f926a673b98a03f79a7c8fd3f55890b71f1b4c4
external/lvgl/src/widgets/led/lv_led.c	5908	1d1a3b3f296289d0530461544fb2cc1eb6042f1ee50d14c50e712
external/lvgl/src/widgets/led/lv_led.h	1983	bd8f1531b0df411eeb04c146189879a75835e185c41df38ffda07
external/lvgl/src/widgets/led/lv_led_private.h	787	5ddc7b9581f7813bc3c03a07b20c06e31bee20e44ef9d1e958e14
external/lvgl/src/widgets/line/lv_line.c	7277	e3f0318f970cdbf33d4f0a151323bb1a47c3afa0fec83bec0b505
external/lvgl/src/widgets/line/lv_line.h	3409	fbd0140da04c4c90c0f9dcf2671820bb4708897571ab8b4460d5e
external/lvgl/src/widgets/line/lv_line_private.h	1153	bfd93ccc72f5769b03748c14323e95a87a25ee7b4cf7e67869766
external/lvgl/src/widgets/list/lv_list.c	3364	5d3a883118cf688f14683b63e13dcc648da1334de2658979aed01
external/lvgl/src/widgets/list/lv_list.h	2284	0c2d57924af52e016372523c9f43f89269c180f487a96000d661b
external/lvgl/src/widgets/lottie/lv_lottie.c	7397	9e014662a251372fe9c40c4028fc76bf7b7178f6b42034b6b83b3
external/lvgl/src/widgets/lottie/lv_lottie.h	2646	8eeb596ffcf969b25954db0fe7aa68b72d84cfe8b4e4e01552b30
external/lvgl/src/widgets/lottie/lv_lottie_private.h	995	5e18bb1ee0349716d4ab02c420ccd6ee1ab03666662f8b5c42726
external/lvgl/src/widgets/menu/lv_menu.c	26353	d71f2f6db8c65c3ba9a92791104a20204f39a001c1a642772db4f
external/lvgl/src/widgets/menu/lv_menu.h	6851	e15b496883a185d508467913856a26d11d3f08848c0c7d12aecd9
external/lvgl/src/widgets/menu/lv_menu_private.h	1774	83b80efa18dc9f46bbaba2cb864f42c4b083341774bda30fd61a5
external/lvgl/src/widgets/msgbox/lv_msgbox.c	8862	62bd69088fbdbe20ee4b56a529d209fe0ef9d8de136a275e76da6
external/lvgl/src/widgets/msgbox/lv_msgbox.h	3731	b94c9f7dc23c39b70d085919f27968cfd8692d6dc433fd6f1f120
external/lvgl/src/widgets/msgbox/lv_msgbox_private.h	859	78088a5fd5982fa15b82593193152bc6b92c96184edfb1085be15
external/lvgl/src/widgets/objx_templ/lv_objx_templ.c	3338	a94e852c094ff6c2c783b3ae987f4ac86459b6656c600f4823e99
external/lvgl/src/widgets/objx_templ/lv_objx_templ.h	1584	8861758b3c67c6836a53f69d186a746a1bd4d9ad19f8715819828
external/lvgl/src/widgets/property/lv_animimage_properties.c	694	855918a1be03919099a0394f89b9cb0cecb617c4f4b154de3ce31
external/lvgl/src/widgets/property/lv_dropdown_properties.c	1000	c46c467ae40faba6e7e6daaf7af2b1988613f4f0c1c43435acaf0
external/lvgl/src/widgets/property/lv_image_properties.c	1071	b7ce94f64743a8b26d74a31fca92f2c9d9ca54ef1c6ae2b785438
external/lvgl/src/widgets/property/lv_keyboard_properties.c	686	42db2b3f7e82d0b02aec7b11a55b089d9caf14b63b4db0746e383
external/lvgl/src/widgets/property/lv_label_properties.c	669	41a77ebb1f97559d8ebd37c627751ca52eba4cbdca083e1b69ff3

external/lvgl/src/widgets/property/lv_obj_properties.c	5139	9fb7af523a72a3fb8f7ebc939def3af1d7c01c34781d6afc85e3a
external/lvgl/src/widgets/property/lv_obj_property_names.h	901	129be4260ac3d94c7637169a60fcd76b63cd8a30d3206af72a23f
external/lvgl/src/widgets/property/lv_roller_properties.c	607	bf76dc63a7ddd0ad401c6e327d4f6d672569cc2ef29887aad74b7
external/lvgl/src/widgets/property/lv_slider_properties.c	911	6e3189f68180dd5b90c6a9a7f73ca32b048a4dd325d33a76743da
external/lvgl/src/widgets/property/lv_style_properties.c	8255	e40d9ddba9701a2a93b8fbe702a1517d9971d099cc1b8f097a62b
external/lvgl/src/widgets/property/lv_style_properties.h	12998	825c4877999765928e72c95992e8355ae7577a3b6d92e8af596ec
external/lvgl/src/widgets/property/lv_textarea_properties.c	1453	45c88147f26cbb85a28a7a87dfc2ac70345ba55e908a2c112f444
external/lvgl/src/widgets/roller/lv_roller.c	34299	91dda7dc1cfab908b9f5f665f098cbfdc2caf3fd3518c346f52ed
external/lvgl/src/widgets/roller/lv_roller.h	4890	5780babd0eff95c207a48480ffc7f20cb94cee9bd5bc6860abe84
external/lvgl/src/widgets/roller/lv_roller_private.h	1071	11a1aabae67e273839a47b00c56ab8726a44746600616f5254982
external/lvgl/src/widgets/scale/lv_scale.c	67684	c25d05846b0d02e93b0a2b1c2dffa19547d28e432d4bcd060f80f
external/lvgl/src/widgets/scale/lv_scale.h	12220	68af8395c779997cd022fc26aa002d83f42ca06755edb44750900
external/lvgl/src/widgets/scale/lv_scale_private.h	4169	f53ea6ffcea11e323b373813ae450d24b3ef1985565199ebd6f2f
external/lvgl/src/widgets/slider/lv_slider.c	24686	04f85af6888db4ebcd1f27b9187c242f3395361025b0ced47eb83
external/lvgl/src/widgets/slider/lv_slider.h	6033	407170590001420262cb00847316969f286c68eac3531d43a0aa9
external/lvgl/src/widgets/slider/lv_slider_private.h	1082	5e260e41b42cf3e29a852dd4c86e828a6af733972ef72c60ef0f1
external/lvgl/src/widgets/span/lv_span.c	48073	f596e2284c87a2108e232bf80a61a83d8e41bb22d5036916375d7
external/lvgl/src/widgets/span/lv_span.h	11701	bac0ef29b4202c37603260d66f8d2f3b66b84b20d152d7c31ba46
external/lvgl/src/widgets/span/lv_span_private.h	1375	6a69d21b844a8061f03b04af3e00e0134d0448884efd567a2f3a9
external/lvgl/src/widgets/spinbox/lv_spinbox.c	16494	364ef15e72d0cd555166422ae2245008bd489c80305c849fd8477
external/lvgl/src/widgets/spinbox/lv_spinbox.h	5238	ec782e8b4d9ce4ff4c1d2be2ecfefac1a687b8bdddbcfb35018ff
external/lvgl/src/widgets/spinbox/lv_spinbox_private.h	1215	c369823ecae7dbc56cbaa603ba6306dfc3c050d80d931e3c859d0
external/lvgl/src/widgets/spinner/lv_spinner.c	2631	2481c33fbdbb5b9e960fb906636bb021f388bab62ebadfd58be87
external/lvgl/src/widgets/spinner/lv_spinner.h	1307	d921ef516b71dcaa4c6b22e9b4cee0ef97656061026ae4de462a1
external/lvgl/src/widgets/switch/lv_switch.c	10460	042cb210c0ae31de121c228620db34d4e0ef09a77c0a6846540a4
external/lvgl/src/widgets/switch/lv_switch.h	1775	9e63c9ac9d3ff66f2cd3d4678f4dec98241c4f54857792f22bbda
external/lvgl/src/widgets/switch/lv_switch_private.h	875	4779d1a427c11cf549b2dae16d7e1b6e01cd6eda6bceb2bc784a9
external/lvgl/src/widgets/table/lv_table.c	39006	a4c33f085ea91e6520dfd991d9fdd2bb687e582f8d5672e016156
external/lvgl/src/widgets/table/lv_table.h	6898	c175832c138c0bfd686cbba880e21ced000bdfd37e9591a62c7c3

external/lvgl/src/widgets/table/lv_table_private.h	1061	bb00879178b9e1ee5ed436ddafbbfb1931bd8096a2479036ad44e
external/lvgl/src/widgets/tabview/lv_tabview.c	11298	b60bab81372de5c65cb1f3b97b374fe95bd7e8747b774961c778e
external/lvgl/src/widgets/tabview/lv_tabview.h	3321	27ab978c44ef10a9de1b797a0841a525168997c4cf83238422c65
external/lvgl/src/widgets/tabview/lv_tabview_private.h	819	f2b48583ddc5a0f548c408833791fb165dafe29abab8c834a18a6
external/lvgl/src/widgets/textarea/lv_textarea.c	47048	a25877928de5cfd36185d03bd8b47871b7172af526cff8d6e45ca
external/lvgl/src/widgets/textarea/lv_textarea.h	11124	e4d4fbc93a9ef1f967920caddfe67f93baeccc4eaefa7397c2aac
external/lvgl/src/widgets/textarea/lv_textarea_private.h	2510	f698207fc5a265310c7831f0be8f56a36bd4b1e4f229c8b314952
external/lvgl/src/widgets/tileview/lv_tileview.c	5448	d170b80b605bd160a4a26e289305c647c08653dcabacf0e2f2447
external/lvgl/src/widgets/tileview/lv_tileview.h	2346	4a57c8b7b68506b248d02e48b4b51d16de812f3ccaa652a74f649
external/lvgl/src/widgets/tileview/lv_tileview_private.h	876	9f66c4d9e7556d6580772c618500a3519854264b159b426608597
external/lvgl/src/widgets/win/lv_win.c	2785	b134782018d2c18dcce20c2158ad194eee9eb1aafae7696ecaa08
external/lvgl/src/widgets/win/lv_win.h	1739	8ff6fdd0a92320f6ffb6933c62ad4c94ca3e4d48a1657ab652208
external/lvgl/src/widgets/win/lv_win_private.h	742	c3c7d3eb71d15f58980eb028eb719957501a34af5cf2fb00da54d
external/optiga-trust-m/config/mbedtls_default_config.h	44543	5d2d35534e9e4b30148baf3d2349f137c010e3a0f4f7d6d813245
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/Hacl_Curve25519.h	1096	463d56088a31cbacf694eb20a7397e1689da0ba5fc0b37445784c
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/everest.h	8235	ddc6c96ec9598fe10c389c155467d4a2cdedf488da8d218630344
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlib.h	985	05db27dda38b43f0d9745a36e20f6ec5ce18fd6f8c130d64c3b8f
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlib/FStar_UInt128.h	7003	4b1ee5b71376af6e6d2f80303b7fdda4427cdf0be8c153dd68212
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlib/FStar_UInt64_FStar_UInt32_FStar_UInt16_FStar_UInt8.h	10800	32ba176eeced425da0d539d3a406275bb880cc78fa67c5e3a4937
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/c_endianness.h	6747	98dfac843cae9138c543d90e900029a04cab0a990c714e5d9edc5
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/builtin.h	471	836d849eb5f824531d5c4e9d3bec50d203461b672b868d17f8f61
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/callconv.h	1416	d62185807f5b016b46f9fd10bcf5ec1debdaba721909e4c59c8b1
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/compat.h	1346	7304c71089f76077ae2b5cbcc9493512bb71c8a615bbec5eb3823
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/debug.h	3056	f154ee91dee1dba31c1039d28b3b39edd9d08b20abac62c33ef6c
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/target.h	3672	923a6430d90e681115c7467dc553a54750292da2d5119b40e2d14

external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/types.h	2009	c3def1c71d571f97be579e535ad79f892be998b1b5702bd686089
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/kremlin/internal/wasmsupport.h	237	6e1b236e46ba8f6df8b503a0ec4283b795db56950eb24623c9680
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/vs2010/Hacl_Curve25519.h	1078	c2e5a6ee616f8410ce8581dd1740a5b3d56dd036904a24ec431bb
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/vs2010/inttypes.h	1029	485a92f0e19ee0f9a9ea3033f995162d4083a065ef73b004ab57f
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/vs2010/stdbool.h	960	a3e67d667976f94bda628738440fd2b8110eccb49cac509ab4bd2
external/optiga-trust-m/external/mbedtls/3rdparty/everest/include/everest/x25519.h	6225	da28aebd8413ebfbe7b0d8069d40ab7cebf24725b5c41c83a3dbd
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/Hacl_Curve25519.c	22405	ff662cfae5e65d968d487a687db5fdc05c357ec9b2035ac9e380c
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/Hacl_Curve25519_joined.c	1261	3a208016cbdb5a5368562c59a346f3b742ea8adf00f809dfafdf7
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/everest.c	3718	2d7f6fccc2e11f3eafea2e606f23ef57d55fcc83f346b8b178a4c
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/kremlib/FStar_UInt128_extracted.c	14010	b75508632b4729497ae042851e3e3b996a93e2ef5ad0688785931
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/kremlib/FStar_UInt64_FStar_UInt32_FStar_UInt16_FStar_UInt8.c	5314	9a3504b115dcc34581dff4dd1f4adb96f7119d2d4d5e0612fc034
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/legacy/Hacl_Curve25519.c	24161	fd769ba72bed1ca61a4e286d7ef1751931fda5707881df514335f
external/optiga-trust-m/external/mbedtls/3rdparty/everest/library/x25519.c	5937	b4dc100b030f55689fdcde10014dfb3f0a57f6d1f3362ef2e5f0f
external/optiga-trust-m/external/mbedtls/configs/config-ccm-psk-dtls1_2.h	3142	c4f842bf5525957ea482df6003c742766fde695574a27c221d51f
external/optiga-trust-m/external/mbedtls/configs/config-ccm-psk-tls1_2.h	2725	8e6f83df978d5fe30a8c1872ce957bb4f448d10f6d8886c8a7d4e
external/optiga-trust-m/external/mbedtls/configs/config-mini-tls1_1.h	1978	9bfb7c3305fce663a9f1109ae39e05ce7375bb1d694c4be212e63
external/optiga-trust-m/external/mbedtls/configs/config-no-entropy.h	2016	f4641d2141b18ab6ab000079d4ac4e7d1b5b9c4205d54737efbea
external/optiga-trust-m/external/mbedtls/configs/config-suite-b.h	3418	1804e1026e204cc0b913577fa4253d14fbf651473f3ea31c5f609
external/optiga-trust-m/external/mbedtls/configs/config-symmetric-only.h	2179	436d2d57c0204a0abffab65b7713baf8f79bd339f106c54d83cc8
external/optiga-trust-m/external/mbedtls/configs/config-thread.h	2221	e96fc915b82f3553e00af693c64712db4e13e4e8cf23acee4258b

external/optiga-trust-m/external/mbedtls/docs/conf.py	1179	ddb943e9ccdb4892b01b9b798519604def81196e1dd94097495d9
external/optiga-trust-m/external/mbedtls/docs/requirements.txt	1344	dce4f3d684dc1e1270a8b811bbff57e50f6f293bfa448e2d725b2
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_encdec.h	2728	6c812f9c228d9cdc6b37748beeeabded85e85857f8c8f00d4c1d4
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_hashing.h	952	55fe54656b1ad1ab2d4a12e6ec41d5019475a23f4b7df4b7c9fbd
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_mainpage.h	497	4f948452f8539361ab323bf050f8981cf27dd1e0a70bd235094a9
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_rng.h	1174	47237b1eeb5093bdc06ff0373b3170efef8b860d910429894fcc5
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_ssltls.h	1425	3023fe21b3b398afd527ff00a38d142e883b1256bd510e140a364
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_tcpip.h	1143	21355833cbc6d797b675fbe968f4a86066dcd9dbbeaa3684822be
external/optiga-trust-m/external/mbedtls/doxygen/input/doc_x509.h	1165	ade5c11ce2be981f62d75d737a5327da1ad7a7742ecbf702c4902
external/optiga-trust-m/external/mbedtls/include/mbedtls/aes.h	29146	cdadaa287185b4ed689b8190c3e82f2e2bdb20bdaeb1afa184979
external/optiga-trust-m/external/mbedtls/include/mbedtls/aesni.h	5771	eafbf804daa73bd2916bd87f6fe4744d61a9260161f03db7fdb6b
external/optiga-trust-m/external/mbedtls/include/mbedtls/arc4.h	3514	d66d70b8dfc0b53452c7617fe07bff4f57c8f9f1994924361443f
external/optiga-trust-m/external/mbedtls/include/mbedtls/aria.h	15462	20664462c7aa9d3607fa5b517ed0acc2cddf9b73ca1a7d25c5ece
external/optiga-trust-m/external/mbedtls/include/mbedtls/asn1.h	25621	833a64871d59694686e23bd85e655c0bcbb004f0775bb5b91a906
external/optiga-trust-m/external/mbedtls/include/mbedtls/asn1write.h	15838	9cf3e48c2ca600d45360df23c97f66074661df52bfeecb635494a
external/optiga-trust-m/external/mbedtls/include/mbedtls/base64.h	2685	dc7ef7c15b67e935cd01ceee229aa45ef0381c4e16999b9961c4f
external/optiga-trust-m/external/mbedtls/include/mbedtls/bignum.h	45717	b756e8734133affa7c6eee0012e1d7ea09f991385327269add333
external/optiga-trust-m/external/mbedtls/include/mbedtls/blowfish.h	12081	d21f1b322d233db8cddc2c486a0d075be4adc106a477bff2de42d
external/optiga-trust-m/external/mbedtls/include/mbedtls/bn_mul.h	39354	268324971674f7187be79de77d1e711ce1df112f328ab104d3eec
external/optiga-trust-m/external/mbedtls/include/mbedtls/camellia.h	13657	367b12fc0a88e140f33c9048c4e2fa2d80b8b6fa2628a6e83dc1d

external/optiga-trust-m/external/mbedtls/include/mbedtls/ccm.h	13561	b8f5d5163cfa884d038c53db38d5466d484080bd76728a1b1656d
external/optiga-trust-m/external/mbedtls/include/mbedtls/certs.h	8777	f3750cf0092afc527c9cf18cdf0f3db63a6f37e1031416b22862c
external/optiga-trust-m/external/mbedtls/include/mbedtls/chacha20.h	8152	c30b39f660fed23d810072271da7b398686b52af01fe9f34a6102
external/optiga-trust-m/external/mbedtls/include/mbedtls/chachapoly.h	15531	8da9cde93f46936690b869e7810531e610b37b514ac92ca29300a
external/optiga-trust-m/external/mbedtls/include/mbedtls/check_config.h	38401	7896c05cd6f28fda00548e6dc2ff70434e6db8f660dc70f6117a8
external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher.h	53820	58fcbbcc35638a6ad1a71232f406441f1530ab15c257046436655
external/optiga-trust-m/external/mbedtls/include/mbedtls/cipher_internal.h	4317	a2f3b2fb6347c613ba9eff1b5e16e2ccfe0b921874051a38430e4
external/optiga-trust-m/external/mbedtls/include/mbedtls/cmac.h	9798	a20ad14a7b8bf5c8277461d60725103b4fb3fef9ce51f0c887625
external/optiga-trust-m/external/mbedtls/include/mbedtls/config_psa.h	26677	52c260c3883770d2fff61ec760f936d3b20e584afbeb85e0c133e
external/optiga-trust-m/external/mbedtls/include/mbedtls/constant_time.h	954	0e4eb0021fe1f271edb21a2fc80bad0c3e25bc48000b789e5986b
external/optiga-trust-m/external/mbedtls/include/mbedtls/ctr_drbg.h	25108	39dd6377a415897d0e26e81ca6bf1f70f1de5898d5528a73a94f9
external/optiga-trust-m/external/mbedtls/include/mbedtls/entropy_poll.h	2692	82def6b820fa4127de1274e04c7e5deb344187acbc3d2bc7efbb5
external/optiga-trust-m/external/mbedtls/include/mbedtls/havege.h	1312	13a8f93d9b584e280e0a7dbc776c4efba7353e97e33ba5214bc04
external/optiga-trust-m/external/mbedtls/include/mbedtls/net.h	619	02c9a6ed25f9361bd5b3ee2df5a82a85848d45b1e603a396c5f7e
external/optiga-trust-m/external/mbedtls/scripts/mbedtls_dev/__init__.py	146	b4c5c80120c7707ab1c6e27adc8b6dbab55388d5451547422263e
src/rust/bitbox-secp256k1/depend/secp256k1-zkp/src/int128_impl.h	376	5683c02dfc495639da259d80624711c1c449bbaee6186729e399e